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Feasibility Study Report

Client Name: Asymchem
Client Project Name: PDF Building Extension
Project Location: Sandwich, Kent
Scitech-Ekium Project Number: 300291

A1	22/05/2026	Feasibility Report Issue	ME/NA/JB/GJ/CG/AB/AW/DS/GS	MC	MC
ISSUE	**DATE**	**DESCRIPTION**	**BY**	**CHECKED**	**APPROVED**

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1.0 Introduction

This report has been prepared by Scitech-Ekium, under contract to Asymchem, to provide a feasibility study on the proposed PDF building extension to the existing B902 building in Sandwich on the old Pfizer campus.

Asymchem has recently taken operation of the PDF building (Building 902) at Discovery Park in Sandwich, Kent.

This report was produced to identify the possible options for a new 600m², four-story extension to the rear/east of the existing building, designed to accommodate 10 new reactor vessels, three new filter dryer workshops, a new material handling area (including milling and packaging), a hydrogenation suite, and a continuous manufacturing equipment suite, along with all required HVAC and utility spaces.

The existing B902 PDF building was originally designed to allow for an extension to the rear/east of the building. To allow for a new extension, knock-out panels have been installed within the existing facade / structure at the east end of the building, adjacent to the existing fire escape staircase. These existing pre-formed knock-out panels will need to be surveyed/investigated to evaluate their suitability for connecting to the new extension. The existing knock-out panels may not be at the correct floor heights/levels to match the required floor heights of the new extension. Therefore, provisions may be needed to be incorporated (additional stairs or elevators) to allow for differences within floor levels between the existing building and the new extension.

Given that the existing PDF building functions as a development facility and pilot plant for new drug manufacturing, the existing processing requirements demands flexibility. The new extension/facility must be designed to safely handle a range of hazardous materials, from highly flammable solvents and gases to highly toxic active pharmaceutical powders.

The study was performed to meet the requirements as detailed with the '*700181 - Scitech-EKIUM Asymchem PDF Extension Feasibility Study Proposal (A1)*' document, issued to Asymchem.

1.1 Report Caveat

To enable the Scitech-Ekium team to create this report, several requests for existing data, regarding the operation of the existing B902 has been made to Asymchem.

Unfortunately, some of the data that has been requested is not available.

To enable the Scitech-Ekium team to finalise this report, many assumptions on flow rates, quantities and peak load have been made.

To enable the Scitech-Ekium team to size the equipment needed for the new equipment and processes, data gathered from the existing building for flows, peak loads and quantities has been used and scaled down to estimate the new building extension requirements

1.2 Description of the Development

Scitech-Ekium was approached to investigate to feasibility of adding a new 600m², four-story extension to the rear/east of the existing B902 PDF building. The new extension will need to accommodate the following:

- 3 off filter dryer workshops
- A materials handling area

- 7 off glass lined reactors
- 3 off Hastelloy® reactors
- A continuous equipment area
- A hydrogenation suite
- Associated service rooms / voids
- New HVAC / AHU equipment
- New utilities for servicing the extension
- Upgraded electrical systems
- Associated storage areas (gas cylinders etc...)
- Upgraded / increase in welfare facilities (WCs, changing rooms etc...)
- Addition of new means of escape (fire escape staircase)

1.3 Feasibility Report Considerations

To enable the Scitech-Ekium team to create this feasibility report, the team will need to consider the following aspects:

- Ground conditions and civil works
- Flood defence systems (regarding the River Stour)
- Space planning (massing and stacking)
- Floor levels between the existing building and the new extension
- Utility capacity and condition assessment including power (inc. generator), thermal fluid, chillers, process gases, scrubbers, control system etc
- British Standards compliance
- Hazard review & reduction
- Constructability
- Coordination with their adjacent site and existing control system (Delta V).
- Pre-application planning report

1.4 Meetings Undertaken

To enable the Scitech-Ekium team to undertake this feasibility report, required to meet Asymchem requirements, the following meetings / data gathering sessions / workshops have been undertaken with Asymchem

09/03/2026 - Client Kick Off Meeting
16/03/2026 - 17/03/2026 - Site Visit
17/03/2026 - Site Utility Discussion
20/03/2026 - Client Progress Meeting #1
31/03/2026 - Client Progress Meeting #2
07/04/2026 - Client Progress Meeting #3
14/04/2026 - Client Progress Meeting #4

21/04/2026 - Client Progress Meeting #5
22/04/2026 - Process and Building Services Workshop
24/04/2026 - Initial Feasibility Risk Register Review
28/04/2026 - Client Progress Meeting #6
05/05/2026 - Client Progress Meeting #7
11/05/2026 - BREEAM Pre-Assessment Workshop
12/05/2026 - Client Progress Meeting #8
14/05/2026 - Electrical Workshop
19/05/2026 - Client Progress Meeting #9

2.0 Core Scope

2.1 RIBA Stage 1 – Feasibility Study

Generally, the scope of service will be in accordance with the RIBA Plan of Work for RIBA 1 - Feasibility.

The Core Tasks are therefore:

- Definition of Spatial Requirements and Quality Standards – To be achieved through the definition of an accepted development option that may proceed to RIBA 2, subject to Outline Business Case approval.
- Definition of Project Outcomes – To be achieved through the definition of outcomes achieved, when measured against Design Standards, to be confirmed by Asymchem.
- Definition of Sustainability Outcomes – To be achieved through the definition of outcomes achieved, when measured against Sustainability Targets, having regard to Asymchem Sustainability Strategy and to be recommended by the Consultant.
- Definition of the Project Budget – To be achieved through the assessment of costs associated with Feasibility Studies.
- Undertake Feasibility Studies – To be achieved by defining and assessing each of the Project Options defined in the next section.
- Review Site Information and carry out Surveys – To be achieved through the review of O+M and other information provided by Asymchem and by carrying out surveys defined in the Scope.
- Prepare the Project Programme – To be achieved through the scheduling of each of the options explored in the Feasibility Studies.
- Prepare the Project Execution Plan – To be achieved through the preparation of an implementation plan for a preferred development option, describing the tasks and activities necessary at RIBA 2, to support the delivery of the project to schedule.
- Establish a Procurement Strategy – To be achieved through dialogue with the Client's procurement team, to ensure alignment between the Client's preferred procurement strategy, the Project Budget and the Project Programme.

2.2 RIBA Stage 1 – Feasibility Studies

Hydrogenation Building - Options

During the feasibility study, the following options involving the existing Hydrogenation Building, to the rear of the existing PDF building, were investigated:

Option 1: The existing Hydrogenation Building will be demolished/relocated.

- If the function of the Hydrogenation building is required, a suitable location for a new Hydrogenation Building could be investigated under a new Feasibility study
- If the existing Hydrogenation Building can be removed, this will enable the footprint of the proposed new extension to align with the existing building, and maximise the floor area of the new extension

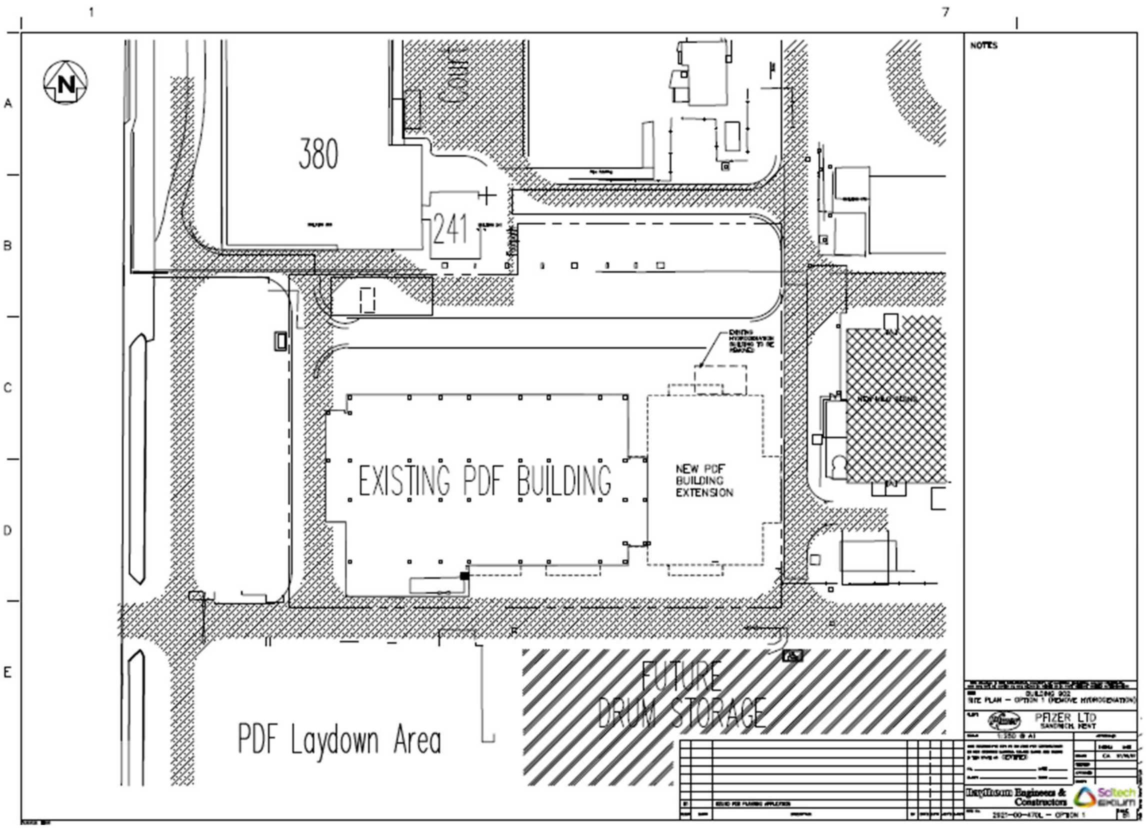


Fig 2.2.1 - Existing Hydrogenation removed/relocated

Option 2: Existing Hydrogenation Building to remain – **New Extension to be offset**

If the existing Hydrogenation Building is required and cannot be relocated to an alternative position on the Asymchem site, then there are knock-on effects to the proposed PDF extension building:

The south facade of the existing Hydrogenation Building aligns with the north facade of the PDF building, which will affect the following:

- Due to the hazardous nature of the Hydrogenation building, the new PDF building will require a set-back from the Hydrogenation building.
- To enable constructability of the new PDF extension, the new PDF building will require a set-back from the Hydrogenation building.

- The north facade of the new PDF Extension may need to be robust construction (i.e. thick reinforced concrete), due to the hazardous operations taking place within the Hydrogenation building. This could have a cost implication.

To enable the above requirements, the entire new proposed extension can be moved south to allow for the required offset – see sketch below Fig 2.2.1.2 - Existing Hydrogenation to remain – Offset Extension

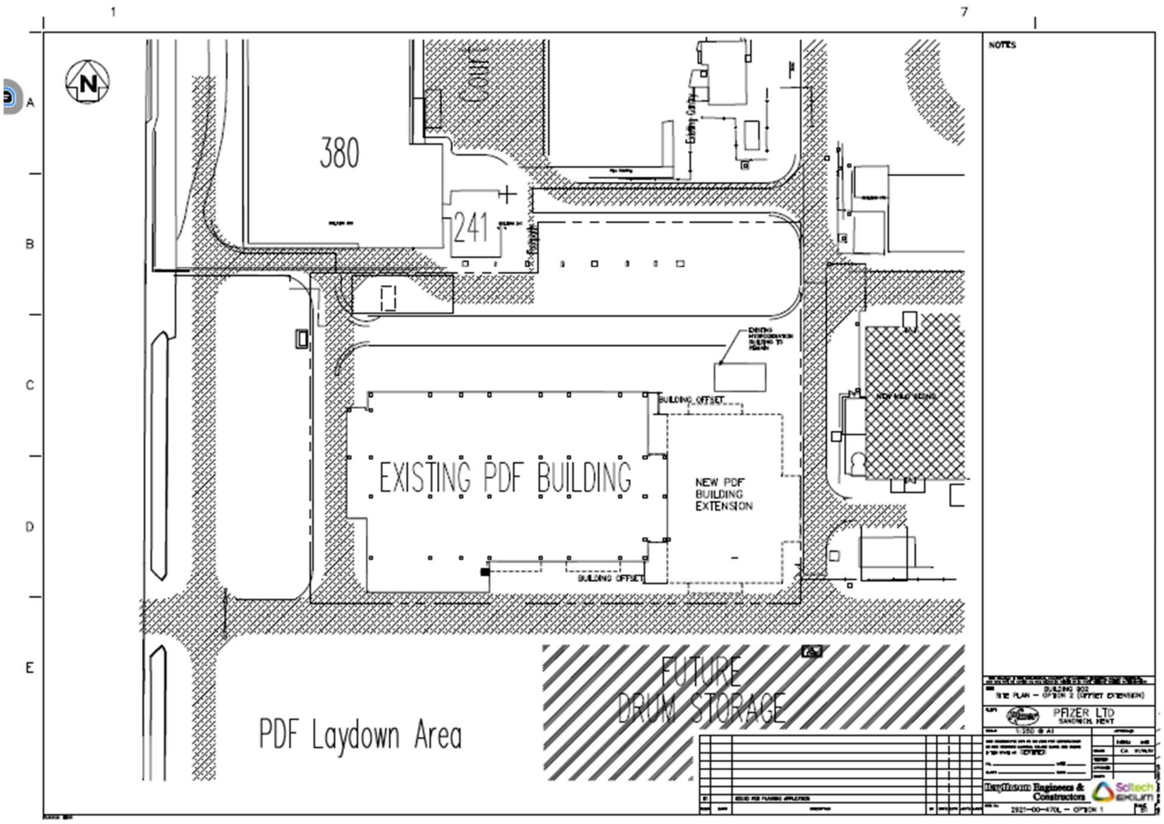


Fig 2.2.2 - Existing Hydrogenation to remain – Offset Extension

If the proposed new extension is offset there are 2No options for the internal layout of the extension building:

Option 2A – The new central circulation corridor with the new extension building will align with the existing central corridor of the PDF building,

- This aligned corridor will produce an uneven split along the building, with the areas north of the corridor being smaller than the areas south of the corridor.

Option 2B – The new central corridor is in the centre of the building (as original client sketch).

- This will introduce a step/kink between the existing central corridor and the new central corridor
- Moving equipment between the new and existing building will now involve manoeuvring around corners.

Option 3: Existing Hydrogenation Building to remain – **New Extension to be reduced in size**

If the existing Hydrogenation Building is required and cannot be relocated to an alternative position on the Asymchem site, then there are knock-on effects to the proposed PDF extension building:
The south facade of the existing Hydrogenation Building aligns with the north facade of the PDF building, which will affect the following:

- Due to the hazardous nature of the Hydrogenation building, the new PDF building will require a set-back from the Hydrogenation building.
- To enable constructability of the new PDF extension, the new PDF building will require a set-back from the Hydrogenation building.
- The north facade of the new PDF Extension may need to be robust construction (i.e. thick reinforced concrete), due to the hazardous operations taking place within the Hydrogenation building. This could have a cost implication

- To enable the above requirements, the new proposed extension floorplate can be reduced to allow for the required offset – see sketch below Fig 2.2.1.3 - Existing Hydrogenation to remain – Reduced Floorplate

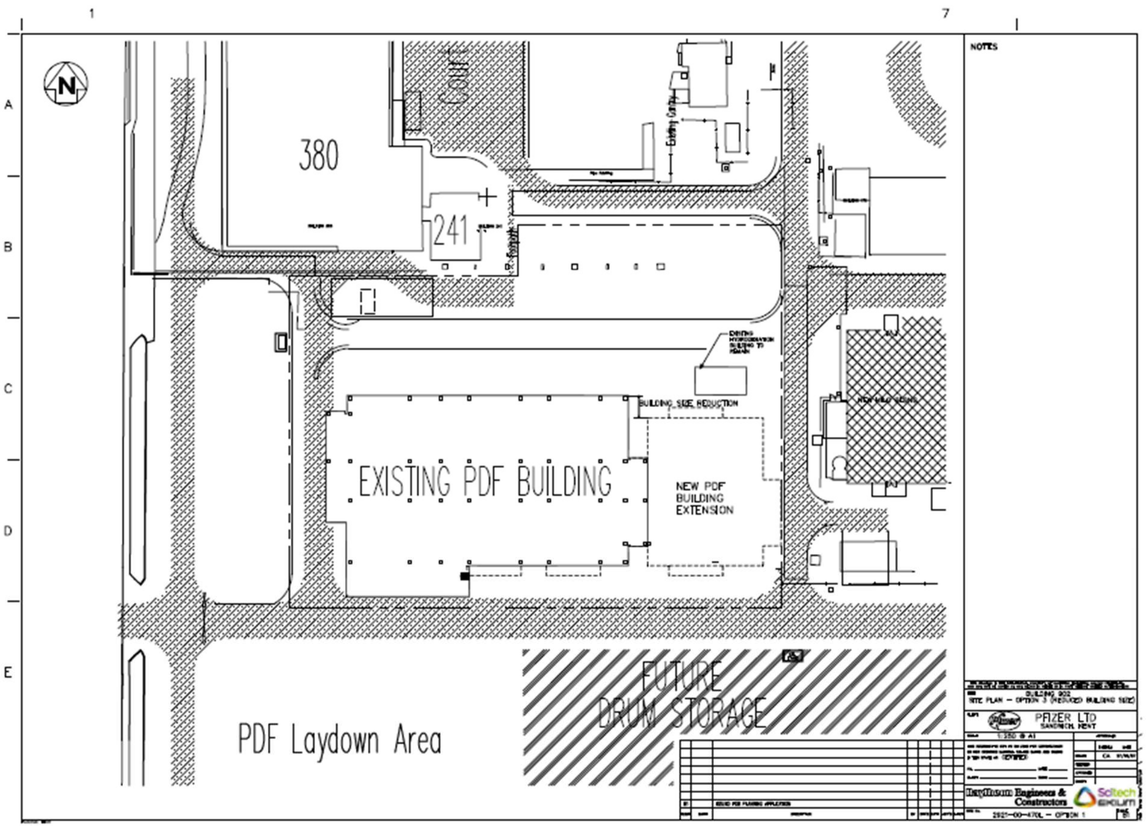


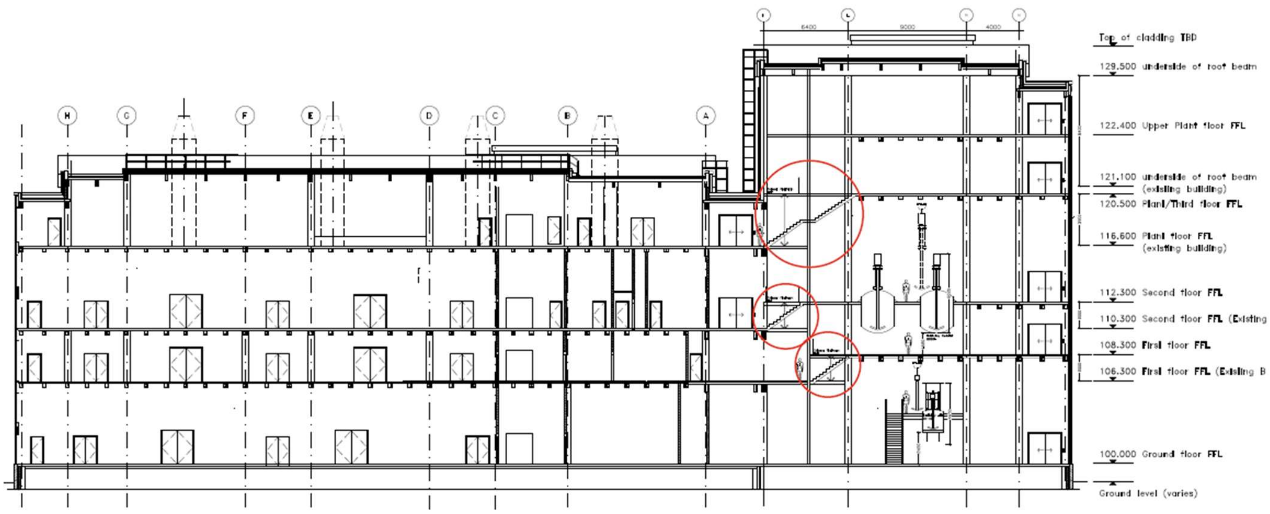
Fig 2.2.3 - Existing Hydrogenation to remain – Reduced Floorplate

- The new central circulation corridor with the new extension building will align with the existing central corridor of the PDF building,
- This aligned corridor will produce an uneven split along the building, with the areas north of the corridor being smaller than the areas south of the corridor.
- The reduced floorplate may not allow for all the required equipment to be located within the new extension.
- Equipment requirements will need to be investigated if this option is chosen.

Building Section – Ground Floor Ceiling Height
In addition to the options regarding the existing Hydrogenation Building, an option was investigated regarding the floor-to-floor height on the ground floor to enable the installation and operation of the 2No Hastelloy ADF DN1400s and the 1No Hastelloy ADF DN1200.

To enable the operation of the Hastelloy units, an increase in the floor-to-floor height is needed.

An option was investigated, where the ground floor ceiling height was raised by 2000mm – See image below



TYPICAL LONGITUDINAL SECTION THROUGH CENTRAL CORRIDOR

Fig 2.2.4 - Building Section with increased Ground Floor ceiling height

The knock-on effect of raising the first-floor slab is that the first floor, second floor, and third floor slabs do not align with the floor slabs in the existing building.

To access the new extension, staircases and scissor platform lifts would need to be introduced at the junction between the existing building and the new extension at 1st, 2nd & 3rd floor levels.

The above section sketch was presented to Asymchem on the 11/05/2026, explaining the effects of raising the first-floor slab, and the option was rejected.

To enable seamless connection between the existing building and the new extension it was agreed that the floor slabs need to align between the existing building and the proposed extension.

To enable the operation of the 3No Hastelloy units on the ground floor, options will be investigated with the RIBA Stage 2 Concept design phase.

Two possible options to enable the operation of the Hastelloy units are:
- Locate the Hastelloy units on a floor with a higher ceiling height
- Lower the floor slab on the ground floor, locally to where the Hastelloy units are positioned – see sketch below

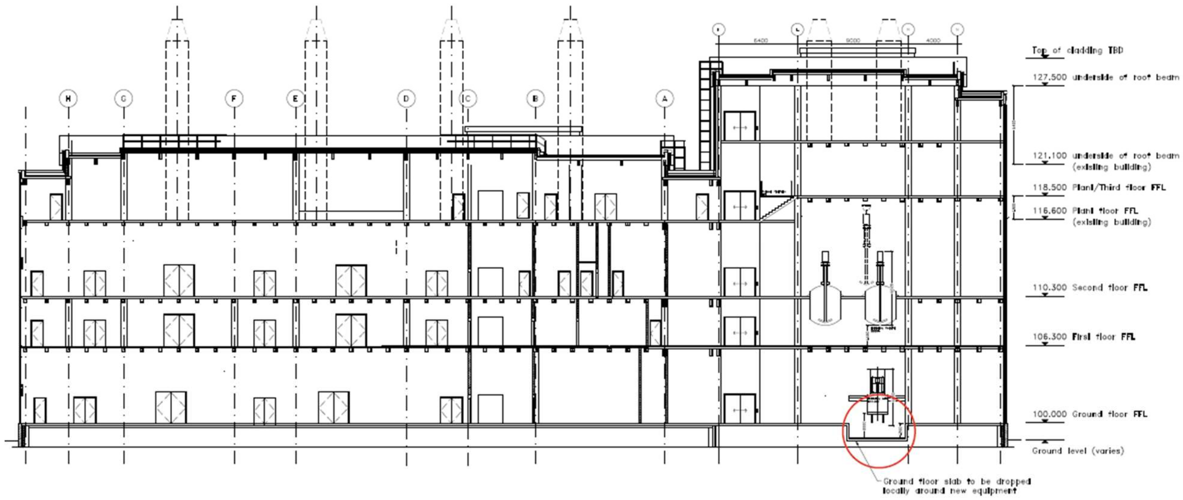


Fig 2.2.5 - Building section with lowered floor slab, locally to Hastelloy equipment

2.3 Option Appraisal

Each of the Feasibility Study Options will be assessed against the following evaluation criteria. These will include

- **Milestone achievability**: Can the new facility be operational by Q2 2028
- **Impact on site operations**: Can the existing B902 continue to operate while the new works are undertaken
- **Project Complexity**: How complex is the project
- **Construction Safety:** Health & Safety risk
- **Requirements compliance**: Identify where the options do not meet the client's requirements

- Below is the evaluation matrix, showing the scores for each option:
  - See Appendix 1 for an enlarged version of the chart below

R001	Asymchem PDF Extension Project																								
Option	Description	Existing Hydrogenation Building to remain	New PDF Building Extension to maximise available area	Central circulation corridor to allign	North & South scientific areas above and below central corridor are symetrical	Milestone Achievability (score 1 to 10) weighting 25%. Score 1-5 is highly complex, lots of interfaces, enabling works etc. Score 6-10 is low complexity, straightforward.			Impact on site operations (score 1 to 10) weighting 20% Score 1-5: Major impact requiring building shutdown and impact to science Score 6-10: Low impact and manageable planned shutdowns			Project Complexity (score 1 to 10) weighting 20%. Score 1-5 is highly complex, lots of interfaces, enabling works etc. Score 6-10 is low complexity, straightforward.			Construction safety (score 1 to 10) weighting 15% 1-5 Low score - High H&S risk 6-10 High score - Low H&S risk			Requirements Compliance (score 1 to 10) weighting 20% Score 1-5: Major impact - User requirements Not met Score 6-10: Low impact - Users requirements met			Total Score (Out of 10) Higher Score - Better Solution	URS Compliant	Pros	Cons	Cost
						Score	Weight	Total	Score	Weight	Total	Score	Weight	Total	Score	Weight	Total	Score	Weight	Total					
Existing Hydrogenation Building - Removed/Relocated																									
1	1. Existing Hydrogenation Building to be removed / relocated 2. New feasibility study required if Hydrogenation Building operations are still required - Where can the Hydrogenation be relocated to? 3. New PDF Extension building meets client requirements - Equipment & central corridor allignment	No	Yes	Yes	Yes	8	25%	2.00	8	20%	1.60	8	20%	1.60	9	15%	1.35	8	20%	1.60	8.15	Yes	1. Removal of Hydrogenation Building will improve constructability on new extension building 2. New PDF extension meets client's requirements	1. Existing Hydrogenation to be removed or relocated 2. Cost for Hydrogenation removal / relocation to be considered	£££
Existing Hydrogenation Building to Remain - New Extension Building to be Offset - Central Corridor Allignment																									
2A	1. Existing Hydrogenation builing to remain 2. New PDF Extension building to be offset from existing Hydrogenation building - Constructability & safety 3. New & existing central circulation corridors to allighn 4. New extension building will not be symetrical - North of the central corridor will be smaller than the south area	Yes	Yes	Yes	No	8	25%	2.00	10	20%	2.00	5	20%	1.00	5	15%	0.75	4	20%	0.80	6.55	No	1. Existing Hydrogenation to remain - No cost required to remove building	1. Hydrogenation building will increase the cost of the façade of the new PDF extension 2. Keeping the Hydrogeantion building, moves the new PDF extension - to enable construction & for safety	£££
Existing Hydrogenation Building to Remain - New Extension Building to be Offset - Central Corridor Offset																									
2B	1. Existing Hydrogenation builing to remain 2. New PDF Extension building to be offset from existing Hydrogenation building - Constructability & safety 3. New & existing central circulation corridors do NOT allign 4. New extension building will be symetrical around central corridor	Yes	Yes	No	Yes	8	25%	2.00	10	20%	2.00	5	20%	1.00	5	15%	0.75	5	20%	1.00	6.75	No	1. Existing Hydrogenation to remain - No cost required to remove building	1. Hydrogenation building will increase the cost of the façade of the new PDF extension 2. Keeping the Hydrogeantion building, moves the new PDF extension - to enable construction & for safety	£££
Existing Hydrogenation Building to Remain - New Extension Building to be Reduced in Size																									
3	1. Existing Hydrogenation builing to remain 2. New PDF Extension building to be offset from existing Hydrogenation building - Constructability & safety 3. New & existing central circulation corridors do allign 4. New extension building will be smaller than required	Yes	No	Yes	No	8	25%	2.00	10	20%	2.00	5	20%	1.00	5	15%	0.75	3	20%	0.60	6.35	No	1. Existing Hydrogenation to remain - No cost required to remove building	1. Keeping the Hydrogenation building will increase the cost of the façade of the new PDF extension - Protection agaist blasts/explosions from the Hydrogenation building 2. Hydrogeantion building reduces the size of the new PDF extension	££

Fig. 2.3.1 - Options Appraisal Matrix

Using the scoring from the Option Appraisal Matrix, all design information within this Feasibility Study is based on Option 1 – The removal / relocation of the existing Hydrogenation building

3.0 Statutory Authorities

3.1 Planning Consent

It is anticipated that planning consent will be required for this project.

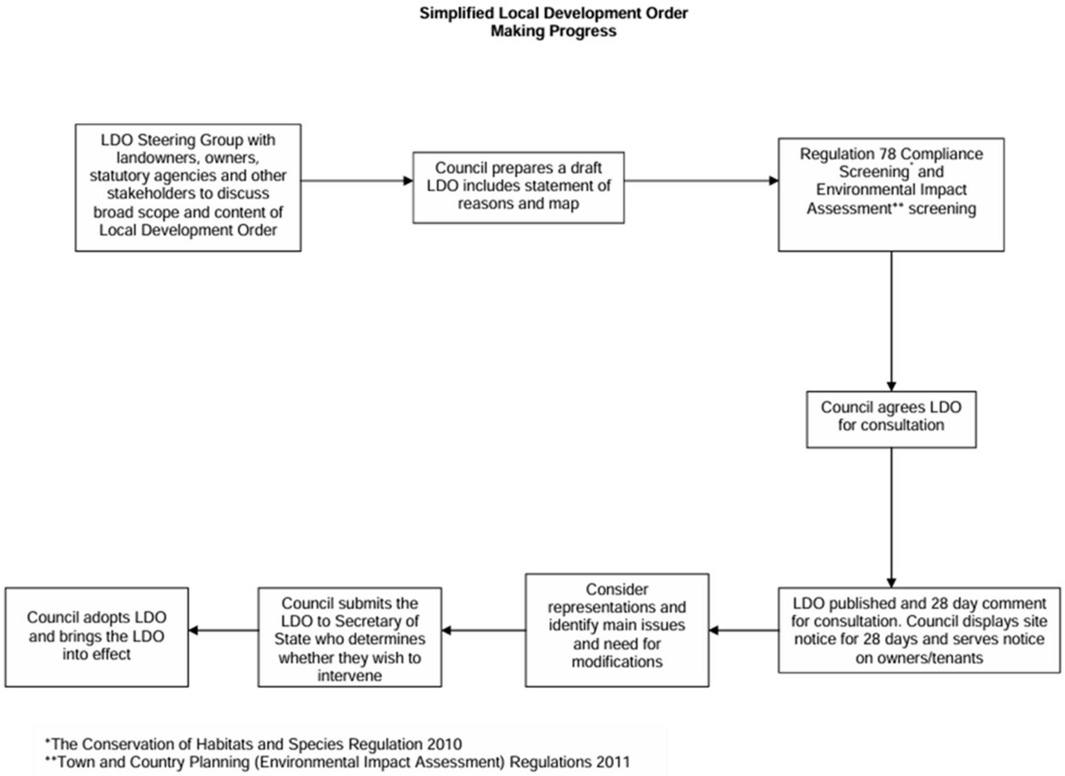
However, the old Pfizer campus in Sandwich, Kent is designated as a Life Science Enterprise Zone (LSEZ), which simplifies the planning application process.

The following quote is from the Dover District Council web page:

> *Simplified planning Local Development Orders grant automatic planning permission for specified types of development within Enterprise Zones. Locating to Discovery Park enables you to potentially develop to accommodate the needs of your business, saving considerable time and money.*
>
> *(Discovery-Park-General-Brochure.pdf)*

One of the ways that the planning system can be simplified within Enterprise Zones is through the creation of a Local Development Order (LDO). This LDO identifies that certain types of development subject to limitations and provisions may be carried out within the EZ without the need for planning permission. The allowances that have been included in this LDO have been specifically drafted to avoid triggering the need for an Environmental Impact Assessment, Habitat Regulations Assessment, Transport Impact Assessment etc. Proposals for development that are beyond the provisions of this LDO would require planning permission or approval from the local planning authority.

The process for agreeing this LDO is set out in Figure 3.1 below:



Planning consent will need to be assessed during the next stages and Scitech-Ekium will produce the required drawings and Design and Access Statement to support a pre-planning application submission.

It’s recommended to use a Planning Consultant to liaise with the Local Planning Authority as this would help expedite the application process.

3.2 BREEAM

As part of the feasibility study the project is targeting a BREEAM rating of ‘Very Good’. To facilitate this, Scitech-Ekium recommended that a pre-assessment be undertaken to ensure a thorough understanding of the credits required during the design process. It was also noted that certain credits can only be obtained at early design stages, providing Asymchem with the opportunity to capitalise on these.

Scitech-Ekium appointed SRE as the BREEAM Accredited Professional. The pre-assessment workshop took place on 11 May 2026, with the initial report issued on 12 May 2026. This document is to be reviewed and agreed by Asymchem prior to the next stage of design.

The report outlines a targeted 62.93% targeted credits with 20.47% potential, noting “Very Good” rating requires 55%.

The initial report has been appended to this document (Appendix 2) and represents the workshop outputs. It is not a live document and, as such, will not be subject to revision control.

3.3 Building Regulations & Building Safety Act

There were informal contacts with a Building Control Approved Inspector, and the initial feedback confirms that the new extension and the alterations to the existing building will require Building Regulation consent.

A fuller assessment of Building Safety Act will be undertaken in the design stage. There are not thought to be any material considerations beyond those needed for compliance with the Building Regulation.

3.4 Fire Strategy

The existing fire strategy within the existing building will need to be updated to suit the new extension.

3.5 Access to and Use of the Building: Approved Document M (ADM)

It has been confirmed with the Building Control Officer that the proposed areas in project scope are not required to comply with ADM.

It is noted in Part M that the approval of proposed works by a building control body or exemption does not necessarily indicate compliance with employer duties under the Equalities Act 2010

3.6 Unexploded Ordnance

Due to the location of Sandwich, it is recommended that an Unexploded Ordnance survey is undertaken, prior to the commencement of any ground works, as there may be unexploded World War II bombs within the boundary of the proposed development.

4.0 Architecture

4.1 Overview

This section of the Feasibility Study establishes the architectural and structural design philosophy to support the new PDF extension, and any improvement works required to the existing facilities, to ensure compliance with Building Regulations / British Standards / Approved Documents.

4.2 Design Assumptions

Scitech-Ekium has made some assumptions during the design process that will have to be clarified during the next stages. The principal assumptions are discussed below

4.2.1 Building Hazard Group

The design was developed assuming that the existing building is a Group H or High hazard Occupancy industrial building, therefore considered as high hazard. The addition of the new extension will not change that classification.

4.2.2 Fire Strategy

As part of the documentation issued to Scitech-Ekium, the existing Fire Strategy drawings have been provided. The existing Fire Strategy will be investigated and adapted to reflect the new extension to the building, during the next stage of design (RIBA Stage 2 – Concept).

Considerations will need to be made for the usage of hydrogen and other flammable materials within the building.

The new extension will require 2no means of escape. It may be possible to utilise the existing fire escape staircase at the east end of the existing B902 building, if Building Control are happy with the width of the staircase. A new fire escape staircase will be required at the east end of the new extension building.

Facade / roof blast-panels may be required within the hydrogenation suite and other high-risk environments, to allow for pressure relief in the event of an explosion. All blast panels should be connected to the building with restraints (i.e. chains) to prevent injuring personnel that may be outside of the building, when an explosion occurs. The restraints will prevent the panels from flying off the building.

Fire truck access around the building will need to be checked with the local fire service.

Scitech-Ekium recommends that a fire risk assessment should be carried out during the next stages to clarify any assumptions and understand if there are any risks to the existing facilities that require attention.

4.3 Foundations

It is assumed that the new extension will use a similar piled foundation solution, to match the piling used for the existing building. A suitable structural grid will be utilised to provide the stability and loadings required within the building, while providing a suitable grid to enable proposed functions and future proofing of the building.

4.4 Floor Slabs – Ground Floor

It is assumed that the new floor slab on the ground floor of the new extension will finish flush at the same level as the existing ground floor within the PDF building.

The new extension ground floor will be raised above grade level to provide flood defence against the River Stour, similar to the existing B902 building, and other buildings on the campus.

The raised floor slab will enable drainage pipework to run within the under-croft. This will allow the drainage pipe to be inspectable before entering the below ground drainage system

4.5 Floor Slabs – First & Second Floors

The First & Second floor slabs will be constructed at heights that suit the new required operations. The new floor slabs may not align with the existing floor slabs within the existing building. To enable connectivity between the new extension and the existing building there may be a requirement for steps/ramps/lifts.

The option shown within this report shows that the connections between the ground, first and second floors between the existing building and the proposed extension are flush with each other. The only floor where additional stairs and a platform lift have been added is on the third floor.

The addition of a new large goods-lift/elevator that can stop at all building levels (new extension and existing building) should be considered, to enable trolleys / carts / equipment to be transported between the new extension and the existing building.

4.6 Steel Frame

It is assumed that the main structure for the new extension will be a steel frame, to match the construction of the existing building.

Using a similar construction (foundations, floor slabs & steel frame), for the new extension, as the existing building will help reduce the movement and expansion/shrinkage between the two buildings.

4.7 Link Bridge

Where possible, the existing knock-out panels within the existing building will be utilised to connect from the existing building into the proposed new extension.

As stated above, new stairs and/or a new lift between the existing building and the proposed new extension will be required, as the slab-to-slab heights within the proposed extension do not align with the existing floor slabs within the existing building.

4.8 Building Facade

The Northeast facade of the proposed extension building, that is positioned adjacent to the existing external hydrogenation compound, may need special attention, due to the flammable and explosive properties of hydrogen.

Options for this facade if the hydrogenation building is preserved are:

- Offset the building line away from the hydrogenation compound – This will reduce the overall footprint of the building
- Increase the fireproofing properties of the facade
- Increase the structural properties of the facade (i.e. thicker reinforced concrete)

If the existing hydrogenation building is removed / relocated, the north facade of the proposed extension building can match the facade of the existing building.

4.9 Facilities Access

Access to existing facilities on the campus will need to be maintained, including, but not limited to:

- Gas cylinder deliveries
- Liquid Nitrogen deliveries

4.10 Welfare Facilities

With the increase in staff numbers, the existing welfare facilities (WCs & Changing Rooms) within the existing building will need to be investigated to ensure that the provisions meet the current regulations.

It is understood that the existing staff numbers within the existing PDF building is 45No staff.

The new extension will take staff numbers to 60No total (Increased by 15No)

The existing PDF Building has the following welfare facilities:
1No Ladies WC
1No Ladies shower
2No Gentleman's WCs
1No Gentleman's urinal
4No Gentleman's showers
1No Disabled WC
1No Disabled WC & Shower room

Additional toilets / showers / changing rooms will need be incorporated within the proposed design – either within the existing building, or within the new extension.

4.11 Contractor's Welfare Facilities

Provision will be required near / adjacent to the site, to enable the main contractor to provide the required welfare facilities / site compound for their staff.

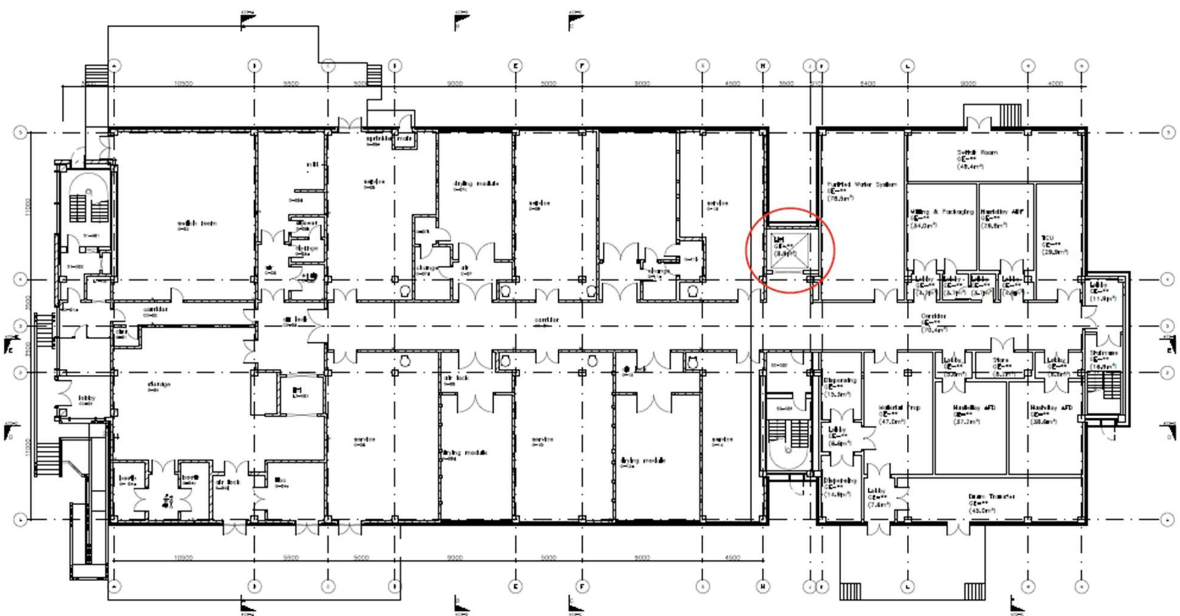
5.0 Architecture – Proposed Layouts

5.1 Overview

The architectural design for this project is based on the strategy documents that have been provided by Asymchem.

Using the scoring from the Option Appraisal Matrix (See section 2), all design information within this Feasibility Study is based on Option 1 – The removal / relocation of the existing Hydrogenation building

5.2 Proposed Ground Floor Layout

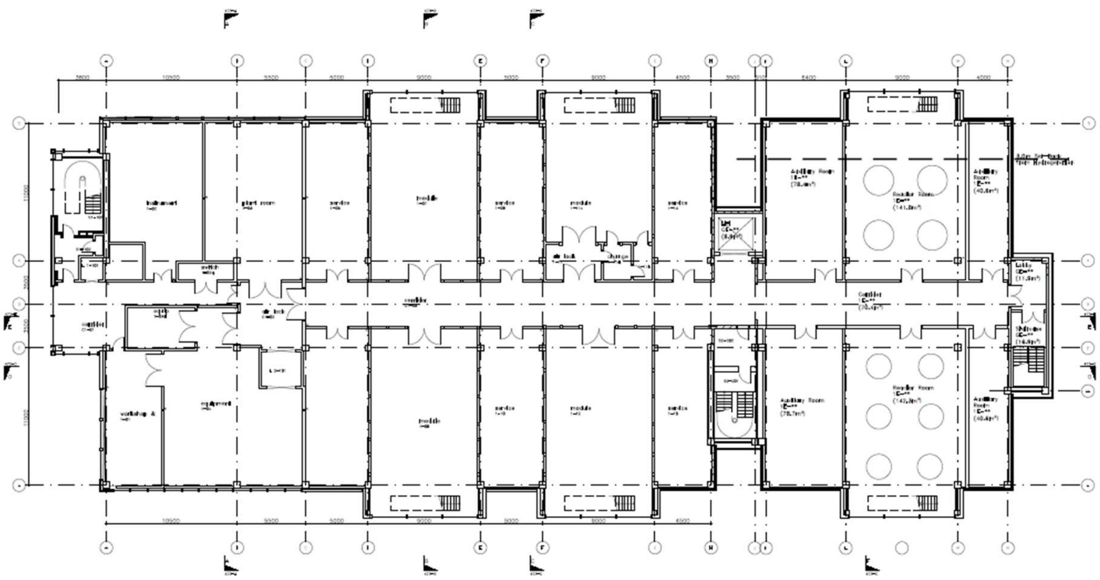


The only change that has been made to the floor plans, that are different from the documents that were provided from Asymchem is the location of the new main goods lift, as indicated by the red circle on the sketch above

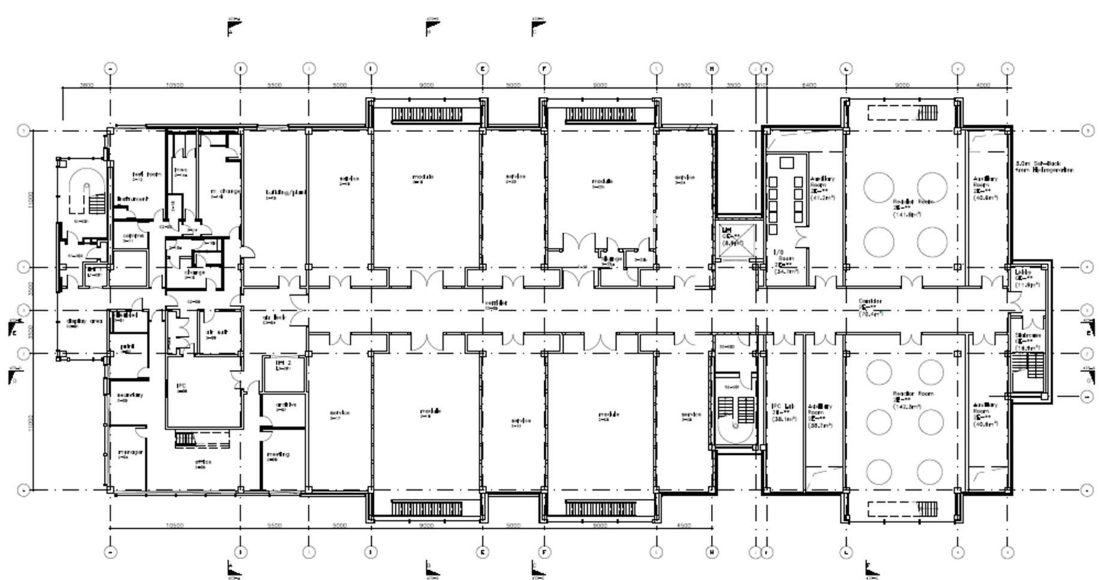
The goods lift has been removed from the floor plate of the new extension building, and has been relocated opposite the existing main stairs, within the gap between the existing building and the new extension.

The repositioning of the goods lift allows for addition space within the new extension, particularly in the I/O room and the plant room, where equipment positioning was getting tight.

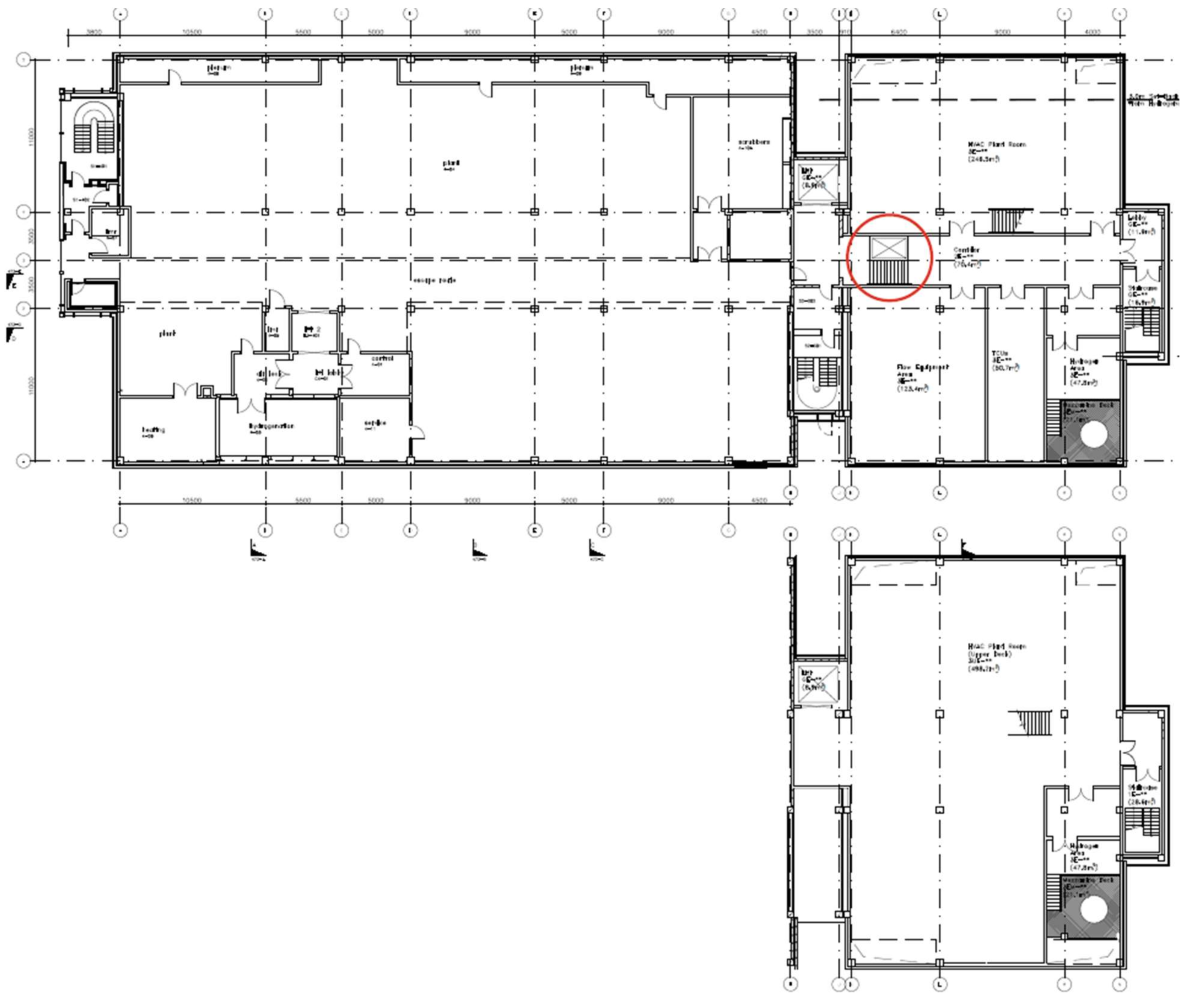
5.3 Proposed First Floor Layout



5.4 Proposed Second Floor Layout



5.5 Proposed Third Floor Layout

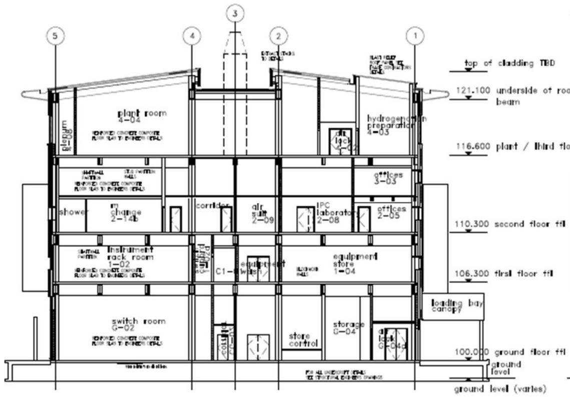


On the third floor, there has been an amendment to the floor plan, which is different from the initial sketch that was provided to Scitech-Ekium.

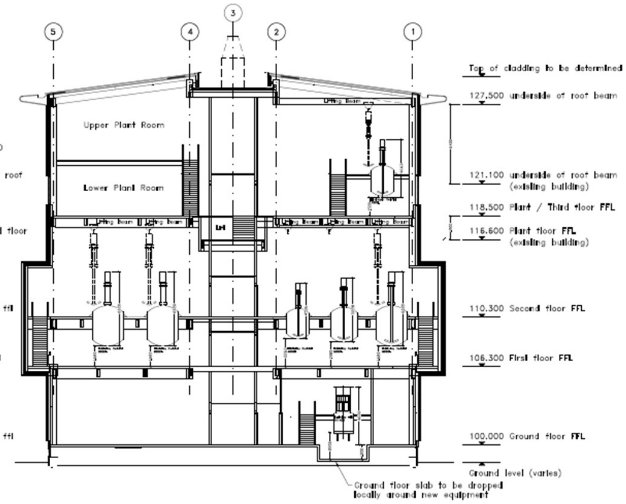
The red circle indicates a new staircase and platform lift, that is required to link the 3rd floor of the existing PDF building to the new facilities within the extension building.

The 3rd floor of the new extension building has been raised by approximately 2000mm, to allow for the removal of the stirrers from the reactors located on the second floor – See Building Section for more details of the proposed floor levels.
Due to the height increase on the 3rd floor, a new mezzanine plant room floor has been added – see above sketch. The new mezzanine floor plate will <u>not</u> go over the new hydrogenation room.

5.6 Proposed Cross Section

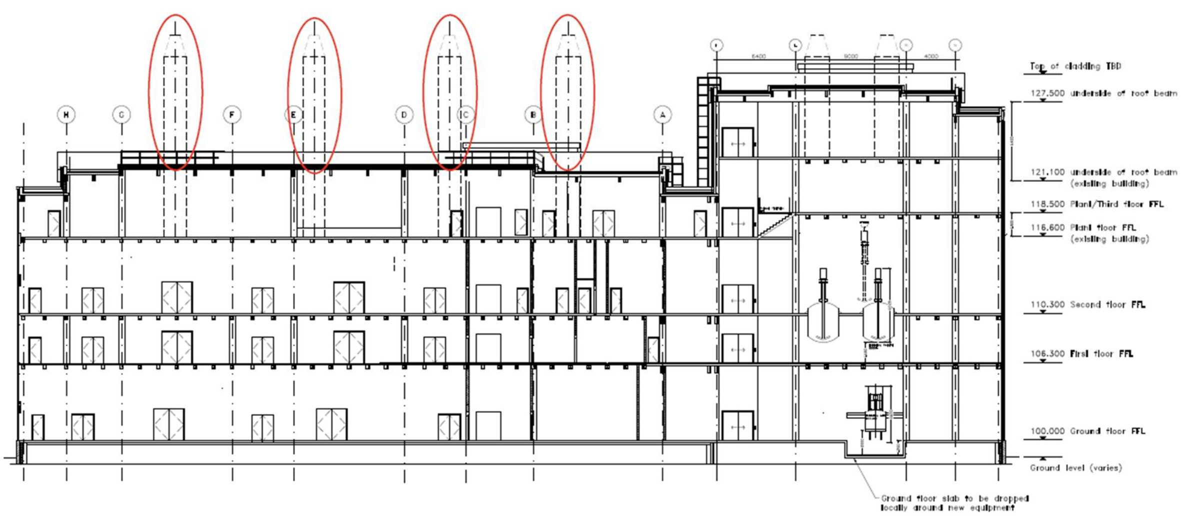


TYPICAL SECTION THROUGH OFFICE AREA
EXISTING BUILDING



TYPICAL SECTION THROUGH REACTOR SPACE
PROPOSED EXTENSION

5.7 Proposed Long Section

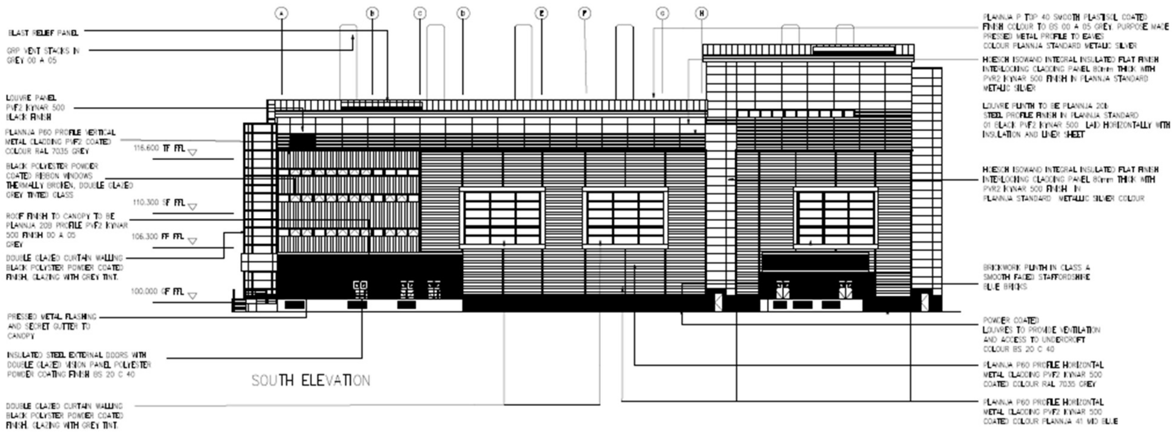


As the new extension will be higher than the existing building, due to the height needed on the 3rd floor, to enable the installation and operation of the equipment, the existing extract flues on the existing building will need to be extended, to ensure compliance with the Building Regulations. If an operative on the roof of the new extension is working/carrying out an inspection, they cannot be higher than the extract flues on the existing building.

During the next stage of design, investigations will be required on how the extended extract flues can be supported, due to their longer length.

Or a possible solution may be to re-route the existing flue stacks into the new extension and only have extract flues on the new extension roof.

5.8 Proposed Elevation



6.0 Architecture – Alterations to existing Building – Welfare Facilities

6.1 Toilets

With the increase in staff numbers, the existing welfare facilities (WCs & Changing Rooms) within the existing building will need to be increased to ensure that the provisions meet the current regulations.

It is understood that the existing staff numbers within the existing PDF building is 45No staff.

The new extension will take staff numbers to 60No total (Increased by 15No)

The existing PDF Building has the following welfare facilities:
1No Ladies WC
1No Ladies shower
2No Gentleman's WCs
1No Gentleman's urinal
4No Gentleman's showers
1No Disabled WC
1No Disabled WC & Shower room

With the increase in staff numbers, the toilet facilities required for the entire building are:

3No Ladies WC – An increase of 2No WCs
3No Ladies wash-hand-basin – An increase of 2No WHBs
2No Gentleman's WCs – No additional WCs required
2No Gentleman's urinal – An increase of 1No urinal
2No Gentleman's wash-hand-basin – No additional WHBs required

To allow for the increase in welfare facilities, it is advised that 2No 'Super-loos' be added to the existing building. A 'Super-loo' is a unisex / shared toilet that includes a WC and Wash-hand-basin, within a single room.

The addition of 2No super-loos will provide sufficient welfare facilities for the ladies, while also allowing for an unbalanced ratio of men to women (as pharmaceutical manufacturing facilities are generally operated by more men than women).

Within the existing PDF building there is an underutilised Disabled WC & Shower room. As this room already has hot-water & cold-water services and drainage, it is recommended to convert the disabled WC & shower room into 2No new separate unisex super-loos.

The existing Disabled WC will be preserved to keep the required disabled facilities within the building

6.2 Showers

Further studies will need to be carried out, during the concept design stage, to understand if an increase in shower facilities will be required.

6.3 Lockers

The additional 15No staff will require lockers for their outside clothes and belongings.

Additional storage of working overalls / clean room clothes will need to be provided.

6.4 Breakout Area

There is currently a Break-Out Room, which has tea & coffee making facilities, fridges to store food, and microwaves to heat pre-prepared food, plus tables and chairs.

Further studies will need to be carried out to see if additional break out facilities will be required to accommodate the additional 15No staff.

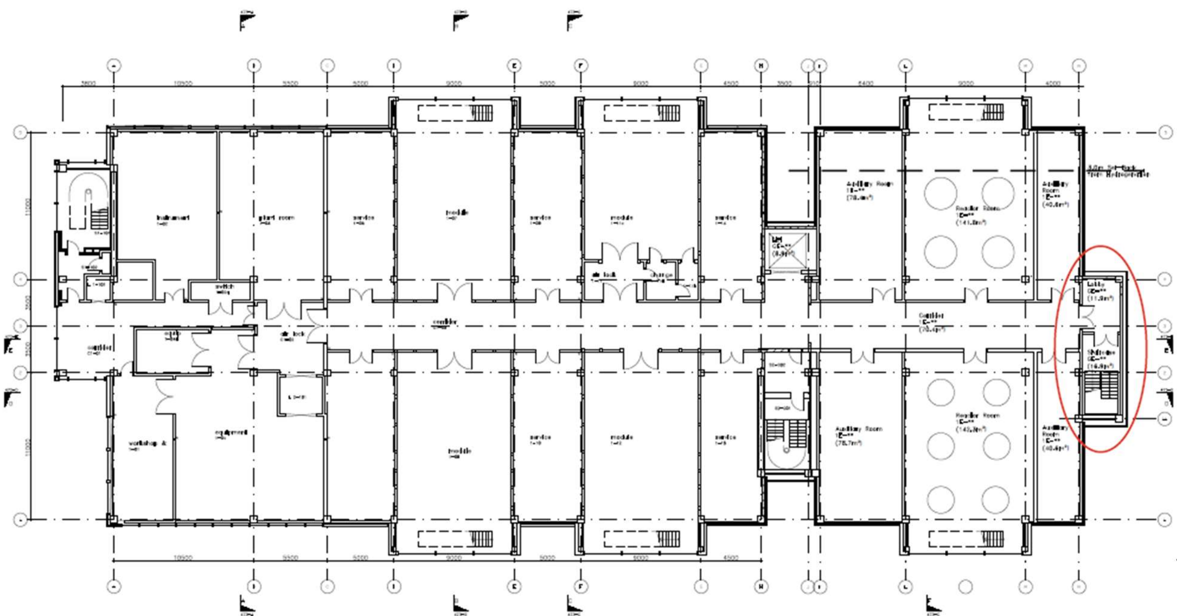
7.0 Fire Strategy

7.1 Overview

The existing fire strategy within the existing building will need to be updated to suit the new extension.

A ‘Fire Consultant’ will need to be appointed during the next stage of design (RIBA Stage 2 – Concept), to review the existing Fire Strategy, for the existing building, and update/revise the existing fire strategy to suit the new extension.

To help with the fire strategy, a new fire escape staircase has been added to the end of the new extension, to enable 2No means of escape out of the new extension.



8.0 Process Engineering

8.1 Overview

The Asymchem PDF Extension Facility adds 10 new reactors (R25–R34, range 2,500–8,000 L), a 2,500 L hydrogenation reactor (R35) and three new filter dryers (FD01–03) to the existing pilot plant. The largest reactor in the existing PDF facility is 2,500 L (R12, R17), so the extension represents a significant step-change in both reactor size and aggregate volume.

This feasibility report establishes whether existing utility infrastructure can accommodate the extension, identifies where new or upgraded systems are required, and documents the basis for recommendations to support the next stage of design.

Key process utilities required for the extension facility are:

- Hot HTF
- Cold HTF
- Hydrogen (H_2)
- Liquid Nitrogen (LN_2)
- Breathing air
- Compressed air (CA)
- Purified water (PW)
- Bulk solvents
    - Acetone
    - Methanol
- Solvent waste
- Aqueous waste

Other utilities are covered in other sections of this feasibility report.

8.2 Basis of Assessment

8.2.1 Reactor Volume Scale-up

Detailed utility usage data for the extension equipment is not yet available. A reactor volume scale-up factor has therefore been applied to existing pilot plant facility usage as a conservative feasibility-level estimator.

- **Existing total equipment volume: 25,065 L**
- **Extension total equipment volume: 63,690 L**
- **Scale-up ratio: 3.54 x**

Note – this 3.54 scale-up ratio takes into account both existing pilot plant and extension facility usage. The ratio is based on volume of reactors and filter dryers across both facilities.

This ratio has been applied to LN_2, bulk solvent and waste usages where data is absent. HTF, H_2 and PW demands have been derived from first-principles calculation using known reactor volumes, fill ratios and process conditions. PW tank and generation sizing was also based on the scale-up ratio.

8.2.2 Reference Documents

Data on existing pilot plant equipment is based on Pilot Plant Equipment List, Rev 15th March 2026. Other usage data has been obtained from RFIs and existing schematics.

For the extension facility, equipment sizing was obtained from the SW-902 Expansion Equipment List, 20th March 2026.

The following documents are to be read alongside this feasibility document:

- 300291-PR-CA-0001 Utility Capacity Assessment Calculations Rev A1
- 300291-PR-SC-0001 Equipment List Rev A1

8.2.3 Key Assumptions

- HTF fluid is Syltherm XLT. Hot HTF supply 130 °C; cold HTF supply −25 °C.
- Solvents modelled on worst-case basis: methanol heat capacity (2.8 kJ/kg·°C) and acetone density (785 kg/m³).
- Process temperature control 0.5–1 °C/min; reactor fill 100% for process (HTF process case), 25% for cleaning (HTF cleaning case). Diversity was agreed with Asymchem
- Simultaneous operation: 6 existing trains + 3 extension trains (2 × 5,000 L + 1 × 8,000 L) for HTF diversity case.
- PW rinse at 44% of reactor working volume; 30 min fill time. PW only used for rinsing reactors.
- Solvent/aqueous waste at 44% of reactor volume per cleaning cycle.
- Current operation for cold and hot HTF utility control in the existing facility will be maintained with the new extension facility.

8.3 Utility Assessments

For proposed locations of new equipment for the extension facility, refer to 300291-SCI-SI-XX-SK-PR-001 – New Plant Setting Out and proposed extension building layout.

8.3.1 Hot Heat Transfer Fluid (HTF)

HTF (Syltherm XLT) is supplied at 130 °C to reactor jackets for heating and temperature control during process operations and solvent cleaning cycles. The existing system comprises a 510-kW heater fed from the HTF tank, with a fixed 50 m³/hr supply pump. Maximum ΔT across the heater is ~24.8 °C, giving a minimum return temperature of ~105 °C.

Existing Capacity	510 kW heater; 50 m³/hr supply; 130 °C supply / 105 °C return.
New Demand – Processing	Worst-case simultaneous load 1,077 kW based on 6 existing trains + 3 extension trains (2× 5,000 L + 1× 8,000 L) operating at 1 °C/min temperature control. Utilisation 211%.
New Demand – Cleaning	Simultaneous load 269 kW (reactor at 25% fill). Utilisation 53%. Within existing capacity.
Recommendation	A larger dedicated 900 kW hot HTF heater (utilising site steam) and dedicated hot HTF loop for the extension is required to meet process heating duty. Existing 510 kW unit to be retained for the existing facility only. Final sizing to be confirmed once extension reactor utility data is available. Refer to 300291-

	SCI-SI-XX-SK-PR-001 for proposed location for new hot HTF heater and circulation pumps for extension facility HTF loop.

8.3.2 Cold Heat Transfer Fluid (HTF)

Cold HTF (Syltherm XLT) is supplied at −25 °C via a 780-kW ammonia chiller, with a 100 m³/hr supply pump. Returned at −20 °C (ΔT 5 °C). Used for exothermic temperature control in process and cleaning cycles.

Existing Capacity	780 kW NH_3 chiller; 100 m³/hr supply; −25 °C supply / −20 °C return.
Demand – Process & Cleaning	Steady-state simultaneous load 538.5 kW (6 existing + 3 extension trains at 0.5 °C/min process, 1 °C/min cleaning). Utilisation 69%; nominal spare capacity 241 kW.
Crash Cooling Case	In this scenario, the inlet valve on the cold HTF supply line is fully opened to flood the reactor jacket with cold HTF. As no data was available on reaction exotherms or crash cooling durations, it is assumed that the current system has sufficient capacity for this with the new extension reactors. As no data is available on reaction exotherms or crash cooling durations, this scenario has not been quantified at feasibility stage. For the purposes of this assessment, the steady-state demand is assumed to govern; the crash cooling case is to be confirmed at the next stage once a detailed assessment of the exothermic reactions can be carried out.
Recommendation	Retain existing 780 kW chiller as basis of design. If crash cooling duty exceeds available spare, a refrigeration top-up (booster chiller or uprate) will be required. This will be confirmed at the next stage once more information on reaction exotherms and crash cooling durations is available. Note – this current HTF system achieves temperature control by combining -25°C with 130°C which is an energy intensive process. At the next stage of design, Asymchem may want to consider a higher temperature cooling fluid or standalone TCUs from a company such as Huber.

8.3.3 Hydrogen (Hydrogenation Suite)

Hydrogen is supplied to the 250L hydrogenation reactor in the existing facility via cylinders, with a 250 L reactor consuming 1 cylinder per batch at 3 batches/week. The extension introduces a 2,500L hydrogenation reactor (R36), scaled 10× on cylinder demand.

Existing Capacity	3 cylinders/week (250 L reactor, 1 cyl/batch × 3 batches/week).
Extension Demand	30 cylinders/week based on 10× scale-up (2,500 L reactor, 3 batches/week).
Proposed Supply	BOC Multi-Cylinder Packs (MCPs) — 15 cylinders / 108.15 m³ per MCP. 2 MCPs in use at peak + 1 spare on standby (3 MCPs total on site). Footprint ~4.87 m × 3.52 m × 2.21 m (H) including 500 mm separation and 400 mm manifold allowance. Will require forklift access
COMAH	H_2 storage ~27 kg across 3 MCPs (at 175 barg, 15 °C, Z = 1.1 — real gas behaviour). Well below the 5 te (tonnes equivalent) COMAH lower-tier threshold for H_2 at this location; a full site-wide inventory review is still recommended to confirm against combined dangerous substances.
Recommendation	Install MCP storage area to the required footprint; 3rd MCP may be held elsewhere and moved in at changeover if footprint is tight. Include fire/gas detection and segregation per DSEAR/ATEX. Refer to 300291-SCI-SI-XX-

	SK-PR-001 for proposed location for new H_2 MCPs. These will need forklift access to change over and install new MCPs. It is proposed that this storage area could also store cylinders for the existing store.

8.3.4 Liquid Nitrogen (LN_2)

LN_2 is supplied from a single existing storage tank serving cryogenic reactors, pressure-test nitrogen, inert blanketing and boil-off. An associated nitrogen supply package / evaporator delivers up to 150 Nm³/hr at 6 barg. Existing tank is refilled weekly at peak demand.

Existing Capacity	Working volume ~19.8 m³; gross volume ~21.6 m³ (1,900 mm D × 7,480 mm O/A). Topped up weekly at peak.
Extension Demand	Scaled working volume ~70 m³ / gross ~76 m³ (based on 3.54× reactor scale-up).
Option 1	Increase fill frequency of the existing tank to every 3–4 days (est. 100 deliveries per year). No capital investment but increases tanker traffic and logistics risk.
Option 2	Replace with a larger 35m³ tank. Concrete plinth in existing tank area will not handle new larger tank so new LN2 tank will need to be relocated elsewhere on site. New tank will require 6.5m x 10.5m concrete plinth for tank and associated equipment. New tank will also allow articulated lorries to be used for refilling the tank which will save ~£50,000/ year on LN_2 delivery costs.
Recommendation	Preferred approach is a new larger tank (Option 2) — removes dependency on tanker availability (articulated vs rigid lorry) and reduces operational risk. New concrete plinth of 6.5m x 10.5m will need to be installed on site. Refer to 300291-SCI-SI-XX-SK-PR-001 for proposed location for new LN_2 tank.

8.3.5 Breathing Air

Breathing air supplies RPE loose hoods to personnel in hazardous reactor rooms via a dedicated supply package and receiver (CMP01-GR1). Worst-case per-person demand is 200 L/min (per BCGA GN11 range of 170–200 L/min for loose hoods).

Existing Capacity	Supply package 100 Nm³/hr; 11 m³ receiver operating at ~9 barg (SV set 10 barg); downstream regulator 6.5 barg. Ref P&ID SAN-902-007-002-000-00099.
New Demand – Normal Operation	Up to 8 persons simultaneously supported at 200 L/min (12 Nm³/hr per person).
New Demand – Supply Outage	Receiver stored energy (ΔP driving force 2.5 barg) supports ~27.1 m³ usable air — equivalent to 4 persons for 30 min evacuation window.
Recommendation	Existing supply adequate provided peak hazardous-area occupancy across both facilities does not exceed 8. If breathing air demand during extension operations exceeds this, uprating of the supply package will be required. This was confirmed at the utilities review workshop — peak occupancy will not exceed 8, and the existing receiver is appropriately sized.

8.3.6 Compressed Air

Compressed air (instrument and plant air) is supplied to the existing facility from a 375 Nm³/hr @ 10 barg compressor package. 100 Nm³/hr is dedicated to breathing air, leaving 275 Nm³/hr available for plant and instrument air. The existing facility CA usage has been estimated from the Asymchem valve and instrument list, with 3 existing diaphragm pumps included as intermittent loads.

Existing Capacity	375 Nm³/hr @ 10 barg compressor package; 100 Nm³/hr dedicated to breathing air, leaving 275 Nm³/hr for plant and instrument air.
Demand – Existing Facility	Peak demand ~174 Nm³/hr based on 194 positioners (continuous bleed at 0.3 Nm³/hr each), purges and tools allowance, on/off valve actuator strokes (531 valves × 10 NL × 3 strokes/hr), and 3 diaphragm pumps. Utilisation 63%.
Demand – Combined Facility	Peak demand ~219 Nm³/hr applying a CA-specific scale-up of 1.5× to valve and positioner loads (reflecting fewer-but-larger equipment in the extension rather than the 3.54× volumetric ratio), with 2 diaphragm pumps running at the same time in the extension. Diversity factors of 0.5 (valves) and 0.7 (pump coincidence) applied. Utilisation 80%. Spare capacity 56 Nm³/hr (~20% margin).
Recommendation	Retain existing 375 Nm³/hr compressor package as basis of design — sufficient capacity to serve both existing and extension facilities at feasibility level, within typical pharma design margin. Consider a modest additional compressor (~100 Nm³/hr) at detailed design for operational redundancy and futureproofing. Refine CA estimate once the extension valve and instrument list is available, replacing the 1.5× scale-up with directly counted equipment data. Consideration for operating at duty/standby/assist.

8.3.7 Purified Water (PW)

PW is used for reactor rinses, typically 44% of reactor working volume. The existing PW generation, storage and distribution loop is not sized for combined existing + extension demand, and simply replacing the supply pump is not sufficient (existing loop diameter is the constraint).

Existing Capacity	Existing PW system insufficient for combined demand — loop not sized for both facilities.
Extension Demand	Worst-case coincidental rinse flow 14.1 m³/hr (2.2 + 4.4 + 7.0 m³/hr) for simultaneous rinse of three largest reactors over 30 min fill time.
Proposed New System (Extension)	Fully independent PW system for the extension facility comprising: ozonated PW generation package (4.0 m³/hr); 16 m³ storage tank with vent filter and ozone destruct unit; 25 m³/hr supply pump on a 2.5 in (65 mm) ASME-BPE distribution loop; UV de-ozonation point-of-use. Final loop velocity ~2.46 m/s at 25.2 m³/hr combined flow within loop. This also includes a heat exchanger for trim cooling (utilising cooling water). A new package chiller is required to serve this duty (see 8.3.8) as the existing cooling tower system has no spare capacity.
Recommendation	Install dedicated extension PW system as above. Maintains independence from the existing loop, avoids cross-facility demand conflict and minimises contamination/cross-contamination risk during changeover.

8.3.8 Cooling Water

General cooling within the existing facility is provided by Jaeggi T1T1E1 Type HTK 2.4/5.45 hybrid cooling towers (893 kW, 32 °C supply / 26 °C return) which serve existing heat exchanger duties and have no spare capacity for additional services. The extension introduces a new purified water (PW) distribution system requiring trim cooling to remove continuous heat gain from the PW recirculation pump and pipework, maintaining the PW setpoint temperature.

Existing Capacity	Existing Jaeggi cooling towers (893 kW, 32 °C / 26 °C) — fully allocated to existing facility duties and supply temperature exceeds PW setpoint. Not suitable for PW trim cooling.
Extension Demand	PW loop continuous heat gain ~10 kW (pump shaft work + insulated pipework gain). Trim cooling heat exchanger duty 12 kW (incl. 25% design margin). Chiller rated capacity 15 kW. CW supply 10 °C / return 15 °C; flow rate ~2.6 m^3/hr.
Proposed New System	New dedicated air-cooled package chiller (~15 kW) with integrated buffer vessel (50–200 L) and CW circulation pump, serving the PW trim cooling heat exchanger only. Located outdoors; ~30% propylene glycol fill on the CW side if pipework is exposed to freezing conditions.
Recommendation	Install dedicated extension cooling water chiller as above. Indicative budget figure ~£12–15k all-in for the chiller package. Confirm during detailed design whether the chiller duty should also accommodate periodic PW loop sanitisation cool-down (could push sizing to 30–50 kW), or whether sanitisation is handled by an alternative strategy. This is only relevant if the PW system design is not an ozonated system. Final installation location to be confirmed at concept stage.

8.3.9 Bulk Solvents (Acetone & Methanol)

Bulk solvents are stored in two existing tanks: BT01 methanol (60 m^3) and BT02 acetone (40 m^3). Current delivery frequency is confirmed by RFI as 4 acetone tankers/yr (~23.25 te each) and 6 methanol tankers/yr (~27.25 te each), based on 4–6 batches/week.

Existing Capacity	BT01 methanol 60 m^3 (delivery every 2 months); BT02 acetone 40 m^3 (delivery every 3 months).
Extension Demand	Scaled by 3.54×: acetone ~15 deliveries/yr; methanol ~22 deliveries/yr.
Assessment	Existing tanks have sufficient storage volume per delivery. Limitation is tanker sizing — increasing tank size does not reduce delivery frequency.
Recommendation	Retain existing BT01/BT02 and increase delivery frequency: methanol 3 deliveries every 2 months (1 for existing + 2 for extension) and acetone at equivalent uplift. Bulk solvent storage quantity is not changing so no further COMAH assessment is required.

8.3.10 Solvent Waste

Solvent waste is collected in ST01 (10 m^3 immiscible), ST02 (20 m^3 miscible) and ST03 (30 m^3) waste tanks. ST01 is transferred to ST03 every 3–4 days; ST02 transfers in alternate weeks; ST03 is emptied approximately biweekly.

Existing Capacity	ST01 10 m^3; ST02 20 m^3; ST03 30 m^3. ST01 transfer pump 15 m^3/hr @ 56 m TDH; ST03 transfer pump 25 m^3/hr.
Extension Demand	Combined solvent waste per cleaning cycle ~36.2 m^3 (existing 10.0 m^3 + extension 26.2 m^3), based on 44% of reactor volumes used for cleaning.
Assessment	ST03 (30 m^3) alone is insufficient to buffer a full combined cleaning cycle. A buffer volume of ~6 m^3 must be held across ST01/ST02 between ST03 discharges.
Recommendation	Retain existing ST01/ST02/ST03 with a scheduled reactor cleaning regime: ST03 discharged after each cleaning cycle (multiple times per week as required). Review ST01 effluent transfer pump capacity against worst-case

	extension discharge rates; a larger ST01 pump may be required. Coordinate with effluent testing/treatment turnaround to ensure ST03 is not overfilled when both facilities are concurrently undergoing solvent cleans.

8.3.11 Aqueous Waste (Trade Effluent)

Aqueous trade waste is generated by PW reactor rinses and collected in ET01/ET02 effluent tanks (each 4 m³) before transfer to TT01/TT02 trade effluent tanks (each 12 m³). Approximately 2 × 10 m³ is collected weekly from TT01/TT02. PW use averages 44% of reactor volume (20% baseline rinse, ~80% occasional acid/base treatment applied ~30% of the time).

Existing Capacity	ET01/02 4 m³ each (8.8 m³/hr @ 19.8 m TDH); TT01/02 12 m³ each (22 m³/hr @ 36.4 m TDH).
Extension Demand	Rinse of the largest extension reactor (8,000 L at 44%) produces 3.52 m³ — sufficient to fill ET01 alone. Coincidental cleaning of multiple extension reactors would overfill ET01.
Recommendation	Install dedicated extension aqueous effluent system to suit larger extension waste flowrate: • ET03 (10–12 m³) with dedicated transfer pumps • TT04/TT05 (35 m³ each) with dedicated discharge pumps sized for the larger extension waste flowrates Maintain existing emptying frequency (equivalent to ET01/TT01/TT02 cycle). Existing ET01/ET02 and TT01/TT02 to serve existing facility only. Refer to 300291-SCI-SI-XX-SK-PR-001 for proposed location for new aqueous effluent tank system. Note – currently, aqueous waste is collected via tanker and removed on site. In future, this may move to a pumped discharge system. A pumped discharge system would allow for increased flexibility in operations and effluent discharge during cleaning cycles.

8.4 Site Layout

The proposed design includes changes to the current site layout. This is based on disinvestment of some existing external equipment and installation of new equipment in those vacated areas as well as other areas on-site. For more details, refer to 300291-SCI-SI-XX-SK-PR-001 – New Plant Setting Out.

It should be noted that the North Road area (North of the existing effluent tanks) reserved for future Asymchem projects has been excluded as a suitable location for any new external equipment. Should this area be available, this would allow better use of available space and less congestion of new equipment around the existing and extension facility buildings.

8.4.1 Disinvestment

Loop Hydrogenator
The existing loop hydrogenation system is not currently in use. It is proposed to disinvest this system to allow installation of new external equipment in its place.

Liquid Nitrogen (LN_2) Tank
As per 8.3.4 the existing LN_2 tank is unable to supply sufficient N_2 to both the existing and extension facilities without more frequent tanker top-ups (every 3-4 days).

It is proposed to install a new larger LN_2 tank on-site which will serve both the existing and extension facilities. BOC have informed us that the existing tank slab is unsuitable for a larger tank to be installed there. There are also access issues to the existing tank location. BOC have stated that an articulated lorry will not be able to access the existing LN_2 tank and that rigid lorries will need to be used. Rigid lorries will add significantly more cost per delivery of liquid nitrogen and more deliveries compared to articulated lorries (~£50,000 per year).

A new tank should be installed at another location and the existing LN_2 tank disinvested once the new tank skid and pipework connections are in place. The disinvested LN_2 tank area can then be utilised for other equipment.

8.4.2 New External Equipment

New equipment that will need to be installed externally on site for the extension includes:

- 900 kW hot HTF heater and distribution pumps – to suit larger heat load required by extension. These will be installed near the existing HTF tank and pumps. This is to minimise pipe runs to the pumps and to localise the HTF distribution system.
- 3x Hydrogen Multi-Cylinder Packs (MCPs) – to serve Hydrogenation reactors in existing and extension facilities. 3x Hydrogen MCPs are required per week with a 3rd as a spare. It is proposed to install these by the extension building near the loading bays. This also allows for forklift access when replacing empty MCPs.
- Liquid nitrogen 35m^3 tank + skid – to serve both existing and extension facilities and suit larger demand for LN_2. This will require a 6.5m x 10.5m concrete plinth which will house the tank and associated skid equipment. This is to be installed near the existing 902 building by the access road.
- Scrubbers SC05/ SC06 – new alkaline gas/ acid gas scrubbers to serve extension facility
- 1x 8000L reactor relief catch-tank – to serve extension facility only. It is proposed to install this near the air compressor building.
  Note – the hydrogenator catch tank would be installed in the hydrogenation room on the third floor of the extension facility building.
- Aqueous effluent system (ET03/TT04/TT05 and associated pumps) to suit larger waste flowrate from extension facility. It is proposed to install this system in the existing LN_2 tank and loop hydrogenator area. This bunded effluent tank system will consist of:
  - 1x 10m^3 trade effluent tank ET03
  - 2x duty/standby effluent transfer pumps
  - 2x 35m^3 effluent tests tanks TT04/TT05
  - 2x duty/standby effluent transfer pumps
  - 1x sump pump

For proposed locations of the new external equipment, refer to 300291-SCI-SI-XX-SK-PR-001.

8.4.3 Utility Tie-ins

For some of the external utilities, existing piped systems will still be utilised as they have sufficient spare capacity. These include:

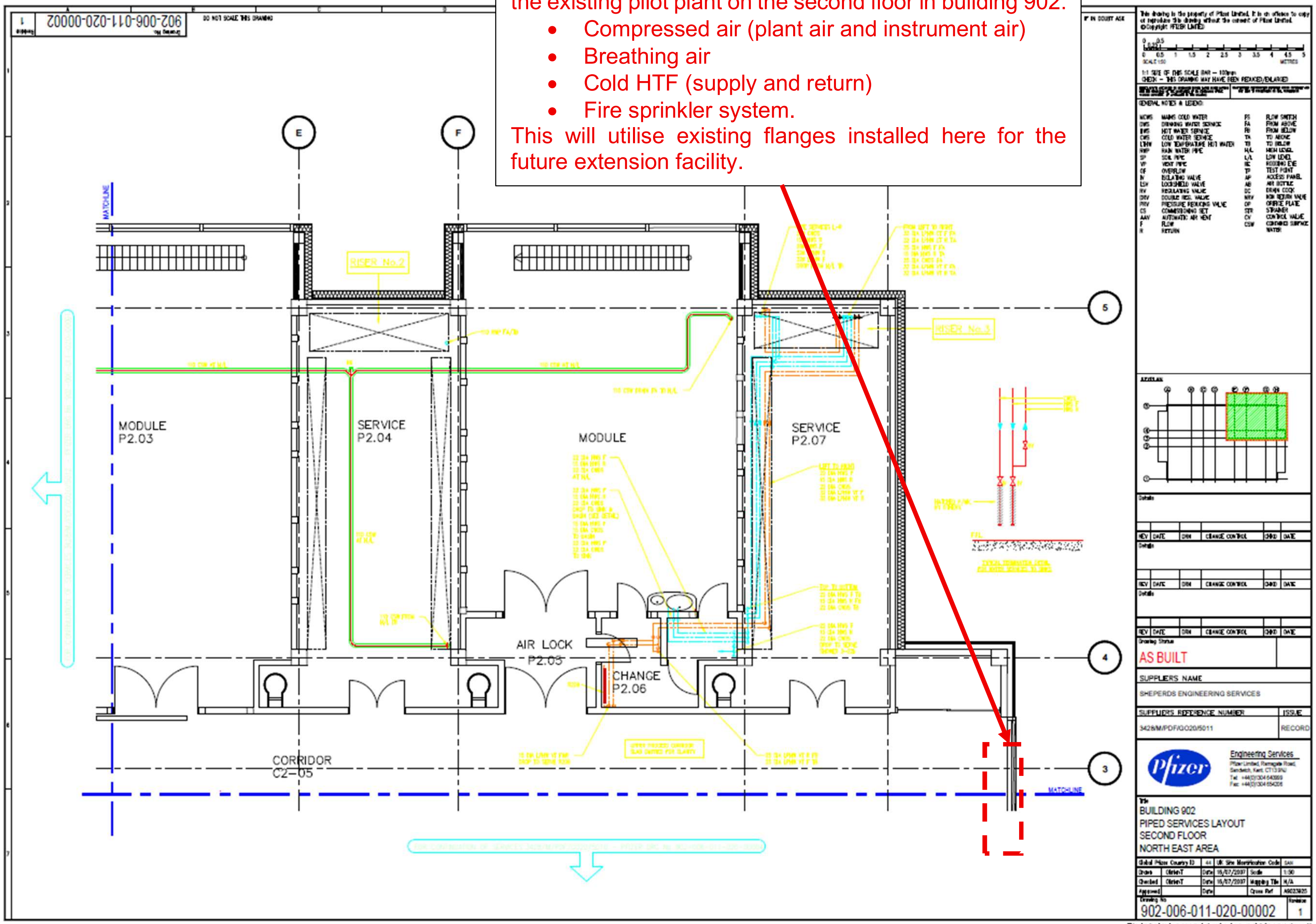
- Cold HTF Supply/ Return
- Fire Sprinkler System
- Breathing Air
- Plant Air
- Instrument Air

These utilities currently enter the existing facility at the second floor of Building 902 at the services void before distributing through the access corridor ceiling. There are blanked flange connections at the end of the corridor through the Northeast side of the existing 902 building. These have been left in anticipation of the future extension in mind. It is recommended to tie in at those locations and connect to the new extension pipework through the ceiling access corridor. Refer to the markup on the following page for proposed tie-in locations.

Tie-ins here in services void for existing services inside the existing pilot plant on the second floor in building 902:

- Compressed air (plant air and instrument air)
- Breathing air
- Cold HTF (supply and return)
- Fire sprinkler system.

This will utilise existing flanges installed here for the future extension facility.



9.0 Building Services

9.1 Overview

The mechanical services systems described below will be required for the new extension building are listed below:

- Heating Ventilation Air Conditioning (HVAC)
    - Supply Air Handling Units (AHUs)
    - Extract Air Fan systems including:
        - General extract systems
        - Local Extract Ventilation systems (LEV)
        - Fume cupboard extract systems
- Chilled Water
- Steam and Condensate steam
- Low Temperature Heating Water (LTHW)
- Towns water service
- Drainage
- Sprinkler Installation
- Building Management System (BMS)

9.2 Mechanical Drawings

The Mechanical Sketch Layouts are included at Appendix 3 and listed below:

Drawing Number	Title / Description
300291-BS-SK-0001	Pressure Regime & AHU Zoning Sketch Layout – Ground Floor
300291-BS-SK-0002	Pressure Regime & AHU Zoning Sketch Layout – First Floor
300291-BS-SK-0003	Pressure Regime & AHU Zoning Sketch Layout – Second Floor
300291-BS-SK-0004	Pressure Regime & AHU Zoning Sketch Layout – Third Floor & Mezzanine
300291-BS-SK-0005	Pressure Regime & AHU Zoning Sketch Layout – Roof
300291-BS-SK-0006	Pressure Regime & AHU Zoning Sketch Layout – Section

9.3 Basis of Design

9.3.1 Operation times:

The hours of operation for the ventilation plant shall be 24/7 to maintain critical air flow direction and containment but with potential for night setback of heating and cooling during out of production hours or periods when areas are un-occupied.

9.3.2 External design Temperatures and Humidity:

External design conditions are based on ASHRAE Climatic Design Conditions 2025, 20-year return period:

- Summer 33.8°C DB 22.9°C WB
- Winter -6.7°C DB -7.5°C WB

9.3.3 Room Pressure and Air Pressure Differential:

Refer to Building Services HVAC Strategy Sketch Layouts: 300291-BS-SK-0001-0006

Process areas will be maintained at a negative pressure to the adjacent by adopting a notional differential pressure of 10Pa-15Pa.

A stable reference point for establishing air flow direction will be provided.

9.3.4 Internal design Temperatures and Relative Humidity:

Internal design conditions need to satisfy product, and occupant demands and have been taken as follows:

- Dispensing, Material Prep and Drum Transfer
    - Temperature Range: 18-25°C
    - Relative Humidity: Uncontrolled, target RH≤70%
- Filter Drying modules
    - Temperature Range: 18-25°C
    - Relative Humidity: Uncontrolled, target RH≤70%
- Reactor rooms
    - Temperature Range: 18- 25°C
    - Relative Humidity: Uncontrolled, target RH≤70%
- Service areas, Auxiliary rooms and technical spaces:
    - Target Temperature Range: ≥10°C
    - Target Relative Humidity: Uncontrolled
- IPC laboratory
    - Temperature Range: 18- 25°C
    - Relative Humidity: Uncontrolled, target RH≤70%
- Hydrogenation suite
    - Temperature Range: 18- 25°C
    - Relative Humidity: Uncontrolled, target RH≤70%
- Corridors:
    - Target Temperature Range: ≥18°C
- Stairs
    - Target Temperature Range: ≥16°C
    - Target Relative Humidity: Uncontrolled
- Plantroom spaces
    - Target Temperature Range: ≥16°C
    - Relative Humidity: Uncontrolled

9.3.5 Noise Criteria:

Noise criteria will be:

- Process areas 85 dB (A)
- Plantroom areas 85 dB (A)
- External areas 85 dB (A)- TBC at the next stage
  External Noise criteria shall be in accordance with the requirement of the predictive Noise Assessment which will need to be confirmed at the next design stage.

9.3.6 Air Change rate per Hour (ACH)

The following are the minimum airflow rates adopted for the purpose of the study:

- 12 ACH in Process areas, associated auxiliary rooms and technical spaces using once through full fresh air HVAC system with run around heat recovery where possible.
- 3 ACH in main plantroom spaces via natural ventilation or/ and mechanical ventilation.

At this stage, it is assumed that the 12-air change rate will offset any potential heat gains in the spaces. This will need to be assessed in detail at the next design stage and local cooling may need to be provided in areas with very high heat gains.

9.4 General Supply AHUs and Extract Fans

Generally, supply air handling units shall be full fresh air and sized to provide a minimum of 12 ACH to the spaces they serve. Air handling units shall be located on the 3rd floor Lower Level and Mezzanine Level plantrooms and shall generally comprise of the following functional sections:

- Inlet section with actuated isolation Damper
- Steam frost coil
- Short bag Filter (G4)
- Run around heat recovery Coil (TBC)
- Bag Filter (M6)
- Heating Coil fed with Steam
- Cooling Coil fed with CWS
- Re-Heat Coil fed with Steam
- Supply fan (fan wall, EC motors)
- Bag Filter (F8)
- Discharge section

Generally fresh air will be drawn in via air inlet louvres on the north side elevation, air conditioned within the AHU then distributed to the areas via galvanised steel ductwork.

For systems where multiple spaces are served by the same AHU, local reheat coils may be required to avoid over cooling lower heat gain rooms.

Dedicated extract fans shall be provided to serve each zone. Typically, duty and standby fans have been proposed for all extract systems.

Refer to HVAC Pressure Regime & AHU Zoning Sketch Layouts drawings 300291-BS-SK-0001-0006 in Appendix 3. The following HVAC systems are proposed to serve the different areas within the new extension:

9.4.1 AHU 1

Located in the Third Floor Lower-Level Plantroom, serving the following rooms:

- Purified Water System
- Switch Room
- TCU
- Auxiliary Rooms
- Reactor Rooms
- I/O Room
- Flow Equipment Area
- TCUs

Extract air Fan System EF 1A & 1B

Extract air from the areas (or from adjacent service area) will be routed to the 3rd floor mezzanine floor where it is proposed to locate extract fans EF 1A & 1B which discharge air to atmosphere via the General Exhaust high velocity discharge stack.

9.4.2 AHU 3

Located in the Third Floor Lower-Level Plantroom, serving the following rooms:

- Dispensing
- Material Prep
- Drum Transfer
- Lobbies

Extract air Fan System EF 3A & 3B

Extract air from the areas will be routed to the 3rd floor mezzanine floor where it is proposed to locate extract fans EF 3A & 3B which discharge air to atmosphere via the General Exhaust high velocity discharge stack.

9.4.3 AHU 4

Located in the Third Floor Mezzanine Plantroom, serving the Hydrogenation Suite.

Extract air Fan System EF 4A & 4B

Extract air from the fume cupboard, LEV points and room extract will be from this single system. Extract air will be routed from the Hydrogenation Suite to the 3rd floor mezzanine floor where it is proposed to locate extract fans EF 4A & 4B which discharge air to atmosphere via a dedicated high velocity discharge stack.

It has been assumed that the exhaust from this system will not require scrubbing.

9.4.4 AHU 5

Located in the Third Floor Lower-Level Plantroom, serving the Process Corridors and Stairs' airlocks.

Extract air Fan System EF 5A & 5B

Extract air from the areas will be routed to the 3rd floor mezzanine floor where it is proposed to locate extract fans EF 3A & 3B which discharge air to atmosphere via the General Exhaust high velocity discharge stack.

9.4.5 AHU 6

Located in the Third Floor Lower-Level Plantroom, serving the IPC Laboratory.

Extract air Fan System EF 5A & 5B

Extract air from the areas will be routed to the 3rd floor mezzanine floor where it is proposed to locate extract fans EF 3A & 3B which discharge air to atmosphere via the General Exhaust high velocity discharge stack.

9.5 Specialist Supply AHUs and Extract Fans

9.5.1 AHU 2 Filter Dryer Areas, Milling & Packing

Incorporating HEPA filtration at the AHU, this system serves the following rooms:

- Milling & Packaging
- Hastelloy AFDs
- Lobbies
- Stores

The air handling unit will be located on the 3rd floor Lower-level plantroom and shall comprise of the following functional sections:

- Inlet section with actuated isolation Damper
- Steam frost coil
- Short bag Filter (G4)
- Run around heat recovery Coil (TBC)
- Bag Filter (M6)
- Heating Coil fed with Steam
- Cooling Coil fed with CWS
- Re-Heat Coil fed with Steam
- Supply fan
- Bag Filter (F9)
- HEPA Filter H13
- Discharge section

Extract air Fan System EF 2A & 2B

Extract air from the areas will be routed to the 3rd floor mezzanine floor where it is proposed to locate extract fans EF 2A & 2B which discharge air to atmosphere via the General Exhaust high velocity discharge stack. This system includes HEPA filtration upstream of the fans.

It has been assumed that the exhaust from this system will not require scrubbing.

9.5.2 EF 7A & 7B Local Extract Ventilation (LEV) System:

Extract air from local extracts will be routed to the 3rd floor mezzanine floor, where it is proposed to locate extract fans EF 2A & 2B which discharge air to atmosphere via the fume cupboard and LEV high velocity discharge stack. The fans shall use variable speed drives and will operate 24/7.

It has been assumed that the exhaust from this system will not require scrubbing.

9.5.3 EF 8A & 8B Solvent Transfer Fume Extract System

Extract air from fume cupboards located in the Solvent Transfer rooms (within the First Floor Reactor rooms) will be routed to the 3rd floor mezzanine floor, where it is proposed to locate extract fans EF 2A & 2B which discharge air to atmosphere via the fume cupboard and LEV high velocity discharge stack.

It has been assumed that the exhaust from this system will not require scrubbing.

9.5.4 Heat Recovery

Where viable, it is proposed to recover energy from exhaust air via extract system heat exchangers and runaround coils within AHUs.

At this stage it is envisaged that heat recovery may be possible on the following systems, which discharge via the General Exhaust stack:

- EF1
- EF2
- EF3
- EF4
- EF5
- EF6

All heating and cooling systems and coils will be sized without taking into account any heat recovery.

9.6 Utilities Systems Description

9.6.1 Plant Steam

Steam shall be used to serve all frost coils and main heating coils within the AHUs. The estimated Steam load is summarised as follows:

System	Estimated Heating Load (kW)
AHU 1	1040
AHU 2	340
AHU 3	380
AHU 4	170
AHU 5	170
AHU 6	80
Total Heating Demand	**2180**

9.6.2 Low Temperature Heating Water (LTHW)

A packaged plant steam to LTHW plate heat exchanger (HX) skid shall provide 80/60°C flow/return to serve zone re-heat coils and any space heaters and fan coil units that may be required (to be fully determined at the next stages).

The system shall consist of a plate HX, secondary distribution pumps, necessary pressurisation system, expansion vessel and carbon steel pipework distribution. The proposed LTHW system shall be variable flow via variable speed pumps and two-port pressure independent Balancing Control valves at the terminal units.

Estimated zone heating loads are summarised in the following table:

System	Estimated Heating Load - (kW)
AHU 1	260
AHU 2	85
AHU 3	70
AHU 4	45
AHU 5	45
AHU 6	20
Total Heating Demand	**520**

The HXs will be located on the Third Floor Mezzanine level along with the pressurisation unit, dosing pot and primary and secondary pumps.

9.6.3 Chilled Water (HVAC)

Chilled Water shall be required for the cooling coils within the air handling units and to serve any fan coil units that may be required in subsequent stages of design. The estimated loads are based on chilled water at 6/12°C flow/return temperatures and are summarised in the following table:

System	(kW)
AHU 1	635
AHU 2	210
AHU 3	170
AHU 4	105
AHU 5	105
AHU 6	50
Total Cooling Demand	**1300**

It is proposed to provide three high efficiency air-cooled chillers to cater for the extension's cooling load. It is proposed that each chiller shall be selected at 33% of the total cooling demand (duty/duty/duty). If a chiller is down due to maintenance or failure, then cooling to non-critical AHUs can be stopped, to provide a level of redundancy.

CHW Buffer tank, pressurisation unit, dosing pot and expansion vessel shall be located locally to the chillers, on the proposed chiller gantry, along with the cooling water pumps and necessary controls.

A reduction in chilled water load should be considered in the next stage; this could be achieved by limiting cooling to critical areas only. This feasibility study makes allowance for the worst-case level of cooling, by allowing for all spaces.

9.6.4 Towns Water

A new Towns water supply shall be taken to a new storage tank in the Third Floor Mezzanine plantroom, which will be dedicated to serve the new extension. The tank shall be appropriately sized to provide an uninterrupted supply to the new purified water generation equipment, once the usage is established (see Process section) and any other consumers identified in the subsequent stages of design, such as plantroom sinks, safety showers and pressurisation units.

Water will be distributed by adjacent multi-stage pumps located on a packaged skid complete with accumulator.

Any domestic hot water requirement shall be via local electric heaters.

9.6.5 Above Ground Drainage

All non-process drainage (condensate from AHUs, drains from sinks, mechanical plant, etc) shall be taken by gravity to the site foul drainage system.

9.6.6 Sprinkler Installation

Existing installation shall be modified to serve the new extension. It is assumed that there is sufficient capacity on the existing infrastructure (pipework, circulating pumps, water storage, etc.)

9.6.7 Building Management System (BMS)

New dedicated controls shall be provided to serve the new extension to allow the operation and the monitoring of the HVAC and the Mechanical systems serving the new facility. The new BMS system will be designed so it is stand-alone albeit it will need to be interfaced with the existing BMS system which will need to be modified to include the new extension interface and main alarms.

10.0 Electrical Engineering

10.1 Introduction

This section of the feasibility report outlines the proposed electrical and ancillary services requirements for the proposed PDF building extension B902 building in Sandwich on the old Pfizer campus.

This section shall be read in conjunction with the architectural, electrical and building services drawings referenced elsewhere in this report, which describe the internal building reconfiguration, and the works required to incorporate all new equipment serving the new extension areas.

It is proposed that new electrical power, data, door access control and interlocking systems, public address systems, fire detection and alarm systems, and other associated ancillary services shall be provided to serve all areas in the new extension.

The existing electrical installation is supplied by 3 transformers, T1 1600kVA, T2 1600kVA and T3 1500kVA. The existing main LV switchboard (Switchboard 902) is approximately 28 years old, which is at or beyond the typical life expectancy for LV switchgear. The switchboard is arranged as two independent sections:

- Section 1 supplied from Transformer 1 via a 2,500A ACB
- Section 2 supplied from Transformer 2 via a 2,500A ACB

The two sections are interconnected via a 2,500 A bus coupler, normally operated normally open. Current operating philosophy allows either transformer to supply both sections, providing resilience.

Transformer 3 is fully dedicated to the HTF system.

There is a 300kVA standby generator connected to section 2 adding a few years ago to supply the Essential loads. Also, there are two independent UPS systems, supplied from this generator for the Emergency lighting system and the PLC system.

The existing Process and HVAC equipment are currently supplied from the main LV switchboard. The existing main LV switchboard has spare capacity to supply the new extension but not suitably rated spare outgoing ways. The electrical strategy options will be discussed later in this report.

New dedicated electrical panels shall be installed to supply the new extension.

New lighting installations, socket outlets, data outlets, wiring systems, fire alarm devices and ancillary equipment connection points shall be installed throughout the defined scope of works.

The engineering principles adopted within this feasibility design may be subject to refinement following the results of surveys, commissioning activities, and the ongoing development of plant and equipment selections.

The following sections provide an overview of the electrical and ancillary services that will be modified by the works.

The electrical scope shall include, but not be limited to:

- Low Voltage (LV) Main Switchboard

- Standby Generator / Uninterruptible Power System (UPS)
- LV Power Distribution and Cabling
- Containment Systems
- Disinvestment / Strip-Out / Decommissioning Works
- Lighting and Emergency Lighting
- Small Power
- Fire Detection and Alarm Systems
- Data and Communications
- Security, CCTV, Access Control and Door Interlocking
- Public Address System
- Earthing and Bonding
- Lightning Protection System

All electrical installations shall be designed and installed in accordance with the latest version of BS 7671 – Requirements for Electrical Installations (IET Wiring Regulations).

10.2 **Design Assumptions**

The following design assumptions have been adopted in developing the concept electrical design.

10.2.1 Power Supplies

- It has been assumed that the existing site electrical infrastructure has sufficient spare capacity to accommodate the additional loads associated with the proposed works but not suitably rated spare outgoing ways. Based on information provided by the Client, the present peak loads at current production levels are Section 1 ~350A and Section 2 ~600A.
- It has been assumed that a new dedicated standby generator rated at 300 kW with a minimum autonomy of 8 hours shall be required to supply the new essential equipment for the new extension.
- It has been assumed that existing UPS systems for emergency lighting and PLC control systems are in suitable condition and have enough spare capacity to supply the new installations for the new extension building.
- It has been assumed that no new UPS system shall be required for the new extension.
- It has been assumed that a new Power Factor unit shall be required for the new extension, but not any Harmonised Filter units.

10.2.2 Loads

- Electrical loads referenced within this document are assumed to be sufficiently accurate for concept design purposes.
- Final equipment loads shall be confirmed during detailed design once equipment selections are finalised and approved.
- Load classifications and ratings shall be reviewed and updated as the design progresses.

The existing site power factor shall be confirmed by the Client.

10.2.3 Unforeseen Loads

Additional electrical loads may arise as equipment requirements are refined. Subject to their magnitude, such loads may be accommodated within the assumed spare capacity or shall be reviewed where they materially impact the site electrical demand.

10.2.4 Diversity Factors

At this concept stage, refer below to the diversity factors applied across all loads.

Assumed diversity factors are:

- Lighting: 0.7
- General Small power: 0.5
- HVAC plant: 1.0
- Process equipment: 0.5

10.2.5 General Items

- Cables: Maximum operation temperature assumed not to exceed 70°C for derating.
- ATEX hazardous areas: It has been identified that there will be ATEX hazardous areas.
- Data cabling: Category 6 assumed.
- Fire alarm system: Existing system assumed capable of extension.
- IT systems: Existing infrastructure assumed capable of expansion.
- Asbestos: Existing surveys assumed complete.

10.3 **Preliminary Electrical Load Assessment**

The preliminary electrical load assessment for the new extension has been developed based on early information provided by the Process and Building Services teams. No manufacturer data is available at this stage.

The load values presented are indicative and shall be used solely to inform initial electrical demand and infrastructure strategy. Further detailed load assessments shall be undertaken during following stages and subsequent design phases.

The table below summarises the current estimated electrical load for the new extension building. These values may change as equipment specifications are finalised.

Overall Estimated Load

Item reference	Power (kW)	Description
Lighting and Small power	34kW	55A, 3 phase TPN (cos ϕ=0.9)
HVAC plant – Non-Essential	518kW	830A, 3phase TPN (cos ϕ=0.9)
Process plant – Essential	180kW	290A, 3phase TPN (cos ϕ=0.9)
Process plant – Non-Essential	158kW	255A, 3phase TPN (cos ϕ=0.9)
Estimated load	890kW	1430A 3phase TPN
Contingency	220kW (25%)	350A 3phase TPN (25%)
Total Estimated load	**1110kW**	**1780A 3phase TPN**

The final electrical load shall be reviewed and updated during subsequent design stages.

10.3.1 Load Classification

Electrical loads shall be classified as follows:

- **Non-Essential Loads**

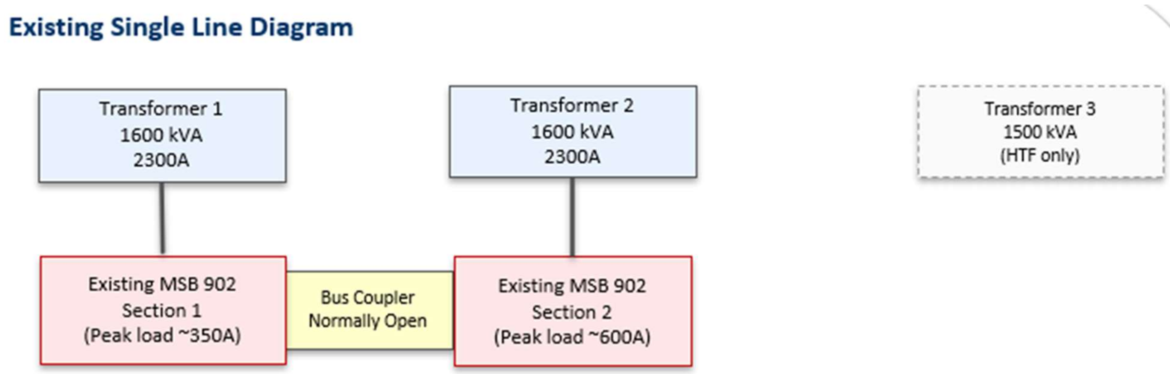
  Supplied from non-essential distribution boards subject to interruption during mains failure, as they are not supported by the standby generator.
- **Essential Loads**

  Supplied from essential distribution boards and maintained during mains failure via standby generation.
- **Critical Loads**

  Supplied from critical distribution boards and maintained during mains failure via UPS systems.

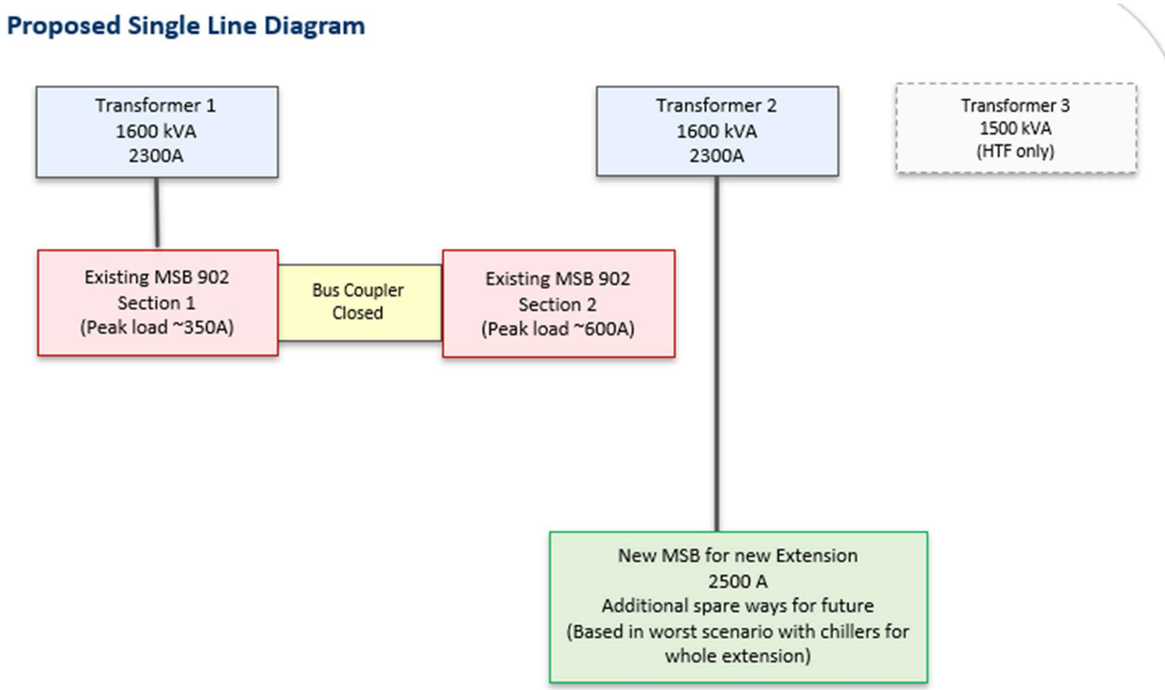
10.4 **Electrical Strategy**

As mentioned above, the existing main LV Switchboard has spare capacity but not suitable rated spare outgoing ways to supply the new extension. Thus, based on this constraint, for this feasibility study has been studied different options, selecting 3 main and preferred ones to be further developed in the next stage (Concept) design. Asymchem shall consider these options and confirm which one shall be adopted.

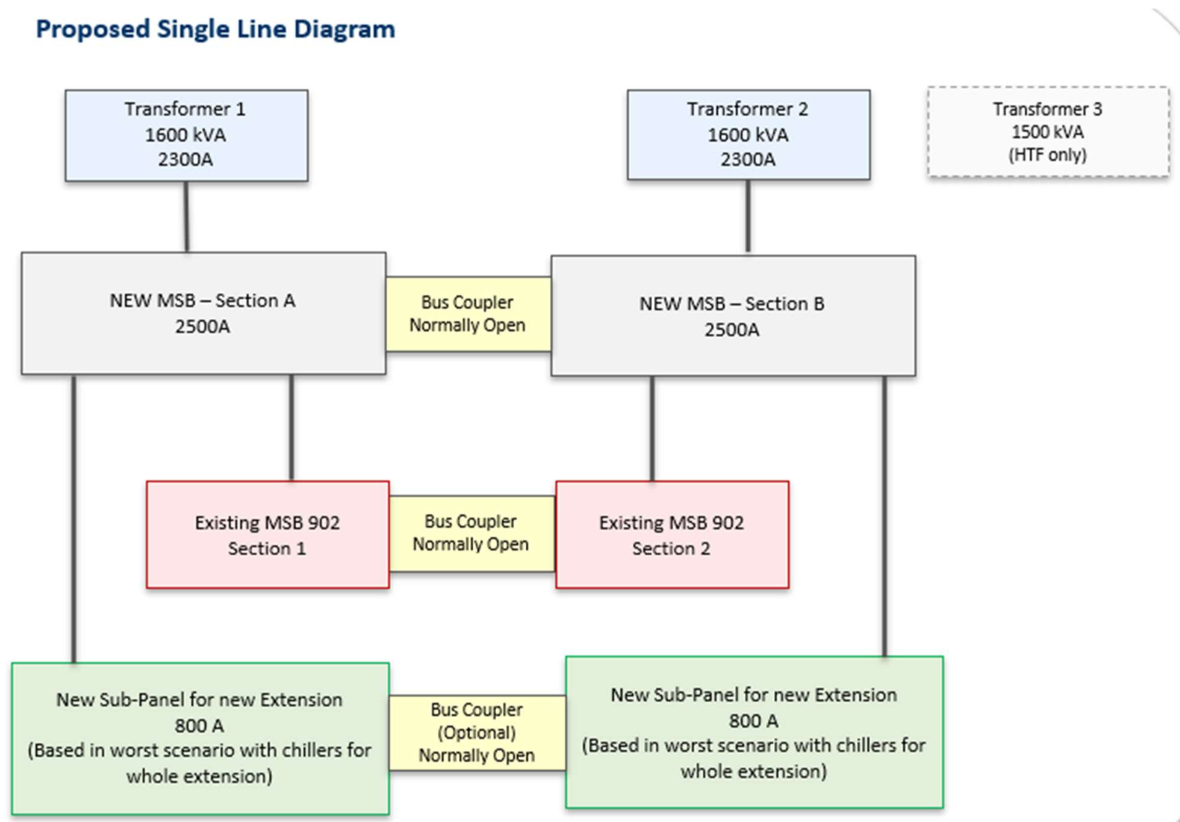


The initial proposed options to supply the new extension building shall be:

- Option 1. One Transformer for existing main LV switchboard, and one transformer for the new extension.
    - Pros
        - Simpler option with lower initial capital costs
        - Minimal disruption to existing site operations
        - Avoids intrusive modification of main LV switchboard
        - Shorter programme and simpler construction phasing
    - Cons
        - Permanent loss of transformer resilience for existing building
        - No resilience provided for the new extension



- Option 2. New upstream main LV Switchboard
    - Pros
        - Robust option, maintaining existing transformer resilience and electrical philosophy.
        - Little disruption to existing site operations
    - Cons
        - Higher initial capital cost compared with reduced-intervention options.
        - Requires physical space for new switchboard installation.
        - Requires protection coordination and planned outages.
        - A little more intrusive works with existing main LV switchboard



- Options 3. Fully replacement of existing main LV Switchboard
  - Pros
    - Most robust option, maintaining existing transformer resilience and electrical philosophy.
  - Cons
    - High disruption to site and longer programme for the works
    - Higher initial cost, for the additional electrical works, e.g. the need to hire a temporary generator, additional extensions, junction boxes, etc.
    - Complex programme for the works to re-use existing incomers and outgoing ways with the likely requirement to new extensions and/or junction boxes.

These are some options discussed during the feasibility stage and the client shall confirm during the Concept stage which option is selected to supply the new extension.

Independently the option selected to supply the new extension, the internal electrical strategy to supply the new extension shall be:

- New LV switch panel for the new extension building supplying:
  - New dedicated essential electrical panel for the Process essential equipment

- New dedicated non-essential electrical panel for the Process non-essential equipment
  - New dedicated electrical panel for the HVAC equipment
  - Dedicated distribution board for lighting and for general small power.
- New standby generator for essential supplies for the new extension building.
- Reusing existing UPS systems to supply the new emergency lighting systems and new PLC control systems for the new extension building.
  - New critical local distribution board for emergency lighting.
  - New critical distribution board for new PLC control system for the new extension building.
- New PV system for the new extension building

10.5 Low Voltage (LV) Distribution

The LV distribution strategy is proposed as follows.

An electrical services room shall be provided at ground floor level in the Northeast corner of the extension building.

The electrical services room shall accommodate:

- The new LV panel board to supply the new extension
- A Power factor correction equipment (if required)
- Harmonic mitigation equipment (if required)
- Lighting, emergency lighting and small power distribution boards for the Ground Floor
- Dedicated process equipment panel boards for essential and non-essential supplies
- The generator changeover panel from the new generator.
- PV inverter and associated distribution (if applicable)

New dedicated electrical panels for HVAC equipment for the new extension shall be located in the new plantroom.

All new panel and distribution boards shall be metered to allow monitoring of energy consumption.

10.6 Standby Generator / Uninterruptible Power System (UPS)

There is an existing 300 kW standby generator installed on site. During the site visit, it was confirmed that the generator is currently operating at approximately 75–80% of its rated capacity under existing load conditions. Based on this information, the available spare capacity of the existing standby generator is insufficient to accommodate the additional essential electrical loads associated with the proposed extension.

Accordingly, the existing standby generator is not capable of supporting the new proposed essential equipment for the extension building without exceeding acceptable loading limits. It is therefore proposed that a new standby generator be provided to supply the essential electrical systems associated with the new extension.

The following systems are considered essential loads for the proposed extension and shall be supported by the new standby generation system:

- Power supplies for new reactors
- Power supplies for the purified water skid
- Power supplies for the SC03.

There are currently two independent UPS systems serving Building 902:

- One UPS system providing critical power to the emergency central lighting system

- A second UPS system providing power to the PLC and control systems

The Client has advised that both existing UPS systems are currently operating at approximately 20–30% of their rated capacities. On this basis, it is considered that sufficient spare capacity is available within the existing UPS systems to support the additional critical and control loads associated with the proposed extension, subject to confirmation during the detailed design stage and final load assessments.

10.7 LV Cable Installation

New LV submain cabling shall be provided from the main LV switchboard to supply the new LV panel board serving the new lighting, general small power and essential and non-essential Process equipment.

The submain cabling shall comprise XLPE insulated, steel wire armoured, low smoke and fume (XLPE/SWA/LSF/PVC) copper cables, rated for continuous operation at 70°C. Cables shall be three-core or four-core as required and sized in accordance with BS 7671, fault level calculations, voltage drop limits and project design criteria.

Cable routes shall be fully coordinated during detailed design. For the purposes of this report, submain cabling is assumed to be routed at high level on cable tray within ceiling voids.

10.8 LV Containment Systems

New electrical containment systems shall be provided to facilitate the extension and connection of existing services from Building 902 to the proposed extension building. This shall include containment for power, data, PLC/control systems, security, fire alarm and associated low-current systems, as required to maintain continuity of services between the existing and new facilities.

Within the extension building, new containment systems installed within ceiling voids shall form the primary distribution backbone for electrical services. These systems shall accommodate lighting, small power and ancillary electrical services. Subject to segregation and electromagnetic compatibility (EMC) requirements, the primary containment routes may also be utilised for control, data, fire alarm and BMS/FMS cabling.

Secondary containment systems shall be provided from the main containment routes to serve individual rooms and equipment locations. This shall include the provision of all necessary trunking, cable tray, cable basket and conduit systems to ensure complete and coordinated distribution of electrical and low-current services throughout the extension.

In general, the following containment arrangements shall be adopted:

- Galvanised steel trunking and conduit systems shall be used for lighting, emergency lighting and small power distribution.
- Cable tray shall be use for submain cables, fire alarm and HVAC and Process (essential and non-essential) equipment.
- Cable basket shall be used for data and control cabling.

All containment systems shall be manufactured from medium to heavy-duty mild steel, hot-dip galvanised in accordance with the relevant British Standards and installed in accordance with EAMA best practice guidelines.

Conduits serving equipment, outlets and fixed connections shall generally be galvanised steel, sized to suit the installed cabling, draw-in requirements and to allow for future capacity and system expansion.

10.9 Lighting Installation Incorporating Emergency Lighting

Lighting installations shall be designed, installed, tested and commissioned in accordance with relevant British Standards, CIBSE Lighting Guides and recognised industry best practice.

All new lighting circuits shall be supplied from the new non-essential LV MCB distribution boards, one per floor, as appropriate, using single-core low smoke and fume (LSF) insulated copper conductors installed within galvanised steel trunking and conduit systems.

Lighting shall be provided using new luminaires incorporating energy-efficient, long-life light sources, typically LED technology. Luminaire layouts shall support safe circulation, effective task lighting and appropriate visual comfort, with due consideration given to both horizontal and vertical illuminance.

Luminaires shall be selected to suit the environmental conditions of the areas served, including appropriate IP ratings and ATEX areas for reactors, laboratory and plant environments.
Lighting levels shall be designed to achieve the following typical maintained illuminance values, unless otherwise defined by room data sheets or Client standards:

- Lobbies: 200–300 lux
- Laboratory cleanrooms: 500 lux
- Changing rooms: 200–300 lux
- Corridors and stairways: 200 lux
- Plantrooms and switchrooms: 200–300 lux
- Service distribution zones: 200 lux

Emergency lighting shall be provided throughout the works area in full compliance with BS 5266, EN 1838 and the requirements of the Fire Officer and Local Authority, to ensure safe means of escape in the event of failure of the normal power supply.

All new emergency lighting circuits shall be supplied from new critical LV MCB distribution boards, one per floor, following the same strategy than building 902. The distribution boards shall be supplied from a new 250A MCCB panel board which is supplied from the existing critical panel board, UPS backed up, located in the electrical service room in building 902.

Final emergency lighting circuits shall be installed using single-core low smoke and fume (LSF) insulated copper conductors installed within galvanised steel trunking and conduit systems.

Emergency lighting shall typically comprise self-contained battery or inverter modules integrated within normal luminaires, providing maintained or non-maintained operation as required. The minimum emergency duration shall be three hours.

Illuminated emergency exit signage shall be provided where required to clearly indicate escape routes and exits.

Lighting control systems shall be provided to suit the functional requirements of each area. Where appropriate and safe to do so, occupancy sensing controls (e.g. PIR) shall be considered to enhance energy efficiency.

Wall-mounted lighting switches shall be provided within plantrooms and technical spaces.

10.10 Small Power Installation

All new LV small power circuits shall be derived from new non-essential LV MCB distribution boards, one per floor.

Small power wiring shall comprise single-core LSF insulated copper conductors installed within galvanised steel trunking and conduit systems, or steel wire armoured cables installed on cable tray, as appropriate to the installation environment.

Residual current protection shall be provided to all small power circuits and distribution boards, including RCBOs or RCDs, as required to comply with BS 7671 and project-specific risk assessments.

General-purpose power supplies shall be provided via socket outlets located to suit room layouts, equipment arrangements and operational requirements.

Supplies to fixed equipment shall terminate at local fused connection units or lockable isolators, as appropriate to the equipment duty and maintenance requirements.

10.11 Power Supplies to Mechanical Equipment

All new HVAC equipment for new extension building shall be supplied from existing spare ways in existing MCC panel located in plantroom on third floor in building 902.

Local means of isolation shall be provided to all mechanical equipment to facilitate safe maintenance and emergency shutdown. Isolation shall typically be provided via lockable; multipole switch disconnectors located adjacent to the equipment served.

Where required to interrupt power supplies under emergency conditions, emergency power off (EPO) devices shall be provided. EPO devices shall be clearly labelled, readily accessible, and located in positions of relative safety in accordance with the site fire strategy.

10.12 Power Supplies to Process Equipment

A new dedicated LV panel board located within the electrical service room shall be provided to supply all new non-essential process equipment, supplied from the new LV Panel board for the new extension building.

A separate dedicated LV panel board located within the electrical service room shall be provided to supply all new essential process equipment. This new panel board shall be supplied from an Automated Transfer Switch (ATS) panel supplied from:

- Primary source. The new LV Panel board for the new extension building.
- Secondary source. The new standby generator for the new extension building.

Final panel board ratings, ways and spare capacity shall be confirmed during detailed design in coordination with mechanical, process and equipment schedules.

Local means of isolation shall be provided to all process equipment to facilitate safe maintenance and emergency shutdown. Isolation shall typically be provided via lockable; multipole switch disconnectors located adjacent to the equipment served.

Where required to interrupt power supplies under emergency conditions, emergency power off (EPO) devices shall be provided. EPO devices shall be clearly labelled, readily accessible, and located in positions of relative safety in accordance with the site fire strategy.

10.13 Fire Alarm and Detection Installation

The scope of works area shall be provided with fire detection and alarm systems incorporating automatic detection and manual alarm initiation, designed and installed in accordance with BS 5839, relevant standards and the site fire strategy.

It is assumed that the existing fire alarm and detection system has sufficient capacity to be extended to serve the new extension building. This assumption shall be confirmed during detailed design.

The fire alarm system shall be category L1.

The system shall comprise new multisensor (smoke and heat) detectors, sounders, visual alarm devices and manual call points, as a minimum.

A VESDA aspirating smoke detection system shall be provided, if required, within ceiling voids exceeding 800 mm above cleanroom areas, to minimise operational disruption during testing and maintenance.

The fire alarm system shall be interfaced with access control systems, door interlocking systems and HVAC plant, as required by the fire strategy.

All fire alarm works shall be installed, tested, commissioned and certified by a suitably approved and registered BS 5839 contractor.

10.14 Data and Communications Systems

There are two independent IT networks.

- One internal IT network
- One external Ethernet network for office.

The new extension building shall be provided with new data and communications outlets, including Wi-Fi coverage, to serve all new rooms and areas.

A new data hub shall be provided for the new extension located in a dedicated Data room. The new data hub shall be connected to the existing data hub in existing building 902 (or maybe with the main data room in building 530. The data and communications system shall originate from the new IT infrastructure, utilising new network switches and equipment racks.
New data cabling shall be routed via new containment systems.

Data cabling shall typically comprise Category 6 UTP copper cabling installed within dedicated cable basket and conduit containment systems, terminating at RJ45 outlets.

10.15 Security, CCTV, Access Control and Door Interlocking Systems

Security systems shall be provided in accordance with company and site standards.

New CCTV cameras shall be installed at strategic locations, to be defined during subsequent design stages in conjunction with the Client.

Access control and door interlocking systems shall be provided for areas associated with cleanrooms and controlled environments.

All access control and door interlocking systems shall be compatible with existing site-wide systems and shall be interfaced with the fire alarm system as required.

Power supplies for security, access control and door interlocking systems shall be derived from new local LV distribution boards.

Final system configuration, coverage and performance requirements shall be confirmed during later stages of design development.

10.16 Public Address System (PA)

The new extension shall be provided with a new Public Address System (PA) connected to the existing building 902.

The final system configuration shall be developed in conjunction with the Client and relevant stakeholders during the later stages of design development.

10.17 Earthing and Bonding Installation

The new extension building shall be provided with a new earthing and bonding installation in line with the existing arrangements already in place for existing building 902.

All new lighting installations, power outlets, panel boards and ancillary equipment shall be earthed and bonded in accordance with BS 7430, BS 7671, the Electricity at Work Regulations, the Electricity Safety, Quality and Continuity Regulations, Building Regulations and all other relevant statutory requirements.

New earthing and bonding conductors shall be copper with green and yellow LSF insulation.

Where single-core conductors are installed, a separate circuit protective conductor (CPC) shall be provided. For armoured cables, the cable armour may be utilised as the CPC, subject to calculation, with additional supplementary CPCs provided where required.

All armoured cable terminations shall be bonded to local earth bars, which shall be connected back to the main building earth bar.

All electrical equipment and cabling shall be clearly labelled in accordance with site standards.

10.18 Lightning Protection System (LPS)

The new extension building shall be provided with a lightning protection system designed, installed, tested and certified by a specialist contractor operating under a Contractor Design Portion (CDP).

The LPS shall be designed in accordance with the relevant British Standards and integrated with the site earthing system.

10.19 Solar Photovoltaic (PV) System

A solar photovoltaic system shall be provided as part of the project Net Zero Carbon strategy for the new extension building.

The PV system shall be sized to support the electrical demand of the building or area served and shall not export energy to the wider site network unless specifically agreed with the Client.

The system shall comprise solar panels, inverters and associated LV distribution equipment, with no provision for energy storage unless otherwise instructed.

Solar panels shall be located on the roof, subject to structural and coordination constraints.

The design, installation, testing and certification of the PV system shall be undertaken by a specialist contractor under a Contractor Design Portion (CDP).

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11.0 Control System

11.1 Overview

The existing B902 Building contains process equipment similar in nature to that proposed for the extension as outlined in section 8.0. This existing process equipment is controlled by an Emerson DeltaV system with both Process Automation System (PAS) and Safety Instrumented System (SIS) elements. The intention is to extend the control scope of the existing DeltaV PAS and SIS systems to cover the control of the new process equipment.

11.2 Implementation of DeltaV Control

The I/O Racks and servers for the existing DeltaV PAS and SIS systems are located in the “Instrument Room” 1-02. To extend the control scope to the new process equipment additional I/O racks are required. These will be located in the new “I/O Room” on the Second Floor of the Extension.

An assessment has been made of the space required to accommodate the necessary Controller Cabinet and CHARM I/O Cabinets and whilst there would not be a lot of spare space the I/O Room is able to house the panels as shown in Figure 1 below.

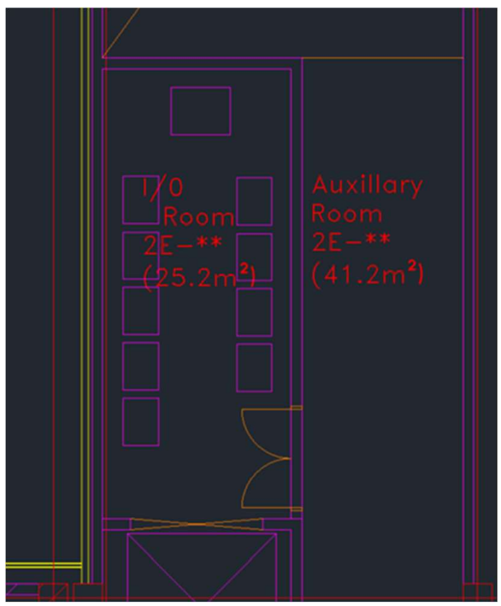

Figure 1 – Plan of Proposed Second Floor I/O Room with DeltaV Controller and I/O Cabinets Shown

Should additional control cabinets ultimately need to be catered for then other space is available in the plant room areas. Alternatively, Remote I/O panels could potentially be used to locate some or all of the I/O in the field more locally to the process equipment (subject to assessment), freeing up space in the I/O Room.

The I/O within the I/O Cabinets will be linked back to the overall DeltaV system via network connection to allow a complete overview of all the Process Equipment to be accessible via the SCADA screens as applicable, with centralised alarming and event logging etc.

Additional new SCADA Screens will be provided as required in the Extension to allow the necessary operator control functions to be carried out.

12.0 Constructability

A constructability review of the current feasibility design was held with the design team on 14 May 26. Items listed below where discussed;

12.1 Site Compound

The construction will require a suitable sized area, ideally close to the works, to accommodate offices, laydown space, parking and waste management.

At present there is a large hard standing area to the east of site (at the end of Haig Drive) that has been used historically for Principal Contractor compounds. The intent would be to utilise this area for the proposed building extension.

Delivery of site set up infrastructure is not expected to cause any undo interruptions to the clients ongoing activities.

12.2 Enabling Works

Various enabling works will need to be completed prior to commencement of groundworks. This is expected to include decommissioning/disinvestment of the existing hydrogenation building and assessment/diversion of existing underground services where these interface with the proposed extension footprint or foundation arrangement. Localised foundation adaptations may also be required depending on confirmed underground utility locations.

12.3 Foundations

Due to the nature of ground, it is assumed that the new building will require piled foundations. This will require excavation for a piling mat. There appears adequate space to do this, with some local traffic management. Once the design is confirmed the following must also be considered:

- Ground water level (pumping may be required)
- Contaminated ground
- Rerouting of underground services
- Depending on depth of excavation, any temporary works to ensure stability of adjacent structures
- UXO (unexploded Ordnance) checks
- Depending on site traffic from other nearby facilities – Just In Time deliveries may also need to be considered

12.4 Structural Frame & Envelope

The frame is relatively standard in design and construction, but will require perimeter space for material delivery, crane and other plant. The final site planning will need to determine access and build sequence to allow construction, but although space is limited on some sides, this is workable with appropriate planning.

Floors are intended to be holorib type with concrete. There appears adequate space but again some local traffic management may be required for delivery lorries, this will require a more detailed plan once during final planning to max number of deliveries per can be considered along with space for delivery lorry washouts.

Perimeter cladding will likely be installed via scissor lift which will require a firm level perimeter hardstanding.

It is also noted that the location is close to the coast and could be subject to high winds which may impact ability to use a crane at certain times.

12.5 Equipment & Plant installation

Timing of the placement of new vessels into the new building will need further planning and may be dependent on lead times of equipment relative to a construction start. Adequate space through the building with provision for moving vessels via lifting beams or other means will need to be determined.

External space for a new chillers may be overly constrained; new chillers could be located on the new building roof or if height is an issue on the existing building roof via installation of new steel stub columns onto existing (all subject to structural review).

An external loading platform may be required for installation of new AHU's, and space for this must be considered in the next stage of design

12.6 Internal fit out

With various working areas, the installation can be broken down into various work fronts that run in parallel. The installation would generally be following a soffit to floor sequence

System ITP would be developed prior to installation to ensure the correct quality checking is carried out during construction

12.7 Integration with Existing Building

It is understood that provision was made in the existing east facade for a relatively simple fabric integration with little or no impact. Any integration with other utilities neds to be assessed at the next stage

12.8 Commissioning

A commission plan will be developed prior to construction so installation can be tailored to enable a smooth commission phase

12.9 Risks

Working in proximity to rooms with blast panels and flammable atmospheres. Depending on exact construction work and any works ongoing in the existing building, the methodology will need to be access and agreed with all stakeholders

Ground vibration during piling/ground works – will this impact on any adjacent buildings or process/work within

13.0 Outline Programme

13.1 Overview

A draft next stage programme has been developed to provide an indicative roadmap for progression of the project following completion of the Feasibility Study, as shown in ***APPENDIX X***.

As is typical at Feasibility stage, the programme beyond the immediate next stage remains high level and will continue to evolve as the design develops, project scope is refined and procurement/delivery strategies are confirmed. The programme should therefore be considered a live document and updated throughout subsequent stages as project definition and delivery strategy mature.

Greater detail has been focused within the RIBA Stage 2 - Concept Design activities to establish the anticipated design development approach, key surveys and investigations, and project interfaces. Additional detail has also been incorporated into the construction section as and outcome of constructability review discussions and associated sequencing considerations identified during the Feasibility Study.

13.2 Stage Gates and Client Review Periods

The programme incorporates a one-month client review and action period between key project stages. These allowances are intended to provide sufficient time for:

- Client review and approval activities;
- Stakeholder engagement and alignment;
- Commercial approvals and purchase order placement; and
- Formal instruction to proceed to the subsequent project stage.

13.3 Third-Party Consultants and Surveys

A consolidated programme activity has been included within the Concept Design phase for third-party consultant inputs and surveys anticipated to be required to support design development, statutory requirements, planning activities and BREEAM assessments. Based on the current Feasibility assumptions, these activities are anticipated to include, but are not limited to:

- 3D internal and external scan surveys;
- Ground investigation and contaminated land assessments;
- Ecological surveys;
- Acoustic assessments;
- Fire strategy and fire engineering activities;
- Transport assessments;
- Flood risk assessments;
- Lifecycle assessment and costing activities;
- Security assessments;
- Underground utility surveys;
- Stack consultant / CFD studies;
- Asbestos surveys; and
- Unexploded ordnance (UXO) assessments.

13.4 Programme Optimisation Opportunities

Whilst the current programme indicates an overall project completion date of Q2 2030, opportunities remain to reduce the overall duration as the project progresses. Potential programme reduction measures may include:

- Early contractor involvement and Design & Build procurement from Stage 4 onwards;
- Overlap of selected design and procurement activities;
- Optimisation of construction sequencing and site working arrangements;
- Reduction of stage gate/client review durations where appropriate;
- Acceleration of design stage activities through increased resource allocation and parallel working arrangements, subject to agreement of the associated commercial implications;
- Further programme efficiencies identified during future design development and contractor engagement.

14.0 Risks

14.1 Overview

A project risk register has been developed during the Feasibility Study to identify and manage key risks associated with the proposed extension works, as shown in ***APPENDIX X***. Risks have been assessed based on likelihood, programme impact and cost impact, with mitigation measures identified to reduce overall project exposure as the design progresses.

The risk register is intended to remain a live document and should continue to be reviewed and updated throughout subsequent design stages.

14.2 Key Project Risk Themes

At Feasibility stage, the project remains subject to a number of typical early-stage uncertainties:

- **Existing site conditions** — including unknown structural/foundation design, asbestos, drainage clashes, uncertain HV cable locations, limited available utility space, landlord/site constraints and existing infrastructure limitations (Risks 2, 3, 4, 14, 20, 21, 27).

- **Statutory requirements** — including BREEAM compliance requirements, planning approval risks, building height implications and potential future planning constraints (Risks 1, 8, 11).

- **Utility infrastructure** — including electrical infrastructure capacity, process utility capacity, cooling requirements, emergency power strategy, waste handling arrangements and reactor cooling philosophy assumptions (Risks 2, 22, 23, 24, 25, 26).

- **Procurement strategy and supply chain** — including long lead equipment procurement, supplier lead times, equipment cost escalation and unclear purchasing responsibilities/strategy (Risks 7, 9, 16, 18).

- **Programme constraints** — including late RFI responses, communication barriers, scope changes and construction sequencing (Risks 5, 6, 13, 19).

14.3 Risk Exposure Summary

The current pre-mitigation total identified exposure is approximately:

- 182 weeks cumulative programme exposure (131.8 weeks factored exposure)
- £2.64m cumulative cost exposure (£1.76m factored exposure)

Following the mitigation measures and assumptions incorporated within the Feasibility Study, the post-mitigation exposure reduces to approximately:

- 107 weeks cumulative programme exposure (60.6 weeks factored exposure)
- £1.13m cumulative cost exposure (£626k factored exposure)

14.4 Residual Risk and OOM Allowances

Several identified risks have been partially mitigated through inclusion of allowances within the Order of Magnitude (OOM) cost estimate, including electrical infrastructure upgrades, process utility capacity upgrades, drainage diversions, cooling strategy requirements and emergency power provisions (Risks 2, 20, 22–26).

Whilst these risks remain live from a technical, coordination and design-development perspective, their potential capital cost impact has been partially accounted for within the current feasibility cost plan.

15.0 Cost Estimate

15.1 Overview

The feasibility cost estimate has been prepared using the information available at this stage of design and should be regarded as an order-of-magnitude assessment prepared broadly in accordance with NRM1 principles. The estimate is based on the current feasibility drawings, outline design criteria, utility assessments, spatial requirements and the proposed scope of works for the extension and associated alterations to the existing facility. At this stage, the estimate necessarily reflects a developing level of definition and is therefore reliant on stated assumptions, design allowances, benchmark data and professional judgement where detailed information is not yet available.

The estimate has been informed by benchmark cost data, recent project experience, preliminary supplier or specialist input where available, and the technical recommendations developed through this feasibility study. Appropriate allowances have been included for preliminaries, contractor overheads and profit, design development, coordination and identified scope interfaces, but the estimate should not be interpreted as a fixed tender sum. The cost plan is intended to support feasibility-stage decision-making, option comparison and budget setting, and will require progressive review and reconciliation as surveys are completed, risks are retired or refined, and the design develops through subsequent RIBA stages.

In line with best practice, the cost estimate should be read together with the project risk register. Risks identified during the feasibility study should continue to inform the level of risk allowance and contingency carried within the estimate, with particular regard to existing site conditions, statutory approvals, utility capacity, procurement strategy, programme interfaces and construction logistics. Where risks have already been allowed for within measured or provisional allowances, these should be clearly distinguished from residual risk contingency to avoid duplication. As the project progresses, the risk register and cost estimate should be reviewed in tandem so that emerging risks, closed risks, transferred risks and mitigation actions are reflected transparently in the developing cost plan.

15.2 Contingency and Risk Allowance

At feasibility stage, contingency and risk allowance should be applied in a structured and transparent manner. The base estimate should reflect the defined scope of works and measured or benchmarked allowances that can reasonably be identified from the current design information. Separate allowance should then be made for project risk, informed by the live risk register and calibrated to the level of uncertainty associated with design development, existing asset interfaces, ground conditions, statutory approvals, utility diversions, procurement route, logistics and programme. This is important so that foreseeable scope items, design development allowances and specific provisional assumptions are not conflated with residual risk contingency.

The contingency carried within the cost estimate should be subject to active governance and periodic review as the project progresses. As surveys are completed, scope is defined, procurement decisions are confirmed and mitigation measures are implemented, the allowance should be adjusted to reflect the changing risk profile rather than remain as a static percentage. Drawdown of contingency should be controlled against defined risk events and approved changes, with clear visibility of what has been committed, what remains available and which risks continue to sit outside the estimate as exclusions or client-held risks. This approach will provide a more robust basis for budget management, business-case approval and subsequent cost plan development through later RIBA stages.

	Description	Rate £/m² / Rate	GIFA (m²) / Nr	Total £
0	**Facilitating Works**			**£254,960**
1	**Substructure**			**£1,499,680**
2	**Superstructure**			**£3,942,636**
3	**Internal Finishes**			**£1,270,672**
4	**Fittings & Equipment**			**£154,960**
5	**BUILDING SERVICES**			**£9,532,716**
5.6	**PROCESS EQUIPMENT**			**£23,249,083**
6	**Prefabricated Buildings**			**£0**
7	**Work to Existing Buildings**			**£204,000**
8	**External Works**			**£1,859,520**
	WORKS COST SUBTOTAL			**£41,968,226**
9	Preliminaries	8%		£3,357,458
	Site management, welfare, temporary works			
10	Contractor OH&P	5%		£2,266,284
	BASE CONSTRUCTION COST			**£47,591,969**
OTHER PROJECT COSTS				
	Design Team Fees	8%	of construction	£3,807,357
	Surveys & Investigations	£80,000		£80,000
	BREEAM	£150,000		£150,000
	Planning & Statutory Fees	£25,000		£25,000
	Client Direct Costs	excl.		excl.
	Other Project Costs Subtotal			**£4,062,357**
RISKS				
	Risk Register (factored costs)			£626,600
	Risk Register Subtotal			**£626,600**
GENERAL RISK & CONTINGENCY ALLOCATIONS AT FEASIBILITY STAGE				
	Design Development Allowance	15%	Feasibility stage	£7,748,149
	Construction Risk and equipment Risk	25%		£12,913,581
	Client Contingency	10%		£5,165,433
	Risk & Contingency Subtotal			**£25,827,163**
COST ESCALATION				
	Tender Date			
	Inflation Rate (annual)			excl.
	Escalation Allowance			excl.
	TOTAL PROJECT COST (Order of Magnitude)			**£78,108,089**

16.0 Executive Summary

Taking into consideration all of the information that has been provided to Scitech-EKIUM from Asymchem, and from all the disciplines, that have inputted into this Feasibility Report, Scitech-EKIUM agree that the proposed option of extending the existing Building 902 **is feasible**, on the understanding that approval from the local authority planning department is granted.

However, during the next stage of design (RIBA Stage 2 – Concept Design), further interaction with Asymchem will be required, through multiple design/user workshops, to ensure that all user requirements are being met and understood, and that all major items of equipment (Hastelloy Reactors etc.) are located within the most optimal locations, to ensure the most efficient operation of the existing & new extension building.

In addition to this document, Scitech-EKIUM will produce a ‘Pre-Planning’ document, that can be used to issue to the local authority, to gauge their acceptance of the proposed extension, that is detailed within this report.

It has been a pleasure working with Asymchem, and Scitech-EKIUM would like to wish Asymchem all the very best with their proposal of extending their existing Building 902, and hope that we can work together again in the very near future.

17.0 Appendices

Appendix 1 - Options Appraisal Matrix

R001	Asymchem PDF Extension Project																								
Option	Description	Existing Hydrogenation Building to remain	New PDF Building Extension to maximise available area	Central circulation corridor to allign	North & South scientific areas above and below central corridor are symetrical	Milestone Achievability (score 1 to 10) weighting 25%. Score 1-5 is highly complex, lots of interfaces, enabling works etc. Score 6-10 is low complexity, straightforward.			Impact on site operations (score 1 to 10) weighting 20% Score 1-5: Major impact requiring building shutdown and impact to science Score 6-10: Low impact and manageable planned shutdowns / no shutdowns			Project Complexity (score 1 to 10) weighting 20%. Score 1-5 is highly complex, lots of interfaces, enabling works etc. Score 6-10 is low complexity, straightforward.			Construction safety (score 1 to 10) weighting 15% 1-5 Low score - High H&S risk 6-10 High score - Low H&S risk			Requirements Compliance (score 1 to 10) weighting 20% Score 1-5: Major impact - User requirements Not met Score 6-10: Low impact - Uers requirement met			Total Score (Out of 10) Higher Score = Better Solution	URS Compliant	Pros	Cons	Cost
						Score	Weight	Total	Score	Weight	Total	Score	Weight	Total	Score	Weight	Total	Score	Weight	Total					
Existing Hydrogenation Building - Removed/Relocated																									
1	1. Existing Hydrogenation Building to be removed / relocated 2. New feasibility study required if Hydrogenation Building operations are still required - Where can the Hydrogenation be relocated to? 3. New PDF Extension building meets client requirements - Equipment & central corridor allignment	No	Yes	Yes	Yes	8	25%	2.00	8	20%	1.60	8	20%	1.60	9	15%	1.35	8	20%	1.60	8.15	Yes	1. Removal of Hydrogenation Building will improve constructability on new extension building 2. New PDF extension meets client's requirements	1. Existing Hydrogenation to be removed or relocated 2. Cost for Hydrogenation removal / relocation to be considered	£££
Existing Hydrogenation Building to Remain - New Extension Building to be Offset - Central Corridor Allignment																									
2A	1. Existing Hydrogenation builing to remain 2. New PDF Extension building to be offset from existing Hydrogenation building - Constructability & safety 3. New & existing central circulation corridors to allighn 4. New extension building will not be symetrical - North of the central corridor will be smaller than the south area	Yes	Yes	Yes	No	8	25%	2.00	10	20%	2.00	5	20%	1.00	5	15%	0.75	4	20%	0.80	6.55	No	1. Existing Hydrogenation to remain - No cost required to remove building	1. Hydrogenation building will increase the cost of the façade of the new PDF extension 2. Keeping the Hydrogenation building, moves the new PDF extension - to enable construction & for safety	£££
Existing Hydrogenation Building to Remain - New Extension Building to be Offset - Central Corridor Offset																									
2B	1. Existing Hydrogenation builing to remain 2. New PDF Extension building to be offset from existing Hydrogenation building - Constructability & safety 3. New & existing central circulation corridors do NOT allign 4. New extension building will be symetrical around central corridor	Yes	Yes	No	Yes	8	25%	2.00	10	20%	2.00	5	20%	1.00	5	15%	0.75	5	20%	1.00	6.75	No	1. Existing Hydrogenation to remain - No cosdt required to remove building	1. Hydrogenation building will increase the cost of the façade of the new PDF extension 2. Keeping the Hydrogenation building, moves the new PDF extension - to enable construction & for safety	£££
Existing Hydrogenation Building to Remain - New Extension Building to be Reduced in Size																									

3	1. Existing Hydrogenation builing to remain 2. New PDF Extension building to be offset from existing Hydrogenation building - Constructability & safety 3. New & existing central circulation corridors do allign 4. New extension building will be smaller than required	**Yes**	**No**	**Yes**	**No**	**8**	25%	2.00	**10**	20%	2.00	**5**	20%	1.00	**5**	15%	0.75	**3**	20%	0.60	**6.35**	**No**	1. Existing Hydrogenation to remain - No cost required to remove building	1.Keeping the Hydrogenation building will increase the cost of the façade of the new PDF extension - Protection agaist blasts/explosions from the Hydrogenation building 2. Hydrogenation building reduces the size of the new PDF extension	**££**
4	The first floor slab to be raised by 2000mm to enable the installation & operation of the 3No Hastelloy units on the ground floor	NA	**NA**	**Yes**	**Yes**	**8**	25%	2.00	**1**	20%	0.20	**2**	20%	0.40	**8**	15%	1.20	**1**	20%	0.20	**4.00**	**No**	Hastelloy units can be operated	1. Floor slabs between the extsing building and the new extension do not align 2. Additional stairs and lifts are required 3. New goods lift need to be reloacted to the far end of the new extension	**£££**

BREEAM Pre-Assessment

BREEAM UK New Construction V7

Scitech Ekium

Discovery Park
Sandwich
CT13 9FF

Dover District Council

Version	Revision	Date	Author	Reviewer	Project Manager
1	A	12.05.2026	Samantha May	Emily Glyde	Samantha May

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Executive Summary

Executive Summary

SRE exists to ensure that the built environment enhances life without costing the earth. This BREEAM[1] Pre-Assessment has been developed by SRE on behalf of Scitech Ekium (The Client) to demonstrate the likely BREEAM score of Discovery Park, Haig Drive, Sandwich (the Proposed Development).

This Pre-assessment has been based on details supplied by the Client and wider design team to date, in addition to further desk-based study, best practice and historical data. SRE has been a licensed BREEAM organisation for over 10 years and has extensive knowledge of the methodology, which is used to produce an accurate assessment score.

Project Name	Discovery Park
Lead Assessor	Samantha May
Target Rating	'Very Good' (≥ 55%)
BREEAM Version	BREEAM UK New Construction Version 7
Assessment Type	Industrial Unit – Light manufacturing
BREEAM Type	Fully Fitted

Scenario	Min Standards met?	Score	BREEAM Rating
Targeted credits	Yes	**62.93%**	'Very Good'

The potential credits have not been thoroughly investigated at this stage of the BREEAM Assessment. They are to remain potential until the original sought out credits are deemed unachievable by the Project Team and Assessor, and are used to secure the score in case the targeted rating is at risk.

Figure 1 - BREEAM Score Chart for the Proposed Development.

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Assessment Overview

1.0 Assessment Overview

1.1. Development Overview

The Proposed Development – Haig Drive, Discovery Park, Sandwich, CT13 9FF – consists of the new construction of a four-storey extension to accommodate increased capacity for manufacturing capacity and to integrate more modern equipment to create an efficient manufacturing capability. The resulting development will comprise of preparation rooms, plant and machinery rooms and associated works. The Site is situated within an established industrial area and the eastern boundary is defined by a through road with derelict hard landscaping directly to the south of the site.

The Proposed Development will cover a GIA of approximately 2,400m^2. The ground floor plan for the proposed development is shown in Figure 2.

1.2. Assessment Criteria

The BREEAM Pre-Assessment has shown that the Proposed Development can currently achieve an 'Very Good' rating under BREEAM UK New Construction Version 7 (SRE Targeted) based on the assumptions and information agreed to-date.

The BREEAM Assessment will be assessed as an 'Industrial Unit – process/ manufacturing, Fully-Fitted' assessment to reflect the scope of the works. The credits included within the assessment are therefore those relevant to the scope of a 'Fully Fitted' assessment.

1.3. BREEAM Score

The target score for the scheme is currently **62.93%,** which will deliver a **'Very Good'** rating. The credits targeted are considered by SRE and the design team to be realistic and deliverable on-site. The potential credits will be investigated at a later stage if the targeted credits are deemed unachievable.

BREEAM standards can be challenging to achieve, and the Pre-Assessment report should be carefully reviewed by the design and construction teams to ensure all targeted credits are delivered. Sections 2.0 and 3.0 list the specific credits proposed as part of the BREEAM 'Very Good' rating.

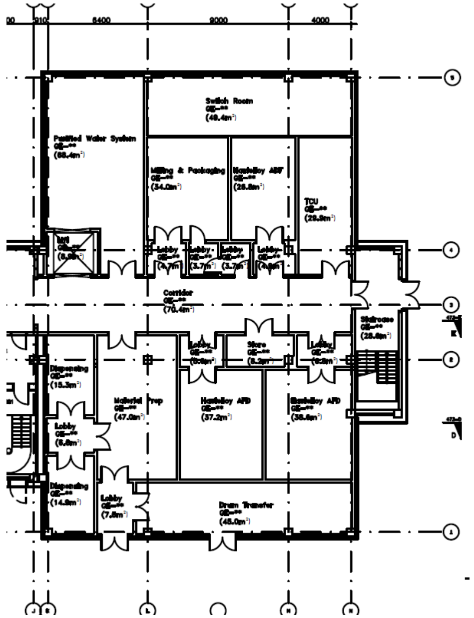

Figure 2 – Floor plan for the Proposed Development.

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1.4. BREEAM Minimum Standards

In addition to the minimum score required, compliance with specific credits (known as minimum standards for each targeted rating) is summarised below. If the required minimum standards are not met, the targeted rating will not be achieved regardless of the overall score attained by the project.

Issue	Targeted
Man 04 - Commissioning and handover	✓
Hea 04 - Indoor air quality	✓
Ene 03 - Energy monitoring	✓
Wat 01 - Water consumption	✓
Wat 02 - Water monitoring	✓
Mat 03 - Responsible sourcing of construction products	✓

Table 1 – BREEAM UK New Construction Version 7 Minimum Standards for a ‘Very Good’ rating

1.5. Site Assumptions

Site and project assumptions made at the current stage are shown in Table 2. These assumptions are to be confirmed by the client and the design team so the BREEAM Design Assessment and achieved score will be updated accordingly.

Assessment Information	Selection
Building Type	Industrial
Building Type sub-group	Process/manufacturing
Does this industrial building have an office area?	No
Assessment stage	Design (Interim)
Project scope	Fully Fitted
Building Net internal floor area m^2	2,400 m^2
Building Gross internal floor area m^2	2,400 m^2
Is the building designed to be untreated?	No
Building services - heating system type	Air system – AHU with steam heating
Building services - cooling system type	Chillers
Safe access: Are there relevant areas relating to this credit?	Yes
Outside space: Are there external areas relating to this credit?	No
Are there any water demands present other than those assessed in Wat 01?	No
Are there statutory requirements, or other issues outside of the control of the project, that impact the ability to provide outdoor space?	No
Is demolition occurring under the developer's ownership for the purpose of enabling the assessed development?	No
Are WC facilities only provided within the residential areas of a long-term residential accommodation?	N/A
Is this a speculative development?	No
Is the project required to connect to a District Heating system, and it supplies all heating and hot water demands to the building?	No
Assessment route for Le 03	Comprehensive route
Does the assessment area have exterior lighting?	Yes
Number of flexible demand response points available (Ene 07)	1-3
Do you want to filter out Ene 08 as per issue specific notes 3.0 and 3.1?	No

Assessment Information	Selection
In-scope unregulated energy use by equipment/regulated energy use (%) (Ene 05)	No in-scope equipment
In-scope unregulated energy using systems / regulated energy use (Ene 06)	No in-scope equipment
In-scope non-sanitary water consumption / sanitary water consumption (%) (Wat 04)	No in-scope equipment

Table 2- Site & Project assumption for BREEAM Assessment

Credit Summary

2.0 Credit Summary

2.1. Summary score sheet

BREEAM Category		Available	Targeted	Potential
Management (0.52 % per Credit)				
Man 01	Project brief and design	4	2	2
Man 02	Life cycle cost and service life planning	4	1	3
Man 03	Responsible construction	6	5	1
Man 03 - Exemplary(1)	Responsible construction	1	0	0
Man 04	Commissioning and handover	4	4	0
Man 05	Aftercare	3	3	0
Man 05 - Exemplary(1)	Aftercare	1	0	0
	Standard Management Credit Total:	**21**	**15**	**6**
	Exemplary Management Credit Total:	**2**	**0**	**0**
	% Management Total (Standard + Exemplary):	**12.92**	**7.80**	**3.12**
Health & Wellbeing (1.25 % per Credit)				
Hea 01	Natural light	3	3	0
Hea 01 - Exemplary(1)	Natural light	1	0	0
Hea 01 - Exemplary(2)	Natural light	1	0	0
Hea 02	Artificial light	2	1	1
Hea 02 - Exemplary(1)	Artificial light	1	0	0
Hea 03	Non-visual effects of light	1	1	0
Hea 04	Indoor air quality	1	1	0
Hea 04 - Exemplary(1)	Indoor air quality	2	2	0
Hea 06	Acoustic performance	3	3	0
Hea 07	Safe and accessible design	1	0	1

BREEAM Category		Available	Targeted	Potential
Hea 08	Security	1	0	1
Hea 08 - Exemplary(1)	Security	1	0	0
	Standard Health & Wellbeing Credit Total:	**12**	**9**	**3**
	Exemplary Health & Wellbeing Credit Total:	**6**	**2**	**0**
	% Health & Wellbeing Total (Standard + Exemplary):	**21.00**	**13.25**	**3.75**
Energy (0.53 % per Credit)				
Ene 01	Energy and carbon performance for regulated energy uses	9	5	2
Ene 01 - Exemplary(1)	Energy and carbon performance for regulated energy uses	3	0	0
Ene 02	Prediction of operational energy and carbon	12	0	12
Ene 02 - Exemplary(1)	Prediction of operational energy and carbon	1	0	0
Ene 02 - Exemplary(2)	Prediction of operational energy and carbon	1	0	0
Ene 02 - Exemplary(3)	Prediction of operational energy and carbon	1	0	0
Ene 02 - Exemplary(4)	Prediction of operational energy and carbon	1	0	0
Ene 03	Energy monitoring	2	1	1
Ene 04	Low carbon design	5	0	2
Ene 06	Energy efficient systems	0	0	0
Ene 07	Flexible demand response	1	0	0
Ene 08	Installed controls	2	1	0
	Standard Energy Credit Total:	**31**	**7**	**17**
	Exemplary Energy Credit Total:	**7**	**0**	**0**
	% Energy Total (Standard + Exemplary):	**23.43**	**3.7**	**9.01**
Transport (0.71 % per Credit)				
Tra 01	Transport assessment and travel plan	2	2	0
Tra 02	Sustainable transport measures	10	0	2
	Standard Transport Credit Total:	**12**	**2**	**2**
	Exemplary Transport Credit Total:	**0**	**0**	**0**

BREEAM Category		Available	Targeted	Potential
	% Transport Total (Standard + Exemplary):	**8.52**	**1.42**	**1.42**
Water (0.83% per Credit)				
Wat 01	Water consumption	5	3	0
Wat 01 - Exemplary(1)	Water consumption	1	0	0
Wat 02	Water monitoring	1	1	0
Wat 03	Water leak detection and prevention	2	2	0
Wat 04	Water efficient equipment and systems	0	0	0
Wat 05	Prediction of operational water consumption	1	0	1
Wat 05 - Exemplary(1)	Prediction of operational water consumption	1	0	0
	Standard Water Credit Total:	**9**	**6**	**1**
	Exemplary Water Credit Total:	**2**	**0**	**0**
	% Water Total (Standard + Exemplary):	**9.47**	**4.98**	**0.83**
Materials (1.14 % per Credit)				
Mat 01	Building life cycle assessment	7	5	2
Mat 01 - Exemplary(1)	Building life cycle assessment	1	0	0
Mat 01 - Exemplary(2)	Building life cycle assessment	1	0	0
Mat 01 - Exemplary(3)	Building life cycle assessment	1	0	0
Mat 02	Environmental product declarations	1	1	0
Mat 03	Responsible sourcing of construction products	4	2	0
Mat 03 - Exemplary(1)	Responsible sourcing of construction products	1	0	0
Mat 04	Durability and resilience	1	1	0
Mat 05	Material efficiency	1	1	0
	Standard Materials Credit Total:	**14**	**10**	**2**
	Exemplary Materials Credit Total:	**4**	**0**	**0**
	% Materials Total (Standard + Exemplary):	**19.96**	**11.40**	**2.28**
Waste (0.78 % per Credit)				

BREEAM Category		Available	Targeted	Potential
Wst 01	Construction waste management	4	4	0
Wst 01 - Exemplary(1)	Construction waste management	1	0	0
Wst 02	Use of recycled and sustainably sourced aggregates	1	0	0
Wst 02 - Exemplary(1)	Use of recycled and sustainably sourced aggregates	1	0	0
Wst 03	Operational waste	1	1	0
Wst 05	Adapting to climate change	1	1	0
Wst 05 - Exemplary(1)	Adapting to climate change	1	0	0
Wst 06	Disassembly and adaptability	2	2	0
	Standard Waste Credit Total:	**9**	**8**	**0**
	Exemplary Waste Credit Total:	**3**	**0**	**0**
	% Waste Total (Standard + Exemplary):	**10.02**	**6.24**	**0**
Land Use & Ecology (0.81 % per Credit)				
Lue 01	Site selection	2	0	0
Lue 02	Ecological risks and opportunities	2	2	0
Lue 02 - Exemplary(1)	Ecological risks and opportunities	1	0	0
Lue 03	Managing impacts on ecology	3	3	0
Lue 04	Ecological change and enhancement	4	2	0
Lue 04 - Exemplary(1)	Ecological change and enhancement	1	0	0
Lue 05	Long term ecological management	2	2	0
	Standard Land Use & Ecology Credit Total:	**13**	**9**	**0**
	Exemplary Land Use & Ecology Credit Total:	**2**	**0**	**0**
	% Land Use & Ecology Total (Standard + Exemplary):	**12.53**	**7.29**	**0**
Pollution (0.62 % per Credit)				
Pol 01	Impact of refrigerants	3	2	0
Pol 02	Local air quality	3	3	0
Pol 03	Flood and surface water management	5	4	0

BREEAM Category		Available	Targeted	Potential
Pol 04	Reduction of nighttime light pollution	1	1	0
Pol 05	Reduction of noise pollution	1	1	0
	Standard Pollution Credit Total:	**13**	**11**	**0**
	Exemplary Pollution Credit Total:	**0**	**0**	**0**
	% Pollution Total (Standard + Exemplary):	**8.06**	**6.82**	**0**
Innovation (1 % per Credit)				
Inn 01	Innovation			
	AI - Innovation Credit Total:	**1**	**0**	**0**
	% Innovation Total (Standard + Exemplary):	**1**	**0**	**0**
	OVERALL TOTALS		**79**	**31**
	% OVERALL SCORE TOTALS		**62.93**	**20.47**

Detailed Pre-Assessment

3.0 Detailed Pre-Assessment

	Credit	Available	Targeted	Potential	Comments
Management (0.52 % per Credit)					
Man 01 - Project brief and design					
1	Project planning	1	1	0	**Project Delivery Planning (1 credit)** **Completed BEFORE end of RIBA Stage 2** **Targeted - Yes** A clear brief is developed prior to completion of the concept design, to include input from all design team members and this information is shared with all relevant parties. This clear briefing document sets out the particular ambitions for the project: • Client requirements, e.g. internal environmental conditions required. • Sustainability objectives and targets, including BREEAM performance targets and any wider business or project sustainability objectives. • Timescales and budget. • List of consultees and professional appointments that may be required, e.g. suitably qualified acoustician (SQA) etc. • Constraints for the project, e.g. technical, legal, physical, environmental. Prior to completion of the concept design, the project team (Client, Design Team and Principal Contractor) meet specifically to discuss, identify and define the roles, responsibilities and contributions for each key phase of the project. When defining responsibilities, the following must be considered: • Building user requirements. • Aims of the design and design strategy. • Particular installation and construction requirements and limitations. • Design and construction risk assessments, e.g. national health and safety regulations or best practice, legionella risk assessment. • Legislative requirements, e.g. local building regulations, heritage requirements. • Sustainability objectives for the project. • Ecological protection measures (where required). • Procurement and supply chain â€“ responsible and ethical sourcing. • Identifying and measuring project success in line with project brief objectives.

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	Credit	Available	Targeted	Potential	Comments
					• Building manager's or occupiers' budget and technical expertise in maintaining any proposed systems. • Maintainability and adaptability of the proposals. • Requirements for the production of project and end user documentation. • Requirements for commissioning, training and aftercare support. The project team demonstrate how the project delivery stakeholders' contributions and their input in the design process have influenced the following: • Initial Project Brief. • Project Execution Plan. • Communication Strategy. • Concept Design. • Final Design. **Comments & Actions** • *Design Team to provide a completed copy of the SRE Man 01 meeting plan, the project execution plan, and the communication strategy.*
2	Stakeholder consultation	1	1	0	**Stakeholder Consultation (1 credit)** **Completed BEFORE end of RIBA Stage 2** **Targeted – Yes** Prior to completion of the Concept Design, the design team consult with all interested parties on matters that covers the minimum consultation content. The project must demonstrate concisely how the interested party contributions and outcomes of the consultation exercise have influenced or changed the initial project brief and concept design; and these changes were implemented within the final design at post-construction. Prior to completion of the detailed design (Technical Design or equivalent), consultation feedback has been given to, and received by, all relevant parties. **Comments & Actions**

	Credit	Available	Targeted	Potential	Comments
					• *Design Team to provide evidence of meetings between third-party stakeholders and how the outcomes of the meetings have influenced the Project Brief and early design options.*
3	BREEAM AP – Concept Design	1	0	1	**BREEAM AP – Concept Design (1 credit)** **Completed BEFORE end of RIBA Stage 2** **Targeted – No (Potential)** A BREEAM AP has been appointed to facilitate the setting and achievement of BREEAM performance targets for the project. Their involvement must be early in the design process. The defined BREEAM performance targets have been formally agreed between the client and design/project team no later than the concept design stage. The BREEAM AP must demonstrate targets have been achieved vis a BREEAM Assessor's design stage assessment report. **Comments & Actions** • *SRE can provide this for an additional fee*
4	BREEAM AP – Monitoring Progress	1	0	1	**BREEAM AP – Monitoring Progress (1 credit)** **Completed BEFORE end of RIBA Stage 4** **Targeted – No (Potential)** The credit for BREEAM AP – Concept Design must be achieved. A BREEAM AP has been appointed to monitor progress against the BREEAM performance targets throughout the design process and formally report progress to the client and design team. The BREEAM AP must attend key project and design team meetings during the concept design, developed design and technical design stages. Reporting must be carried out for each stage, as a minimum. **Comments & Actions** • *SRE can provide this for an additional fee*

	Credit	Available	Targeted	Potential	Comments
Man 02 - Life cycle cost and service life planning					
1	Elemental LCC options appraisal	2	0	2	**Elemental LCC options appraisal (2 credits)** **Completed BEFORE end of RIBA Stage 2** **Targeted – No (Potential)** A competent person carries out an outline, entire asset elemental life cycle cost (LCC) options appraisal (at least two) at the Concept Design stage in line with ISO 15686-5:2017. The elemental LCC plan: a. Provides an indication of future replacement costs over a period of analysis as required by the client (e.g. 20, 30, 50 or 60 years); b. Includes service life, maintenance and operation cost estimates. *The study period should ideally be agreed by the client, in line with the design life expectancy of the building. However, where the life expectancy of the building is not yet formally agreed (due to being at very early design stages), the default design life of 60 years should be used for modelling purposes (in line with the UK default).* Demonstrate, using appropriate examples provided by the design team, how the elemental LCC plan has been used to influence building and systems design and specification to minimise life cycle costs and maximise critical value. **Comments & Actions** • *SRE have been appointed as the suitably qualified individual to perform Elemental LCC calculations before the end of Concept Design (RIBA Stage 2).* • *The ELCC report will be required prior to the end of Concept Design, as well as evidence to confirm how the ELCC has influenced building and systems design and specifications.*
2	Component level LCC options appraisal	1	0	1	**Component level LCC options appraisal (1 credit)** **Completed BEFORE end of RIBA Stage 4** **Targeted – No (Potential)**

Credit	Available	Targeted	Potential	Comments
				A competent person develops a component level LCC options appraisal (at least two) by the end of Technical Design stage in line with ISO 15686-5:2017, and includes the following component types (where present): a. Envelope, e.g. cladding, windows, or roofing b. Services, e.g. heat source, cooling source, or controls, also intelligent lighting controls against standard controls. c. Finishes, e.g. walls, floors or ceilings d. External spaces, e.g. alternative hard landscaping, boundary protection. *The component level LCC option appraisal should review all of the above component types (where present). However, you do not need to consider every single example cited under each component; only a selection of those most likely to draw valued comparisons. This is to ensure that a wide range of options are considered and help focus the analysis on components which would benefit the most from appraisal.* Demonstrate, using appropriate examples provided by the design team, how the component level LCC options appraisal has been used to influence building and systems design and specification to minimise life cycle costs and maximise critical value. Component Level LCC: This is commonly used for cost planning specification choices of systems or component levels during the design development. Component level LCC options appraisal for service life planning requires the environment of the building and other local conditions to be identified and the fundamental requirements to be met in planning the service life of the building. Decisions should be made on: • The likely design of the building (rather than the contractual design itself). • Minimum functional performance criteria for each component over the buildings design life. • Components that must be repairable, maintainable or replaceable within the design of the building. Only the key differentiators between components and systems need to be comparatively modelled. **<u>Comments & Actions</u>** • *Client or Project Team to appoint a competent individual to undertake a component level LCC options appraisal.*

	Credit	Available	Targeted	Potential	Comments
					• *SRE can provide this service for an additional fee, if required.*
3	Capital Cost Reporting	1	1	0	**Capital Cost Reporting (1 credit)** **Targeted – Yes** Report the capital cost for the building in pounds per square meter (£/m^2) of gross internal area (GIA). Written confirmation of the anticipated cost is to be provided during the Design Stage, with final confirmation of the actual cost provided at the end of the project during the Post-Construction Stage assessment. The capital cost for the building includes the expenses related to the initial construction of the building: • Construction, including preparatory works, materials, equipment and labour • Site management • Construction financing • Insurance and taxes during construction • Inspection and testing Costs relating to land procurement, clearance, design, statutory approvals and post occupancy aftercare are not included. **Comments & Actions** • *Quantity Surveyor to confirm that written confirmation of the final capital cost of the project in £k/m^2 will be provided.*
Man 03 - Responsible construction					
Pre-req	Prerequisite Legal and sustainable site timber	✓	✓	✗	**Prerequisite: Legal and sustainable site timber** All temporary site timber and timber-based products (excluding reused timber) used during the construction process of the project must be legal and sustainable timber.
1	Environmental management	1	1	0	**Environmental management (1 credit)** **Targeted – Yes** All parties who manage the construction site at any stage must operate an EMS covering their main operations. The EMS must be third-party certified to ISO 14001, EMAS (EU Eco-Management and Audit Scheme), or an equivalent standard.

	Credit	Available	Targeted	Potential	Comments
					All parties who manage the construction site at any stage must implement best practice pollution prevention policies and procedures on site in order to minimise air and water pollution during construction works. **<u>Comments & Actions</u>** • *Client to ensure that the Main Contractor and Demolition Contractors appointed have ISO 14001/EMAS certification, and the commitment to implementing Pollution Prevention Guideline (PPG6) has been made.*
2	BREEAM Advisory Professional – Site	1	0	1	**BREEAM Advisory Professional - Site (1 credit)** **Targeted – No (Potential)** The client must formally agree with the contractor the performance targets relating to the BREEAM certification that have been set for the project. A BREEAM AP must be involved in the project at an appropriate time and level to ensure that the agreed performance targets are met. Specifically, the AP must: a. Work with the project team, including the client, to help them maximize performance against the agreed targets throughout construction, handover and closeout. This support will include considering links between the criteria that need to be met across all project areas to be assessed by BREEAM (e.g., Management, Energy, Ecology etc), going beyond the design intent wherever possible. b. Monitor progress against the agreed performance targets throughout all stages of the construction, paying particular attention wherever decisions are made that could critically impact the BREEAM performance, and keeping all records needed as evidence for the assessment. c. Proactively identify risks and opportunities related to the procurement and construction processes and the achievement of the targets agreed above. d. Feedback promptly and regularly to the contractor and project team helping the project stay on track to meet its targets and supporting any necessary corrective actions. Where appropriate, coordinate and help the project team to provide the assessor with the necessary evidence to achieve the credits.

	Credit	Available	Targeted	Potential	Comments
					To achieve this credit at the final post-construction stage of assessment, the submitted assessors report must show that a BREEAM AP was appointed and that the agreed performance targets have been demonstrably achieved by the project. **<u>Comments & Actions</u>** • *This credit is not sought at this stage.* • *Main Contractor to ensure that a BREEAM AP is appointed during the construction stage to ensure that performance standards are achieved.* • *SRE can provide this service for an additional fee.*
3	Responsible construction management	2	2	0	**Responsible construction management (2 credits)** **Minimum standard to achieve one credit for 'Excellent' target rating OR two credits for 'Outstanding' target rating** **Targeted – Yes (2 credits)** For small-scale projects: a. The contractor must achieve items below: i. Implement practices to ensure the development footprint is safe, clean and organised at all times. This includes, but is not limited to, facilities, materials and waste storage (f). ii. Provide processes and equipment required to respond to medical emergencies (h). iii. Identify and implement initiatives to promote and maintain the health and wellbeing of all site operatives within the development footprint. This can involve site facilities, site management arrangements, staff policies etc (i). iv. Establish management practices and facilities encouraging equality, fair treatment and respect of all site operatives and visitors (j). v. Minimising nuisance and intrusion on the lives of the local community, and engaging with the community and neighbourhood to ensure that any unavoidable impacts of construction are communicated regularly and sensitively (m)

Credit	Available	Targeted	Potential	Comments
				b. Where the contractor achieves those items checked as required for one credit (for standard projects), the small-scale project may achieve two credits. All buildings not defined as small-scale projects are considered to be standard projects and credits are awarded as follows: a. **1 credit**: All items below achieve (for standard projects) must be achieved for one credit to be awarded**: (Targeted)** i. Manage the construction site entrance to minimise the impacts (e.g. safety, disruption) arising from vehicles approaching and leaving the development footprint (a). ii. Minimise the risks of air, land and water pollution (for EU Taxonomy compliance (d). iii. Minimise the risks of nuisance from vibration, light and noise pollution (e). iv. Implement practices to ensure the development footprint is safe, clean and organised at all times. This includes, but is not limited to, facilities, materials and waste storage (f). v. Ensure clear and safe access in and around the buildings at the point of handover (g). vi. Provide processes and equipment required to respond to medical emergencies (h). vii. Identify and implement initiatives to promote and maintain the health and wellbeing of all site operatives within the development footprint. This can involve site facilities, site management arrangements, staff policies etc (i). viii. Establish management practices and facilities encouraging equality, fair treatment and respect of all site operatives and visitors (j). ix. Minimising nuisance and intrusion on the lives of the local community, and engaging with the community and neighbourhood to ensure that any unavoidable impacts of construction are communicated regularly and sensitively (m) x. All visitor, workforce and community accidents, incidents and near misses are recorded and action is taken to reduce the likelihood of them reoccurring (r). b. **2 credits:** Projects achieving the criterion above PLUS six additional items from below, may be awarded a second credit: i. Ensure the development footprint is accessible for delivery vehicles fitted with safety features (e.g. side under run protection) to remove

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Credit	Available	Targeted	Potential	Comments
				or limit the need for on-street loading or unloading. Where on-street loading is unavoidable, this should be appropriately managed (b). ii. Identify access routes to the development footprint, including for heavy vehicles to minimise traffic disruption and safety risks to others (c). iii. Provide secure, clean and organised facilities (e.g. welfare, changing and storage facilities) for site operatives within the development footprint (k). iv. Minimise risks of the site becoming a focus for antisocial behaviour in the local community (e.g. robust perimeter fencing, CCTV, avoid creating dark corners etc.) (l). v. Ensuring ongoing and up-to-date training is provided for personnel and visitors (covering items a to l, as appropriate.) (n). vi. Ensuring all site operatives are trained for the tasks they are undertaking (including any site-specific considerations) (o). vii. Promoting safety on and off-site by ensuring driving and operator training and safety awareness have been undertaken for any company vehicles (cars and vans), movable plant and equipment (such as dumpers, diggers, cranes), trucks, lorries or waste removal wagons whether owned, leased or hired by the principal contractor (p). viii. Any vehicular accidents, incidents and near misses, on-site or in the vicinity of the site entrance and relating to site activities, are recorded, investigated and analysed and that learnings are incorporated in updated policies and training (q). ix. Processes are in place to facilitate collecting and recording feedback from the community and to address any concerns related to the development footprint (s). If the activities listed above are not relevant for the development, then the project team should provide adequate evidence to the assessor to demonstrate why the project is compliant without these items having been achieved. **<u>Comments & Actions</u>** • *Project team to provide signed letter of commitment demonstrating specifically which responsible management items will be fulfilled.*

	Credit	Available	Targeted	Potential	Comments
4	Site monitoring	2	2	0	**Site monitoring (2 credits)** **Targeted – Yes** Assign responsibility for site monitoring to a named individual. This is a prerequisite for the award of any of the site monitoring credits as set out below. The responsible individual must have the appropriate authority and experience to access the data required. They may be the BREEAM AP, where one has been appointed. They will be required to monitor, record and report energy use, water consumption and transportation data (where measured) resulting from all on-site construction processes (and dedicated off-site manufacturing) throughout the build programme. Report the project value. **One credit – Monitoring utility consumption** *Energy consumption* Set targets at design stage for the site energy consumption in kWh (and where relevant, litres of fuel used) resulting from the use of construction plant, equipment (mobile and fixed) and site accommodation. Monitor and record data for the energy consumption and show that the consumption targets have been achieved. Report the total carbon dioxide emissions (total $kgCO_2e$) and energy consumption (electricity in kWh or litres of fuel) resulting from the construction process. *Water consumption* Set targets at design stage for the potable water consumption (m^3) arising from the use of construction plant, equipment (mobile and fixed) and site accommodation. Monitor and record data for the potable water consumption and show that the consumption targets have been achieved. Use the collated data to report the total net water consumption resulting from the construction process (m^3), i.e. consumption minus any recycled water use. **One credit – Monitoring transport of construction materials and waste** Set targets at design stage and monitor and record data relating to transport movements and other impacts resulting from delivery of the majority of construction

	Credit	Available	Targeted	Potential	Comments
					materials to the site and removal of construction waste from the site. As a minimum this must cover: a. Transport of materials from the point of supply to the building site, including any transport, intermediate storage and point of supply. b. The scope of this monitoring must cover the materials used within the following as a minimum: i. Superstructure ii. Internal finishes iii. Substructure and hard landscaping iv. Core building services c. Transport of construction waste from the construction gate to waste disposal processing or recovery centre gate. Monitoring must cover the construction waste groups outlined in the project's resource management plan. Using the collated data, report separately for materials and waste, the total transport-related carbon dioxide emissions (kgCO„e), plus total distance travelled (km) and amount of fuel used (litres). **Comments & Actions** • *Project Team to ensure the appointed Main Contractor commits to monitoring energy and water (**one credit – Targeted**) and transport of materials on site and transport of waste (**one credit – Targeted**).*
e1	Responsible construction management - Exemplary credit	1	0	0	**Comments & Actions** • *Credit not sought.*
Man 04 - Commissioning and handover					
Pre-req	Minimum standard - Handover	✓	✓		**Prerequisite: Minimum standard to achieve 'Very Good' target rating – Handover** Prepare two training schedules timed appropriately around handover and proposed occupation plans for the following users: a. A non-technical training schedule for the building occupiers. b. A technical training schedule for the premises facilities managers.

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	Credit	Available	Targeted	Potential	Comments
1	Schedule and responsibilities	1	1	0	**Schedule and responsibilities (1 credit)** **Minimum Standard to achieve 'Very Good' target rating** **Targeted – Yes** Prepare a schedule of commissioning and testing. The schedule identifies and includes a suitable timescale for commissioning and re-commissioning of all complex and non-complex building services and control systems and for testing and inspecting building fabric. The schedule identifies the appropriate standards for all commissioning activities to be conducted, where applicable, in accordance with: a. Current Building Regulations . b. BSRIA guidelines(5). c. CIBSE guidelines(6). d. Other appropriate standards (see CN1 on page 68). Any process or manufacturing-related equipment specified as part of the project can be excluded from the assessment, except where it forms an integral part of building HVAC services, such as some heat recovery systems. Where a building management system (BMS) is specified: a. Carry out commissioning of air and water systems when all control devices are installed, wired and functional. b. Include physical measurements of room temperatures, off-coil temperatures and other key parameters, as appropriate, in commissioning results. c. The BMS or controls installation should be running in auto with satisfactory internal conditions prior to handover. d. All BMS schematics and graphics (if BMS is present) are fully installed and functional to user interface prior to handover. e. Fully train the occupier or facilities team in the operation of the system. Appoint an appropriate project team member to monitor and programme pre-commissioning, commissioning and testing. Where necessary include re-commissioning activities on behalf of the client. The principal contractor accounts for the commissioning and testing programme, responsibilities and criteria within their budget and the main programme of works. Allow the required time to complete all commissioning and testing activities prior to handover.

	Credit	Available	Targeted	Potential	Comments
					<u>Comments & Actions</u> • *Project Team to ensure the appointed Main Contractor will account for a commissioning and testing schedule within the project's budget and ensure that these are provided at the post-construction stage.* • *Prepare and deliver a commissioning schedule (where available).*
2	Commissioning role	1	1	0	**Commissioning role (1 credit)** **Targeted – Yes** Achieve the schedule and responsibilities credit. During the design stage, the client or the principal contractor appoints an appropriate project team member, provided they are not involved in the general installation works for the building services systems, with responsibility for: a. Undertaking design reviews and giving advice on suitability for ease of commissioning. b. Providing commissioning management input to construction programming and during installation stages. c. Management of commissioning, performance testing and handover or post-handover stages. For buildings with complex building services and systems, this role needs to be carried out by a specialist commissioning manager. **<u>Comments & Actions</u>** • *Project Team to deliver signed letter confirming appointment of specialist commissioning manager.* • *Provide the relevant documentation confirming the appointed commissioning manager meets the definition for the role. i.e. membership of the Commissioning Specialists Association (CSA).*
3	Envelope testing	1	1	0	**Envelope testing (1 credit)** **Targeted – Yes**

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	Credit	Available	Targeted	Potential	Comments
					Rectify any defects identified during post-construction testing and inspection prior to building handover and close out. Any remedial work must meet the required performance characteristics for the building or element as defined at the design stage. Achieve the schedule and responsibilities credit. Complete post-construction testing and inspection to quality-assure the integrity of the building fabric, including continuity of insulation, avoidance of thermal bridging and air leakage paths (this is through **airtightness testing and a thermographic survey**). A suitably qualified professional undertakes the survey and testing in accordance with the appropriate standard. For EU taxonomy alignment, please see. **Comments & Actions** • *Project team to ensure main contractor commits to produce a thermographic survey and air leakage report at post-construction stage.*
4	Handover	1	1	0	**Handover (1 credit)** **Targeted – Yes** Prior to handover, develop two building user guides for the following users: • A non-technical user guide for distribution to the building occupiers. • A technical user guide for the premises facilities managers. A draft copy is developed and discussed with users first (where the building occupants are known) to ensure the guide is most appropriate and useful to potential users Prepare two training schedules timed appropriately around handover and proposed occupation plans for the following users: • A non-technical training schedule for the building occupiers. • A technical training schedule for the premises facilities managers. **Comments & Actions** • *BUGs and training schedules will be required from the Main Contractor at the Post-construction stage, before handover takes place. The Main Contractor must commit to producing and discussing the BUG with building users, and*

	Credit	Available	Targeted	Potential	Comments
					providing the relevant training to building users in line with the training schedule, prior to handover. • *Draft BUGs and training schedules should be provided at the design stage, where available.* • *SRE can provide this service for an additional fee.*
Man 05 - Aftercare					
1	Aftercare Support	1	1	0	**Aftercare support (1 credits)** **Targeted – Yes** Provide aftercare support to the building occupiers through having in place operational infrastructure and resources. This includes as a minimum: a. A meeting between the aftercare support team or individual, and the building occupier or management team (prior to initial occupation, or as soon as possible thereafter) to: i. Introduce the aftercare support available, including the content of the building user guide (where it exists) and training schedule. ii. Present key information about features of the building including the design intent and how to use the building to ensure it operates as efficiently and effectively as possible. b. On-site facilities management training including: i. a walkabout of the building. **AND** ii. introduction to and familiarisation with the building systems, their controls and how to operate them in accordance with the design intent and operational demands. c. Provide initial aftercare support for at least the first month of building occupation, e.g. weekly attendance on-site, to support building users and management (the level of frequency will depend on the complexity of the building and building operations). d. Provide longer term aftercare support for occupiers for at least the first 12 months from occupation, e.g. a helpline, nominated individual or other appropriate system to support building users and management.

	Credit	Available	Targeted	Potential	Comments
					Establish operational infrastructure and resources to coordinate the collection and monitoring of energy and water consumption data for a minimum of 12 months, once the building is substantially occupied. This facilitates analysis of discrepancies between actual and predicted performance, with a view to adjusting systems and user behaviours accordingly. **<u>Comments & Actions</u>** • *Project Team to ensure the appointed Main Contractor commits to the above.*
2	In-use-commissioning	1	1	0	**In Use Commissioning (1 credit)** **Targeted – Yes** Complete the following commissioning activities over a minimum 12-month period, once the building becomes substantially occupied: a. Complex systems: The specialist commissioning manager will: i. Identify changes made by the owner or operator that might have caused impaired or improved performance. ii. Test all building services under full and part-load load conditions. iii. Where applicable, carry out testing during periods of extreme (high or low) occupancy. iv. Interview building occupants (where they are affected by the complex services) to identify problems or concerns regarding the effectiveness of the systems. v. Compare sub-metered energy performance to the predicted energy performance. vi. Identify inefficiencies and areas in need of improvement. vii. Re-commission systems (following any work needed to serve revised loads) and incorporate any revisions in operating procedures into the operations and maintenance (O&M) manuals. viii. b. Simple systems: The external consultant, aftercare team, or facilities manager will: i. Review thermal comfort, ventilation, and lighting, at three, six and nine month intervals after initial occupation, either by measurement or occupant feedback. ii. Identify deficiencies and areas in need of improvement. iii. Re-commission systems and incorporate any relevant revisions in operating procedures into the O&M manuals.

	Credit	Available	Targeted	Potential	Comments
					Comments & Actions • *Project Team to ensure the appointed Main Contractor commits to the above.*
3	Post-occupancy evaluation	1	1	0	**Post Occupancy Evaluation (POE) (1 credit)** **Targeted – Yes** Complete to carry out a post-occupancy evaluation (POE) exercise one year after the building is substantially occupied which gains comprehensive in-use performance feedback and identifies gaps between design intent and in-use performance. The aim is to highlight any improvements or interventions that need to be made and to inform operational processes. An independent party should carry out the POE to cover: a. A review of the design intent and construction process (review of design, procurement, construction and handover processes). b. Feedback from a wide range of building users including facilities management on the design and environmental conditions of the building covering: i. Internal environmental conditions (light, noise, temperature, air quality). ii. Control, operation and maintenance. iii. Facilities and amenities. iv. Access and layout. v. Energy and water consumption (see criterion 2 and M1). vi. Other relevant issues, where appropriate. The independent party provides a report with lessons learned to the client and building occupiers. The client or building occupier commits funds to pay for the POE in advance. This requires an independent party to be appointed to carry out the POE as described above. **Comments & Actions** • *Project Team to ensure the appointed Main Contractor commits to the above.*

	Credit	Available	Targeted	Potential	Comments
e1	Zero carbon transition planning	1	0	0	**Comments & Actions** • *Credit not sought.*
		21	15	6	**Standard Management Credit Total**
		2	0	0	**Exemplary Management Credit Total**
		12.92	7.80	3.12	**% Management Total (Standard + Exemplary)**
Health & Wellbeing (1.25 % per Credit)					
Hea 01 - Natural light					
Pre-req	Minimum standard - Natural light	✔	✔		**Prerequisite: Minimum standard for 'Outstanding' target rating - Natural light** Achieve one of the following: a. At least 95% of all *relevant building areas* for daylight, have vertical and/or inclined windows with a glazed area 20% of the surrounding wall area. OR b. At least 80% of *relevant building areas* for daylight have a glazed area as a percentage of floor area 7%. **Comments & Action** It is assumed the glazing will meet requirements.
1	Glare Control	1	1	0	**Glare control – credit targeted by default as there are no relevant areas** Identify areas at risk of glare using a glare control assessment. The glare control assessment also justifies any areas deemed not at risk of glare. Where risk has been identified within a **relevant building area**, a glare control strategy is used to design out the potential for glare. The glare control strategy does not increase energy consumption used for lighting. This is achieved by:

	Credit	Available	Targeted	Potential	Comments
					a. Maximising daylight levels in all weather (cloudy or sunny). AND b. Ensuring the use or location of shading does not conflict with the operation of lighting control systems. The **relevant building areas** are defined as the below: Glare control criteria apply to areas of the building where resultant glare could be problematic for users. This includes, but not exclusively: • Areas that have been designed to contain or use workstations or projector screens (such as offices, study bedrooms, libraries, facility management offices and reception desks). • Areas where people must spend time in fixed locations and cannot change their viewing direction (such as classrooms and hospital wards). • Sports halls. • Areas where occupants are expected to spend a significant amount of time in the day (such as care home bedrooms). **<u>Comments & Action</u>** It is assumed the required glare control will not be required as there are no relevant building areas.
2	Daylighting	1	1	0	**Daylighting (1 credit)** **Targeted – Yes** Daylighting criteria have been met for Industrial developments using either one of the options outlined below: **OPTION 1** • The *relevant building areas meet good practice average and minimum point daylight illuminance criteria as outlined below:

	Credit	Available	Targeted	Potential	Comments
					- Internal association or atrium aria: 80% of the area achieves an average daylight illuminance of at least 300 lux for 2650 hours per year or more **AND** achieves minimum point daylight illuminance of at least 210 lux for 2650 hours per year or more. - Teaching, lecture and seminar spaces: 80% of the area achieves an average daylight illuminance of at least 300 lux for 2000 hours per year or more **AND** achieves minimum point daylight illuminance of at least 90 lux for 2000 hours per year or more. - All occupied spaces: 80% of the area achieves an average daylight illuminance of at least 300 lux for 2000 hours per year or more **AND** achieves minimum point daylight illuminance of at least 90 lux for 2000 hours per year or more. **OPTION 2** • The relevant building areas meet any level of recommendation using the daylight illuminance criteria in EN 17037. **OPTION 3** • The relevant building areas achieve at least 'nominally accepted' daylight sufficiency as per IES LM-83. *Relevant building area – Areas within the building where good daylighting is of benefit to the building users, if it is an occupied space (i.e a room or space within the assessed building that is likely to be occupied for 30 minutes or more by a building user). **Comments & Actions** *SRE can perform the daylight modelling (ideally before planning) to confirm the occupied spaces of the development achieve the above daylighting levels[EG1].*
4	View out	1	1	0	**Glare control – credit targeted by default as there are no relevant areas** Relevant areas are spaces where close work in a fixed position is carried out for sustained periods of time. The view out criteria is therefore not applicable to occupied areas such as meeting rooms, or other spaces where such close work is not being carried out. BREEAM defines relevant building areas requiring a view out to include spaces where close work in a fixed position is carried out for sustained periods of time: 1. There are or will be workstations or benches or desks for building users. 2. Close work will be undertaken or visual aids will be used.

sre

Credit	Available	Targeted	Potential	Comments
				3. Occupants spend a significant amount of time, and a view out is deemed to be of benefit to the building occupants, this includes: a. Bedrooms (Residential care home) b. Living rooms (Residential institution – long-term stay, and Residential) Excluded areas for view out might include: • Conference rooms, meeting rooms, lecture theatres, sports halls, acute SEN rooms, kitchens. • Any spaces where the exclusion or limitation of natural light is a functional requirement (such as laboratories or media spaces). • Individual booths or enclosed workstations that are intended for temporary non-permanent work. • Bedrooms where occupants mainly use the space for sleeping, such as in hospitality, residential and residential institutions (long- and short-term stay) and bedrooms that are not specified above and in building type specific note 2.1. If the scope of the project changes, these are the requirements (for information): At least 95% of the floor area in 95% of spaces for each relevant building area are within the distances of a window or permanent opening that provides an adequate view out in line with the below:

Distance from window to workspace or desk, X (in metres)	**Window or opening size (as % of surrounding wall area)**
X < 8	20%
8 ≤ X ≤ 11	25%
11 < X ≤ 14	30%
X > 14	35%

OR

At least 75% of the floor area in 95% of spaces for each relevant building area can meet both of the following criteria:

a. Horizontal sight angle 28°

	Credit	Available	Targeted	Potential	Comments
					AND b. The landscape layer and at least one additional layer (sky or ground) are visible through a permanent opening. **<u>Comments & Actions</u>** • *Architect and SRE to confirm whether the above criteria are achievable for the Proposed Development.* **<u>Comments & Action</u>** It is assumed there are no relevant building areas and therefore this credit is met by default.
e1	Evaluating glare from daylight – Exemplary credit	1	0	0	**Glare control - Exemplary credit (1 credit)** The above credit for 'Glare control' should be achieved. An evaluation should be undertaken to demonstrate compliance with either of the following and M6 from the BREEAM manual: a. The Daylight Glare Probability (DGP) does not exceed 0.40 over more than 5% of annual operating hours at all workstations in all relevant building areas. OR b. Annual Sunlight Exposure ASE1000, 250 ≤ 10% of an occupied space. This means no more than 10% of the occupied space in all relevant building areas can receive a daylight illuminance of ≥ 1000 lux for ≥ 250 hours per year. **<u>Comments & Actions</u>** • *Credit not sought.*
e2	Daylighting - Exemplary credit	1	0	0	**Daylighting - Exemplary credit (1 credit)** **Targeted – No (Potential)** Daylighting criteria have been met using EITHER one of the following options:

	Credit	Available	Targeted	Potential	Comments
					OPTION 1 a. Relevant building areas meet exemplary criteria for average and minimum point daylight illuminance as outlined below: **i.** <u>All occupied spaces</u>: 80% of the area achieves an average daylight illuminance of at least 300 lux for 2650 hours per year or more **AND** achieves minimum point daylight illuminance of at least 90 lux for 2650 hours per year or more. **OPTION 2** b. Relevant building areas meet the highest rating in EN 17037:2018+A1:2021 OR IES LM-83-23 and the relevant criteria outlined above (a.i): i. Highest rating in standard is met. **<u>Comments & Actions</u>** • *This credit is not sought at this stage due to high benchmarks.*
Hea 02 - Artificial light					
Pre-req	Prerequisite Flicker	✓	✓		**Prerequisite: Flicker** All lighting types should be designed to avoid flicker and stroboscopic effects, including across their full dimming and colour tuning range if applicable. This is applicable to all internal and external lighting.
1	Internal and external lighting levels	1	1	0	**Internal and external lighting levels (1 credit)** **Targeted – Yes** *Internal lighting* Internal lighting in all relevant areas of the building is designed to provide illuminance (lux) levels and colouring rendering index in accordance with the SLL Code for Lighting 2022(24) and any other relevant industry standard. Internal lighting should be appropriate to the tasks undertaken, accounting for building user concentration and comfort levels. Compliance criteria should include, as applicable for each type of application: a. Illuminance levels and uniformity b. Glare limits c. Colour appearance (where standard recommendations exist)

sre

	Credit	Available	Targeted	Potential	Comments
					d. Colour rendering For areas where computer screens are regularly used, the lighting design complies with CIBSE Lighting Guide 7 Sections 2.5, 2.15 to 2.17, 2.22, and 6.10 to 6.20.(25) This gives recommendations highlighting: a. Limits to the luminance of the luminaires to avoid screen reflections. (Manufacturers data for the luminaires should be sought to confirm this.) b. Any area where a surface is used to reflect light in to a space, such as uplighting, the recommendations refer to the luminance of the lit ceiling rather than the luminaire; a design team calculation is usually required to demonstrate this. c. Recommendations for direct lighting, ceiling illuminance, and average wall illuminance. *External lighting* All external lighting located within the construction zone is specified in accordance with BS 5489-1:2020(26) and BS EN 12464-2:2024(27). External lighting should provide illuminance levels that enable users to perform outdoor visual tasks efficiently and accurately, especially during the night. Compliance criteria should include, as applicable for each type of application: a. Illuminance / luminance levels and uniformity b. Glare limits c. Colour rendering **Comments & Actions** • *Project Team building services engineer or lighting designer to confirm internal lighting is designed in accordance with SLL code for lighting, and all external lighting is designed to comply with BS 5489-1:2020(26) and BS EN 12464-2:2024(27).* • *Deliver design drawings and lighting schedules in line with the standards.*
2	Zoning and occupant control	1	0	1	**Zoning and control (1 credit)** **Targeted – No (Potential)**

sre

	Credit	Available	Targeted	Potential	Comments
					Internal lighting is zoned to allow for occupant control in accordance with the criteria below for relevant areas present within the building, as applicable: a. Workstations adjacent to windows or atria and other building areas are separately zoned and controlled b. In office areas where furniture layout is known, zones of no more than four workplaces c. Seminar and lecture rooms: zoned for presentation and audience areas d. Library spaces: separate zoning of stacks, reading and counter areas e. Teaching space or demonstration area f. Whiteboard or display screen g. Auditoria: zoning of seating areas, circulation space and lectern area h. Dining, restaurant, café areas: separate zoning of servery and seating or dining areas i. Retail: separate zoning of display and counter areas j. Bar areas: separate zoning of bar and seating areas k. Wards or bedded areas: zoned lighting control for individual bed spaces and control for staff over groups of bed spaces l. Treatment areas, dayrooms, waiting areas: zoning of seating and activity areas and circulation space with controls accessible to staff. Where automatic lighting controls are used for energy efficiency reasons, manual override must be provided. **Comments & Actions** • *Project Team building services engineer or lighting designer to confirm internal lighting is designed in accordance with the above – zoned appropriately[EG2].*
e1	Lighting control – Exemplary credit	1	0	0	Lighting in each control zone with an occupied space can be manually dimmed by occupants down to no more than 20% of the maximum light output using lighting control interfaces positioned in accessible locations. Lighting in each control zone within an occupied space can be colour tuned over a correlated colour temperature range from 2700K or less to 4000K or more, with automatic controls set to limit the correlated colour temperature in the afternoon and evening to 2700K or less. **Comments & Actions**

sre

	Credit	Available	Targeted	Potential	Comments
					• *Credit not sought.*
Hea 03 Non-visual effects of light					
1	Non-visual effects of light	1	1	1	**Non-visual effects of light – not applicable as this building does not have any relevant areas. This credit is awarded by default.** If it becomes relevant, the criteria are: Achieve at least one credit for 'Daylighting' in Hea 01 Natural light. All relevant areas achieve one of the below, as applicable: a. Where occupant positions are known and fixed (see M1.1): All positions in all relevant building areas receive a vertical illuminance of at least 250 lux melanopic EDI at occupant eyes for at least four hours during the daytime (beginning by noon at the latest). b. Where occupant positions are unknown or variable (see M1.2): Cylindrical illuminance at eye height across each relevant building area corresponds to at least 250 lux melanopic EDI for at least four hours during the daytime (beginning by noon at the latest). Electric lighting systems and controls are in place to adjust the flux and spectral output of the emitted light over time (colour tuning) so that melanopic EDI at occupant eye level can be reduced in the afternoon and evening, as relevant for each type of application. **Comments & Actions** • *Credit is awarded by default as there are no relevant areas.*
Hea 04 - Indoor air quality					
1	Indoor air quality plan	1	1	0	**Indoor air quality plan – not applicable as this building does not have an office or other occupied space, therefore the credit is awarded by default.** If the credit becomes applicable (addition of occupied areas), the below criteria will apply: **Minimum standard to achieve 'Very Good' target rating** **Targeted – Yes**

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<table>
<tr><th></th><th>Credit</th><th>Available</th><th>Targeted</th><th>Potential</th><th>Comments</th></tr>
<tr><td></td><td></td><td></td><td></td><td></td><td>A site-specific indoor air quality plan has been produced in accordance with Guidance Note 6 (GN06). This plan informs the design, specification, and installation to maximise improvement of indoor air quality during occupation. The plan must consider the following:

a. Removal of highly contaminating sources
b. Dilution and control of other contaminant sources, including:
i. Air quality requirements of specialist areas such as laboratories, where present
c. Procedures for pre-occupancy flush out and purge ventilation
d. Third party testing and analysis
e. Maintaining indoor air quality in-use
f. Any relevant local or national authority plans or policies (for example, Air Quality Management Areas or Local Air Quality Action Plans)

<u>Comments & Actions</u>

• SRE can provide a compliant Indoor Air Quality Plan (IAQP).</td></tr>
<tr><td>2</td><td>Ventilation</td><td>0</td><td>0</td><td>0</td><td>Ventilation – not applicable as this building does not have an office or other occupied space.

Not applicable
The building has been designed to minimise the concentration and recirculation of pollutants in the building as follows:
Provide fresh air into the building in accordance with one of the best practice standards for ventilation:

<u>Clinical areas with controlled environmental conditions</u>
a. BS ISO 17772-1:2017 (excluding naturally ventilated buildings), whichever requires the higher performance.

<u>All other buildings (mixed mode or mechanical ventilation)</u>
a. BS ISO 17772-1:2017 Annex I.

<u>All other buildings (naturally ventilated)</u>
a. CIBSE AM10.

Ventilation pathways are designed to minimise the ingress and build-up of air pollutants inside the building:</td></tr>
</table>

Credit	Available	Targeted	Potential	Comments
				Mechanically ventilated The design of air-conditioned and mixed mode buildings must minimise the build-up of air pollutants. Locations of ventilation intakes and airflow pathways should be designed in accordance with one or a combination of the following methods: **METHOD 1** 1. Locating the building's air intakes and exhausts, in relation to each other and sources of external pollution, in accordance with the following best practice as appropriate: a. PDCEN/TR 16798-4:2017 Sections 8.8.1 to 8.8.4 b. b. BRE FB 30 Ventilation for healthy buildings: Reducing the impact of urban air pollution (2011) c. c. BRE IP 9/14 Locating ventilation inlets to reduce ingress of external pollutants into buildings, as appropriate d. CIBSE TM21 **METHOD 2** 2. Pollutant dispersion modelling can be used to inform the location of the building's air intakes and exhausts in relation to each other and sources of external pollution. This can be achieved using either **wind tunnel modelling** or **numerical modelling**. Undertaken by a competent person. **METHOD 3** 3. Positioning the building's air intakes and exhausts at least 10 m of horizontal distance apart. Positioning intakes at least 10 m horizontal distance from sources of external pollution (including the location of air exhausts from other buildings). There is no minimum vertical distance set in this approach. The building's air intakes and exhausts should be located to reduce the risk of the intake air being contaminated by the exhausts. Exhausts or other pollutant sources should not be discharged into enclosed spaces, such as courtyards, in which intakes are also located. Where significant levels of gaseous pollutants such as nitrogen dioxide are identified in the outdoor air, as in an Air Quality Management Area (AQMA), the use of appropriate gas phase filtration in the building ventilation system should be considered. Design teams must ensure that filter performance is appropriate for the pollutant conditions experienced at the site

sre

	Credit	Available	Targeted	Potential	Comments
					Where present, HVAC systems must incorporate suitable filtration to minimise external air pollution, as defined in BS EN 16798-3:2025 Section B.4.2. The specified filters should achieve supply air classification of at least SUP 2. The level of filtration required is dependent on the quality of the outdoor air. a. In mechanically ventilated buildings or spaces: sensors are linked to the mechanical ventilation system and provide demand-controlled ventilation to the space **(Assumed).** b. In naturally ventilated buildings or spaces: sensors either have the ability to alert the building owner or manager when CO2 levels exceed the recommended set point or are linked to controls with the ability to adjust the quantity of fresh air, i.e. automatic opening windows or roof vents. c. The total number of sensors, and the net internal area of relevant areas covered by the sensors, is reported via the BREEAM Platform. The ventilation strategy provides adequate ventilation rates throughout the year, including sufficient airflow rates in summer to prevent overheating and maintain required thermal comfort conditions, in accordance with: a. CIBSE AM10 (for naturally ventilated buildings) b. CIBSE AM13 (for mixed-mode buildings) **(Assumed)** **<u>Comments & Actions</u>** • *This credit is not sought at this stage.* • *Project team/MEP to confirm the ventilation system specified for the Proposed Development and that the above criteria will be met.*
3	Emissions from construction products	2	2	0	**Emissions from construction products** **Targeted – Yes** Three of the five product types meet the emission limits, testing requirements and any additional requirements listed by the BRE: • Interior paints and coatings 1. Formaldehyde: ≤ 0.06 mg/m³ 2. TVOC: ≤ 1.0 mg/m³ 3. Category 1A and 1B carcinogens: ≤ 0.001 mg/m³ • Wood-based products (including wood flooring)

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	Credit	Available	Targeted	Potential	Comments
					1. Formaldehyde: ≤ 0.06 mg/m³ (Non-MDF) / ≤ 0.06 mg/m³ (MDF) 2. TVOC: ≤ 1.0 mg/m³ 3. Category 1A and 1B carcinogens: ≤ 0.001 mg/m³ • Flooring materials (including floor levelling compounds and resin flooring) 1. Formaldehyde: ≤ 0.06 mg/m³ (Non-MDF) / ≤ 0.06 mg/m³ (MDF) 2. TVOC: ≤ 1.0 mg/m³ 3. Category 1A and 1B carcinogens: ≤ 0.001 mg/m³ • Ceiling materials, wall materials, acoustic insulation and thermal insulation materials 1. Formaldehyde: ≤ 0.06 mg/m³ (Non-MDF) / ≤ 0.06 mg/m³ (MDF) 2. TVOC: ≤ 1.0 mg/m³ 3. Category 1A and 1B carcinogens: ≤ 0.001 mg/m³ • Interior adhesives and sealants (including flooring adhesives) 1. Formaldehyde: ≤ 0.06 mg/m³ (Non-MDF) / ≤ 0.06 mg/m³ (MDF) 2. TVOC: ≤ 1.0 mg/m³ **3.** Category 1A and 1B carcinogens: ≤ 0.001 mg/m³
Hea 05 Thermal Comfort					
1	Thermal comfort	0	0	0	**Industrial (without offices)** Where an industrial unit contains no office space and only an operational or storage area, this BREEAM issue does not apply. **Comments & Actions** • Credit not applicable.
Hea 06 - Acoustic performance					
1	Acoustic performance	3	3	0	**Acoustic performance** **Targeted – Yes** **Pre-requisite** A suitably qualified acoustician (SQA) is appointed to by the client at an appropriate stage to provide early advice on a. External sources of noise impacting the chosen site. b. Site layout and zoning of the building for good acoustics.

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Credit	Available	Targeted	Potential	Comments
				c. Acoustic requirements for users with special hearing and communication needs. d. Acoustic treatment of different zones and façades. The building meets the appropriate acoustic performance standards and testing requirements for the acoustic principles of: (See 'Acoustic criteria' below for further details) a. Sound insulation b. Indoor ambient noise level c. Reverberation times – not applicable as there are no areas 'used for speech' <u>Indoor ambient noise levels:</u> Achieve indoor ambient noise levels that comply with national building regulations or 'good practice' levels outlined by the BRE. Testing: A programme of pre-completion acoustic testing is carried out by a compliant test body. <u>Sound Insulation:</u> Achieve sound insulation levels that comply with national building regulations or 'good practice' levels outlined by the BRE. Testing: A programme of pre-completion acoustic testing is carried out by a compliant test body. <u>Reverberation times – not applicable where there are no areas used for speech:</u> Achieve the requirements relating to reverberation times that comply with national building regulations or 'good practice' levels outlined by the BRE. Testing: A programme of pre-completion acoustic testing is carried out by a compliant test body. **<u>Comments & Actions</u>** • *A suitably qualified acoustician is appointed to undertake an acoustics assessment to show the indoor ambient noise level, sound insulation and*

	Credit	Available	Targeted	Potential	Comments
					reverberation times (where applicable) are in accordance with the relevant guidelines.
Hea 07 - Safe and accessible design					
1	Safe access	1	0	1	**Safe access (1 credit)** **Targeted – No (potential)** Where external site areas form part of the assessed development the following apply: Dedicated and safe cycle paths are provided from the site entrance to any cycle storage, and connect to off-site cycle paths where applicable. Dedicated and safe footpaths are provided on and around the site providing suitable links for the following: a. The site entrance to the building entrance b. Car parks (where present) to the building entrance c. The building to outdoor space d. Connecting to off-site paths where applicable. Pedestrian drop-off areas are designed off, or adjoining to, the access road and should provide direct access to other footpaths. Where vehicle delivery access and drop-off areas form part of the assessed development, the following apply: Delivery areas are not accessed through general parking areas and do not cross or share the following: a. Pedestrian and cyclist paths b. Outside amenity areas accessible to building users and general public. There is a dedicated parking or waiting area for goods vehicles with appropriate separation from the manoeuvring area and staff and visitor car parking. Parking and turning areas are designed for simple manoeuvring according to the type of delivery vehicle likely to access the site, thus avoiding the need for repeated shunting.

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	Credit	Available	Targeted	Potential	Comments
					Comments & Actions • *Project Team and Architect to deliver design drawings highlighting safe pedestrian and cycle routes, and other compliant features.*
2	Outside space	0	0	0	Where it is not possible to provide outdoor space, due to statutory requirements, or other issues outside of the control of the developer, then these criteria will be filtered out. **Comments & Actions** • *Credit not applicable*
Hea 08 - Security					
1	Security of site and building	1	0	1	**Security of site and building (1 credit)** **Completed BEFORE end of RIBA Stage 2** **Targeted – No (Potential)** **Assessment type specific note – For speculative developments, if the SQSS cannot make complete recommendations due to the speculative nature of the assessment, the credit is still achievable. The SQSS must confirm that they have addressed all parts of the project where feasible based on the information available to them at the time of the assessment. Regarding the occupiers' influence on security, the SQSS shall document their assumptions in the SNA.** A suitably qualified security specialist (SQSS) must conduct an evidence-based security needs assessment (SNA) during or before Concept Design (RIBA Stage 2). The purpose of the SNA is to identify any security-relevant attributes of the proposed development, site context, and surrounding environment that may influence the security strategy. Based on the SNA findings, the SQSS must develop a set of security controls and design recommendations. These must directly address the specific threats and assets identified in the SNA. The agreed security controls and recommendations shall be incorporated into proposals and implemented in the as-built development. Any deviation from those controls and recommendations shall be justified and agreed with the SQSS. **Comments & Actions**

	Credit	Available	Targeted	Potential	Comments
					• *This credit is a potential at this stage until further information is reviewed.*
e1	Security rating scheme - Exemplary credit	1	0	0	**Security rating scheme - Exemplary credit (1 credit)** **Targeted – No (Potential)** A compliant risk-based security rating scheme has been used. The performance against the scheme has been confirmed by independent assessment and verification. *The Security of site and building credit must first be achieved for the exemplary credit to be awarded.* **Comments & Actions** • *This credit is not sought at this stage.* • *Project Team to confirm whether a compliant security rating scheme has been used.* • *Provide a copy of the security rating scheme certificate from design stage.*
		12	9	3	**Standard Health & Wellbeing Credit Total**
		6	0	2	**Exemplary Health & Wellbeing Credit Total**
		21.00	13.25	3.75	**% Health & Wellbeing Total (Standard + Exemplary)**
Energy (0.53 % per Credit)					
Ene 01 - Energy and carbon performance for regulated energy uses					
1	Calculated energy performance	9	5	2	**Calculated energy performance (9 credits)** **Minimum Standard to achieve 'Excellent' target rating the Energy Performance Ratio (EPR_{NC}) must be ≥ 0.4** **Targeted – Yes (5 credits, 2 potential)** Calculate the overall Energy Performance Ratio (EPR_{NC}). Must achieve ≥ 0.4 EPR_{NC}. BREEAM uses the following three metrics to determine the overall energy performance improvement: a. Demand – the building's demand for heating and cooling per unit floor area (kWh/m² per year)

	Credit	Available	Targeted	Potential	Comments
					b. Primary energy – energy consumption measured in primary energy per unit floor area (kWh/m² per year) c. Carbon emissions – carbon emissions arising from energy use per unit floor area (kgCO_2e/m² per year) Generate a BRUKL.inp file at the design stage, using an approved energy calculation software: - For Scotland: Non-domestic Technical Handbook 2023 - Section 6, Energy. **Comments & Actions** • *SBEM energy modelling is to be undertaken by an energy specialist, to deliver the BRUKL file to determine the credits.* • *SRE can provide this service for an additional fee, if required.*
e1	Beyond zero net regulated carbon - Exemplary credit	3	0	0	**Comments & Actions** • *This credit is not sought due to high benchmarks.*
Ene 02 - Prediction of operational energy and carbon					
Pre-req	Minimum standard - No fossil fuels on-site	✓	✓		**Prerequisite: Minimum standard - No fossil fuels on-site** No fossil fuels will be used by the building in operation.
1	Predictive energy modelling process	5	0	5	**Predictive energy modelling process (5 credits)** **Targeted – No (potential)** *One credit – Energy strategy* **Completed BEFORE the end of RIBA Stage 2** Define an energy strategy for the building at the concept design stage that describes qualitatively how the operational energy and carbon performance of the building will be minimised. *Three credits – Predictive energy modelling* **Completed BEFORE the end of RIBA Stage 4** Estimate unregulated energy uses and occupancy patterns for the building:

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Credit	Available	Targeted	Potential	Comments
				Significant energy end uses that will be present in the building must be identified. As a minimum the following end uses should be considered: a. Energy use for lighting not covered by building regulations b. Energy use for lifts and escalators c. Energy use for small power d. Energy use for catering e. Energy use for server rooms f. Energy use for domestic hot water g. Energy use for any other types of unregulated plant and equipment – e.g., commercial refrigeration, medical equipment etc. h. Energy use for industrial process load Construct a detailed energy model during the technical design stage and undertake sensitivity analysis and scenario modelling to explore the impact of key parameters and assumptions on energy consumption: The following operational assumptions should be considered when selecting input variables for the sensitivity analysis: a. The efficiency of major energy consuming components b. Alternative system configurations and control strategies c. Occupancy density and hours of use d. Management and occupant behaviour e. Current and future weather scenarios The results of the sensitivity analysis should be used to inform the development of scenarios that reflect the likely energy performance range for the building. The scenarios must include: a. A central case scenario which represents the most reasonable assumptions about occupancy, fabric and building system efficiencies and control strategies. b. A high energy consumption scenario based on uncertainties relating to key input parameters which result in higher energy consumption. c. A low energy consumption scenario based on uncertainties relating to key input parameters which result in lower energy consumption.

Credit	Available	Targeted	Potential	Comments
				Predict annual operational energy consumption in use by fuel type for the building based on the scenario modelling. The modelling must be updated at the post-construction stage to account for any changes to the building specification: The predicted annual operational energy consumption for the building must be based on the scenario modelling results and be higher than that for central case scenario. How high the predicted energy performance should be set above the central case scenario should be informed by the range of the scenario modelling results, the extent to which risks of underperforming in use have been mitigated, and previous experience with other similar projects. An uncertainty margin of 25% over the central case scenario should be used as a default unless a lower modelling margin is recommended by a suitably qualified energy modeller in which case the margin used must be justified in the energy modelling report. An energy modelling report must be produced covering the following: a. The approach to modelling the building fabric, HVAC systems, and control strategies b. Sources of the information for the assumptions and inputs about occupancy patterns and unregulated energy loads c. The levels of uncertainty that can be ascribed to the key assumptions and inputs d. The selection of input parameters for the sensitivity analysis and scenario modelling e. The energy modelling margin and any justification. *One credit – Risk assessment and mitigation* Achieve the predictive energy modelling credits. Highlight any significant design, technical, and process risks that might prevent the buildings achieving energy performance target and indicate how these risks will be mitigated throughout the construction and commissioning process.

Credit	Available	Targeted	Potential	Comments
				The risk assessment should consider situations where the design assumptions and operational situation could have a significant negative impact on energy consumption in use. Risks to be considered (where relevant) include: a. Product specification and the management of value engineering decisions b. Construction quality c. System installation considerations, particularly for complex or novel systems d. Appropriate commissioning of servicing systems e. Ensuring that end users understand how to operate the building as intended. Options for mitigating these risks include: a. Product specification and the management of value engineering decisions –Contractual requirements relating to changes to the design specification. b. Construction quality – Achieving the Testing and inspecting building fabric credit in Man 04. c. System installation considerations, particularly for complex or novel systems – The use of experienced installers and contractors. d. Appropriate commissioning of servicing systems – Achieving the Commissioning building services credit in Man 04 and/or the In-use commissioning credit in Man 05. e. Ensuring that end users understand how to operate the building as intended – Achieving the Handover credit in Man 04. **<u>Comments & Actions</u>** • *Project Team to deliver an energy strategy for the building.* • *Project Team to appoint a suitably qualified energy specialist to undertake Predictive Energy modelling in line with CIBSE TM 54.* • *The energy modelling report must include a risk assessment.*

	Credit	Available	Targeted	Potential	Comments
2	Predicted operational energy performance score	7	0	7	**Predicted operational energy performance score (7 credits)** **Minimum Standard to achieve 'Excellent' target rating the Operational Energy Performance score (OEP_{NC}) must be ≥ 0.4286** **Targeted – No (potential)** Achieve the predictive energy modelling credits. Credits are awarded based on the Operational Energy Performance score for New Construction (OEP_{NC}). **Comments & Actions** • *Project Team and energy specialist to deliver the predictive energy modelling report to determine the credit.* • *SRE can provide this service for an additional fee, if required.*
e1	Disclosing an in-use energy performance target - Exemplary credit	1	0	0	**Disclosing an in-use energy performance target - Exemplary credit (1 credit)** **Completed BEFORE practical completion on site** **Targeted – No** An in-use energy performance target has been disclosed to stakeholders and been made publicly available prior to practical completion. The energy target should be expressed as an annual energy intensity normalised by floor area. The scope of energy uses included in the target must be explicitly stated, for example, whole building or core building services and common areas only. Where a target is set in units other than delivered energy (for example, carbon emissions or primary energy), factors to convert these to delivered energy must be provided. In all cases the floor metric must be defined (for example, GIA or NIA). This information must be made available to building stakeholders (for example, prospective owners and occupiers of the building) and be accessible to members of the public. **Comments & Actions** • *This credit is not sought at this stage.*

	Credit	Available	Targeted	Potential	Comments
e2	Third-party validation of predictive energy modelling - Exemplary credit	1	0	0	**Third-party validation of predictive energy modelling - Exemplary credit (1 credit)** **Targeted – No (Potential)** The predicted energy consumption modelling results have undergone third-party validation. The energy prediction model and the predicted operational energy values should be validated by a suitably qualified energy modelling engineer or accredited professional who is not involved with the project team or the client. A validation report must be produced that presents an unbiased analysis of the energy model carried out and recommendations for improving the modelling. The recommendations for improving the modelling must either be implemented, or justification for not implementing recommendations provided. **Comments & Actions** • *This credit is not sought at this stage.*
e3	Preparing for in-use measurement of energy consumption - Exemplary credit	1	0	0	Achieve maximum available credits in Ene 03 Energy monitoring. The energy modelling outputs include a breakdown of the energy use by sub-meter that aligns with the energy monitoring strategy for the building. **Comments & Actions[EG3]** • *This credit is not sought at this stage.*
e4	Commitment to measure energy consumption in-use - Exemplary credit	1	0	0	The client or building occupier commits funds to pay for the in-use energy consumption to be measured by a suitably qualified energy assessor once the building reaches the occupancy threshold. The funding for in-use energy consumption measurement must be sufficient to cover monitoring of energy consumption for the first 12 months of normal occupancy (≥ 80% full occupancy) for all relevant end uses and compare these to the predicted energy consumption and any energy targets set for the building[EG4].

<table>
<tr><th></th><th>Credit</th><th>Available</th><th>Targeted</th><th>Potential</th><th>Comments</th></tr>
<tr><td></td><td></td><td></td><td></td><td></td><td>This could be demonstrated by undertaking a BREEAM In-Use assessment of operational energy or for office buildings in the UK, NABERS UK Energy for Offices.

Where the building will be sold or let a green lease agreement must include clauses that guarantee access to energy consumption data once the building is occupied.

<u>Comments & Actions</u>

• This credit is not sought at this stage.</td></tr>
<tr><td colspan="6">Ene 03 - Energy monitoring</td></tr>
<tr><td>1</td><td>Energy monitoring by end-use</td><td>1</td><td>1</td><td>0</td><td>Energy monitoring by end-use (1 credit)
Minimum standard to achieve ‘Very Good’ target rating
Targeted – Yes
Identify the energy sources that will be used by the building and estimate the annual energy consumption for the following:

a. Grid supply electricity
b. Grid supply gas
c. District heating
d. District cooling
e. Renewable electricity generated on-site.

Energy monitoring is installed that allows 90% of the estimated annual energy consumption for each energy source to be assigned to an end-use category.

As a minimum the following end-uses must be considered, where present:
a. Space heating and cooling generators
b. Hot water generator
c. Fans for mechanical ventilation, space heating or cooling
d. Pumps for distribution heating or cooling
e. Commercial scale refrigeration or cold storage
f. Internal lighting
g. External lighting
h. Controls and telecommunications
i. IT equipment and small plug-in loads
j. Internal transport (lifts and escalators)</td></tr>
</table>

	Credit	Available	Targeted	Potential	Comments
					Renewable electricity generated on-site and exported from the building must be separately monitored. There is an energy metering system appropriate for the size of the building: a. For buildings with a gross internal area >1000 m², the energy meters are connected to an energy monitoring and management system. b. For buildings with a gross internal area < 1000 m², the energy meters are either: i. Connected to an appropriate energy monitoring and management system. OR ii. Are accessible meters with pulsed outputs or other open protocol communication outputs. Those responsible for monitoring the buildings energy consumption can clearly identify the end-use category covered by each meter. **<u>Comments & Actions</u>** • *MEP to confirm that energy metering systems will be specified and installed in line with the above requirements and that an appropriated energy monitoring and management system will be installed.* • *MEP to deliver the specification for the energy monitoring system and/or meters.*
2	Energy monitoring for different occupiers and functional areas	1	0	1	**Energy monitoring for different occupiers and functional areas (1 credit)** **Targeted – Yes** Energy monitoring is installed that allows energy consumption to be assigned to separately tenanted or occupied areas and to relevant functional areas within the building. Identify units or areas in the building with different tenants or occupiers, OR, where the building is designed for a single occupancy, different functional areas. Devise an energy metering strategy that ensures that different occupants or functional areas are separately monitored where it is technically feasible and cost effective to do so.

	Credit	Available	Targeted	Potential	Comments
					For this issue metering is considered cost effective where the cost of installing a sub-meter is lower than the energy cost savings arising from improved energy management over the expected lifetime of the building. There is an energy metering system appropriate for the size of the building: a. For buildings with a gross internal area > 1000 m², the energy meters must be connected to an energy monitoring and management system. b. For buildings with a gross internal area < 1000 m², the energy meters are either: i. Connected to an appropriate energy monitoring and management system. OR ii. Are accessible meters with pulsed outputs or other open protocol communication outputs. Those responsible for monitoring the buildings energy consumption can clearly identify the functional or occupier area covered by each meter. **Comments & Actions** • *MEP to confirm that energy metering systems will be specified and installed in line with the above requirements and that an appropriated energy monitoring and management system will be installed.* • *MEP to deliver the specification for the energy monitoring system and/or meters.* • *MEP to deliver floor plans showing functional or tenanted areas, where applicable.*
Ene 04 - Low carbon design					
1	Building form optimisation	1	0	0	**Building performance optimisation (2 credits)** **Targeted-No** The project team carries out an analysis of the site and proposed development during the Concept Design stage to explore the effect of building form on energy demand. Demonstrate that the analysis has informed the building design and will lead to a reduction in building energy consumption in use.

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	Credit	Available	Targeted	Potential	Comments
					<u>**Comments & Actions**</u> • *Credit not sought due to orientation of building.*
2	Fabric performance optimisation	2	0	2	**Fabric performance optimisation (2 credits)** **Targeted – No (potential)** The project team carries out an analysis of the proposed development to explore the impact of alternative fabric specifications on the building's demand for heating and cooling. **One credit:** 5% reduction in building energy demand AND implement fabric performance measures which reduce the overall building heating and cooling energy demand by ≥5% compared to minimum values set by building regulations. OR **Two credits**: 10% reduction in building energy demand AND implement fabric performance measures which reduce the overall building heating and cooling energy demand by ≥10% compared to minimum values set by building regulations **(Targeted).** OR **Two credits**: Fabric performance has been optimised. Demonstrate that the fabric specification has minimised the demand for heating and cooling **(Targeted).** The analysis to reduce energy demand from heating and cooling must consider the following: i. Fabric insulation ii. Thermal mass to store heat or coolth iii. Air tightness iv. Solar gains <u>**Comments & Actions**</u> • *Project Team to confirm they will undertake an analysis of fabric performance.* • *Energy modelling to demonstrate the reduced energy demand for heating and cooling arising from the proposed development.*

	Credit	Available	Targeted	Potential	Comments
3	Low and zero carbon energy generation technologies	2	0	0	**Low and zero carbon energy generation technologies (2 credits)** **Targeted – No – Biomass is not an eligible LZC energy generation** *One credit – Installation of LZC energy generation technologies* One or more eligible LZC energy generation technologies have been installed by competent installers. *One credit – Meeting LZC energy generation targets* Achieve the Installation of LZC energy generation technologies credit. The energy generated by the installed LZC energy generation technologies meet the following targets: a. Where photovoltaic (PV) panels are the only LZC energy generation technology installed, the electricity generated must: i. Be equal to, or exceed, that generated by reference PV panels covering 50% of the ground floor area. OR ii. Account for ≥ 60% of the buildings annual electricity consumption. b. Where only LZC energy generation technologies other than PV panels are installed, the energy generated must: i. Be equal to, or exceed, 137 kWh/year/m² of ground floor area. OR ii. Meet ≥ 60% of the buildings annual demand for the relevant energy source. c. Where the installed LZC energy generation technologies include both PV panels and other LZC energy generation technologies, selected generation targets as listed in 11.a and 11.b should be used in conjunction scaled proportionally to the amount of electricity generated by each technology.

	Credit	Available	Targeted	Potential	Comments
					<u>Comments & Actions</u> • *SRE can provide this study for an additional fee.* • *Drawings and technical specifications of LZC technologies proposed for the development required.*
Ene 05 - Energy efficient equipment					
1	Energy efficient equipment	0	0	0	Items[EG5] of equipment included in the project specification are considered to be in-scope unless one or more of the following conditions apply: 1. They are part of an industrial process. This credit only applies to unregulated energy equipment. It is assumed there is none in this development. **<u>Comments & Actions</u>** *Credit not applicable – it is assumed there is no equipment in scope apart from equipment used for industrial uses.*
Ene 06- Energy efficient systems					
1	Energy efficient systems	0	0	0	**<u>Credit not applicable – it is assumed there are no systems in scope apart from systems used for industrial uses.</u>** **Energy efficient systems** **Targeted – Not in scope[EG6]** Identify in-scope systems of unregulated energy demand. i.e. Building energy consumption resulting from building services and fittings that are not generally covered by building regulations and are not included in the scope of the Ene 01 assessments, unless one of the following apply: 1. They are part of an industrial process. 2. They provide temporary services e.g., external lighting for theatrical, stage, and local display installations. 3. They are only used when maintenance work is being carried out. 4. They are only used in an emergency.

Credit	Available	Targeted	Potential	Comments
				5. There are overriding safety, security or accessibility requirements that prevent energy efficiency criteria being met by any relevant compliance routes. 6. Existing systems are used in place of new systems (due to the reduction in embodied carbon). Estimate the expected annual energy consumption by in-scope systems. Some include: a. External lighting **(Assumed)** b. Cold storage c. Lifts and escalators **(Assumed)** d. Swimming pools e. Commercial laundries f. Cooling systems in server rooms Determine the number of credits available using Table 6.7:

In-scope unregulated energy using systems / regulated energy use (%)	**Credits available**
No in-scope equipment OR Energy consumption of in-scope equipment is not significant	Issue filtered out
< 10%	1
≥ 10% - 20%	2
≥ 20% - 30%	3
≥ 30% - 40%	4
≥ 40% - 50%	5
≥ 50% - 60%	6
≥ 60% - 70%	7
≥ 70% - 80%	8
≥ 80%	9

Determine in-scope systems that are energy efficient by one of the following:

a. Where the system is more energy efficient than that of a standard version of the system
b. Where the systems meet specific "deemed to comply" energy efficiency requirements

OPTION A: Exceeding the standard version

	Credit	Available	Targeted	Potential	Comments
					You must demonstrate that the energy performance of the actual system achieves a meaningful reduction in energy consumption compared to a standard version. A standard system is defined as being a system that: 1. Has energy using components with energy efficiency levels that are typical for the component types. 2. Has not been designed to optimise energy consumption. 3. Has only energy efficient features and controls that are typical for the system type **<u>OPTION B: "Deemed to comply"</u>** This is an alternative compliance route that is available for the following types of system: g. External lighting **(Assumed)** h. Cold storage i. Lifts and escalators **(Assumed)** j. Swimming pools k. Commercial laundries l. Cooling systems in server rooms Deemed to comply requirements: *External lighting* 1. The average initial luminous efficacy across all in-scope external lighting systems within the construction zone achieves ≥ 70 luminaire lumens per circuit watt. 2. All external light fittings are automatically controlled to prevent operation during daylight hours and are fitted with presence detection in areas of intermittent pedestrian traffic **<u>Comments & Actions</u>** • *Project Team to confirm that a meaningful reduction in energy reduction will be demonstrated for all relevant energy efficient systems on site.*
Ene 07- Flexible demand response					
1	Flexible demand response	1	0	0	**Flexible demand response** **Targeted – No (potential)**

	Credit	Available	Targeted	Potential	Comments
					Where the following systems are present, they must meet the associated criteria: - Electric space heating - Operation can be optimised based on signals from the electricity supplier or local renewable sources. - Electric space cooling - Operation can be optimised based on signals from the electricity supplier or local renewable sources. - Electric domestic hot water - Operation can be optimised based on signals from the electricity supplier or local renewable sources. - Other systems or equipment with flexible demand capability - The battery energy storage system is capable of charging based on signals from the electricity supplier or local renewable sources (1 point), The battery energy storage system is capable of feeding electricity back into the grid (1 point). **<u>Comments & Actions</u>** *This credit is not sought at this stage.*
Ene 08 - Installed controls					
1	Installed controls	2	1	0	**Installed controls (2 credits)** **Targeted – Yes (1 credit)** Determine the number of installed controls points available based on the installed HVAC services and relevant system details. Determine the number of installed controls points achieved based on the functionality of the controls. The number of credits awarded is determined by the proportion of available installed controls points that are achieved. ***One credit strategy: 40-55% available points*** *Space Heating (Air system)* - *Room / Zone Level Temperature Control - Heating output controlled at room/zone level.* - *Variable Speed Fans - Variable speed control of fans* - *Outside Temperature Sensor – Weather compensation* *Space Cooling (Air conditioning / Comfort cooling)*

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	Credit	Available	Targeted	Potential	Comments
					- *Room / Zone Level Cooling Control - Cooling output controlled at room/zone level* - *Variable Control Cooling Plant - Inverter / variable capacity compressors* - *Variable Speed Fans - Cooling distribution airflow modulation* *Mechanical Ventilation* - *Central Air Quality Sensors (CO_2) - Ventilation controlled centrally based on IAQ* - *Multistage AHU Control – Basic staged control* - *Heat Recovery (Sensor Controlled) - Heat recovery linked to exhaust sensors* **Comments & Actions** • *Project Team and MEP to confirm the specification for HVAC systems and controls will support the available points required for 1 credit.* • *Relevant contracts and drawings will be provided in support.*
		31	7	17	**Standard Energy Credit Total**
		7	0	0	**Exemplary Energy Credit Total**
		23.43	3.71	9.01	**% Energy Total (Standard + Exemplary)**
Transport (0.71 % per Credit)					
Tra 01 - Transport assessment and travel plan					
1	Transport assessment and travel plan	2	2	0	**Transport assessment and travel plan (2 credits)** **Completed BEFORE end of RIBA Stage 2** **Targeted – Yes** Conduct a site-specific transport assessment by the end of the Concept Design stage. Develop a site-specific travel plan following the transport assessment recommendations by the end of the Concept Design stage. Demonstrate that the travel plan can guide the appropriate implementation of sustainable transport measures during the design, construction, and occupancy stages and be supported by the building's management during operation. **Comments & Actions**

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	Credit	Available	Targeted	Potential	Comments
					• *Project Team to confirm Transport Assessment and Travel Plan will be provided before the end of Concept Design.*
Tra 02 - Sustainable transport measures					
Pre-req	Prerequisite Transport assessment and travel plan	✓	✓		**Prerequisite: Transport assessment and travel plan** Complete a transport assessment and travel plan in line with Tra 01. If these documents are completed after Concept Design, they can still be used to meet this prerequisite.
1	Transport options implementation	10	0	2	**Transport options implementation (10 credits)** **Targeted – No (Potential - 2 credits)** Building type: Building type A Credits are awarded based upon the site's Accessibility Index (AI) and the implementation of any of the following: 1. The existing PTAI calculated is ≥ 8 for all other building types **(Assumed).** 2. Improve the existing PTAI by at least 1.00 **(Potential).** 3. Provide up-to-date local public transport information in a convenient, accessible area to help building users plan journeys by public transport **(Assumed).** 4. Provide electric vehicle charging infrastructure a. **1 Point:** 10% parking spaces are fitted with ≥ 7 kW EV charging points AND 20% parking spaces have necessary infrastructure for future demand b. **2 Points:** Requirements for 1 Point are met AND, EITHER EV charging score of ≥20 is achieved, OR ≥30% of parking spaces must be active EV charging (each with a charging point ≥7kW) 5. Car sharing: a. Set up a car sharing group or facility to promote car sharing among building users. b. Actively market the car sharing scheme and provide communication channels that make it easy to join and use.

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Credit	Available	Targeted	Potential	Comments
				c. Provide priority spaces for car sharers, allocating ≥ 5% of the total staff car parking capacity. d. Ensure that car share spaces are placed near the building entrance as the requirements of other types of priority parking (disabled, etc.) allow. 6. Pedestrian/cycle route improvements: a. Engage with the local authority at the earliest (brief) stage to review public cycle and pedestrian routes. b. Agree on possible improvements to the local active travel network that supplement existing plans and are relevant to the site/project. c. Select at least one such improvement and carry it out. 7. **POTENTIAL:** Install at least the minimum number of cycle storage spaces. 8. **POTENTIAL:** Having supplied cycle storage space as per Option 7 above, provide building users with at least two of the following facilities: a. Showers b. Changing facilities c. Lockers d. Drying spaces *(Where facilities are combined – e.g. showers and changing facilities – it must be possible for different people to use them at the same time if they are to count as more than one facility).* 9. Existing amenities - Ensure at least three relevant amenity types within less than or equal to 500 m of proximity. 10. New or enhanced amenities - Provide relevant and accessible new amenities within ≤500m of the building entrance. These can be an enhanced existing amenity, or a new type of amenity. a. **2 Points:** at least one new or enhanced amenity type. b. **3 Points:** Two or more new or enhanced amenities type. 11. Carry out one site-specific improvement measure in line with the travel plan not listed as an option in Tra 02. Submit this for BRE review. **<u>Comments & Actions</u>** *We have assumed cycle storage and cycle facilities will be available as part of the wider site. Therefore 2 credits have been targeted.*
	12	2	2	**Standard Transport Credit Total**
	0	0	0	**Exemplary Transport Credit Total**
	8.52	1.42	1.42	**% Transport Total (Standard + Exemplary)**

	Credit	Available	Targeted	Potential	Comments
Water (0.83 % per Credit)					
Wat 01 - Water consumption					
1	Water consumption	5	3	0	**Water consumption (5 credits)** **Targeted – Yes (3 credits)** This issue assesses the efficiency of the domestic water-consuming components specified for the development. Determine the water efficiency of all in-scope water-consuming components. Determine the amount of recycled water expected from approved blackwater, greywater, and rainwater systems. The number of credits awarded is determined by the percentage improvement in potable water use in litres per person per day for the actual building compared to that of a baseline building. Information on the types of water-using components present in the building, and the number of each type of water-using component should be provided by the project team. For this assessment, it is assumed a **≥ 45% improvement over baseline** is achievable. Table 8.3: Baseline performance levels for water-consuming components

Water consuming component	Baseline performance level
WCs/toilet	6.0 litres effective flush volume
Urinal	8.75 litres/bowl/hour
Washbasin taps	10 litres/minute
Shower	12 litres/minute
Kitchen tap	10 litres/minute
Bath	200 litres
Dishwasher – Domestic sized	17 litres/cycle
Dishwasher – Commercial sized	8 litres/rack
Washing machine – Domestic sized	90 litres/use
Washing machine – Commercial sized	14 litres/kg
Waste disposal unit	17 litres/minute

Comments & Actions

	Credit	Available	Targeted	Potential	Comments
					• *MEP to confirm domestic sanitaryware proposed and deliver specifications in line with the below recommendations.* • *Assessor must be informed if water fittings quantity and/or flow rate will change.* *Consider the following performance levels when proposing Sanitaryware:* - *WCs/toilet ~ ≤ 4.5/3 L dual flush capacity* - *Urinals ~ ≤ N/a* - *Washbasin taps ~ ≤ 5.0 L* - *Showers ~ ≤ N/a* - *Kitchen tap (kitchenette) ~ N/a*
e1	Water consumption - Exemplary credit	0	0	0	**Comments & Actions** • *Credit not targted.*
Wat 02 - Water monitoring					
Pre-req	Prerequisite - Water monitoring	✔	✔		**Prerequisite - Water monitoring** **Minimum Standard to achieve 'Good' target rating** A water meter with a pulsed or other open protocol output is installed on all potable water supplies entering the building.
1	Water monitoring	1	1	0	**Water monitoring (1 credit)** **Minimum Standard to achieve 'Good' target rating** **Targeted – Yes** A water meter with a pulsed or other open protocol output is installed on all potable water supplies entering the building. Water-consuming plant or building areas with significant water demand must be fitted with easily accessible sub-meters or have water monitoring equipment that is integral to the plant or area. A water monitoring system that is appropriate for the size of the building has been installed in accordance with the below:

	Credit	Available	Targeted	Potential	Comments
					a. For buildings with a gross internal area ≥ 1000 m², the water meters are connected to a monitoring and management system. b. For buildings with a gross internal area < 1000 m², the water meters are either: i. Connected to an appropriate water monitoring and management system. ii. Accessible meters with pulsed outputs or other open protocol communication outputs Appropriate labelling is used to clearly identify the water uses and areas covered by each meter. **<u>Comments & Actions</u>** • *MEP to confirm that water meters will be specified in line with the above requirements.* *Water uses deemed to be significant and to be fitted with a sub-meter:* - *Common areas (e.g., covering the supply to toilet blocks).* - *Service areas (covering the supply to outlets within storage, delivery, waste disposal areas etc.).* - *Ancillary or separate buildings to the main development with a water supply.* - *All water consuming systems or building areas that are expected to account for more than 10% of the buildings total water demand must be considered when determining significant water uses.* *It is not recommended to install a sub-meter in the following cases:* - *Where a building has only one or two small sources of water demand (e.g. an office with sanitary fittings and a small kitchen).* - *Where the building has two sources of water demand, one significantly larger than the other, and the water consumption for the larger demand is likely to mask the smaller demand.*
Wat 03 - Water leak detection and prevention					
1	Leak detection system	1	1	0	**Leak detection system** • *Credit not applicable – as there are no in-scope water uses*

	Credit	Available	Targeted	Potential	Comments
					Targeted – Yes Install a leak detection system capable of detecting a major water leak: a. On the utilities' water supply within the building, to detect any major leaks within the building. AND b. Between the building and the utilities' water supply, to detect any major leaks between the utilities' supply and the building under assessment. The leak detection must be either: a. A permanent automated water leak detection system that alerts the building occupants to any leaks. OR b. An inbuilt automated diagnostic procedure for detecting leaks. **<u>Comments & Actions</u>** • *MEP to confirm that a leak detection system will be installed in line with the above requirements.*
2	Flow control devices	1	1	0	**Flow control devices (1 credit)** **Targeted – Yes** Flow control devices that regulate the supply of water to each WC area or facility according to demand are installed. The following types of flow devices meet the criteria: a. A time controller, i.e. an automatic time switch device to switch off the water supply after a predetermined interval. b. A programmed time controller, i.e. an automatic time switch device to switch water on or off at predetermined times. c. A volume controller, i.e. an automatic control device to turn off the water supply once the maximum pre-set volume is reached. d. A presence detector and controller, i.e. an automatic device that switches water on and off, depending on the detection of occupancy or movement.

	Credit	Available	Targeted	Potential	Comments
					e. A central control unit, i.e. a dedicated computer-based control unit for the overall management of a water control system, utilising some or all the types of control elements listed above. **Comments & Actions** • *MEP to confirm whether flow control devices will be specified for each WC area, in line with the above.*
Wat 04 – Water efficient equipment and systems					
1	Water efficient equipment and systems	0	0	0	**Comments & Actions** • *Credit not applicable – as there is no in-scope water uses*
Wat 05 - Prediction of operational water consumption					
1	Predict operational water consumption	1	0	1	**Predict operational water consumption (1 credit)** **Targeted – No (potential)** Estimate the annual operational water consumption for the building. This includes both sanitary and non-sanitary water uses. Calculations should be carried out to confirm the overall estimated annual operational water consumption for the building and water-using components and equipment as identified by the BRE are required. **Comments & Actions** • *This credit is not targeted.*
e1	Disclosing an in-use water performance target and commitment to measure water consumption in-use - Exemplary credit	1	0	0	**Predict operational water consumption (1 credit)** **Targeted- No** Achieve the credit for predicting operational water consumption above and achieve the maximum available credits in Wat 02 Water monitoring. An in-use water performance target has been disclosed to stakeholders and been made publicly available prior to practical completion.

	Credit	Available	Targeted	Potential	Comments
					The client or building occupier commits funds to pay for in-use water consumption measurement and reporting once the building is occupied. This requires a suitably qualified individual or organisation to be appointed and to measure and report on the actual in-use operational consumption. **Comments & Actions** • *This credit is not targeted.*
		9	6	1	**Standard Water Credit Total**
		2	0	0	**Exemplary Water Credit Total**
		9.47	4.98	0.83	**% Water Total (Standard + Exemplary)**
Materials (1.14 % per Credit)					
Mat 01 - Building life cycle assessment					
1	Early design LCA and embodied carbon reporting	2	2	0	**Early design LCA and embodied carbon reporting (2 credits)** **Minimum Standard to achieve 'Excellent' target rating an LCA and embodied carbon reporting must be produced at ANY project stage** **Completed BEFORE end of RIBA Stage 2** **Targeted – Yes** Carry out an early design phase LCA calculation from the materials that the project incorporates into the building. The LCA must: a. Meet the requirements for tools and scope: i. Use an LCA tool recognised by BREEAM. ii. Early stage reporting (Concept Design) must be delivered no later than 20 working days after submitting for planning application. iii. Results must be submitted for all core environmental impact indicators. iv. The following life cycle modules must be covered (based on EN 15978): a. Modules A1–A5: Product sourcing and construction stage. b. Modules B1–B5: Use stage (excluding B6–7). c. ModulesC1–C4: End-of-life stage. d. Module D:Considering benefits and loads beyond the system boundary. This is required unless EN 15804-A2 EPD data is not available for used products.

	Credit	Available	Targeted	Potential	Comments
					v. All building elements must align with RICS Whole Life Carbon Assessment Standard, 2nd Edition. b. Include relevant options appraisals, including a baseline assessment. There are no requirements to carry out a specific number of options appraisals for the project. c. Be used as a basis for choosing design solutions and materials with the aim of achieving a low environmental impact. If an alternative solution with a lower environmental impact has not been chosen, then this must be justified. From the early design LCA results, report the embodied carbon emissions from materials used in the building. **Comments & Actions** • *SRE can provide this service for an additional fee.*
2	Technical design LCA and embodied carbon reporting	1	1	0	**Technical design LCA and embodied carbon reporting (1 credit)** **Completed BEFORE end of RIBA Stage 4** **Targeted – Yes** Carry out a technical design phase LCA calculation from the materials that the project incorporates into the building. The LCA must: a. Meet the requirements for tools and scope as stated above. b. Include at least one assessment showing the evolving design and specification at technical design. From the technical design LCA results, report the embodied carbon emissions from materials used in the building. **Comments & Actions** • *Client/developer to appoint a suitable project team member to undertake an LCA for the Proposed Development during Technical Design.* • *SRE can provide this service for an additional fee.*
3	Post-construction LCA and embodied carbon reporting	2	2	0	**Post-construction LCA and embodied carbon reporting (2 credits)** **Targeted – Yes** Carry out a post-construction phase LCA calculation from the materials that the project incorporates into the building. The LCA must:

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	Credit	Available	Targeted	Potential	Comments
					a. Meet the requirements for tools and scope as stated above. b. Include as-built specification at post-construction. From the post-construction LCA results, report the embodied carbon emissions from materials used in the building. **Comments & Actions** • *Client/developer to appoint a suitable project team member to undertake an LCA for the Proposed Development at Post-Construction.* • *SRE can provide this service for an additional fee.*
4	Embodied carbon benchmark comparison	2	0	2	**Comments & Actions** • *Credit not sought – potential depending on the LCA results where available* • *Client/developer to appoint a suitable project team member to undertake an LCA at Technical Design AND Post-Construction, which confirms the embodied carbon benchmark band of >C.*
e1	Embodied carbon benchmark comparison - Exemplary credit	2	0	0	**Comments & Actions** • *This credit is not sought due to high benchmarks.*
e2	Third party verification - Exemplary credit	1	0	0	**Third party verification - Exemplary credit (1 credit)** **Targeted – Yes** A suitably qualified third party verifies the building LCA work. The suitably qualified third party must: 1. Produce a report describing how they have checked that the building LCA work accurately represents the designs under consideration during the: a. Early design phase, and b. Technical design phase, and c. Post-construction phase (only relevant if criteria for Post-construction LCA and embodied carbon reporting has been met) 2. Itemise in the report the checks made for each LCA option.

	Credit	Available	Targeted	Potential	Comments
					3. Include in the report details of their relevant skills and experience and a declaration of their third-party independence from the project client and design team. **Comments & Actions** • *Client/developer to appoint third party to verify LCA for Concept and Technical Design data.* • *Client/developer to commit that third-party will be appointed to verify LCA at the Post-Construction stage.* • *SRE can offer this service for an additional fee, if required*
e3	Embodied carbon public data disclosure - Exemplary credit	1	0	0	**Comments & Actions** • *This credit is not sought.*
Mat 02 - Environmental product declarations					
1	Product specification	1	1	0	**Product specification (1 credit)** **Targeted – Yes** Specify construction products with EPDs, for the significant material categories listed below, that achieve a total EPD points score of at least 20. Material categories: - Timber and timber-based products - Concrete or cementitious - Metal - Stone or aggregate - Clay-based - Gypsum - Glass - Plastic, polymer, resin, paint, chemicals, and bituminous - Animal fibre or skin, cellulose fibre - Other The score available for each material category is capped at 4. Therefore, you must present evidence which sufficiently covers as many significant materials as possible.

	Credit	Available	Targeted	Potential	Comments
					<u>Comments & Actions</u> • *This credit is not sought at this stage.* • *Prefer products with verified EPDs. Where these products are specified, copies of their EPD certificates to be provided.*
Mat 03 - Responsible sourcing of construction products					
Pre-req	Prerequisite Legally harvested and traded timber	✔	✔		**Prerequisite: Legally harvested and traded timber** **Minimum Standard to achieve ANY target rating** 100% of timber and timber-based products used on the project are 'Legal' and 'Sustainable' as per the UK Government's Timber Procurement Policy (TPP).
1	Enabling sustainable procurement	1	1	0	**Enabling sustainable procurement (1 credit)** **Completed BEFORE end of RIBA Stage 2** **Targeted – Yes** A sustainable procurement plan must be used by the design team to guide specification towards sustainable construction products. The plan must: a. Be in place before the end of Concept Design. b. Include sustainability aims, objectives, and strategic targets to guide procurement activities. Targets do not need to be achieved for the credit to be awarded at post-construction stage but justification must be provided for targets that are not achieved. c. Include a requirement for assessing the potential to procure construction products locally. There must be a policy to procure construction products locally where possible. d. Include details of procedures in place to check and verify the effective implementation of the sustainable procurement plan. e. The documented policy and procedure must be disseminated to all relevant internal and external personnel, and included within the construction contract to ensure that they are enforceable on the assessed project. In addition, if the plan is applied to several sites or adopted at an organisational level it must:

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	Credit	Available	Targeted	Potential	Comments
					a. Identify the risks and opportunities of procurement against a broad range of social, environmental, and economic issues following the process set out in ISO 20400:2017. **<u>Comments & Actions</u>** • *SRE can provide this for an additional fee.*
2	Measuring responsible sourcing	3	1	0	**Measuring responsible sourcing (3 credits)** **Targeted – Yes (1 credit)** Credits for this issue are awarded based on the proportion of responsible sourcing of materials (RSM) points achieved. To assess this issue there are up to four types of data needed for each product: - Material category - Responsible sourcing certification scheme (RSCS) or environmental management system (EMS) certifications - Product or material quantities (optional, route 2 only). To determine the number of credits that are achieved for responsible sourcing of construction products, the Mat 03 calculator must be completed with all the relevant information. For **1 credit** to be awarded, data for building materials must cover the minimum scope level requirements: - Superstructure, AND; - Internal finishes - Substructure and hard landscaping ≥ 10% of available points for RSM must be achieved. Calculation of the above requires completion of the Mat 03 Calculator tool by the BREEAM assessor, based on materials information (including materials schedule, drawings and responsible sourcing certificates) provided by the Project Team (specifically the Main Contractor or Architect).

	Credit	Available	Targeted	Potential	Comments
					<u>Comments & Actions</u> • *Prefer products from verified manufacturers who operate with EMS certification or RSM certificates.*
e1	Measuring responsible sourcing - Exemplary credit	1	0	0	**<u>Comments & Actions</u>** • *This credit is not sought due to high benchmarks.*
Mat 04 - Durability and resilience					
1	Durability and resilience	1	1	0	**Durability and resilience (1 credit)** **Targeted – Yes** *Protecting vulnerable parts of the building from damage* Protection measures are incorporated into the building's design and construction to reduce damage to the building's fabric or materials in case of accidental or malicious damage occurring. These measures must provide protection against: a. Negative impacts of high user numbers in relevant areas of the building (e.g. corridors, lifts, stairs, doors etc.). b. Damage from any vehicle or trolley movements within 1 metre of the internal building fabric in storage, delivery, corridor and kitchen areas. c. External building fabric damage by a vehicle. Protection where parking or manoeuvring areas are within 1 metre of the building façade and where delivery areas or routes are within 2 metres of the façade, i.e. specifying bollards or protection rails. d. Potential malicious damage to building materials and finishes, in public and common areas where appropriate. *Protecting exposed parts of the building from material degradation* Key exposed building elements have been designed and specified to limit long and short term degradation due to environmental factors. This can be demonstrated through one of the following: a. The element or product achieving an appropriate quality or durability standard or design guide. Use the BREEAM manual or refer to BS 7543:2015(93) as the default appropriate standard or use relevant equivalent standards.

Credit	Available	Targeted	Potential	Comments
				OR b. A detailed assessment of the element's resilience when exposed to the applicable material degradation and environmental factors. Produce an access to the roof and façade strategy for cost-effective cleaning, replacement and repair in the building's design. Design the roof and façade to prevent water damage, ingress and detrimental ponding. **<u>Comments & Actions</u>** • *Project team to provide design drawings illustrating vulnerable parts of the building.* • *Architect to prepare drawings/specification confirming the durability measures specified.* • *Architect to prepare drawings/specification measures in paces to limit degradation effects.* • *Project team/Architect to deliver Roof and façade access strategy and design drawings showing measures to prevent water damage, ingress and ponding.* *Examples of suitable durability measures:* - *Bollards, barriers or raised kerbs to delivery and vehicle drop-off areas* - *Robust external wall construction, up to 2m high* - *Corridor walls specified to Severe Duty (SD) as per BS 5234-2* - *Protection rails to walls of corridors* - *Kick plates or impact protection (e.g. trolleys) on doors* - *Hard-wearing and easily washable floor finishes in heavily used circulation areas* - *Door stoppers to prevent door handles damaging walls* - *Designing out the risk without the need for additional materials specification to protect vulnerable areas.* *Designing to prevent water damage* - *Carefully consider the drainage mechanisms of the façade and roof* - *Prevent staining, detrimental oxidation, ponding, rot, ingress, penetration or any other deleterious effect.*

	Credit	Available	Targeted	Potential	Comments
Mat 05 - Material efficiency					
1	Material efficiency	1	1	0	**Material efficiency (1 credit)** **Completed BEFORE end of RIBA Stage 1 and UPDATED at every project stage (RIBA Stage 2, 3, 4, 5 and 6)** **Targeted – Yes** At the Preparation and Brief and Concept Design stages, set targets and report on opportunities and methods to optimise the use of materials. This must be done for each of the following stages: • Preparation and Brief • Concept Design • Developed Design • Technical Design • Construction Develop and record the implementation of material efficiency during: • Developed Design • Technical Design • Construction Report the targets and actual material efficiencies achieved. **Comments & Actions** • *Architect to provide the Material Efficiency report for RIBA stages 1, 2 and 3 and 4. SRE can provide templates and guidance for this report if appointed as the BREEAM AP* • *Main Contractor to supply the report for RIBA Stage 5 (Construction).* *Examples of suitable material efficiency design measures can include:* - *Increasing the utilisation factor of structural members* - *Designing to standard material dimensions to reduce off-cuts and waste on site* - *Removing redundant materials from the design* - *Using materials that can be recycled or reused at the end of their service life* - *Making use of recycled or reclaimed materials* - *Designing for deconstruction and material reuse* - *Using pre-fabricated elements where appropriate to reduce material waste*

	Credit	Available	Targeted	Potential	Comments
					- *Consider using an 'exposed thermal mass' design strategy to reduce finishes* - *Avoiding over-specification of predicted loads* - *Using lightweight structural design strategies* - *Making use of bespoke structural elements where this will reduce overall material use* - *'Rationalisation' of structural elements* - *Optimising the foundation design for embodied environmental impact.*
		14	10	2	**Standard Materials Credit Total**
		4	0	0	**Exemplary Materials Credit Total**
		19.96	11.40	2.28	**% Materials Total (Standard + Exemplary)**
Waste (0.78 % per Credit)					
Wst 01 - Construction waste management					
1	Pre-demolition audit	0	0	0	**Pre-demolition audit – not applicable** **Completed BEFORE end of RIBA Stage 2** **Targeted – No – not applicable** Complete a pre-demolition audit of any existing buildings, structures or hard surfaces being considered for demolition. This must be used to determine whether refurbishment or reuse is feasible and, in the case of demolition, to maximise the recovery of material for subsequent high grade or value applications. The audit must cover the content of M1: Pre- demolition audit scope on the next page and: a. Be carried out at Concept Design stage by a competent person prior to strip-out or demolition works b. Guide the design, consider materials for reuse and set targets for waste management c. Engage all contractors in the process of maximising high-grade reuse and recycling opportunities Make reference to the audit in the resource management plan (RMP). Compare actual waste arisings and waste management routes used with those forecast and investigate significant deviations from planned targets. **Comments & Actions**

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	Credit	Available	Targeted	Potential	Comments
					• *Produce a pre-demolition audit compliant with the criteria[EG7].*
2	Construction resource efficiency	3	3	0	**Construction resource efficiency (3 credits)** **Targeted – Yes** Prepare a compliant Resource Management Plan (RMP) covering: a. Non-hazardous waste materials (from on-site construction and dedicated off-site manufacture or fabrication), including demolition and excavation waste. b. Accurate data records on waste arisings and waste management routes. Meet or improve upon the benchmarks below: - One credit awarded where the amount of waste generated per 100m^2 is ≤13.3 (m^3) or ≤11.1 (tonnes) - Two credits awarded where the amount of waste generated per 100m^2 is ≤7.5 (m^3) or ≤6.5 (tonnes) - **Three credits awarded where the amount of waste generated per 100m^2 is ≤3.4 (m^3) or ≤3.2 (tonnes) – <u>Targeted</u>** - Exemplary level awarded where the amount of waste generated per 100m^2 is ≤1.6 (m^3) or ≤1.9 (tonnes) **<u>Comments & Actions</u>** • *Client/developer to appoint and deliver compliant resource management plan. Where the main contractor is not yet appointed, letter of commitment may be used to award the credits.*
3	Diversion from landfill	1	1	0	**Diversion from landfill (1 credit)** **Targeted – Yes** Meet, where applicable, the diversion from landfill benchmarks in Table 10.2 for non-hazardous construction waste and demolition and excavation waste generated. Table 10.2: Diversion from landfill benchmarks

BREEAM Credits	Type of waste	Amount of waste diverted from landfill

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	Credit	Available	Targeted	Potential	Comments
					One credit: Construction 70% / 80%; Demolition 80% / 90%; Excavation N/A / N/A. Exemplary level: Construction 85% / 90%; Demolition 85% / 95%; Excavation 95% / 95%. Sort waste materials into separate key waste groups, either on-site or through a licensed contractor for recovery. **Comments & Actions** • *Client/developer to appoint and deliver compliant resource management plan, including the relevant commitments/targets to divert waste from landfill. Where the main contractor is not yet appointed, letter of commitment may be used to award the credits.* • *Ensure procedures in place for sorting construction waste into key waste groups.*
e1	Construction waste management - Exemplary credit	1	0	0	**Comments & Actions** • *This credit is not sought at this stage due to high benchmarks.*
Wst 02 - Use of recycled and sustainably sourced aggregates					
1	Project sustainable aggregate points	1	0	0	**Comments & Actions** • *This credit is not sought due to high benchmarks.*
e1	Project sustainable aggregate points - Exemplary credit	1	0	0	**Comments & Actions** • *This credit is not sought due to high benchmarks.*
Wst 03 - Operational waste					
1	Operational waste	1	1	0	**Operational waste (1 credit)** **Targeted – Yes**

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Credit	Available	Targeted	Potential	Comments
				Provide a dedicated space, identified by clear signage, for the segregation and storage of operational recyclable waste. The space is: a. Clearly labelled for long-term durability, to assist with segregation, storage and collection of the recyclable waste streams. b. Accessible to building occupants or facilities operators for the deposit of materials and collections by waste management contractors. c. Of a capacity appropriate to the building type, size, number of units (if relevant) and predicted volumes of waste that will arise from daily or weekly operational activities and occupancy rates Determining the capacity appropriate to the building type & size: **Buildings < 5000 m²(Assumed)** - At least 2 m² per 1000 m² of net floor area OR - 4 m² per 1000 m² of net floor area where commercial-scale catering is provided **Buildings ≥ 5000 m²** - A minimum of total of 10 m² OR - 20 m² where commercial-scale catering is provided Note: The net floor area should be rounded up to the nearest 1000 m². For consistent and large amounts of operational waste generated, provide: a. Static waste compactors or balers; situated in a service area or dedicated waste management space. b. Vessels for composting suitable organic waste OR adequate spaces for storing segregated food waste and compostable organic material for collection and delivery to an alternative composting facility. c. A water outlet provided adjacent to or within the facility for cleaning and hygiene purposes where organic waste is to be stored or composted on site.

	Credit	Available	Targeted	Potential	Comments
					Comments & Actions • *At least 2m² of refuse space per 1000m² should be provided given the development's GIA = < 5000m².* • *Project Team to confirm waste refuse store adequate for the size of the development will be provided.* • *Design drawings to show that this has been incorporated into the Proposed Development are required.* • *Facilities within the existing building can be used to assess compliance. These facilities must cater for the total volume of predicted recyclable waste arising from the new extension and existing buildings.*
Wst 04 – Speculative finishes (offices)					
1	Speculative finishes	0	0	0	**Comments & Actions** *Credit not applicable as there are no office areas*[EG8]
Wst 05 - Adapting to climate change					
1	Resilience of structure, fabric, building services and renewables installation	1	1	0	**Resilience of structure, fabric, building services and renewables installation (1 credit)** **Completed BEFORE end of RIBA Stage 2 and provide an update BEFORE end of RIBA Stage 4** **Targeted – Yes** Conduct a climate change adaptation strategy appraisal using a systematic risk assessment to identify the impact of expected extreme weather conditions arising from physical climate risks and climate change on the building and vulnerabilities over its projected life cycle. The assessment covers the installation of building services and renewable systems, as well as structural and fabric resilience aspects and includes: a. Hazard identification b. Hazard assessment c. Risk estimation d. Risk evaluation e. Risk management

	Credit	Available	Targeted	Potential	Comments
					Develop recommendations or solutions based on the climate change adaptation strategy appraisal, before or during Concept Design, that aim to mitigate the identified impact. Provide an update during Technical Design demonstrating how the recommendations or solutions proposed at Concept Design have been implemented where practical and cost effective. Omissions have been justified in writing by the assessor. **Comments & Actions** • *Project Team/Architect to provide a copy of a compliant climate change adaptation strategy appraisal before the end of Concept Design, AND an update required at Technical Design.* *All adaptation plans should consider the following impacts of climate change and extreme weather events and describe how the design mitigates against them, where appropriate:* - *Flooding* - *Storms (including high winds)* - *Cold events* - *Heat waves (including temperature increases)* - *Drought (including reduced summer rainfall)* - *Milder winters* - *Wetter winters (including increased moisture and driving rain)* - *Warmer summers and increased solar radiation* - *Temperature variation* - *Precipitation, e.g. rain and snow* - *Subsidence or ground movement.*
e1	Responding to climate change - Exemplary credit	1	0	0	**Comments & Actions** • *This credit is not sought due to high benchmarks.*
Wst 06 - Disassembly and adaptability					
1	Design for disassembly and adaptability Recommendations	1	1	0	**Design for disassembly and adaptability Recommendations (1 credit)** **Completed BEFORE end of RIBA Stage 2** **Targeted – Yes**

sre

Credit	Available	Targeted	Potential	Comments
				By the end of Concept Design, conduct a study to explore the disassembly and adaptation potential of different design scenarios. By the end of Concept Design, develop solutions based on the study that allow disassembly and adaptability. *Adaptability principles:* 1. **Versatility**: the ability of spaces to accommodate changes in use with minimal changes to maximise space utilisation. For example, a performance space that can also be used as a meeting or community space. 2. **Convertibility**: related to versatility, the potential of internal spaces to accommodate more significant changes in use without affecting building structure. For example, changing a cellular office to an open plan office through changing the internal layout and partitions. 3. **Expandability**: the ability of a design to accommodate a substantial change that supports or facilitates the addition of new space, features, capabilities, and capacities. It involves designing to allow for either vertical or horizontal additions in floor space. *Disassembly principles:* 1. **Ease of access to components and services**: it allows for a material, component, or connector of an assembly, especially those with the shortest anticipated life cycle, to be easily approached, with minimal damage to and impact on it and adjacent assemblies. 2. **Independence**: it allows for parts, components, modules, and systems to be removed or upgraded without affecting the performance of connected or adjacent systems. 3. **Avoidance of unnecessary treatments and finishes**: choice of finishes can limit the options for reusing or recycling, particularly if potentially hazardous substances are included. To support disassembly, finishes that can prevent components from being re-used or recycled should be avoided. 4. **Supporting re-use (circular economy) business models**: this principle is concerned with supporting the market for re-used, refurbished, remanufactured, and recycled materials and products now and in the future, in support of circular economy business models. 5. **Simplicity**: it refers to the quality of an assembly or system that is designed to be straightforward, easy to understand, and meet performance requirements with the least amount of customization.

	Credit	Available	Targeted	Potential	Comments
					6. **Standardization**: it is concerned with the use of common components, products, or processes to satisfy a multitude of requirements. 7. **Safety of disassembly**: it is concerned with easy access to accurate information on the original materials and assembly methods used for an asset, together with details of any subsequent major renovation. **<u>Comments & Actions</u>** • *Project Team/Architect to deliver a copy of a compliant disassembly and adaptability study before the end of Concept Design.* *Examples:* - *Modular insulated panel systems (demountable & reusable)* - *Bolted structural steel frame* - *Replaceable refrigeration plant & services zones* - *Durable long-life envelope components*
2	Design for disassembly and adaptability Implementation	1	1	0	**Design for disassembly and adaptability Implementation (1 credit)** **Completed BEFORE end of RIBA Stage 4** **Targeted – Yes** Achieve criteria for the Design for disassembly and adaptability Recommendations credit as above. During Technical Design, provide an update on the following: a. How the solutions proposed by Concept Design have been implemented where practical and cost effective. Omissions must be clearly justified. b. Changes to the recommendations and solutions during the development of the Technical Design. Produce a building design for disassembly and adaptability manual to communicate the disassembly and adaptability principles and characteristics to prospective owners or building operators. A disassembly and adaptability manual shall be prepared and submitted, including the following information:

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	Credit	Available	Targeted	Potential	Comments
					a. Design details: Sufficient information shall be generated documenting specific disassembly methods, material composition, recovery methods, and adaptable design features. b. Material constituents and manufacturers: Products and materials should be traceable to a specific manufacturer or supplier, with contact details, such as product manufacturer websites, included in the construction documentation. c. Connection detailing: Connection detailing, especially for reversible connections, should be well documented for the aspect of assembly and disassembly. **Comments & Actions** • *Project Team/Architect to provide an update of a compliant disassembly and adaptability study before the end of Technical Design.* • *Client/developer to provide letter of commitment confirming that they will produce a building design for disassembly and adaptability manual.*
		9	8	0	**Standard Waste Credit Total**
		3	0	0	**Exemplary Waste Credit Total**
		10.02	6.24	0	**% Waste Total (Standard + Exemplary)**
Land Use & Ecology (0.81 % per Credit)					
Lue 01 - Site selection					
1	Previously occupied land	1	0	0	**Previously occupied land (1 credit)** **Targeted – No** At least 75% of the proposed development is on previously occupied land. **Comments & Actions** • *Project Team to confirm whether the Proposed Development is on ≥75% of previously occupied land.*
2	Contaminated land	1	0	0	**Contaminated land (1 credit)** **Targeted – No**

sre

	Credit	Available	Targeted	Potential	Comments
					A contaminated land professional undertakes a site investigation, risk assessment and appraisal, using ISO 18400 or equivalent, which deems that land within the development footprint to be affected by contamination. This report identifies: a. The degree of contamination b. The contaminant sources or types c. The options for remediating sources of contamination which present an unacceptable risk. The client or principal contractor confirms that a remediation strategy will be implemented, in line with the report. The credit can ONLY be awarded where remediation has taken place to enable development of the site for the assessed building. **Comments & Actions** • *Project Team confirmed that a contaminated land professional has undertaken a contaminated land study.* • *To confirm whether land remediation has/will be carried out.* • *Remediation strategy to be provided.*
Lue 02 - Ecological risks and opportunities					
1	Survey and evaluation	1	1	0	**Survey and evaluation (1 credit)** **Targeted – Yes** **Foundation route** The site is evaluated using the BREEAM Ecological Risk Evaluation Checklist (Guidance Note 34) confirming that the foundation route can be used. **Comprehensive route – Targeted** **Completed BEFORE end of RIBA Stage 2 (typically Preparation and Brief stage)** A suitably qualified ecologist (SQE) carries out a survey and evaluation for the site early enough to influence site preparation works, layout and, where necessary, strategic planning decisions (typically Preparation and Brief stage). The SQEs survey and evaluation determines the sites ecological baseline, including: a. Current and potential ecological value and condition of the site and related areas within the zone of influence.

	Credit	Available	Targeted	Potential	Comments
					b. Direct and indirect risks to current ecological value from the project. c. Capacity and feasibility for enhancement of the site's ecological value and, where relevant, areas within the zone of influence. **<u>Comments & Actions</u>** • *Project Team and Ecologist to provide a copy of the Ecological Survey and Evaluation report.*
2	Determining ecological outcomes	1	1	0	**Determining ecological outcomes (1 credit)** **Completed BEFORE end of RIBA Stage 2** **Targeted – Yes** **Foundation route** Achieve the criteria applicable to Foundation route for the Survey and evaluation credit. AND Relevant members of the project team liaise and collaborate, and where appropriate with representative stakeholders early enough to influence key planning decisions, to: a. Identify the optimal ecological outcomes for the site. b. Select measures to meet the optimal ecological outcomes for the site, in line with the mitigation hierarchy for the foundation route: i. Avoidance ii. Protection **Comprehensive route (Targeted)** Achieve the criteria applicable to Comprehensive route for the Survey and evaluation credit. AND The ecologist collaborates with appropriate project team members (e.g. landscape architect, architect etc.), sharing relevant information and recommendations from the survey and evaluation to influence the project design as well as decisions regarding site preparation and construction works.

	Credit	Available	Targeted	Potential	Comments
					The ecologist, with full support from the project team, collaborates with representative stakeholders early enough to influence key planning decisions (typically Concept Design stage). The ecologist identifies the optimal ecological outcomes for the site, and identifies measures to achieve these outcomes in line with: a. The collaborations undertaken with the project team and relevant stakeholders. b. The mitigation hierarchy for the comprehensive route: i. Avoidance ii. Protection iii. Reduction or limitation of negative impacts iv. On-site compensation v. Enhancement, considering the capacity and feasibility within the site, or where this is not viable, off-site. **<u>Comments & Actions</u>** • *Ecologist to provide a statement confirming that they collaborated with all the Project Team members and stakeholders necessary to identify and deliver the project's optimal ecological outcomes, no later than Concept Design.* • *Ecologist's report must identify the optimal ecological outcomes, and measures implemented (in line with the mitigation hierarchy).*
e1	Wider site sustainability - Exemplary credit	1	0	0	**<u>Comments & Actions</u>** • *This credit is not sought.*
Lue 03 - Managing impacts on ecology					
Pre-req	Prerequisite Ecological risks and opportunities	✓	✓		**Prerequisite Ecological risks and opportunities**

	Credit	Available	Targeted	Potential	Comments
					Whether a project is using the foundation route or the comprehensive route, the criteria in Lue 02 for 'Survey and evaluation' and 'Determining ecological outcomes' must have been met to achieve any credits in this issue.
1	Implementation of measures on-site	2	2	0	**Implementation of measures on-site (2 credits)** **Targeted – Yes** **Foundation and comprehensive route** For both routes, handover between the design and construction teams must be sufficiently structured and thorough to ensure that all planned ecological measures are communicated, understood, and adhered to throughout construction. **Comprehensive route (Targeted)** For the comprehensive route only, an ecologist will have been in place throughout planning and design. If an Ecological Clerk of Works has been appointed for the construction and handover stages (RIBA Stages 5 and 6), there must be an additional handover between the two ecologists. The measures required to achieve optimal ecological outcomes must be integrated with site preparation and construction processes. These must be measured and monitored throughout construction and handover (RIBA Stages 5 and 6) as follows: a. **Foundation route**: The contractor must nominate a biodiversity champion with authority to influence site activities and to ensure the necessary measures are implemented. b. **Comprehensive route**: An Ecological Clerk of Works (ECoW) will have been appointed by the contractor to monitor and advise on the protection of site ecology as agreed. If the ECoW requires the input of an additional ecological specialist at any point (for example, an expert on a particular form of wildlife), the contractor must provide for such services as required. **<u>Comments & Actions</u>** • *Project Team to confirm that an Ecologist has been appointed in the Design phase.* • *Project Team to provide tender documents incorporating ecological requirements to ensure ecological outcomes and measures required to achieve have been handed over to the demolition/construction team.*

	Credit	Available	Targeted	Potential	Comments
					• *Where a contractor has not yet been appointed at Design, developer to include the appointment of an Ecologist, and the facilitation of handover between Ecologists, in the tender documents.* • *Ecologist to provide a statement covering:* - *All measures to achieve optimal ecological outcomes.* - *How each measure are to be implemented.* - *What evidence should be provided by the contractor to demonstrate compliance at handover.* - *The management and monitoring actions to be undertaken by the Ecologist (if any).* • *The statement must also state the name of the contractor's appointed on-site biodiversity champion, and their role (including appropriate authority).*
2	No net loss	1	1	0	**No net loss (1 credit)** **Targeted – Yes** The criteria for Implementation of measures on-site credit have been achieved as above. There is no net loss of ecological value on site caused by construction and handover activities. This must be confirmed at project handover by the ecologist. **Comments & Actions** . • *The Ecologist must confirm via report or statement that there will be no net loss of ecological value on site.*
Lue 04 - Ecological change and enhancement					
Pre-req	Prerequisite Managing negative impacts on ecology	✓	✓		**Prerequisite Managing negative impacts on ecology** Achieve Implementation of measures on-site credit in Lue 03.
1	Ecological enhancement	1	1	0	**Ecological enhancement (1 credit)** **Targeted – Yes** **Foundation route** Locally relevant ecological measures have been implemented that enhance the site's ecological value. The measures adopted are based on:

	Credit	Available	Targeted	Potential	Comments
					a. Recommendations from recognised 'local' ecological expertise and specialist input and guidance b. Input from the project team in collaboration with representative stakeholders and data collated as part of 'Determining ecological outcomes' in Lue 02. **Comprehensive route** Measures have been implemented that enhance ecological value, which are based on input from the project team and SQE in collaboration with representative stakeholders and data collated as part of 'Determining ecological outcomes' in Lue 02. Measures are implemented in the following order: a. On-site, and where this is not feasible b. Outside of the site boundary, within the zone of influence Data collated are analysed and where potentially valuable, provided to the local environmental records centres nearest to, or relevant for, the site. **Comments & Actions** • *Ecologist to provide recommendations for ecological enhancement within the report.* • *Drawings and specifications demonstrating that the recommendations have been implemented in the design.*
2	Quantifying change in ecological value	3	1	0	**Quantifying change in ecological value (3 credits)** **Targeted – Yes** **Comprehensive route | Site baseline > 0.1 BU** Up to three credits are available based on the scale in Table 11.4 below. The enhancement measures that contribute to the score for biodiversity net gain should be implemented in the following order: a. On-site within the boundary, and where this is not feasible, b. Enhancements within the zone of influence/wider development area, and where this is not feasible, c. Off-site units following the rules of the Statutory Biodiversity Metric, subject to the following: i. The SQE confirms that better outcomes for nature can be achieved off-site and/or off-site enhancement would be more be effective than

Credit	Available	Targeted	Potential	Comments
				what can be done solely in the development boundary/zone of influence. ii. There has been no net loss of biodiversity on-site. iii. There are arrangements in place for the on-going maintenance of the enhanced land to ensure the habitat is safeguarded over a 30-year period. Table 11.4: Credits awarded based on output from the Statutory Biodiversity Metric tool **Total project biodiversity change** – **Credits** ≥ 10% – 1 ≥ 20% – 2 ≥ 30% – 3 ≥ 40% – 3 + 1 exemplary OR **Comprehensive route | Site baseline ≤ 0.1 BU** Up to three credits are available based on biodiversity units (BUs) per site area (ha) or per 10 m of hedgerow post development as shown in Table 11.5 below. This is only based on the hedgerow-based metric and area-based metric. Table 11.5: Credits awarded based on biodiversity units (BUs) per site area (ha) and per length of hedgerow **Biodiversity units per site area (BU/ha)** – **Biodiversity units per 10m length of hedgerow (BU/10m length)** – **Credits** ≥ 1.50 – ≥ 0.01 – 1 ≥ 2.00 – ≥ 0.03 – 2 ≥ 2.50 – ≥ 0.04 – 3 ≥3.00 – ≥ 0.05 – 3 + 1 exemplary **Comments & Actions** • *Ecologist to provided completed version of the Statutory Biodiversity Metric tool in order to award the credits.*

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	Credit	Available	Targeted	Potential	Comments
e1	Quantifying change in ecological value - Exemplary credit	1	0	0	**<u>Comments & Actions</u>** • *This credit is not sought due to high benchmarks.*
Lue 05 - Long term ecological management					
Pre-req	Prerequisite Ecological change and enhancement	✓	✓		**Prerequisite Ecological change and enhancement** Achieve at least one credit in Lue 04.
1	Handover to building owner or tenant	1	1	0	**Handover to building owner or tenant (1 credit)** **Targeted – Yes** A section on ecology and biodiversity has been included as part of the information supplied to the tenant or building owner. This information must be in a separate document from the landscape and ecology management plan and: a. Inform the building owner or tenant about local ecological features and biodiversity associated with the site including an 'as-built' biodiversity metric calculation (where relevant). b. Include detailed management and maintenance plans and relevant handover information to minimise loss of planting during establishment phase. **<u>Comments & Actions</u>** • *Client/developer to confirm ecological and biodiversity handover information will be provided.* • *Project Team to provide copy of ecological and biodiversity handover information.*
2	Landscape and ecology management plan	1	1	0	**Landscape and ecology management plan (1 credit)** **Targeted – Yes** A landscape and ecology management plan, or equivalent, has been developed in accordance with BS 42020:2013 Section 11.1. This will cover detailed plans for an initial 5-year establishment period and maintenance of the scheme, as well as an additional 25-year management strategy. The whole plan should cover a period of 30 years in total. It should cover the broader management of habitats and other ecological features on site, not just those specific to the BNG measures. The plan includes: a. Actions and responsibilities of relevant individuals before handover.

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	Credit	Available	Targeted	Potential	Comments
					b. The ecological value and condition of the site at handover and how this is expected to develop and change over time. c. Procedures to monitor and review the on-going implementation and effectiveness of measures, incorporating an adaptive management approach. d. Risk register, including trigger for action and remedial measures with consideration on how these feed into monitoring requirements. e. Maintenance and management plans for the retained, enhanced and created habitats, as well as any features for protected species. This should include a detailed work schedule outlining relevant maintenance on a monthly basis for the first five years after completion. f. Identifying opportunities for ongoing alignment with activities external to the asset. g. Clearly defined responsibilities and competencies of those delivering the plan. **Comments & Actions** • *Project Team to provide a draft copy of the Landscape and Ecological Management plan.*
		13	9	0	**Standard Land Use & Ecology Credit Total**
		2	0	0	**Exemplary Land Use & Ecology Credit Total**
		12.53	7.29	0	**% Land Use & Ecology Total (Standard + Exemplary)**
Pollution (0.62 % per Credit)					
Pol 01 - Impact of refrigerants					
1	Impact of refrigerants	3	2	0	**Impact of refrigerants (2 credits)** **Targeted – Yes** **Specific note – credits may only be awarded where clauses are included in a green lease agreement that ensure that the relevant criteria will be met.** Up to two credits can be achieved where the building does not require the use of refrigerants within its installed plant or systems. *Buildings with systems that use refrigerants **(Assumed)***

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Credit	Available	Targeted	Potential	Comments
				Prerequisite: All systems with electric compressors must comply with relevant safety standards for installing refrigerant systems. Safety standards for installation of refrigeration systems: - EN 378-3:2016+A1:2020 – Refrigerating systems and heat pumps. Safety and environmental requirements – Installation site and personal protection - ISO 5149:2014 – Refrigeration systems and heat pumps - ASHRAE 15-2016 – Safety Standard for Refrigeration Systems - Safety Code of Practice for Ammonia Refrigerant R717 – Institute of Refrigeration - GBT9237-2017 Standard for Safety and Environmental Requirements for Refrigeration Systems and Heat Pumps The refrigerants used must have an **ozone depletion potential of zero.** *Impact of refrigerant* **One credit:** Systems using refrigerants must have an average direct effect life cycle (DELC) carbon dioxide equivalent (CO_2e) emission rate of < 1000 kgCO_2e per kW of cooling or heating capacity (Targeted). **Two credits**: Systems using refrigerants must either: a. Have an average direct effect life cycle (DELC) carbon dioxide equivalent (CO_2e) emission rate of < 100 kgCO_2e per kW cooling or heating capacity. OR b. Use a refrigerant with a global warming potential (GWP) of < 10 kgCO_2e/kg. *Leak detection – not targeted* Systems using refrigerants are either: a. Hermetically sealed, OR; b. The refrigerant charge volume is < 500 tonnes of carbon dioxide equivalent (CO_2e), OR; c. Use environmentally benign refrigerants. OR

	Credit	Available	Targeted	Potential	Comments
					There must either be: a. A permanent automated refrigerant leak detection system that is capable of continuously monitoring for leaks, OR; b. An inbuilt automated diagnostic procedure for detecting leakage capable of continuously monitoring for leaks. The leak detection system must facilitate the isolation and containment of the remaining refrigerant charge in response to a leak detection incident. **Comments & Actions** • *MEP confirmed that 2 credits is assumed at this stage.* • *MEP to provide letter of confirmation declaring systems using refrigerants will have an average direct effect life cycle (DELC) carbon dioxide equivalent (CO_2e) emission rate of < 1000 $kgCO_2e$ per kW of cooling or heating capacity.* • *All relevant datasheets for refrigerant type, and systems using refrigerants.* *Examples:* *Ensure that the chosen refrigerant has a GWP of* < 10 $kgCO_2e$/kg. *(e.g. R433B, R441A, R432A, R443A).* *Hermetically sealed refrigerants or, consider using environmentally benign refrigerants (e.g. air, water, carbon dioxide, butane, lithium bromide, solid refrigerants)*
Pol 02 - Local air quality					
1	No or low emission combustion plant	3	3	0	**No or low emission combustion plant (3 credits)** **Targeted – Yes** **Three credits – No installed combustion plant:** Non-combustion systems supply all space heating and domestic hot water generation on-**site (Assumed).** **Up to two credits – Low-emission combustion plant**: Emissions from all installed combustion plant that provide space heating or domestic hot water do not exceed the levels set below:

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	Credit	Available	Targeted	Potential	Comments
					Table 12.2: Credits awarded based on NO_x emission levels (mg/kWh) by population density **Maximum mg NO_x/kWh fuel input in terms of gross calorific value (GCV)** 1 credit (low population locations): 56 1 credit (high population locations): 27 2 credits (low population locations): 27 2 credits (high population locations): 24 **<u>Comments & Actions</u>** • *It is assumed that all electric plant equipment will be installed.* • *MEP design drawings to be provided showing all heat-generating equipment for space heating and domestic hot water in the building.*
Pol 03 - Flood and surface water management					
Pre-req	Prerequisite Appropriate consultant	✓	✓		**Prerequisite Appropriate consultant** An appropriate consultant is appointed to carry out and demonstrate the development's compliance with the all the criteria for this issue.
1	Flood resilience	2	1	0	**Flood resilience (2 credits)** **Targeted – Yes (1 credit)** **Two credits Low flood risk (Not Targeted)** A site-specific flood risk assessment (FRA) confirms the development is in a flood zone that is defined as having a low annual probability of flooding. The FRA takes all current and future sources of flooding into consideration. **One credit Medium or high flood risk (Targeted)** A site-specific FRA, developed by an appropriate consultant, confirms the development is in a flood zone that is defined as having a medium or high annual probability of flooding and is not in a functional floodplain. The FRA must take all current and future sources of flooding into consideration. As the site is **<2000m²** an acceptable FRA could be a brief report carried out by the contractor’s engineer confirming the risk of flooding. This must include the risk from all sources of flooding, and information obtained from the Environment Agency, water company or sewerage undertaker, other relevant statutory authorities, site investigation and local knowledge.

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	Credit	Available	Targeted	Potential	Comments
					To increase the resilience and resistance of the development to flooding, one of the following must be achieved: a. The ground level of the building and access to both the building and the site, are designed (or zoned) so they are at least 600 mm above the design flood level of the site's flood zone. b. The final design of the building and the wider site reflects the recommendations made by an appropriate consultant in accordance with the hierarchy approach outlined in section 5 of BS 8533:2017. **Comments & Actions** • *Desk based study carried out and it is assumed that the site exists in a low flood risk zone.* • *Project Team and Drainage consultant to provide a copy of a compliant Flood Risk Assessment.*
2	Surface water run-off	2	2	0	**Surface water run-off (2 credits)** **Targeted – Yes** **Pre-requisite for surface water run-off credits** Surface water run-off design solutions must be bespoke, i.e. they must take account of the specific site requirements and natural or man-made environment of and surrounding the site. The priority levels detailed below must be followed, with justification given by the appropriate consultant where water is allowed to leave the site. Priority levels: - Priority Level 1: Water is collected for use in the development (e.g. rainwater reuse) - Priority Level 2: Water is infiltrated into the ground - Priority Level 3: Water is discharged to surface water body - Priority Level 4: Water is discharged to the drainage system - Priority Level 5: Water is discharged to a combined sewer **One credit Surface water run-off Rate** For brownfield sites, drainage measures are specified so that the peak rate of run-off from the site to the watercourses (natural or municipal) shows a 30% improvement for

Credit	Available	Targeted	Potential	Comments
				the developed site compared with the pre-developed site. This should comply at the 1-year and 100-year return period event AND For greenfield sites, drainage measures are specified so that the peak rate of run-off from the site to the watercourses (natural or municipal) is no greater for the developed site than it was for the pre-development site. This should comply at the 1-year and 100-year return period events AND Relevant maintenance agreements for the ownership, long term operation and maintenance of all specified Sustainable Drainage Systems (SuDS) are in place AND Calculations include an allowance for climate change. This should be made in accordance with current best practice planning guidance. **One credit Surface water run-off Volume** Flooding of assessed buildings will not occur in the event of local drainage system failure (caused either by extreme rainfall or a lack of maintenance); AND EITHER a. Specify drainage design measures that ensure post-development surface water run-off volume is no greater than the pre- development run-off volume. Show this with calculations for a 100-year 6-hour event adjusted for climate change, timing the event at the end of the developments expected life AND any additional predicted volume of run-off for this event is prevented from leaving the site by using infiltration or other SuDS techniques. OR (only where the above (a.) cannot be achieved): b. Justification from the appropriate consultant indicating why the above criteria cannot be achieved, i.e. where infiltration or other SuDS techniques are not technically viable options. Drainage design measures are specified so that the post-development peak rate of run-off is reduced to the limiting discharge. The limiting discharge is defined as the highest flow rate from the following options: i. The pre-development one-year peak flow rate ii. The mean annual flow rate (Qbar)

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	Credit	Available	Targeted	Potential	Comments
					iii. 2 L/s/ha. For the one-year peak flow rate, the one-year return period event criterion applies. Relevant maintenance agreements for the ownership, long term operation and maintenance of all specified SuDS are in place. For either option, above calculations must include an allowance for climate change; this should be made in accordance with current best practice planning guidance. **<u>Comments & Actions</u>** • *Surface water drainage report and calculations required to show the pre-development, greenfield and post-development peak rate of run-off and volume run-off.* • *Design drawings required to show the drainage design.* *Examples:* *SuDS management to reduce run-off rate and volume:* - *Source control: Porous paving; Rainwater harvesting; Green roofs etc.* - *Site or local control: Infiltration basins; Swales; Detention basins etc.*
3	Minimising watercourse pollution	1	1	0	**Minimising watercourse pollution (1 credit)** **Targeted – Yes** There is no discharge from the developed site for rainfall up to 5 mm (confirmed by the appropriate consultant). Areas with a low risk source of watercourse pollution, have an appropriate level of pollution prevention treatment provided using appropriate SuDS techniques. Areas with a high risk of contamination or spillage of substances, such as petrol and oil, have separators (or an equivalent system) installed in surface water drainage systems. Chemical or liquid gas storage areas have a means of containment fitted to the site drainage system (i.e. shut-off valves). This is to prevent the escape of chemicals to natural watercourses in the event of a spillage or bunding failure.

	Credit	Available	Targeted	Potential	Comments

					All water pollution prevention systems have been designed and installed in accordance with the recommendations of documents such as the SuDS manual and other relevant industry best practice. They must be bespoke solutions taking account of the specific site requirements and natural or man-made environment of and surrounding the site.
					
					A comprehensive and up to date drainage plan of the site will be made available for the building or site occupiers.
					
					Relevant maintenance agreements for the ownership, long term operation and maintenance of all specified SuDS must be in place.
					
					All external storage and delivery areas are designed and detailed in accordance with the current best practice planning guidance.
					
					<u>Comments & Actions</u>
					
					• *Project Team/Structural Engineer confirmed credit is achievable.*
					• *Drainage report to include calculations confirming there will be no discharge from the site for rainfall up to 5mm.*
					• *Design drawings showing the drainage strategy:*
					    - *High/low risk areas on site*
					    - *Specification of SuDS, source control systems, oil or petrol separators and shut-off valves as appropriate.*
					    - *Water pollution prevention systems.*
					    - *External storage and delivery areas.*
					
					Letter of confirmation from the appropriate consultant to declare agreements for ownership, long-term operation and maintenance of all specified SuDS must be in place.

Pol 04 - Reduction of nighttime light pollution					
1	Reduction of light pollution	1	1	0	**Reduction of light pollution (1 credit)** **Targeted – Yes** Light pollution has been eliminated through effective design that removes the need for artificial lighting at night. This does not adversely affect the safety and security of the site and its users.

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<table>
<tr><th></th><th>Credit</th><th>Available</th><th>Targeted</th><th>Potential</th><th>Comments</th></tr>
<tr><td></td><td></td><td></td><td></td><td></td><td>OR

Where the building does use artificial lighting at night the lighting strategy has been designed to reduce obtrusive light. This includes both external lighting and internal lighting in spaces with glazed apertures AND;

All external lighting (except for safety and security) is automatically switched off between 23:00 and 07:00 AND;

Illuminated advertisements are designed to control brightness and minimise visual disturbance during nighttime hours AND;

Lighting is designed to minimise impacts on nocturnal wildlife.

The lighting strategy design should follow the five sets of recommendations outlined in the Institution of Lighting Professionals (ILP) Guidance Note 01/21 – The reduction of obtrusive light (GN01/21):

1. Limitation of illumination on surrounding premises
2. Limitation of bright luminaires in the field of view
3. Limitation of the effects on transport systems
4. Limitation of skyglow
5. Limitations of the effect of over-lit building facades and signs

<u>Comments & Actions</u>

• Project Team/MEP to confirm that the external lighting to be installed will comply with the standards.
• Design drawings and lighting specifications must be provided, where available.

Examples:

Limit vertical illuminance where possible and make use of down lighters/cut-off type and angled luminaires.</td></tr>
<tr><td colspan="6">Pol 05 - Reduction of noise pollution</td></tr>
<tr><td>1</td><td>Noise level</td><td>1</td><td>1</td><td>0</td><td>Noise level (1 credit)
Targeted – Yes</td></tr>
</table>

Credit	Available	Targeted	Potential	Comments
				There are no noise-sensitive areas within the assessed building or within 800m radius of the assessed site OR Where there are noise-sensitive areas within the assessed building or noise-sensitive areas within 800m radius of the assessed site, a noise impact assessment compliant with BS 4142:2014(145) is commissioned. Noise levels must be measured or determined for: a. Existing background noise levels: i. At the nearest or most exposed noise-sensitive development to the proposed assessed site. ii. Including existing plant on a building, where the assessed development is an extension to the building. b. New noise level from the assessed building The noise impact assessment must be carried out by a suitably qualified acoustic consultant. The noise level from the assessed building, as measured in the locality of the nearest or most exposed noise-sensitive development, must be at least 5 dB lower than the background noise throughout the day and night. If the noise sources from the assessed building are greater than the levels described above, measures have been installed to attenuate the noise at its source to a level where it will comply with the noise level limits. **Comments & Actions** • *Project Team to appoint suitably qualified acoustician to produce a compliant external noise impact assessment.* • *Client/developer commits to completing a post-construction testing.* • *Relevant design drawings confirming any attenuation measures or recommendations from the acoustician.* *Examples:*

	Credit	Available	Targeted	Potential	Comments
					Install noise attenuators / sound screening around any external plant equipment if relevant.
		13	11	0	**Standard Pollution Credit Total**
		0	0	0	**Exemplary Pollution Credit Total**
		8.06	6.82	0	**% Pollution Total (Standard + Exemplary)**
Innovation (1 % per Credit)					
Inn 01 - Innovation					
1	Approved innovations	1	0	0	**Approved innovations (1 credit)** **Targeted – No** One innovation credit can be awarded for each innovation application approved by BRE Global, where the building complies with the criteria defined within an approved innovation application form.
		0	0	0	**Standard Innovation Credit Total**
		1	0	0	**Exemplary Innovation Credit Total**
		1	0	0	**% Innovation Total (Standard + Exemplary)**
			79	31	**OVERALL TOTALS**
			62.93	20.47	**% OVERALL TOTALS**

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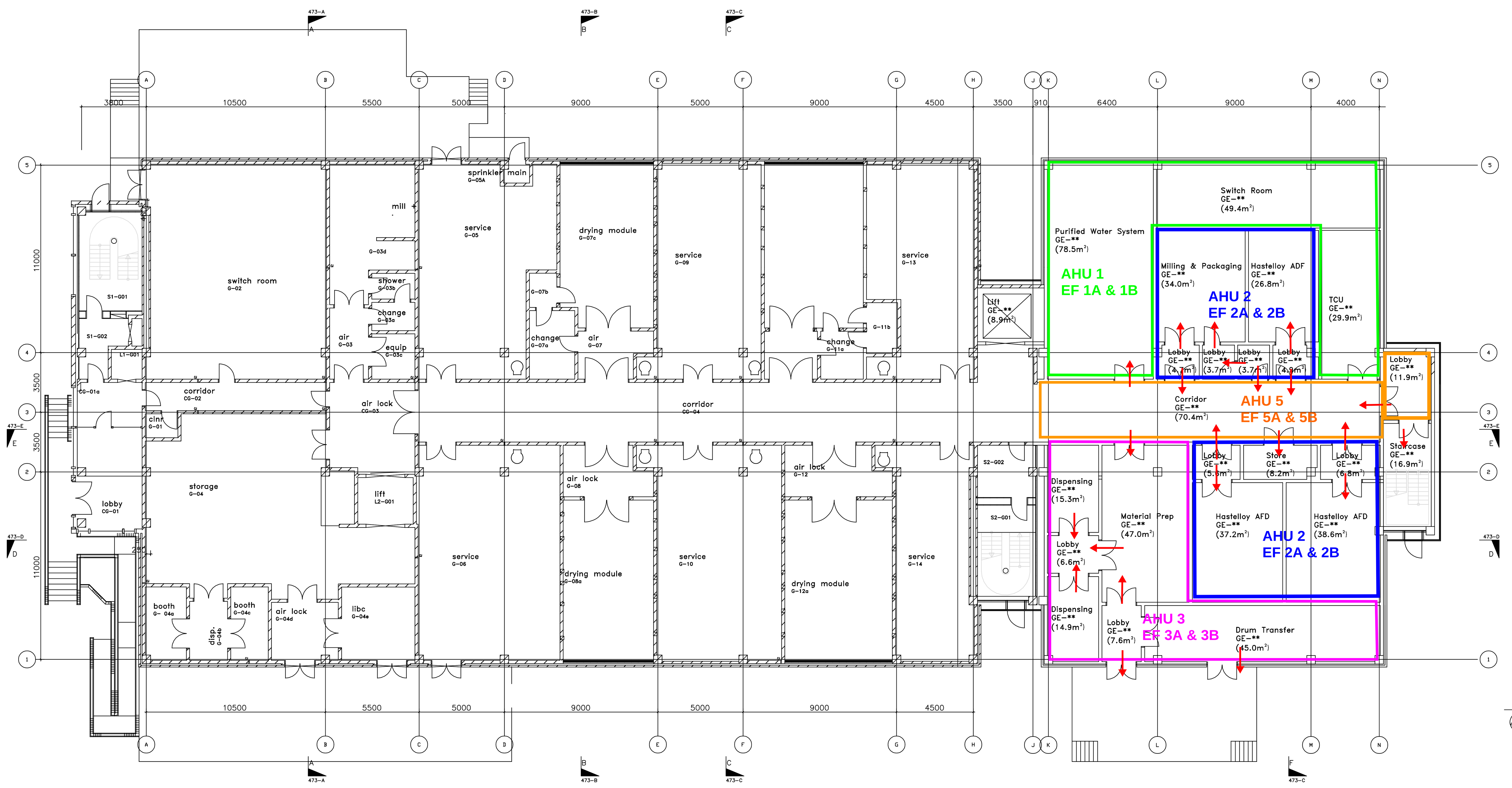
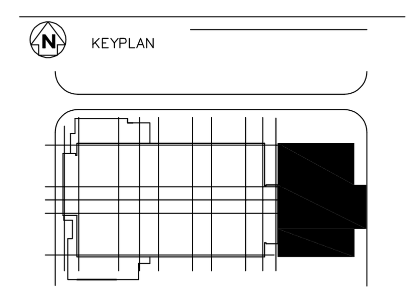
Appendix 3 - Mechanical Drawings



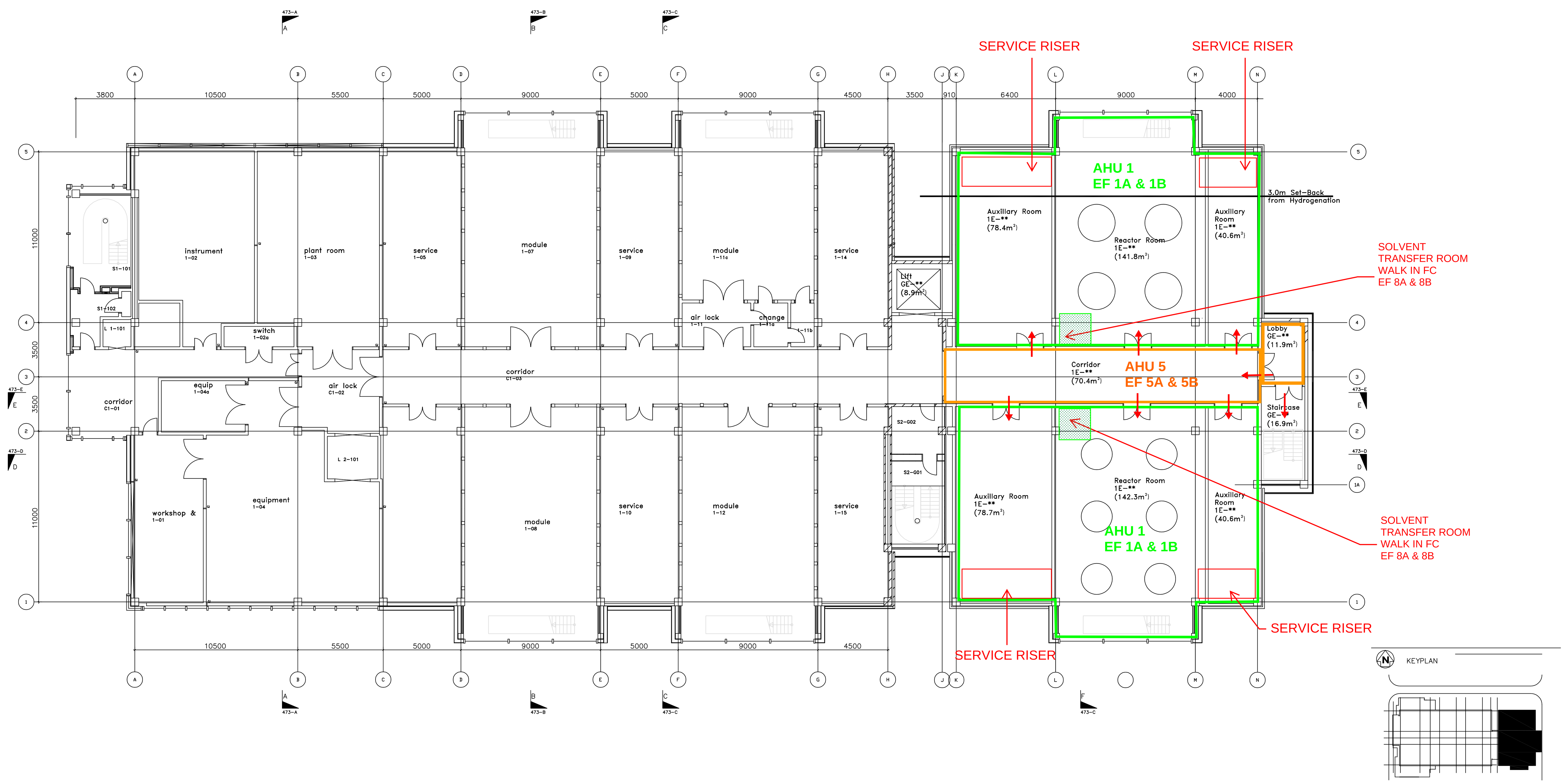


PRESSURE REGIME & AHU ZONING LAYOUT
GROUND FLOOR

Scitech EKIUM

DRAWING No.: 300291-BS-SK-0001

DATE: 22/05/26 SCALE: 1:200 REV: P1

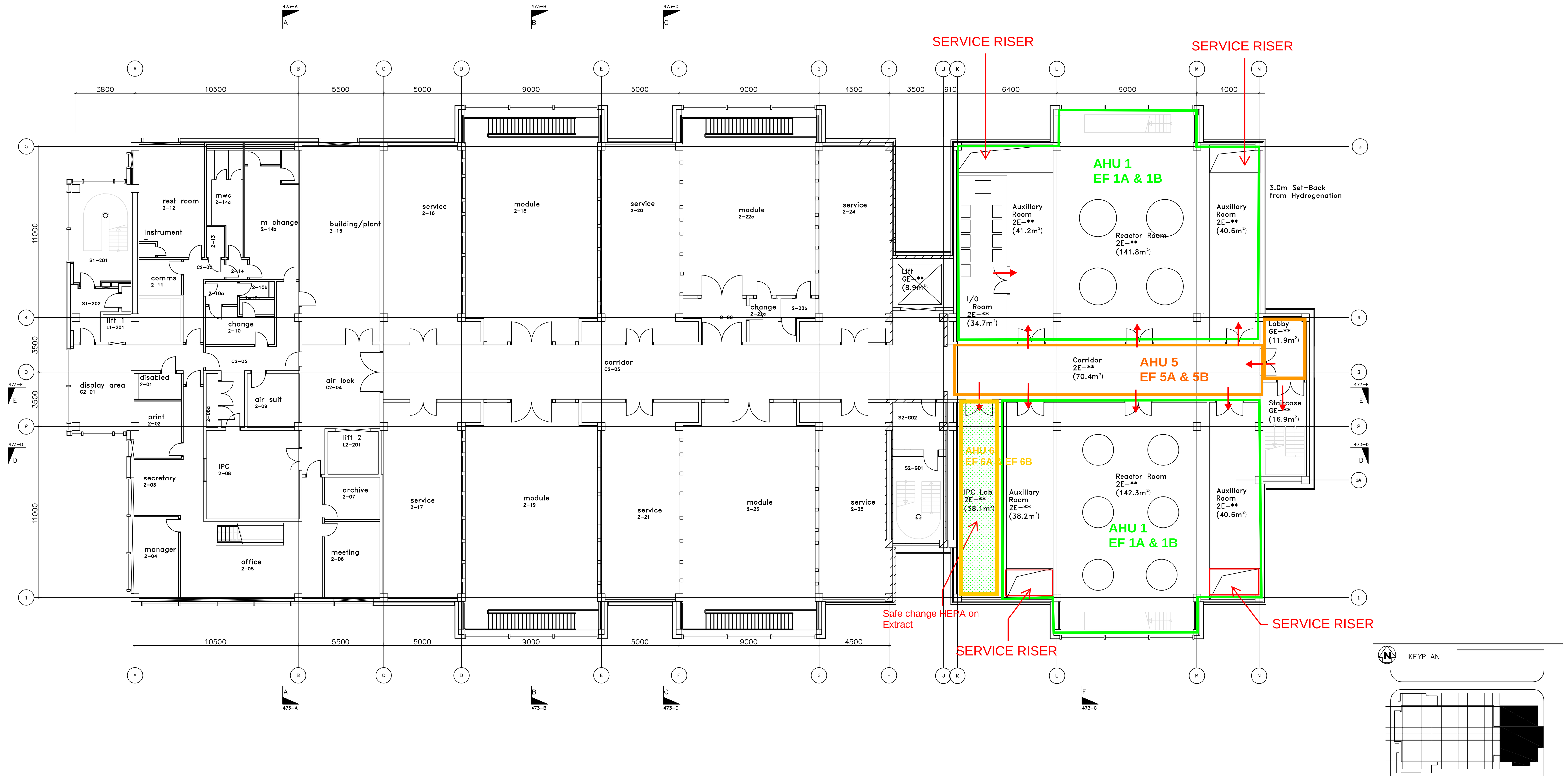


PRESSURE REGIME & AHU ZONING LAYOUT
FIRST FLOOR

Scitech
EKIUM

DRAWING No.: 300291-BS-SK-0002

DATE: 22/05/26 SCALE: 1:200 1 REV: P1



PRESSURE REGIME & AHU ZONING LAYOUT SECOND FLOOR

Scitech EKIUM

DRAWING No.: 300291-BS-SK-0003

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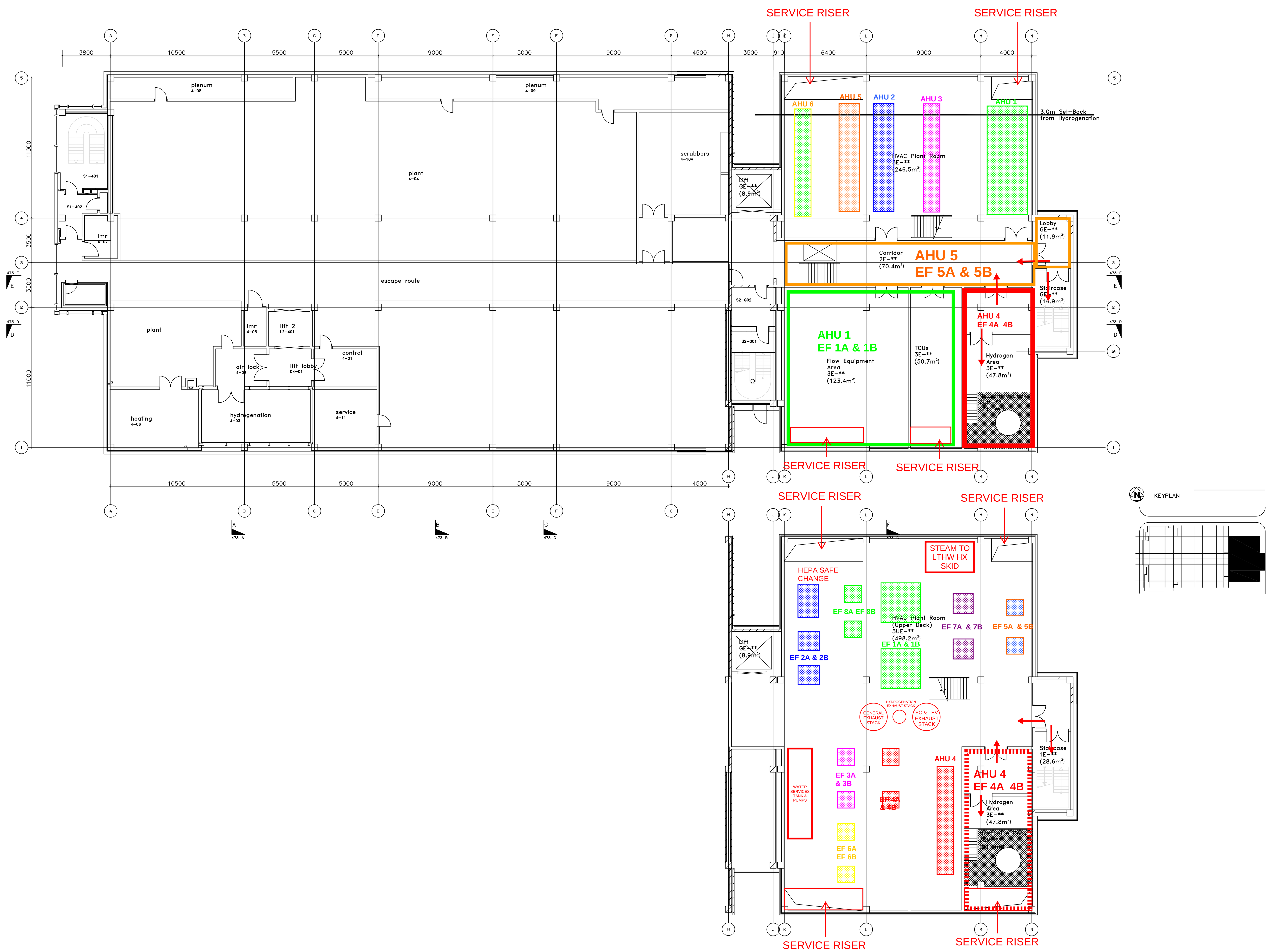
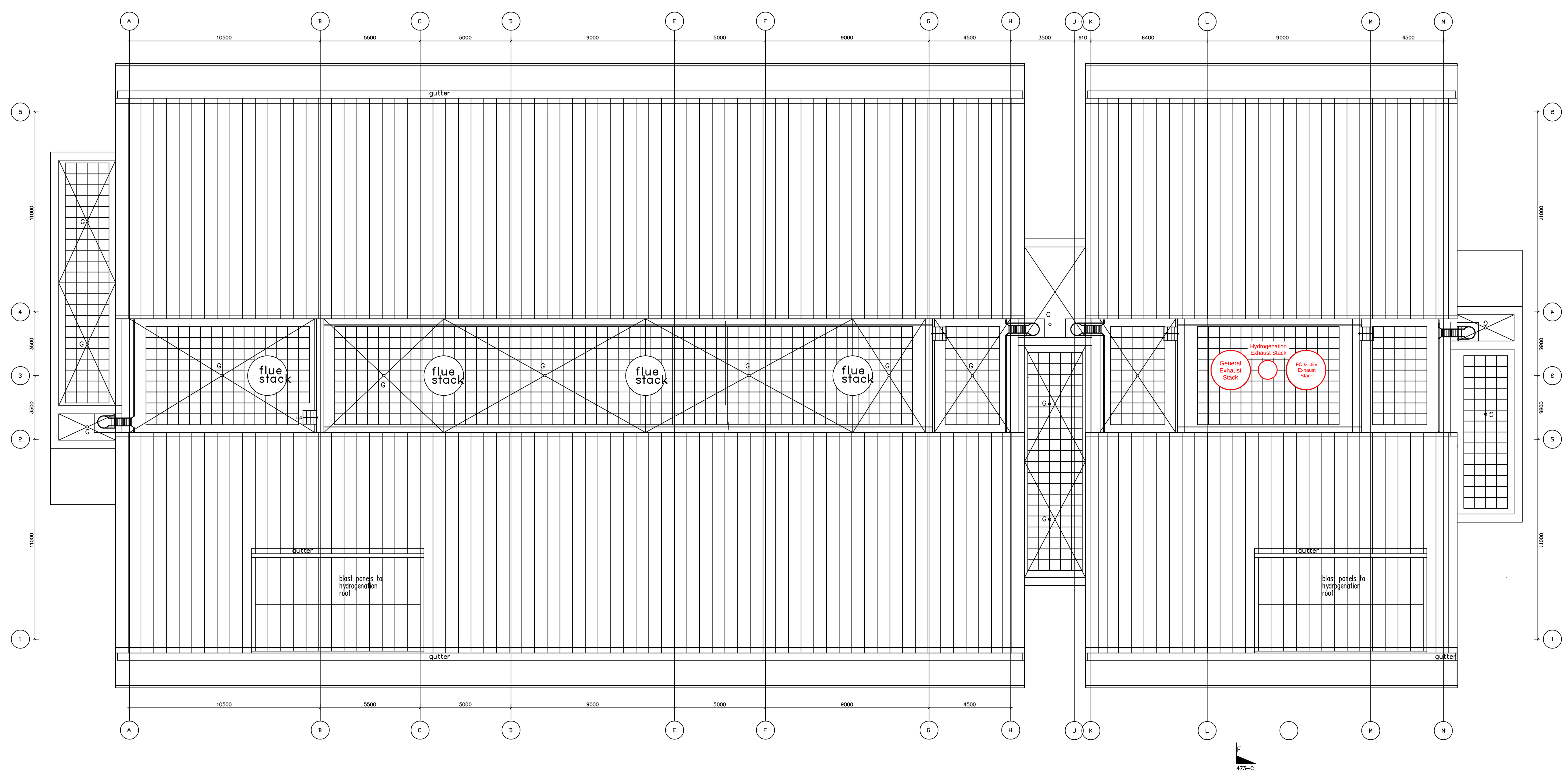


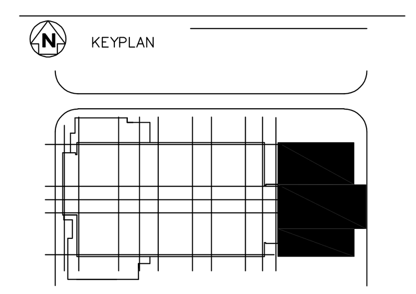
PRESSURE REGIME & AHU ZONING LAYOUT
THIRD FLOOR & PLANT MEZZANINE

Scitech EKIUM

DRAWING No.: 300291-BS-SK-0004

DATE: 22/05/26 SCALE: 1:200 REV: P1



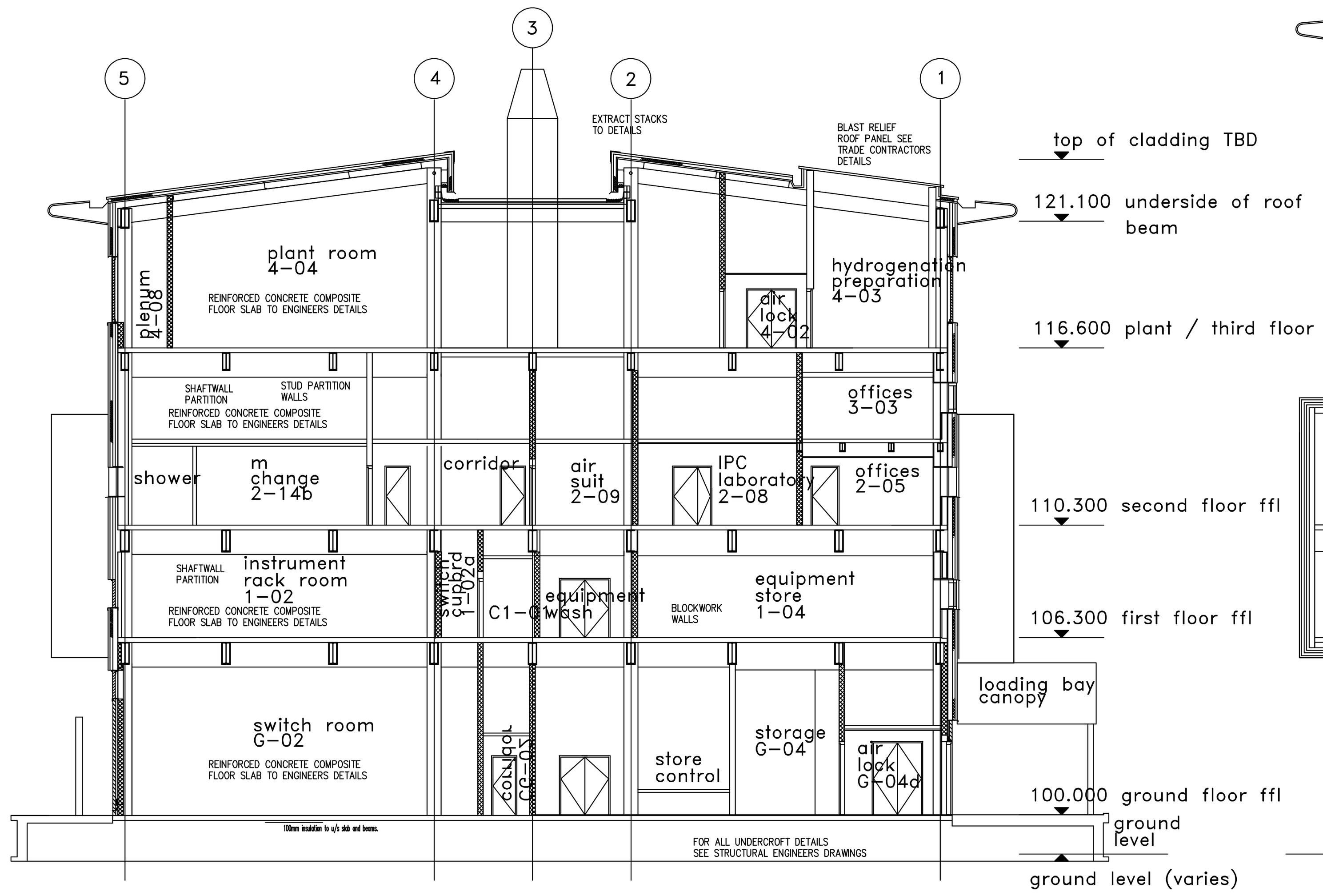


PRESSURE REGIME & AHU ZONING LAYOUT
ROOF FLOOR

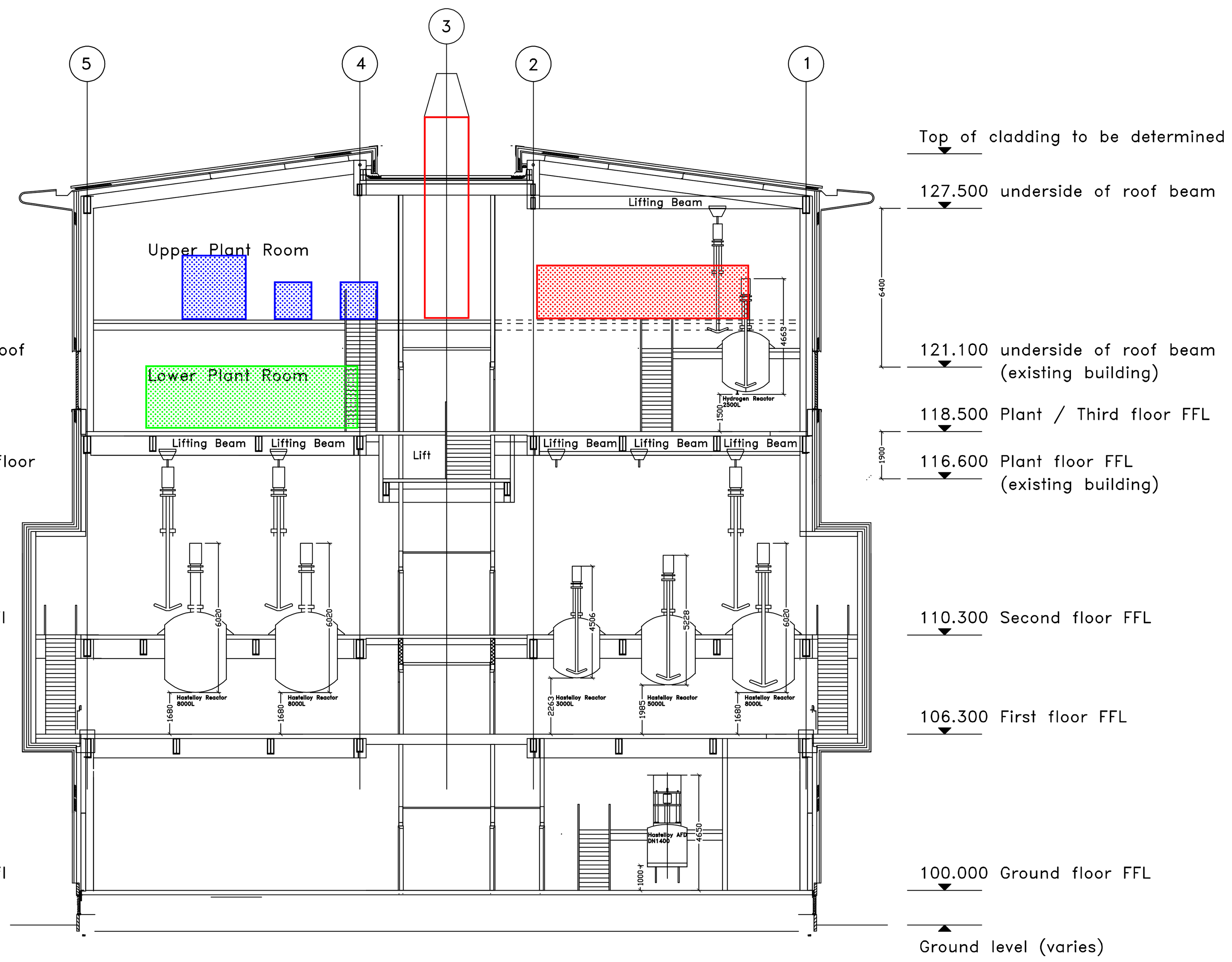
Scitech EKIUM

DRAWING No.: 300291-BS-SK-0005

DATE: 22/05/26 SCALE: 1:200 REV: P1



TYPICAL SECTION THROUGH OFFICE AREA
EXISTING BUILDING



TYPICAL SECTION THROUGH REACTOR SPACE
PROPOSED EXTENSION

PRESSURE REGIME & AHU ZONING SECTION

Scitech EKIUM

DRAWING No.: 300291-BS-SK-0006

DATE: 22/05/26 SCALE: 1:200 REV: P1

Appendix 4 - Utility Capacity Assessment Calculations

		CALCULATION SHEET	
		CALC NO:	**300291-PR-CA-0001**
		ISSUE No:	**A1**
PROJECT NO:	**300291**	**BY:**	**ME**
TITLE:	**Asymchem PDF Extension Facility**	**CHKD:**	**MC**
SUBJECT:	**Utility Capacity Assessment Calculations**	**APPROVED**	**ME**
		DATE:	**21-May-26**

References

Objective

• This assessment determines whether existing utility systems can accommodate the new extension facility, or whether new/upgraded infrastructure is required.
• The extension introduces 10 new reactors (R25–R34) ranging from 2,500L to 8,000L — significantly larger than the existing facility maximum of 2,500L (R12, R17). The extension also has a dedicated hydrogenation reactor (R35; 2500L compared to existing 250L R23 hydrog reactor) and 3x filter dryers (1x 1130L and 2x 1530L)
• Detailed utility usage data for the extension equipment is not yet available. A reactor volume scale-up factor of 3.54x has therefore been applied throughout as a conservative feasibility-level estimate.
• This approach will be refined as process design progresses and usage data becomes available at later design stages.

Summary

See specific calculation sheets

Assumptions

See specific calculation sheets

Project:	Asymchem PDF Extension Facility				
Project No:	300291				
Subject:	Utility Capacity Assessment Calculations			**Document No:**	300291-PR-CA-0001
By:	ME	**Checked:**	MC	**Sheet Name:**	Reactor Volume Scale-up
Approved:	ME	**Date:**	21-May-26	**Revision:**	A1

Calculation	Notes

Scale-up for reactor volume and utility usage assessment

1) To estimate the scale-up in reactor volumes between the existing facility and new extension facility

Scenario: In order to estimate the utility usage in the new extension facility (as this is not known), a scale-up factor will be applied for the process utilities. This will be based on the ratio between the total reactor volumes of the existing facility and the new extension facility. Utility usages for the existing facility will be multiplied by this scale-up factor to estimate the utility usage by the new extension facility.

Notes: Assumption using volume scale-up factor is based on unavailability of utility usage data for equipment in new extension facility

Existing Facility

Notes: Existing facility equipment data obtained from *Pilot Plant Equipment List 15 March 2026*

	Tag No.	Facility Room No.	Volume (L)
Reactors	R01	P2.11	250
	R02	P2.11	250
	R03	P2.11	630
	R04	P2.11	630
	R05	P2.11	1600
	R06	P2.11	1600
	R07	P2.09	250
	R08	P2.09	630
	R09	P2.09	630
	R10	P2.09	1600
	R11	P2.09	1600
	R12	P2.09	2500
	R13	P2.03	630
	R14	P2.03	250
	R15	P2.03	1600
	R16	P2.03	630
	R17	P2.03	2500
	R18	P2.03	1600
	R19	P2.05	630
	R20	P2.05	250
	R21	P2.05	1600
	R22	P2.05	630
Hydrogenation	R23	P2.05	250
Filter Dryers	MFD01	P1.13	125
	MFD02	P1.13	300
	FD02-RC1	PG.05	250
	FD03-RC1	PG.09	750
	FD04-RC1	PG.09	300
	FD05-RC1	PG.03	600
		Total	25065 L

Total volume of vessels in existing facility

New Extension Facility

Notes: New extension facility equipment data obtained from *SW-902Expansion-Equipment_List-20 March*

	Tag No.	Facility Room No.	Volume (L)
Reactors	R25	TBC	5000
	R26	TBC	8000
	R27	TBC	5000
	R28	TBC	8000
	R29	TBC	2500
	R30	TBC	2500
	R31	TBC	5000
	R32	TBC	5000
	R33	TBC	8000
	R34	TBC	8000
Hydrogenation	R35	TBC	2500
Filter Dryers	FD01	TBC	1130
	FD02	TBC	1530
	FD03	TBC	1530
		Total	63690 L

Total volume of vessels in new extension facility

Reactor Volume Scale-up Ratio

Existing facility total reactor volume	25065	L
New extension facility total reactor volume	63690	L
Reactor scale-up ratio	**3.54**	

Project:	Asymchem PDF Extension Facility				
Project No:	300291				
Subject:	Utility Capacity Assessment Calculations			**Document No:**	300291-PR-CA-0001
By:	ME	**Checked:**	MC	**Sheet Name:**	Hot HTF - Process
Approved:	ME	**Date:**	21-May-26	**Revision:**	A1

Line No.	Calculation	Notes

Heading Loads

Information on material properties obtained from engineeringtoolbox

HTF (Heat Transfer Fluid) is the cold/hot utility used. In this case, HTF fluid is Syltherm XLT.

1) To estimate the maximum delta T that can be achieved for hot HTF utility based on the existing heater

Scenario: The existing pilot plant has a 510kW heater which heats HTF from the HTF tank to obtain 130C hot HTF for maintaining the temperature of the facility reactor vessels. The existing hot HTF supply pump has a fixed pumping rate of 50m3/hr which supplies hot HTF to the reactor jackets.

Item	Value	Unit	Remark	Notes
Existing Hot HTF heater heat capacity	510	kW		
Hot HTF heat capacity	2.024	kJ/kg°C	assumption	Hot HTF heat capacity at 130C
Hot HTF density	732	g/cm^3	assumption `	Hot HTF density at 130C
Flowrate	50	m^3/hr		Based on existing supply pump capacity
	10.17	kg/s		
Supply Temperature	130	°C		
Maximum Delta T Hot HTF	**24.78**	°C		
Minimum return T Hot HTF	**105.2**	°C		

2) To calculate the hot HTF flowrate required to achieve desired temperature control in the solvent reactors

Scenario: Hot HTF is used to control temperature of solvent mixture in the reactor to a specific range of 0.5C - 1C/min. Hot HTF is pumped to the reactor jackets to achieve this temperature control where it is supplied at 130C and returns at 105C. Solvents used are acetone and methanol

Item		Value	Unit	Remark	Notes
Temperature control to achieve in reactors		0.5	C/min		
		1.0	C/min		
Existing Hot HTF heater heat capacity		510	kW		
Hot HTF heat capacity		2.024	kJ/kg°C	assumption	Hot HTF heat capacity at 130C
Solvent:	Acetone heat capacity	2.14	kJ/kg°C		Acetone heat capacity at standard conditions
	Methanol heat capacity	**2.8**	kJ/kg°C		Methanol heat capacity at standard conditions
Hot HTF density		732	g/cm^3	assumption	Hot HTF density at 130C
Solvent:	Acetone density	**785**	g/cm^3		Acetone density at standard conditions
	Methanol density	753	g/cm^3		Methanol density at standard conditions
Hot HTF Supply Temperature		130	°C		
Hot HTF Return Temperature		105.2	°C		

Scenario is based on 2500L reactor. This is the largest reactor volume in the existing facility so is considered the worst-case/ largest heating load here.

For this calculation, the worst-case scenario will be based on the largest value for density and heat capacity for the solvents. Methanol has higher heat capacity (2.8 kJ/kg°C) and acetone has higher density at 785 kg/m^3)

Item	Value	Unit				Notes
Reactor volume	2500	L	Vessel filling volume for process	100%		Vessel filled with solvent for processing
Mass of solvent in reactor	1963	kg				Based on acetone density
0.5°C/min reactor temperature control						
Hot HTF cooling load	**45.8**	**kW**				0.5°C/min reactor temperature
Hot HTF flowrate	**0.91**	**kg/s**		**4.5**	**m^3/hr**	
1°C/min reactor temperature control						
Hot HTF cooling load	**91.6**	**kW**				1°C/min reactor temperature
Hot HTF flowrate	**1.83**	**kg/s**		**9.0**	**m^3/hr**	

3) Hot HTF diversity assessment on all reactors in existing and new extension facility

An assessment on the hot HTF diversity is required to determine the operability of the new extension facility with regards to hot HTF used for reactor temperature control. This will be based on the existing facility reactor volumes. A scale-up will be applied for the new extension facility based on the reactor volumes to estimate potential diversity in hot HTF usage.

0.5°C/min reactor temperature control

Existing Facility Reactors			New Extension Facility Reactors		
Tag No.	Reactor Volume (L)	Heat Load (kW)	Tag No.	Reactor Volume (L)	Heat Load (kW)
R01	250	4.6	R25	5000	91.6

R02	250	4.6		R26	8000	146.5
R03	630	11.5		R27	5000	91.6
R04	630	11.5		R28	8000	146.5
R05	1600	29.3		R29	2500	45.8
R06	1600	29.3		R30	2500	45.8
R07	250	4.6		R31	5000	91.6
R08	630	11.5		R32	5000	91.6
R09	630	11.5		R33	8000	146.5

R10	1600	29.3
R11	1600	29.3
R12	2500	45.8
R13	630	11.5
R14	250	4.6
R15	1600	29.3
R16	630	11.5
R17	2500	45.8
R18	1600	29.3
R19	630	11.5
R20	250	4.6
R21	1600	29.3
R22	630	11.5
R23	250	4.6
MFD01	125	2.3
MFD02	300	5.5
FD02-RC1	250	4.6
FD03-RC1	750	13.7
FD04-RC1	300	5.5
FD05-RC1	600	11.0
	Total	**459** kW

R34	8000	146.5
R35	2500	45.8
FD01	1130	20.7
FD02	1530	28.0
FD03	1530	28.0
	Total	**1075** kW

1°C/min reactor temperature control

Existing Facility Reactors

Tag No.	Reactor Volume (L)	Heat Load (kW)
R01	250	9.2
R02	250	9.2
R03	630	23.1
R04	630	23.1
R05	1600	58.6
R06	1600	58.6
R07	250	9.2
R08	630	23.1
R09	630	23.1
R10	1600	58.6
R11	1600	58.6
R12	2500	91.6
R13	630	23.1
R14	250	9.2
R15	1600	58.6
R16	630	23.1
R17	2500	91.6
R18	1600	58.6
R19	630	23.1
R20	250	9.2
R21	1600	58.6
R22	630	23.1
R23	250	9.2
MFD01	125	4.6
MFD02	300	11.0
FD02-RC1	250	9.2
FD03-RC1	750	27.5
FD04-RC1	300	11.0
FD05-RC1	600	22.0
	Total	**918** kW

New Extension Facility Reactors

Tag No.	Reactor Volume (L)	Heat Load (kW)
R25	5000	183.2
R26	8000	293.1
R27	5000	183.2
R28	8000	293.1
R29	2500	91.6
R30	2500	91.6
R31	5000	183.2
R32	5000	183.2
R33	8000	293.1
R34	8000	293.1
R35	2500	91.6
FD01	1130	41.4
FD02	1530	56.0
FD03	1530	56.0
	Total	**2150** kW

Tag	Existing or New Extension?	Reactor Volume	Heat Load @ 0.5°C/min	Heat Load @ 1°C/min	Is Reactor Active?	Desired Temp Control	Active Heat Load (kW)
R01	Existing	250	4.6	9.2	☐	1°C/min	0.0
R02	Existing	250	4.6	9.2	☐	1°C/min	0.0
R03	Existing	630	11.5	23.1	☐	1°C/min	0.0
R04	Existing	630	11.5	23.1	☐	1°C/min	0.0
R05	Existing	1600	29.3	58.6	☐	1°C/min	0.0
R06	Existing	1600	29.3	58.6	☐	1°C/min	0.0
R07	Existing	250	4.6	9.2	☐	1°C/min	0.0
R08	Existing	630	11.5	23.1	☐	1°C/min	0.0
R09	Existing	630	11.5	23.1	☐	1°C/min	0.0
R10	Existing	1600	29.3	58.6	☑	1°C/min	58.6
R11	Existing	1600	29.3	58.6	☑	1°C/min	58.6
R12	Existing	2500	45.8	91.6	☑	1°C/min	91.6
R13	Existing	630	11.5	23.1	☐	1°C/min	0.0
R14	Existing	250	4.6	9.2	☐	1°C/min	0.0
R15	Existing	1600	29.3	58.6	☐	1°C/min	0.0
R16	Existing	630	11.5	23.1	☐	1°C/min	0.0
R17	Existing	2500	45.8	91.6	☑	1°C/min	91.6

Typical reactor operation is:
Existing - 6 equipment trains at same time
Design based on 6 largest reactors in use

New extension - 3 equipment trains at same time
Design based on 2x 5000L and 1x 8000L reactors in use

R18	Existing	1600	29.3	58.6	☑	1°C/min	58.6
R19	Existing	630	11.5	23.1	☐	1°C/min	0.0
R20	Existing	250	4.6	9.2	☐	1°C/min	0.0
R21	Existing	1600	29.3	58.6	☑	1°C/min	58.6
R22	Existing	630	11.5	23.1	☑	1°C/min	23.1
R23	Existing	250	4.6	9.2	☐	1°C/min	0.0
MFD01	Existing	125	2.3	4.6	☐	1°C/min	0.0
MFD02	Existing	300	5.5	11.0	☑	1°C/min	11.0
FD02-RC1	Existing	250	4.6	9.2	☐	1°C/min	0.0
FD03-RC1	Existing	750	13.7	27.5	☐	1°C/min	0.0
FD04-RC1	Existing	300	5.5	11.0	☐	1°C/min	0.0
FD05-RC1	Existing	600	11.0	22.0	☑	1°C/min	22.0
R25	New Extension	5000	91.6	183.2	☑	1°C/min	183.2
R26	New Extension	8000	146.5	293.1	☑	1°C/min	293.1
R27	New Extension	5000	91.6	183.2	☑	1°C/min	183.2
R28	New Extension	8000	146.5	293.1	☐	1°C/min	0.0
R29	New Extension	2500	45.8	91.6	☐	1°C/min	0.0
R30	New Extension	2500	45.8	91.6	☐	1°C/min	0.0
R31	New Extension	5000	91.6	183.2	☐	1°C/min	0.0
R32	New Extension	5000	91.6	183.2	☐	1°C/min	0.0
R33	New Extension	8000	146.5	293.1	☐	1°C/min	0.0
R34	New Extension	8000	146.5	293.1	☐	1°C/min	0.0
R35	New Extension	2500	45.8	91.6	☐	1°C/min	0.0
FD01	New Extension	1130	20.7	41.4	☐	1°C/min	0.0
FD02	New Extension	1530	28.0	56.0	☑	1°C/min	56.0
FD03	New Extension	1530	28.0	56.0	☐	1°C/min	0.0

Total Simultaneous Active Heat Load	1189	kW
Maximum Heater Capacity	510	kW
Spare Capacity	**-679**	**kW**
Capacity Status	**EXCEEDS HEATER CAPACITY**	

Recommendation:
Separate 900 kW heater dedicated for extension alone required. Additionally, new recirculation loop for hot HTF with dedicated mag drive pumps for extension facility

Summary		
Existing reactors active load	474	kW
New extension reactors active load	715	kW
Total simultaneous load	1189	kW
Heater capacity	510	kW
Spare capacity	**-679**	**kW**
% utilisation	**233**	**%**

Project:	Asymchem PDF Extension Facility				
Project No:	300291				
Subject:	Utility Capacity Assessment Calculations			**Document No:**	300291-PR-CA-0001
By:	ME	**Checked:**	MC	**Sheet Name:**	Hot HTF - Cleaning
Approved:	ME	**Date:**	21-May-26	**Revision:**	A1

Line No. | **Calculation** | **Notes**

Heading Loads

Information on material properties obtained from engineeringtoolbox

HTF (Heat Transfer Fluid) is the cold/hot utility used. In this case, HTF fluid is Syltherm XLT.

1) To estimate the maximum delta T that can be achieved for hot HTF utility based on the existing heater

Scenario: The existing pilot plant has a 510kW heater which heats HTF from the HTF tank to obtain 130C hot HTF for maintaining the temperature of the facility reactor vessels. The existing hot HTF supply pump has a fixed pumping rate of 50m3/hr which supplies hot HTF to the reactor jackets.

	Value	Unit		Notes
Existing Hot HTF heater heat capacity	510	kW		
Hot HTF heat capacity	2.024	kJ/kg°C	assumption	Hot HTF heat capacity at 130C
Hot HTF density	732	g/cm^3	assumption `	Hot HTF density at 130C
Flowrate	50	m^3/hr		Based on existing supply pump capacity
	10.17	kg/s		
Supply Temperature	130	°C		
Maximum Delta T Hot HTF	**24.78**	°C		
Minimum return T Hot HTF	**105.2**	°C		

2) To calculate the hot HTF flowrate required to achieve desired temperature control in the solvent reactors

Scenario: Hot HTF is used to control temperature of solvent mixture in the reactor to a specific range of 0.5C - 1C/min. Hot HTF is pumped to the reactor jackets to achieve this temperature control where it is supplied at 130C and returns at 105C. Solvents used are acetone and methanol

		Value	Unit		Notes
Temperature control to achieve in reactors		0.5	C/min		
		1.0	C/min		
Existing Hot HTF heater heat capacity		510	kW		
Hot HTF heat capacity		2.024	kJ/kg°C	assumption	Hot HTF heat capacity at 130C
Solvent:	Acetone heat capacity	2.14	kJ/kg°C		Acetone heat capacity at standard conditions
	Methanol heat capacity	**2.8**	kJ/kg°C		Methanol heat capacity at standard conditions
Hot HTF density		732	g/cm^3	assumption	Hot HTF density at 130C
Solvent:	Acetone density	**785**	g/cm^3		Acetone density at standard conditions
	Methanol density	753	g/cm^3		Methanol density at standard conditions
Hot HTF Supply Temperature		130	°C		
Hot HTF Return Temperature		105.2	°C		

Scenario is based on 2500L reactor. This is the largest reactor volume in the existing facility so is considered the worst-case/ largest heating load here.

For this calculation, the worst-case scenario will be based on the largest value for density and heat capacity for the solvents. Methanol has higher heat capacity (2.8 kJ/kg°C) and acetone has higher density at 785 kg/m^3)

						Notes
Reactor volume	2500	L	Vessel filling volume for process	25%		Vessel 1/4 full for solvent cleaning
Mass of solvent in reactor	491	kg				Based on acetone density
0.5°C/min reactor temperature control						
Hot HTF cooling load	**11.4**	**kW**				0.5°C/min reactor temperature
Hot HTF flowrate	**0.23**	**kg/s**		**1.1**	**m^3/hr**	
1°C/min reactor temperature control						
Hot HTF cooling load	**22.9**	**kW**				1°C/min reactor temperature
Hot HTF flowrate	**0.46**	**kg/s**		**2.2**	**m^3/hr**	

3) Hot HTF diversity assessment on all reactors in existing and new extension facility

An assessment on the hot HTF diversity is required to determine the operability of the new extension facility with regards to hot HTF used for reactor temperature control. This will be based on the existing facility reactor volumes. A scale-up will be applied for the new extension facility based on the reactor volumes to estimate potential diversity in hot HTF usage.

0.5°C/min reactor temperature control

Existing Facility Reactors			**New Extension Facility Reactors**		
Tag No.	Reactor Volume (L)	Heat Load (kW)	Tag No.	Reactor Volume (L)	Heat Load (kW)

R01	250	1.1		R25	5000	22.9
R02	250	1.1		R26	8000	36.6
R03	630	2.9		R27	5000	22.9
R04	630	2.9		R28	8000	36.6
R05	1600	7.3		R29	2500	11.4
R06	1600	7.3		R30	2500	11.4
R07	250	1.1		R31	5000	22.9
R08	630	2.9		R32	5000	22.9
R09	630	2.9		R34	8000	36.6

R10	1600	7.3		R35	8000	36.6
R11	1600	7.3			**Total**	**261** kW
R12	2500	11.4				
R13	630	2.9				
R14	250	1.1				
R15	1600	7.3				
R16	630	2.9				
R17	2500	11.4				
R18	1600	7.3				
R19	630	2.9				
R20	250	1.1				
R21	1600	7.3				
R22	630	2.9				
	Total	**103** kW				

1°C/min reactor temperature control

Existing Facility Reactors

Tag No.	Reactor Volume (L)	Heat Load (kW)
R01	250	2.3
R02	250	2.3
R03	630	5.8
R04	630	5.8
R05	1600	14.7
R06	1600	14.7
R07	250	2.3
R08	630	5.8
R09	630	5.8
R10	1600	14.7
R11	1600	14.7
R12	2500	22.9
R13	630	5.8
R14	250	2.3
R15	1600	14.7
R16	630	5.8
R17	2500	22.9
R18	1600	14.7
R19	630	5.8
R20	250	2.3
R21	1600	14.7
R22	630	5.8
	Total	**206** kW

New Extension Facility Reactors

Tag No.	Reactor Volume (L)	Heat Load (kW)
R25	5000	45.8
R26	8000	73.3
R27	5000	45.8
R28	8000	73.3
R29	2500	22.9
R30	2500	22.9
R31	5000	45.8
R32	5000	45.8
R34	8000	73.3
R35	8000	73.3
	Total	**522** kW

Tag	Existing or New Extension?	Reactor Volume	Heat Load @ 0.5°C/min	Heat Load @ 1°C/min	Is Reactor Active?	Desired Temp Control	Active Heat Load (kW)
R01	Existing	250	1.1	2.3	☐	1°C/min	0.0
R02	Existing	250	1.1	2.3	☐	1°C/min	0.0
R03	Existing	630	2.9	5.8	☐	1°C/min	0.0
R04	Existing	630	2.9	5.8	☐	1°C/min	0.0
R05	Existing	1600	7.3	14.7	☑	1°C/min	14.7
R06	Existing	1600	7.3	14.7	☑	1°C/min	14.7
R07	Existing	250	1.1	2.3	☐	1°C/min	0.0
R08	Existing	630	2.9	5.8	☐	1°C/min	0.0
R09	Existing	630	2.9	5.8	☐	1°C/min	0.0
R10	Existing	1600	7.3	14.7	☑	1°C/min	14.7
R11	Existing	1600	7.3	14.7	☑	1°C/min	14.7
R12	Existing	2500	11.4	22.9	☑	1°C/min	22.9
R13	Existing	630	2.9	5.8	☐	1°C/min	0.0
R14	Existing	250	1.1	2.3	☐	1°C/min	0.0
R15	Existing	1600	7.3	14.7	☐	1°C/min	0.0
R16	Existing	630	2.9	5.8	☐	1°C/min	0.0
R17	Existing	2500	11.4	22.9	☑	1°C/min	22.9
R18	Existing	1600	7.3	14.7	☐	1°C/min	0.0
R19	Existing	630	2.9	5.8	☐	1°C/min	0.0
R20	Existing	250	1.1	2.3	☐	1°C/min	0.0
R21	Existing	1600	7.3	14.7	☐	1°C/min	0.0
R22	Existing	630	2.9	5.8	☐	1°C/min	0.0
R25	New Extension	5000	22.9	45.8	☑	1°C/min	45.8
R26	New Extension	8000	36.6	73.3	☑	1°C/min	73.3
R27	New Extension	5000	22.9	45.8	☑	1°C/min	45.8
R28	New Extension	8000	36.6	73.3	☐	1°C/min	0.0
R29	New Extension	2500	11.4	22.9	☐	1°C/min	0.0
R30	New Extension	2500	11.4	22.9	☐	1°C/min	0.0
R31	New Extension	5000	22.9	45.8	☐	1°C/min	0.0
R32	New Extension	5000	22.9	45.8	☐	1°C/min	0.0
R33	New Extension	8000	36.6	73.3	☐	1°C/min	0.0
R34	New Extension	8000	36.6	73.3	☐	1°C/min	0.0

Typical reactor operation is:
Existing - 6 equipment trains at same time
Design based on 6 largest reactors in use

New extension - 3 equipment trains at same time
Design based on 2x 5000L and 1x 8000L reactors in use

Total Simultaneous Active Heat Load	269	kW
Maximum Heater Capacity	510	kW
Spare Capacity	**241**	**kW**
Capacity Status	**WITHIN CAPACITY**	

Summary		
Existing reactors active load	104	kW
New extension reactors active load	165	kW
Total simultaneous load	269	kW
Heater capacity	510	kW
Spare capacity	**241**	**kW**
% utilisation	**53**	**%**

Project:	Asymchem PDF Extension Facility				
Project No:	300291				
Subject:	Utility Capacity Assessment Calculations			**Document No:**	300291-PR-CA-0001
By:	ME	**Checked:**	MC	**Sheet Name:**	Cold HTF - Process
Approved:	ME	**Date:**	21-May-26	**Revision:**	A1

Line No.	Calculation	Notes

Heating Loads

Information on material properties obtained from engineeringtoolbox

HTF (Heat Transfer Fluid) is the cold/hot utility used. In this case, HTF fluid is Syltherm XLT.

1) To calculate the Chiller load for cold HTF for the existing facility and estimate spare capacity available for the new expansion

Scenario: The existing pilot plant has a 780kW ammonia chiller which cools cold HTF from -20C to -25C for maintaining the temperature of the facility reactor vessels. The existing cold HTF supply pump has a fixed pumping rate of 100m3/hr which supplies cold HTF to the reactor jackets.

				Notes
Existing cold HTF chiller heat capacity	780	kW		
Cold HTF heat capacity	1.7	kJ/kg°C		Cold HTF heat capacity at -20C
Cold HTF density	896.4	g/cm^3		Cold HTF density at -20C
Flowrate	100	m^3/hr		Based on existing supply pump capacity
Supply Temperature	-18	°C	Into chiller	
Return Temperature	-25	°C	From chiller to reactor jackets	
Temperature rise	**7**	°C		
Chiller heat load	**294.2**	**kW**		

2) To calculate the cold HTF flowrate required to achieve desired temperature control in the solvent reactors

Scenario: Cold HTF is used to control temperature of solvent mixture in the reactor to a specific range of 0.5C - 1C/min. This is due to the exothermic nature of the solvent reactors. Cold HTF is pumped to the reactor jackets to achieve this temperature control where it is supplied at -25C and returns at -20C. Solvents used are acetone and methanol

				Notes
Temperature control to achieve in reactors		0.5	C/min	
		1.0	C/min	
Existing cold HTF chiller heat capacity		780	kW	
Cold HTF heat capacity		1.7	kJ/kg°C	Cold HTF heat capacity at -20C
Solvent:	Acetone heat capacity	2.14	kJ/kg°C	Acetone heat capacity at standard conditions
	Methanol heat capacity	**2.8**	kJ/kg°C	Methanol heat capacity at standard conditions
Cold HTF density		896.4	g/cm^3	Cold HTF density at -20C
Solvent:	Acetone density	**785**	g/cm^3	Acetone density at standard conditions
	Methanol density	753	g/cm^3	Methanol density at standard conditions
Cold HTF Supply Temperature		-25	°C	
Cold HTF Return Temperature		-20	°C	

Scenario is based on 2500L reactor. This is the largest reactor volume in the existing facility so is considered the worst-case/ largest cooling load here.

For this calculation, the worst-case scenario will be based on the largest value for density and heat capacity for the solvents. Methanol has higher heat capacity (2.8 kJ/kg°C) and acetone has higher density at 785 kg/m^3)

							Notes
Reactor volume	2500	L	Vessel filling volume for process	100%			Vessel filled with solvent for processing
Mass of solvent in reactor	1963	kg					Based on acetone density
0.5°C/min reactor temperature control							
Cold HTF cooling load	**45.8**	**kW**					0.5°C/min reactor temperature
Cold HTF flowrate	**5.43**	**kg/s**			**21.8**	**m^3/hr**	
1°C/min reactor temperature control							
Cold HTF cooling load	**91.6**	**kW**					1°C/min reactor temperature
Cold HTF flowrate	**10.85**	**kg/s**			**43.6**	**m^3/hr**	

3) To calculate the maximum cold HTF cooling for the chiller based on the worst-case 8000L reactor crash cooling

Currently, crash cooling load is undetermined due to no knowledge of reaction exotherms at this stage. When crash cooling is required, the inlet jacket valve is fully opened and the reaction is flooded with cold HTF until it is sufficiently cooled. It it assumed that there is sufficient cold HTF to maintain this crash cooling load for the new extension reactors. This will be assessed further at the next stage.

4) Cold HTF diversity assessment on all reactors in existing and new extension facility

An assessment on the cold HTF diversity is required to determine the operability of the new extension facility with regards to cold HTF used for reactor temperature control. This will be based on the existing facility reactor volumes. A scale-up will be applied for the new extension facility based on the reactor volumes to estimate potential diversity in cold HTF usage

0.5°C/min reactor temperature control

Existing Facility Reactors

Tag No.	Reactor Volume (L)	Heat Load (kW)	
R01	250	4.6	
R02	250	4.6	
R03	630	11.5	
R04	630	11.5	
R05	1600	29.3	
R06	1600	29.3	
R07	250	4.6	
R08	630	11.5	
R09	630	11.5	
R10	1600	29.3	
R11	1600	29.3	
R12	2500	45.8	
R13	630	11.5	
R14	250	4.6	
R15	1600	29.3	
R16	630	11.5	
R17	2500	45.8	
R18	1600	29.3	
R19	630	11.5	
R20	250	4.6	
R21	1600	29.3	
R22	630	11.5	
R23	250	4.6	
MFD01	125	2.3	
MFD02	300	5.5	
FD02-RC1	250	4.6	
FD03-RC1	750	13.7	
FD04-RC1	300	5.5	
FD05-RC1	600	11.0	
	Total	**412**	**kW**

New Extension Facility Reactors

Tag No.	Reactor Volume (L)	Heat Load (kW)	
R25	5000	91.6	
R26	8000	146.5	
R27	5000	91.6	
R28	8000	146.5	
R29	2500	45.8	
R30	2500	45.8	
R31	5000	91.6	
R32	5000	91.6	
R33	8000	146.5	
R34	8000	146.5	
R35	2500	45.8	
FD01	1130	20.7	
FD02	1530	28.0	
FD03	1530	28.0	
	Total	**1075**	**kW**

1°C/min reactor temperature control

Existing Facility Reactors

Tag No.	Reactor Volume (L)	Heat Load (kW)	
R01	250	9.2	
R02	250	9.2	
R03	630	23.1	
R04	630	23.1	
R05	1600	58.6	
R06	1600	58.6	
R07	250	9.2	
R08	630	23.1	
R09	630	23.1	
R10	1600	58.6	
R11	1600	58.6	
R12	2500	91.6	
R13	630	23.1	
R14	250	9.2	
R15	1600	58.6	
R16	630	23.1	
R17	2500	91.6	
R18	1600	58.6	
R19	630	23.1	
R20	250	9.2	
R21	1600	58.6	
R22	630	23.1	
R23	250	9.2	
MFD01	125	4.6	
MFD02	300	11.0	
FD02-RC1	250	9.2	
FD03-RC1	750	27.5	
FD04-RC1	300	11.0	
FD05-RC1	600	22.0	
	Total	**824**	**kW**

New Extension Facility Reactors

Tag No.	Reactor Volume (L)	Heat Load (kW)	
R25	5000	183.2	
R26	8000	293.1	
R27	5000	183.2	
R28	8000	293.1	
R29	2500	91.6	
R30	2500	91.6	
R31	5000	183.2	
R32	5000	183.2	
R33	8000	293.1	
R34	8000	293.1	
R35	2500	91.6	
FD01	1130	41.4	
FD02	1530	56.0	
FD03	1530	56.0	
	Total	**2150**	**kW**

Tag	Existing or New Extension?	Reactor Volume	Heat Load @ 0.5°C/min	Heat Load @ 1°C/min	Is Reactor Active?	Desired Temp Control	Active Heat Load (kW)
R01	Existing	250	4.6	9.2	☐	0.5°C/min	0.0
R02	Existing	250	4.6	9.2	☐	0.5°C/min	0.0
R03	Existing	630	11.5	23.1	☐	0.5°C/min	0.0
R04	Existing	630	11.5	23.1	☐	0.5°C/min	0.0
R05	Existing	1600	29.3	58.6	☐	0.5°C/min	0.0
R06	Existing	1600	29.3	58.6	☐	0.5°C/min	0.0

Typical reactor operation is:
Existing - 6 equipment trains at same time
Design based on 6 largest reactors in use

New extension - 3 equipment trains at same time
Design based on 2x 5000L and 1x 8000L reactors in use

R07	Existing	250	4.6	9.2	☐	0.5°C/min	0.0
R08	Existing	630	11.5	23.1	☐	0.5°C/min	0.0
R09	Existing	630	11.5	23.1	☐	0.5°C/min	0.0
R10	Existing	1600	29.3	58.6	☑	0.5°C/min	29.3
R11	Existing	1600	29.3	58.6	☑	0.5°C/min	29.3
R12	Existing	2500	45.8	91.6	☑	0.5°C/min	45.8
R13	Existing	630	11.5	23.1	☐	0.5°C/min	0.0
R14	Existing	250	4.6	9.2	☐	0.5°C/min	0.0
R15	Existing	1600	29.3	58.6	☑	0.5°C/min	29.3
R16	Existing	630	11.5	23.1	☐	0.5°C/min	0.0
R17	Existing	2500	45.8	91.6	☐	0.5°C/min	0.0
R18	Existing	1600	29.3	58.6	☐	0.5°C/min	0.0
R19	Existing	630	11.5	23.1	☐	0.5°C/min	0.0
R20	Existing	250	4.6	9.2	☐	0.5°C/min	0.0
R21	Existing	1600	29.3	58.6	☐	0.5°C/min	0.0
R22	Existing	630	11.5	23.1	☐	0.5°C/min	0.0
R23	Existing	250	4.6	9.2	☐	1°C/min	0.0
MFD01	Existing	125	2.3	4.6	☐	1°C/min	0.0
MFD02	Existing	300	5.5	11.0	☐	1°C/min	0.0
FD02-RC1	Existing	250	4.6	9.2	☐	1°C/min	0.0
FD03-RC1	Existing	750	13.7	27.5	☐	1°C/min	0.0
FD04-RC1	Existing	300	5.5	11.0	☐	1°C/min	0.0
FD05-RC1	Existing	600	11.0	22.0	☐	1°C/min	0.0
R25	New Extension	5000	91.6	183.2	☑	0.5°C/min	91.6
R26	New Extension	8000	146.5	293.1	☑	0.5°C/min	146.5
R27	New Extension	5000	91.6	183.2	☑	0.5°C/min	91.6
R28	New Extension	8000	146.5	293.1	☐	0.5°C/min	0.0
R29	New Extension	2500	45.8	91.6	☐	0.5°C/min	0.0
R30	New Extension	2500	45.8	91.6	☐	0.5°C/min	0.0
R31	New Extension	5000	91.6	183.2	☐	0.5°C/min	0.0
R32	New Extension	5000	91.6	183.2	☐	0.5°C/min	0.0
R33	New Extension	8000	146.5	293.1	☐	0.5°C/min	0.0
R34	New Extension	8000	146.5	293.1	☐	0.5°C/min	0.0
R35	New Extension	2500	45.8	91.6	☐	1°C/min	0.0
FD01	New Extension	1130	20.7	41.4	☐	1°C/min	0.0
FD02	New Extension	1530	28.0	56.0	☑	1°C/min	56.0
FD03	New Extension	1530	28.0	56.0	☐	1°C/min	0.0

Total Simultaneous Active Heat Load	519	kW
Maximum Chiller Capacity	780	kW
Spare Capacity	**261**	**kW**
Capacity Status	**WITHIN CAPACITY**	

Summary		
Existing reactors active load	134	kW
New extension reactors active load	386	kW
Total simultaneous load	519	kW
Maximum chiller capacity	780	kW
Spare capacity	**261**	**kW**
% utilisation	**67**	**%**

Project:	Asymchem PDF Extension Facility				
Project No:	300291				
Subject:	Utility Capacity Assessment Calculations			**Document No:**	300291-PR-CA-0001
By:	ME	**Checked:**	MC	**Sheet Name:**	Cold HTF - Cleaning
Approved:	ME	**Date:**	21-May-26	**Revision:**	A1

Calculation					Notes
Heating Loads					**Information on material properties obtained from engineeringtoolbox**
HTF (Heat Transfer Fluid) is the cold/hot utility used. In this case, HTF fluid is Syltherm XLT.					
1) To calculate the Chiller load for cold HTF for the existing facility and estimate spare capacity available for the new expansion					
Scenario: The existing pilot plant has a 780kW ammonia chiller which cools cold HTF from -20C to -25C for maintaining the temperature of the facility reactor vessels. The existing cold HTF supply pump has a fixed pumping rate of 100m3/hr which supplies cold HTF to the reactor jackets.					
Existing cold HTF chiller heat capacity	780	kW			
Cold HTF heat capacity	1.7	kJ/kg°C			Cold HTF heat capacity at -20C
Cold HTF density	896.4	g/cm^3			Cold HTF density at -20C
Flowrate	100	m^3/hr			Based on existing supply pump capacity
Supply Temperature	-18	°C	Into chiller		
Return Temperature	-25	°C	From chiller to reactor jackets		
Temperature rise	**7**	°C			
Chiller heat load	**294.2**	**kW**			
2) To calculate the cold HTF flowrate required to achieve desired temperature control in the solvent reactors					
Scenario: Cold HTF is used to control temperature of solvent mixture in the reactor to a specific range of 0.5C - 1C/min. This is due to the exothermic nature of the solvent reactors. Cold HTF is pumped to the reactor jackets to achieve this temperature control where it is supplied at -25C and returns at -20C. Solvents used are acetone and methanol					
Temperature control to achieve in reactors	0.5	C/min			
	1.0	C/min			
Existing cold HTF chiller heat capacity	0	kW			
Cold HTF heat capacity	1.7	kJ/kg°C			Cold HTF heat capacity at -20C
Solvent: Acetone heat capacity	2.14	kJ/kg°C			Acetone heat capacity at standard conditions
Methanol heat capacity	**2.8**	kJ/kg°C			Methanol heat capacity at standard conditions
Cold HTF density	896.4	g/cm^3			Cold HTF density at -20C
Solvent: Acetone density	**785**	g/cm^3			Acetone density at standard conditions
Methanol density	753	g/cm^3			Methanol density at standard conditions
Cold HTF Supply Temperature	-25	°C			
Cold HTF Return Temperature	-20	°C			
Scenario is based on 2500L reactor. This is the largest reactor volume in the existing facility so is considered the worst-case/ largest cooling load here.					
For this calculation, the worst-case scenario will be based on the largest value for density and heat capacity for the solvents. Methanol has higher heat capacity (2.8 kJ/kg°C) and acetone has higher density at 785 kg/m^3)					
Reactor volume	2500	L	Vessel filling volume for process	50%	Vessel 1/2 full for solvent cleaning
Mass of solvent in reactor	981	kg			Based on acetone density
0.5°C/min reactor temperature control					
Cold HTF cooling load	**22.9**	**kW**			0.5°C/min reactor temperature
Cold HTF flowrate	**2.71**	**kg/s**	**10.9**	**m^3/hr**	
1°C/min reactor temperature control					
Cold HTF cooling load	**45.8**	**kW**			1°C/min reactor temperature
Cold HTF flowrate	**5.43**	**kg/s**	**21.8**	**m^3/hr**	
3) To calculate the maximum cold HTF cooling for the chiller based on the worst-case 8000L reactor crash cooling					
Existing cold HTF chiller heat capacity		kW			Cold HTF heat capacity at -20C
Cold HTF heat capacity	1.7	kJ/kg°C			Cold HTF density at -20C
Cold HTF density	896.4	g/cm^3			Based on supply pump capacity
Flowrate	100	m^3/hr			
Supply Temperature	-20	°C			
Return Temperature	-25	°C			
Temperature rise	**5**	°C			

Chiller heat load		kW

4) Cold HTF diversity assessment on all reactors in existing and new extension facility

An assessment on the cold HTF diversity is required to determine the operability of the new extension facility with regards to cold HTF used for reactor temperature control. This will be based on the existing facility reactor volumes. A scale-up will be applied for the new extension facility based on the reactor volumes to estimate potential diversity in cold HTF usage

0.5°C/min reactor temperature control

Existing Facility Reactors

Tag No.	Reactor Volume (L)	Heat Load (kW)	
R01	250	2.3	
R02	250	2.3	
R03	630	5.8	
R04	630	5.8	
R05	1600	14.7	
R06	1600	14.7	
R07	250	2.3	
R08	630	5.8	
R09	630	5.8	
R10	1600	14.7	
R11	1600	14.7	
R12	2500	22.9	
R13	630	5.8	
R14	250	2.3	
R15	1600	14.7	
R16	630	5.8	
R17	2500	22.9	
R18	1600	14.7	
R19	630	5.8	
R20	250	2.3	
R21	1600	14.7	
R22	630	5.8	
	Total	**206**	**kW**

New Extension Facility Reactors

Tag No.	Reactor Volume (L)	Heat Load (kW)	
R25	5000	45.8	
R26	8000	73.3	
R27	5000	45.8	
R28	8000	73.3	
R29	2500	22.9	
R30	2500	22.9	
R31	5000	45.8	
R32	5000	45.8	
R34	8000	73.3	
R35	8000	73.3	
	Total	**522**	**kW**

1°C/min reactor temperature control

Existing Facility Reactors

Tag No.	Reactor Volume (L)	Heat Load (kW)	
R01	250	4.6	
R02	250	4.6	
R03	630	11.5	
R04	630	11.5	
R05	1600	29.3	
R06	1600	29.3	
R07	250	4.6	
R08	630	11.5	
R09	630	11.5	
R10	1600	29.3	
R11	1600	29.3	
R12	2500	45.8	
R13	630	11.5	
R14	250	4.6	
R15	1600	29.3	
R16	630	11.5	
R17	2500	45.8	
R18	1600	29.3	
R19	630	11.5	
R20	250	4.6	
R21	1600	29.3	
R22	630	11.5	
	Total	**412**	**kW**

New Extension Facility Reactors

Tag No.	Reactor Volume (L)	Heat Load (kW)	
R25	5000	91.6	
R26	8000	146.5	
R27	5000	91.6	
R28	8000	146.5	
R29	2500	45.8	
R30	2500	45.8	
R31	5000	91.6	
R32	5000	91.6	
R34	8000	146.5	
R35	8000	146.5	
	Total	**1044**	**kW**

Tag	Existing or New Extension?	Reactor Volume	Heat Load @ 0.5°C/min	Heat Load @ 1°C/min	Is Reactor Active?	Desired Temp Control	Active Heat Load (kW)
R01	Existing	250	2.3	4.6	☐	1°C/min	0.0
R02	Existing	250	2.3	4.6	☐	1°C/min	0.0
R03	Existing	630	5.8	11.5	☐	1°C/min	0.0
R04	Existing	630	5.8	11.5	☐	1°C/min	0.0
R05	Existing	1600	14.7	29.3	☐	1°C/min	0.0
R06	Existing	1600	14.7	29.3	☐	1°C/min	0.0
R07	Existing	250	2.3	4.6	☐	1°C/min	0.0
R08	Existing	630	5.8	11.5	☐	1°C/min	0.0
R09	Existing	630	5.8	11.5	☐	1°C/min	0.0
R10	Existing	1600	14.7	29.3	☑	1°C/min	29.3
R11	Existing	1600	14.7	29.3	☑	1°C/min	29.3
R12	Existing	2500	22.9	45.8	☑	1°C/min	45.8
R13	Existing	630	5.8	11.5	☐	1°C/min	0.0
R14	Existing	250	2.3	4.6	☐	1°C/min	0.0

Typical reactor operation is:
Existing - 6 equipment trains at same time
Design based on 6 largest reactors in use

New extension - 3 equipment trains at same time
Design based on 2x 5000L and 1x 8000L reactors in use

R15	Existing	1600	14.7	29.3	☑	1°C/min	29.3
R16	Existing	630	5.8	11.5	☐	1°C/min	0.0
R17	Existing	2500	22.9	45.8	☑	1°C/min	45.8
R18	Existing	1600	14.7	29.3	☑	1°C/min	29.3
R19	Existing	630	5.8	11.5	☐	1°C/min	0.0
R20	Existing	250	2.3	4.6	☐	1°C/min	0.0
R21	Existing	1600	14.7	29.3	☐	1°C/min	0.0
R22	Existing	630	5.8	11.5	☐	1°C/min	0.0
R25	New Extension	5000	45.8	91.6	☑	1°C/min	91.6
R26	New Extension	8000	73.3	146.5	☑	1°C/min	146.5
R27	New Extension	5000	45.8	91.6	☑	1°C/min	91.6
R28	New Extension	8000	73.3	146.5	☐	1°C/min	0.0
R29	New Extension	2500	22.9	45.8	☐	1°C/min	0.0
R30	New Extension	2500	22.9	45.8	☐	1°C/min	0.0
R31	New Extension	5000	45.8	91.6	☐	1°C/min	0.0
R32	New Extension	5000	45.8	91.6	☐	1°C/min	0.0
R33	New Extension	8000	73.3	146.5	☐	1°C/min	0.0
R34	New Extension	8000	73.3	146.5	☐	1°C/min	0.0

Total Simultaneous Active Heat Load	539	kW
Maximum Chiller Capacity	780	kW
Spare Capacity	**241**	**kW**
Capacity Status	**WITHIN CAPACITY**	

Summary		
Existing reactors active load	209	kW
New extension reactors active load	330	kW
Total simultaneous load	539	kW
Maximum chiller capacity	780	kW
Spare capacity	**241**	**kW**
% utilisation	**69**	**%**

Project:	Asymchem PDF Extension Facility				
Project No:	300291				
Subject:	Utility Capacity Assessment Calculations			**Document No:**	300291-PR-CA-0001
By:	ME	**Checked:**	MC	**Sheet Name:**	H2 Cylinders
Approved:	ME	**Date:**	21-May-26	**Revision:**	A1

Line No.	Calculation	Notes

Hydrogen cylinders for hydrogenation suite

1) To estimate the hydrogen cylinder usage for the new extension facility and new 2500L hydrogenation reactor

Scenario: Hydrogen cylinders are used to supply the hydrogenation suite and hydrogenation reactors. For the existing facility, a 250L hydrogenation reactor using 1 cylinder per batch and 3 batches are manufactured per week.
For the new extension facility, a 2500L hydrogenation reactor will be installed. Scale-up will be applied for the hydrogen cylinder usage.

Existing facility		
Hydrogenation reactor volume	250	L
Existing facilty H2 cylinder usage	1	H2 cylinders/ batch
Number of batches per week	3	batches/ week
Total H2 cylinders per week	**3**	**H2 cylinders/ week**
Extension facility		
Hydrogenation reactor volume	2500	L
Reactor multiplier factor	10	
Existing facilty H2 cylinder usage	10	H2 cylinders/ batch
Number of batches per week	3	batches/ week
Total H2 cylinders per week	30	**H2 cylinders/ week**

It is recommended to utilise hydrogen MCPs (Multi-Cylinder Packs) for H2 cylinder usage

BOC MCPs house 15 H2 cylinders (108.15m3). To provide enough H2 for the hydrogenation reactors, 2 MCPs are estimated to be used per week at peak usage. It is recommended to have 2 MCPs in the storage location and have 1 MCP available as spare to replace one of the other 2 in use should they deplete

Typical dimensions for each MCP is:

Length	1290	mm	Between MCPs	800	mm
Width	840	mm	Above MCPs	400	mm
Height	1810	mm			

3 MCPs total in storage (2 in use and 1 spare)

Allowing 500mm distance between MCPs and including 400mm height allowance for distribution manifold, total footprint is:

Length	**5470**	**mm**		**Length**	**3380**	**mm**	
Width	**1680**	**mm**	for 3 MCPs	**Width**	**1680**	**mm**	for 2 MCPs
Height	**2210**	**mm**		**Height**	**2210**	**mm**	

2) Estimating hydrogen storage mass per MCP for COMAH consideration

Each MCP storage hydrogen at 15C and 175barg. Cylinder gross weight is 1500kg and volume is 108.15m^3.
At these conditions, hydrogen deviates from ideal gas behaviour and behaves as a real gas.

Per MCP

MCP storage volume	V	108.15	m^3	@ standard conditions (25C, 1atm)
No. of cylinders per MCP		15	cylinders	
H2 storage density	ρ	0.0832	kg/m^3	@ standard conditions (25C, 1atm)
H2 storage mass per MCP	m_{H2}	9.0	kg per MCP	
H2 storage mass per 3 MCPs		27.0	kg per 3 MCPs	54

Summary:
Recommendation to use Multi-Cylinder Packs (MCPs). BOC MCPs house 15 cylinders each.
At peak usage, 2 MPCs in use simultaneously, plus 1 spare on standby = 3 MCPs total on site at any time. Footprint for this needs to be accommodated in hazardous storage design. 3rd/ spare MCP may be located elsewhere and brought in during changeover.
This is based on batch frequency of 3 batches/week.
Note - An assessment into the total H2 storage on-site will need to be done to determine H2 storage for COMAH requirements. It is very unlikely the COMAH lower tier of 5t of H2 storage here.

Project:	Asymchem PDF Extension Facility				
Project No:	300291				
Subject:	Utility Capacity Assessment Calculations			**Document No:**	300291-PR-CA-0001
By:	ME	**Checked:**	MC	**Sheet Name:**	LN2 Tank
Approved:	ME	**Date:**	21-May-26	**Revision:**	A1

Calculation					Notes
Liquid Nitrogen Tank					
1) To estimate the nitrogen usage in the new extension facility and whether the existing liquid nitrogen tank fill-up rate can cope with the added requirement from the extension facility					
Scenario: LN2 is used in the existing facility for the cryogenic reactors. Gaseous N2 is used for pressure testing reactors and for inert blanketing/ boil-off of the reactors. Both are supplied from an existing LN2 storage tank. This also consists of a nitrogen supply package and evaporator which can supply up to 150Nm3/hr of N2 at 6barg. Currently, the LN2 tank is refilled every 1-2 weeks with peak demand requiring fill-up every week. Existing nitrogen supply package can provide 150 Nm3/hr of N2 at 6barg.					
Existing LN2 storage tank					
Diameter	1900	mm	1.9	m	
Overall height	7480	mm	7.48	m	
Tank T-T height (excl skirt)	6980	mm	6.98	m	Assume 0.5m height for skirt and dished ends
Tank working volume	10.00	m^3			
Volume of dished ellipsoidal ends	1.8	m^3			
Tank gross volume	11.8	m^3			
Scale-up LN2 storage tank					
Scaled-up tank working volume	**35**	**m^3**			
Scaled-up tank gross volume	**42**	**m^3**			

Summary:
Existing LN2 tank will be insufficient alone to provide required liquid and gaseous N2 for both the existing and new extension facility with the current tank top-up every week. 2 options are recommended:
1. More frequent top-up of the existing LN2 tank. Currently this is topped up once a week at peak demand. With the extension facility, this will need to be topped up every 3-4 days.
2. Install a new larger LN2 storage tank based on the scale-up in N2 usage. For new tank installation, footprint will require new concrete plinth at 6.5m x 10.5m for all tank sizes at 30m3 - 70m3. New tank will need to be minimum 35m3 working volume. Existing tank area is not suitable for larger tank due to condition of existing plinth (confirmed with BOC). Thus, new tank will need to be located elsewhere then existing tank disinvested once the new tank is installed and is able to supply both existing and extension facilities.

Project:	Asymchem PDF Extension Facility				
Project No:	300291				
Subject:	Utility Capacity Assessment Calculations			**Document No:**	300291-PR-CA-0001
By:	ME	**Checked:**	MC	**Sheet Name:**	Compressed Air
Approved:	ME	**Date:**	21-May-26	**Revision:**	A1

Line No.	Calculation	Notes

Compressed air for plant and instrument air usage

Notes: Information on material properties obtained from engineeringtoolbox

1) To estimate the compressed air usage at the existing facility and whether the existing compressors can cope with the increased usage with the extension facility

Scenario: Compressed air is supplied to the existing facility from on-site compressors. The existing compressed air package supplies 375Nm3/hr @ 10barg to plant air, instrument air and breathing air users. Current breathing air package capacity is 100Nm3/hr. This leaves 275m3/hr available for plant air and instrument air.
Asymchem have provided a valve and instrument list of their existing facility to estimate the CA usage at the current facility.
Additionally, there are 3x diaphragm pumps with intermittent use which will also be considered in the CA use.

This will be used to estimate the existing facility CA use and scale it accordingly for the extension facility. The extension facility will have larger equipment whereas the existing facility has more equipment. As a result, a scale-up ratio of 1.5 will be used to estimate the CA usage in the extension facility compared to the existing one.

Based on the valve and instrument list provided, plant air use for existing facility is:

Item	Value	Unit			Notes
Positioners (continuous bleed)	58	Nm3/hr			Estimate each valve at 0.3 Nm3/hr
Purges & tools (continuous)	10	Nm3/hr			
On/off valve actuators (intermittent)	16	Nm3/hr			Assume 10 NL/store and 3 strokes/hr
Diaphragm pumps (intermittent)	90	Nm3/hr			3 existing diaphragm pumps
Peak demand	174	Nm3/hr			
Existing compressor capacity	375	Nm3/hr @ 10barg			From pilot plant equipment list
Breathing air package capacity	100	Nm3/hr			

Applying 1.5 scale-up factor, plant air use for both existing and new extension facilities is:

Item	Value	Unit			Notes
Scale-up factor	1.5				
Positioners	87	Nm3/hr			
Purges & tools	15	Nm3/hr			
On/off valve actuators (0.5 diversity)	12	Nm3/hr	0.5	diversity	
Diaphragm pumps (0.7 coincidence)	105	Nm3/hr	0.7	diversity	
Peak demand across both existing and extension facilities	219	Nm3/hr			
Spare capacity	**56**	**Nm3/hr**			
Spare margin	**15**	**%**			

Summary:
Existing 375Nm3/hr compressor package has sufficient capacity to provide compressed air to both the existing and extension facilities. The compressed air utilisation for plant air is estimated to be 209Nm3/hr thus, leaving 56Nm3/hr as spare capacity (noting 100Nm3/hr breathing air requirement. An additional compressor would be beneficial to allow larger spare capacity for future upgrades and equipment. However, for the feasibility stage, this assessment determines the existing compressor to be sufficient for the requirement.

Project:	Asymchem PDF Extension Facility				
Project No:	300291				
Subject:	Utility Capacity Assessment Calculations			**Document No:**	300291-PR-CA-0001
By:	ME	**Checked:**	MC	**Sheet Name:**	Breathing Air
Approved:	ME	**Date:**	21-May-26	**Revision:**	A1

Line No.

Calculation					Notes
Breathing air for personnel in hazardous areas					
1) To estimate the breathing air flowrate requirement for personnel handling hazardous materials in the reactor rooms in 30 minutes					
Scenario: Compressed air is supplied to personnel handling hazardous materials for respiratory protection. This is due to potential exposure to toxic chemicals, solvent atmospheres etc. This breathing air is supplied to breathing air points of use in these areas which supplies air to breathing hoods which the personnel must wear to operate in the hazardous areas. For this calculation, it is assumed that personnel will be wearing RPE loose hoods which require 170-200 L/min of breathing air per person. The calculation will estimate the breathing air requirement for 30 minutes.					
RPE Loose Breathing Hoods - 170 - 200 L/min					
Breathing air flow per person	200	L/min			200 L/min assumed as worst-case flow req.
Time requirement	30	min			
Breathing air flow across time frame	**6000**	**L/30 min**			
Number of personnel across both facilities	10				Assume no more than 10 personnel requiring breathing air at one time across both existing and new extension facilities
Total breathing air requirement across both facilities	**60000**	**L/ 30 min**			
	60	**m^3/30 min**			
2) From this, the maximum number of personnel that can be present in the hazardous rooms has been estimated based on the 200L/min air requirement for breathing hoods for 2 scenarios:					
a) Existing breathing air supply package of 100Nm^3/hr					
b) Outage in breathing air supply and available breathing air supply based on existing breathing air receiver					
a) Existing breathing air supply package of 100Nm^3/hr					
Existing breathing air supply package	100	Nm^3/hr			
Breathing air flow requirement per person	200	NL/min			
	12	Nm^3/hr			
Maximum number of personnel that can be in hazardous areas across both existing and new extension facilities	**8**				
b) Outage in breathing air supply and available breathing air supply based on existing breathing air receiver					
Scenario: In case of an outage in the breathing air supply, the maximum breathing air supply capacity will be based on the breathing air receiver (CMP01-GR1). This has a volume of 11m3 and the receiver has an operating pressure of <10barg. The breathing air header has a supply pressure of 6.5barg. In case of an outage, the breathing air supply will be limited to the difference between the receiver pressure and breathing air header pressure (driving force is pressure difference between receiver and header pressure)					Information on breathing air supply obtained from breathing and compressed air P&ID SAN-902-007-002-000-00099
Air receiver volume	11	m^3			
Air receiver pressure	9	barg	10.01	bara	Receiver SV set at 10barg Assume 9barg operating pressure
Downstream regulator set pressure	6.5	barg	7.51	bara	
Pressure difference driving force	2.5	barg			
Breathing air volume available	27.14	m^3			Assuming ideal gas behaviour
Personnel in hazardous areas require	200	L/min			
	6	m^3/30 mins			
Maximum number of personnel that can be present in hazardous areas in case of breathing air outage	**4**				Assume breathing air supply for personnel in hazardous areas is continuous (flexible connection) rather than finite air cylinders
Summary: **Existing 100 Nm³/hr supply is adequate for up to 8 people simultaneously on the basis of 200L/min RPE loose breathing hoods. If peak hazardous area occupancy exceeds this across both facilities, the breathing air supply will need uprating.** **In the event of an outage of breathing air supply, air receiver will be able to provide sufficient usable air for maximum 4 people up to 30 minutes (safe evacuation window).**					

Project:	Asymchem PDF Extension Facility				
Project No:	300291				
Subject:	Utility Capacity Assessment Calculations			**Document No:**	300291-PR-CA-0001
By:	ME	**Checked:**	MC	**Sheet Name:**	Purified Water
Approved:	ME	**Date:**	21-May-26	**Revision:**	A1

Line No.	Calculation					Notes
1	**Purified Water for Reactor Rinses**					
2						
3	**1) To estimate the size of the new purified water system for the extension facility**					
4	Scenario: Purified Water is used for rinsing the reactors based on average 44% water usage of reactor working volume. It is considered that the existing purified water system is unable to suit the water usage for both existing and new extension facility. Replacing the pump would not work either as the existing loop is not sized for both facilities. The recommendation is to install a new purified water system solely for the extension facility. This will consist of:					
5						
6	- Ozonated purified water generation system - 4.0m3/hr generation					
7	- 16m3 storage tank with vent filter and ozone destruct unit					
8	- 25m3/hr supply pump through 2.5" diameter distribution loop size					
9	- UV de-ozonation unit					
10						
11	The above is based on rinsing with 44% of reactor volume and coincidental cleaning of the reactors. Not all reactors will be able to be rinsed at the same time. A design basis of 30 minutes has been assumed to fill up each reactor and achieved the required rinse volume of purified water.					
12	Total rinse flowrate for each reactor size = 2.2+4.4+7 = 14.1 m3/hr					
13						
14						
15	**Purified water loop pipe diameter**	**2.5 in - 65 mm**				
16	Pipe spec	ASME-BPE				
17	Pipe internal diameter	60.2	mm	0.060198	m	
18	Required remaining loop velocity	1.0	m/s			
19	**User flowrate**	**15**	**m^3/hr**			
20						
21	Pipe area	0.00285	m^2			
22	Required remaining loop flowrate	0.00285	m^3/s			
23		10.25	m^3/hr			
24	**Total flowrate**	**25.25**	**m^3/hr**			
25						
26	**Final combined velocity**	**2.46**	**m/s**			
27						
28						
29	**Summary:**					
30	**Install a new, fully independent purified water system for the extension facility.**					
31	**Proposed specification:**					
32	**4.0 m^3/hr generation (ozonated), 16 m^3 storage tank with vent filter and ozone destruct unit, 25 m^3/hr supply pump on a 2.5 in (65 mm) ASME-BPE distribution loop, UV de-ozonation unit.**					
33						
34	**This keeps the extension independent of the existing PW loop, avoiding cross-facility demand conflict and contamination risk.**					
35						
36						
37						

Project:	Asymchem PDF Extension Facility				
Project No:	300291				
Subject:	Utility Capacity Assessment Calculations			**Document No:**	300291-PR-CA-0001
By:	ME	**Checked:**	MC	**Sheet Name:**	Cooling Water
Approved:	ME	**Date:**	21-May-26	**Revision:**	A1

Line No.	Calculation	Notes

Cooling Water for Purified Water Trim Cooling

1) To estimate the cooling water requirement for the extension facility

Scenario: Cooling water is used for general cooling within the facility as a utility for heat exchangers. Within the existing facility, this is supplied via Jaeggi Hybrid cooling towers (893kW) using spray water medium at 32C supply and 26C return. For the extension facility, cooling water is required for trim cooling of the purified water heat exchanger. This is due to heat generated from the purified water distribution pump which can heat up the purified water over time. Cooling water will be supplied to the purified water heat exchanger for trim cooling. This will be from a new chiller designated for this cooling water.
The following assumptions are used for estimating the cooling water requirement and chiller size:
- 10 kW heat gain for purified water based on 25m3/hr loop flowrate
- Trim cooling heat exchanger duty = 12 kW (25% design margin)
- Cooling water chiller capacity = 15 kW

This is then used to estimate the chilled water flow requirement based on 10C supply T and 15C return T.

Purified water heat gain in loop	10	kW
Design margin	20%	
Trim cooling heat exchanger duty	12.0	kW
Chiller duty	**15.0**	**kW**
Cooling water calculation		
Supply temperature	10	°C
Return temperature	15	°C
Specific heat capacity	4.184	kJ/kg°C
Water density	1000	kg/m^3
Cooling water flowrate	**2581**	**kg/h**
	2.58	**m^3/h**

Summary:
Install a new dedicated chiller for the cooling water flowrate to the purified water trim cooling heat exchanger. This will cool down incoming cooling water to the required temperature to then be pumped to the purified water trim cooler. The new chiller will be sized specifically for the purified water trim cooling requirement. Final installation location to be confirmed at the project concept stage.

Project:	Asymchem PDF Extension Facility				
Project No:	300291				
Subject:	Utility Capacity Assessment Calculations			**Document No:**	300291-PR-CA-0001
By:	ME	**Checked:**	MC	**Sheet Name:**	Bulk Solvent
Approved:	ME	**Date:**	21-May-26	**Revision:**	A1

Calculation	Notes
Bulk acetone and methanol solvents	**Information on material properties obtained from engineeringtoolbox**
1) To estimate the usage of acetone and methanol solvents from the bulk solvent tanks and whether the existing tanks and changeover is sufficient with the new extension facility	
Scenario: Acetone and methanol solvents are used on-site in the reactors. These are stored in 2 solvent tanks: BT01 (methanol, 60m3) and BT01 (acetone, 40m3). Current solvent usage and refilling is confirmed via RFI as: - Acetone: 4 tankers ordered per year (~23.25 MT per delivery). Deliveries every 3 months - Methanol: 6 tankers ordered per year (~27.25 MT per delivery). Deliveries every 2 months on average Note this is based on 4-6 batches per week	Info from RFI response Bulk tank volumes estimated from HMI screenshot

Item	Value	Unit	Property	Value	Unit	Notes
Acetone bulk tank volume	40	m^3				Estimated from HMI screenshot
Acetone delivery frequency	4	times per year (quarterly)				
Acetone delivery amount	23.25	MT	Acetone density	785	kg/m^3	
	30	m^3				
Methanol bulk tank volume	60	m^3				Estimated from HMI screenshot
Methanol delivery frequency	6	times per year (bimonthly)				
Methanol delivery amount	27.25	MT	Methanol density	753	kg/m^3	
	36	m^3				

Solvent usage with new extension facility is estimated based on reactor volume scale-up factor. This is a ratio in the total reactor volumes between the existing facility and new extension facility to estimate the utility usage with the extension equipment (usage data unknown)

Item	Value	Unit	Notes
Reactor volume scale-up ratio	3.54		
Acetone delivery frequency with new extension	**15**	**times per year i.e. ~ one delievery one month then two deliveries the next month**	Note - this is based on current solvent refill pattern with existing facility which is based on 4-6 batches per week
Methanol delivery frequency with new extension	**22**	**times per year i.e. ~ two delieveries per month.**	

Summary:
Methanol tanker 27.25 MT deliveries will need to be 2x deliveries for one month then 1x delievery the following month i.e. 3 deliveries every 2 months. 1 delivery for the existing facility operation and 2x delieveries for the new extension facility. The option for installing an additional bulk methanol tank will not be considered as delivery frequency is dependent on tanker sizes and frequent deliveries will still be required even with an additional bulk solvent tank.

Project:	Asymchem PDF Extension Facility				
Project No:	300291				
Subject:	Utility Capacity Assessment Calculations			**Document No:**	300291-PR-CA-0001
By:	ME	**Checked:**	MC	**Sheet Name:**	Solvent Waste
Approved:	ME	**Date:**	21-May-26	**Revision:**	A1

Line No.	Calculation	Notes

Waste Solvents

1) To estimate the acetone and methanol solvent waste generated within the existing facility and whether the existing tanks and changeover is sufficient with the new extension facility

Scenario: Acetone and methanol solvent waste is generated and collected in solvents tanks on-site. There are 3 solvent waste tanks involved:
- ST01: Immiscible solvent waste tank, $10m^3$
- ST02: Miscible solvent waste tank, $20m^3$
- ST03: Waste solvent tank, $30m^3$

When at capacity, ST03 is emptied via tanker every 2 weeks. ST01 is transferred to ST03 every 3-4 days. ST02 is transferred to ST03 following ST01 and the tanks alternate.

ST01 immiscible waste tank volume	10	m^3	Effluent Transfer Pump - 15 m3/hr @ 56m TDH
ST02 miscible waste tank volume	20	m^3	
ST03 waste solvent tank volume	30	m^3	Effluent Transfer Pump - 25 m3/hr

ST01 current emptying frequency to ST03	1	every 3-4 days in first week
ST02 current emptying frequency	1	in second week following ST01
ST03 current emptying frequency	0.5	per week i.e. biweekly

Solvent waste generation with new extension facility is estimated based on reactor volume scale-up factor. This is a ratio in the total reactor volumes between the existing facility and new extension facility to estimate the utility usage with the extension equipment (usage data unknown). Typical solvent waste generated is based on 44% of reactor volumes where solvents are used for cleaning reactors. The scale-up will be applied to the existing facility to estimate solvent waste generated from the extension facility and whether the existing solvent waste tanks can cope with this volume

Existing facility reactors total volume	25065	L
New extension facility reactors total volume	63690	L
Volume scale-up ratio	3.54	L
Existing facility solvent waste generated	11029	L
New extension facility solvent waste generated	**28024**	**L**
Total solvent waste generated	**39052**	**L**

Summary:
Recommendation is to utilise existing ST01/ST02/ST03 tanks for solvent waste collection from both the existing and new extension reactor cleans. This is based on:
- **Scheduled reactor cleaning and solvent waste discharge will be required to prevent ST01 and ST03 from being overfilled. This is to also allow time for effluent testing**
- **ST03 will need to be emptied after every cleaning cycle — if multiple cleans are carried out per week, multiple discharges per week will be required to ensure ST03 is not overfilled**
- **A buffer volume(~ 6 m^3) may need to be held across ST01 and ST02 between collections (ST03 30 m^3 volume; total waste across both facilities is 36 m^3).**
- **A larger ST01 effluent transfer pump may be required depending on the cleaning schedule and discharge flow rates**

Note - currently, solvent waste is collected by tanker every 2 weeks. There may be a desire to change this and move to pumping away the solvent waste instead in the future.

Notes:

Information on material properties obtained from engineeringtoolbox

Info from RFI response
Bulk tank volumes estimated from HMI screenshot

Project:	Asymchem PDF Extension Facility				
Project No:	300291				
Subject:	Utility Capacity Assessment Calculations			**Document No:**	300291-PR-CA-0001
By:	ME	**Checked:**	MC	**Sheet Name:**	Aqueous Waste
Approved:	ME	**Date:**	21-May-26	**Revision:**	A1

Line No.	Calculation	Notes

Aqueous Trade Waste

1) To estimate the aqueous/ effluent waste waste generated within the existing facility and whether the existing effluent tanks and changeover is sufficient with the new extension facility

Scenario: Aqueous trade waste is generated from the process from rinsing the reactor vessels with purified water. This is then collected in Effluent Tanks ET01/ET02 (both 4m^3) before being pumped to Trade Effluent Tanks TT01/ TT02 (both 12m^3). Currently, 2x 10m^3 of aqueous trade waste is generated weekly from TT01/TT02. This is then pumped away offsite.

Typical purified water usage for the reactor rinses is 20% of the reactor volume. Occasioinally, acid or base treatment of the reactors is carried out during cleaning which uses ~80% of the reactor volumes (only done 30% of the time). Average volume of water used during cleaning is ~44% of reactor volumes.

ET01 effluent tank volume	4	m^3	Effluent Transfer Pump - 8.8 m3/hr @ 19.8m TDH
ET02 contained waste effluent tank volume	4	m^3	Effluent Transfer Pump - 8.8 m3/hr @ 19.8m TDH
TT01 trade effluent tank volume	12	m^3	Effluent Transfer Pump - 22 m3/hr @ 36.4m TDH
TT02 trade effluent tank volume	12	m^3	
TT01/TT02 emptying frequency	10000	x2 from each tank weekly	

Aqueous waste generation with new extension facility is estimated based on reactor volume scale-up factor. This is a ratio in the total reactor volumes between the existing facility and new extension facility to estimate the utility usage with the extension equipment (usage data unknown)

Reactor volume scale-up ratio	3.54	
TT01/TT02 collection frequency with new extension	**35410**	**x2 from TT01/TT02 weekly** **Transfer pump sized for 22m^3/hr**

Summary:
The existing ET01 (4 m³) has insufficient capacity to accommodate aqueous waste from purified water (PW) rinses generated by the new extension reactors. The largest single reactor alone has the potential to fill ET01 to capacity, and coincidental cleaning operations across multiple reactors would result in ET01 overfilling.

It is recommended that dedicated aqueous effluent tanks are installed for the extension facility - ET03 and TT04/TT05. ET03 would be 10–12 m³ volume and TT04/TT05 would each be 30-36 m³ volume. These would also have new dedicated pumps to cope with the larger waste flowrates from the extension facility equipment. Waste emptying frequency would be the same as that for the existing facility tanks ET01 and TT01/TT02.

Notes:

Information on material properties obtained from engineeringtoolbox

Info from RFI response
Bulk tank volumes estimated from HMI screenshot

Note - TT03 is for contained waste

Appendix 5 - Equipment List

		Revision	Date	Description	By	Checked	Approved	DOCUMENT NUMBER	REV
Client	Asymchem							300291-PR-SC-0001	A1
Location	Sandwich, UK								
Project Name	PDF Extension Facility								
Project No.	300291	A1	21 May 26	Feasibility Design Issue	ME	MC	ME		

ASYMCHEM PDF EXTENSION FACILITY

EQUIPMENT LIST

Scitech EKIUM

FILE NAME

300291-PR-SC-0001 Equipment List

Scitech EKIUM

EQUIPMENT LIST

		Revision	Date	Description	By	Checked	Approved		DOCUMENT NUMBER	REV
Client	Asymchem							This document contains proprietary information belonging to Scitech and / or affiliated companies and shall be used only for the purpose for which it was supplied. It shall not be copied, reproduced or otherwise used, nor shall such information be supplied to others except with the prior consent of Scitech and shall be returned on request.	300291-PR-SC-0001	A1
Location	Sandwich, UK									
Project Name	PDF Extension Facility									
Project No.	300291	A1	21 May 26	Feasibility Design Issue	ME	MC	ME			

Tag	ID No.	No. Off	DESCRIPTION	TECHNICAL CHARACTERISTICS - SIZE/DUTY	MATERIALS OF CONSTRUCTION	WEIGHT (kg)	DIMENSIONS (mm) W x D x H	LOCATION	Process	Electrical Load	NOTES	REVISION
REACTORS & ASSOCIATED EQUIPMENT												
R25		1	Reactor	5000L Design Temp: -29/200 °C Oper. Temp: -20/150 °C Design P: -0.1~0.6 MPa Oper. P: -0.1~0.3 MPa	Glass Lined Vessel and jacket from ASME SA516 Grade 60	Empty - 6320 Full water - 12170	Internal 1750 D x 2525 H External 1920 D x 5228 H	2nd floor	Y - Transfer from reagent, slurries and liquids & solvents from other process areas	7.5 kW electromotor		A1
R25	PC01	1	Primary Condenser	$30m^2$	Hastelloy C-22			2nd floor			Review use of tantalum as condenser MOC	A1
R25	VC01	1	Vent Condenser	$3m^2$	Hastelloy C-22			2nd floor			Review use of tantalum as condenser MOC	A1
R25	JP01	1	Jacket Circulation Pump		SS316L			1st floor				A1
R26		1	Reactor	8000L Design Temp: -29/200 °C Oper. Temp: -20/150 °C Design P: -0.1~0.6 MPa Oper. P: -0.1~0.3 MPa	Glass Lined Vessel and jacket from ASME SA516 Grade 60	Empty - 8765 Full water - 17965	Internal 2000 D x 3025 H External 2224 D x 6020 H	2nd floor	Y - Transfer from reagent, slurries and liquids & solvents from other process areas	11 kW electromotor		A1
R26	PC01	1	Primary Condenser	$40m^2$	Hastelloy C-22			2nd floor			Review use of tantalum as condenser MOC	A1
R26	VC01	1	Vent Condenser	$5m^2$	Hastelloy C-22			2nd floor			Review use of tantalum as condenser MOC	A1
R26	JP01	1	Jacket Circulation Pump		SS316L			1st floor				A1
R27		1	Reactor	5000L Design Temp: -29/200 °C Oper. Temp: -20/150 °C Design P: -0.1~0.6 MPa Oper. P: -0.1~0.3 MPa	Glass Lined Vessel and jacket from ASME SA516 Grade 60	Empty - 6320 Full water - 12170	Internal 1750 D x 2525 H External 1920 D x 5228 H	2nd floor	Y - Transfer from reagent, slurries and liquids & solvents from other process areas	7.5 kW electromotor		A1
R27	PC01	1	Primary Condenser	$30m^2$	Hastelloy C-22			2nd floor				A1
R27	VC01	1	Vent Condenser	$3m^2$	Hastelloy C-22			2nd floor				A1
R27	JP01	1	Jacket Circulation Pump		SS316L			1st floor				A1
R28		1	Reactor	8000L Design Temp: -29/200 °C Oper. Temp: -20/150 °C Design P: -0.1~0.6 MPa Oper. P: -0.1~0.3 MPa	Glass Lined Vessel and jacket from ASME SA516 Grade 60	Empty - 8765 Full water - 17965	Internal 2000 D x 3025 H External 2224 D x 6020 H	2nd floor	Y - Transfer from reagent, slurries and liquids & solvents from other process areas	11 kW electromotor		A1
R28	PC01	1	Primary Condenser	$40m^2$	Hastelloy C-22			2nd floor				A1
R28	VC01	1	Vent Condenser	$5m^2$	Hastelloy C-22			2nd floor				A1
R28	JP01	1	Jacket Circulation Pump		SS316L			1st floor				A1
R29		1	Reactor	2500L Design Temp: -85/165 °C Oper. Temp: -70/150 °C Design P: -0.1~0.6 MPa Oper. P: -0.1~0.57 MPa	Glass Lined Vessel and jacket from ASME SA516 Grade 60		Internal 1600 D x 2290 H External 1750 D x 4506 H	2nd floor		5.5 kW electromotor	Review option to have this also act as a cryogenic reactor with hydrogen venting capabilities. Cryo reactors should be in close proximity	A1
R29	PC01	1	Primary Condenser	$19m^2$	Hastelloy C-22			2nd floor				A1
R29	VC01	1	Vent Condenser	$1.5m^2$	Hastelloy C-22			2nd floor				A1
R29	JP01	1	Jacket Circulation Pump	$36m^3/h$-20m	SS316L			1st floor				A1
R29	CC01	1	Jacket Cryogenic Cooler		SS304			1st floor				A1
R30		1	Reactor	2500L working volume (3800L gross volume) Design Temp: -29/200 °C Oper. Temp: -20/150 °C Design P: -0.1~0.6 MPa Oper. P: -0.1~0.3 MPa	Glass Lined Vessel and jacket from ASME SA516 Grade 60	Empty - 3972 Full water - 7772	Internal 1600 D x 2290 H External 1750 D x 4506 H	2nd floor	Y - Transfer from reagent, slurries and liquids & solvents from other process areas	5.5 kW electromotor		A1
R30	PC01	1	Primary Condenser	$19m^2$	Hastelloy C-22			2nd floor				A1
R30	VC01	1	Vent Condenser	$1.5m^2$	Hastelloy C-22			2nd floor				A1
R30	JP01	1	Jacket Circulation Pump	$36m^3/h$-20m	SS304			1st floor				A1
R31		1	Reactor	5000L Design Temp: -29/200 °C Oper. Temp: -20/150 °C Design P: -0.1~0.6 MPa Oper. P: -0.1~0.3 MPa	Glass Lined Vessel and jacket from ASME SA516 Grade 60	Empty - 6320 Full water - 12170	Internal 1750 D x 2525 H External 1920 D x 5228 H	2nd floor	Y - Transfer from reagent, slurries and liquids & solvents from other process areas	7.5 kW electromotor		A1
R31	PC01	1	Primary Condenser	$30m^2$	Hastelloy C-22			2nd floor				A1
R31	VC01	1	Vent Condenser	$3m^2$	Hastelloy C-22			2nd floor				A1
R31	JP01	1	Jacket Circulation Pump		SS304			1st floor				A1

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Location	Sandwich, UK									
Project Name	PDF Extension Facility									
Project No.	300291	A1	21 May 26	Feasibility Design Issue	ME	MC	ME			

Tag	ID No.	No. Off	DESCRIPTION	TECHNICAL CHARACTERISTICS - SIZE/DUTY	MATERIALS OF CONSTRUCTION	WEIGHT (kg)	DIMENSIONS (mm) W x D x H	LOCATION	Process	Electrical Load	NOTES	REVISION
R32		1	Reactor	5000L Design Temp: -29/200 °C Oper. Temp: -20/130 °C Design P: -0.1~0.6 MPa Oper. P: -0.1~0.3 MPa	Hastelloy C-22		Internal 1750 D x 2525 H External 1920 D x 5228 H	2nd floor		7.5 kW electromotor		A1
R32	PC01	1	Primary Condenser	30 m^2	Hastelloy C-22			2nd floor				A1
R32	VC01	1	Vent Condenser	3 m^2	Hastelloy C-22			2nd floor				A1
R32	JP01	1	Jacket Circulation Pump		SS304			1st floor				A1
R33		1	Reactor	8000L Design Temp: -85/165 °C Oper. Temp: -70/150 °C Design P: -0.1~0.6 MPa Oper. P: -0.1~0.3 MPa	Hastelloy C-22		Internal 2000 D x 3025 H External 2224 D x 6020 H	2nd floor		11 kW electromotor	Review option to have this capable of venting Hydrogen IIC T4	A1
R33	PC01	1	Primary Condenser	40 m^2	Hastelloy C-22			2nd floor				A1
R33	VC01	1	Vent Condenser	5 m^2	Hastelloy C-22			2nd floor				A1
R33	JP01	1	Jacket Circulation Pump		SS304			1st floor				A1
R33	CC01	1	Jacket Cryogenic Cooler		SS304			1st floor				A1
R34		1	Reactor	8000L Design Temp: -29/200 °C Oper. Temp: -20/150 °C Design P: -0.1~0.6 MPa Oper. P: -0.1~0.3 MPa	Glass Lined Vessel and jacket from ASME SA516 Grade 60	Empty - 8765 Full water - 17965	Internal 2000 D x 3025 H External 2224 D (100+10mm thickness) x 6020 H	2nd floor	Y - Transfer from reagent, slurries and liquids & solvents from other process areas	11 kW electromotor		A1
R34	PC01	1	Primary Condenser	40 m^2	Hastelloy C-22			2nd floor				A1
R34	VC01	1	Vent Condenser	5 m^2	Hastelloy C-22			2nd floor				A1
R34	JP01	1	Jacket Circulation Pump		SS304			1st floor				A1
HYDROGENATION												
R35		1	Hydrogenation Reactor	2500L working volume (3400L gross volume) Design Temp: -29/200 °C Oper. Temp: -20/150 °C Design P: -0.1~2.4 MPa Oper. P: -0.1~2.0 MPa	Hastelloy C-22		Internal 1400 D x 2075 H External 1570 D x 4663 H	3rd floor	Y - Methanol Acetone LIBC?		Hydrogen venting vessel IIC T4	A1
R35	VC01	1	Vent Condenser	10 m^2	Hastelloy C-22			3rd floor			Hydrogen venting vessel IIC T4	A1
R35	TCU01	1	TCU	Welded plate, 92kW	SS304			3rd floor				A1
R35	CU1	1	Hydrogenator Cooler Unit	Welded plate, 210 kW	SS316L			3rd floor				A1
R35	JP01	1	Jacket Circulation Pump		SS304			3rd floor		Y		A1
RECEIVERS & TRANSFER PUMPS												
RC01		1	Distillate Receiver	1000L Design Temp: -29/200 °C Oper. Temp: -20/150 °C Design P: -0.1~0.6 MPa	Glass Lined			1st floor				A1
RC01	TP01	1	Transfer Pump	5.5 m^3/h-20m	PTFE			1st floor		Y		A1
RC02		1	Distillate Receiver	1000L Design Temp: -29/200 °C Oper. Temp: -20/150 °C Design P: -0.1~0.6 MPa Oper. P: -0.1~0.3 MPa	Glass Lined			1st floor				A1
RC02	TP01	1	Transfer Pump	5.5 m^3/h-20m	PTFE			1st floor		Y		A1
RC03		1	Distillate Receiver	1000L Design Temp: -29/200 °C Oper. Temp: -20/150 °C Design P: -0.1~0.6 MPa Oper. P: -0.1~0.3 MPa	Glass Lined			1st floor				A1
RC03	TP01	1	Transfer Pump	5.5 m^3/h-20m	PTFE			1st floor		Y		A1
RC04		1	Distillate Receiver	1000L Design Temp: -29/200 °C Oper. Temp: -20/150 °C Design P: -0.1~0.6 MPa Oper. P: -0.1~0.3 MPa	Glass Lined			1st floor				A1
RC04	TP01	1	Transfer Pump	5.5 m^3/h-20m	PTFE			1st floor		Y		A1
RC05		1	Distillate Receiver	1000L Design Temp: -29/200 °C Oper. Temp: -20/150 °C Design P: -0.1~0.6 MPa Oper. P: -0.1~0.3 MPa	Glass Lined			1st floor				A1
RC05	TP01	1	Transfer Pump	5.5 m^3/h-20m	PTFE			1st floor		Y		A1

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Tag	ID No.	No. Off	DESCRIPTION	TECHNICAL CHARACTERISTICS - SIZE/DUTY	MATERIALS OF CONSTRUCTION	WEIGHT (kg)	DIMENSIONS (mm) W x D x H	LOCATION	Process	Electrical Load	NOTES	REVISION
RC06		1	Distillate Receiver	1000L Design Temp: -29/200 °C Oper. Temp: -20/150 °C Design P: -0.1~0.6 MPa Oper. P: -0.1~0.3 MPa	Glass Lined			1st floor				A1
RC06	TP01	1	Transfer Pump	5.5m^3/h-20m	PTFE			1st floor		Y		A1
FILTER / DRYERS												
FD01		1	Filter/Dryer	DN1200（1.13m^2） Design Temp: -20/150 °C Oper. Temp: -15/130 °C Design P: -0.1~0.6 MPa Oper. P: -0.1~0.3 MPa	Hastelloy C-22			Ground floor				A1
FD01	RC01	1	Filtrate Receiver	1000L	Glass Lined			Ground floor				A1
FD01	TP01	1	Transfer Pump	5.5m^3/h-20m	PTFE			Ground floor				A1
FD01	VC01	1	Vent Condenser	5m^2	Hastelloy C-22			Ground floor				A1
FD01	TCU01	1	TCU		SS304			Ground floor				A1
FD02		1	Filter/Dryer	DN1400（1.53m^2） Design Temp: -20/150 °C Oper. Temp: -15/130 °C Design P: -0.1~0.6 MPa Oper. P: -0.1~0.3 MPa	Hastelloy C-22			Ground floor				A1
FD02	RC01	1	Filtrate Receiver	1000L	Glass Lined			Ground floor				A1
FD02	TP01	1	Transfer Pump	5.5m^3/h-20m	PTFE			Ground floor				A1
FD02	VC01	1	Vent Condenser	7m^2	Hastelloy C-22			Ground floor				A1
FD02	TCU01	1	TCU		SS304			Ground floor				A1
FD03		1	Filter/Dryer	DN1400（1.53m^2） Design Temp: -20/150 °C Oper. Temp: -15/130 °C Design P: -0.1~0.6 MPa Oper. P: -0.1~0.3 MPa	Hastelloy C-22			Ground floor				A1
FD03	RC01	1	Filtrate Receiver	1000L	Glass Lined			Ground floor				A1
FD03	TP01	1	Transfer Pump	5.5m^3/h-20m	PTFE			Ground floor				A1
FD03	VC01	1	Vent Condenser	7m^2	Hastelloy C-22			Ground floor				A1
FD03	TCU01	1	TCU		SS304			Ground floor				A1
VACUUM PUMPS												
-	VP01	1	Vacuum Pump	pumping speed：300L/S limiting vacuum：1Pa	——			1st floor				A1
-	VP02	1	Vacuum Pump	pumping speed：300L/S limiting vacuum：1Pa	——			1st floor				A1
-	VP03	1	Vacuum Pump	pumping speed：300L/S limiting vacuum：1Pa	——			1st floor				A1
-	VP04	1	Vacuum Pump	pumping speed：300L/S limiting vacuum：1Pa	——			1st floor				A1
-	VP05	1	Vacuum Pump	pumping speed：300L/S limiting vacuum：1Pa	——			1st floor				A1
-	VP06	1	Vacuum Pump	pumping speed：300L/S limiting vacuum：1Pa	——			1st floor				A1
SCRUBBER PACKAGES												
			ACID & Alkali IBC Package	Run & standby NaOH & Sulfuric acid storage & pumping system. 4 x 1m3 IBC's , 2 bunds with weather covers , dip tubes, distribution pumps, hoses and piping. Controls to interface with scrubber package PLC's, telemetry to central SCADA contol on level and status	PP/ GRP/ PE/ 316, Painted steel	4500 kg TBC	Nominal 5m W x 1.8m D x 3.0m H TBC	TBC		1 Ph nominal, 1000 W for lights and possible trace heating		A1
SC05		1	Alkaline Gas Scrubber Package	425 m3/h Air flow (variable). Pressure rating +-55mBar @-10 to =40 C. Scrubber performance Inlet 2%w/w max as NH3 Outlet concentration <10mg/m3. Power & Control: Local PLC & MCC with telemetry to central SCADA for status and alarms. Externally Bunded (by others) Include to raised gavanised access platform / skid. ATEX Zone 2 .	PP/GRP/HAST/316SS/PTFE-CS	5000 kg (Flooded) TBC	Nominal 5m W x 2.5m D x 7.0m H TBC	Externally located		Approx 12kW	Acid scrubbing fluid, for alkaline fumes such as NH3. Package shall be cabable of operating as an acid scrubber using an aklaline scrubbing fluid	A1
	CU1	1	Alkaline Gas Scrubber Cooler	Heat exchanger for scrubbing fluid Primary target exit temp <30 C	Hastelloy C-22 process contact & 304/316	SC05 Package	SC05 Package	Externally located				A1
	FN1	1	Alkaline Gas Scrubber Fan	0 to 425m3/h High pressure centrifugal fan design, External Dp 8kPa.	PP / GRP - (TBC)			Externally located				A1
	FN2	1	Stand-by Alkaline Gas Scrubber Fan	0 to 425m3/h High pressure centrifugal fan design, External Dp 8kPa.	PP / GRP - (TBC)			Externally located				A1
	P1	1	Alkaline Gas Scrubber Pump	Centrifugal , sealless mag drive	Glass Filled Furane Resin			Externally located				A1
	P2	1	Alklaine Gas Scrubber Pump	Centrifugal , sealless mag drive	Glass Filled Furane Resin			Externally located				A1
	PS1	1	Alkaline Gas Process Scrubber	Packed bed with demister and cleaning sprays, fluid distribution nozzles and piping to BS EN 13121-3	PP / GRP -			Externally located				A1
	ST1	1	Alkaline Gas Scrubber Storage Tank	Srubbing fluid vessel with appriate nozzles, suppports and manway to BS EN 13121-3	PP / GRP -			Externally located				A1

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Project Name	PDF Extension Facility									
Project No.	300291	A1	21 May 26	Feasibility Design Issue	ME	MC	ME			

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SC06		1	Acid Gas Scrubber Package	425 m3/h Air flow (variable). Pressure rating +-55mBar @-10 to =40 C. Scrubber performance Inlet 2%w/w max as HCl Outlet concentration <10mg/m3. Power & Control: Local PLC & MCC. Externally Bunded (by others) Include to raised gavanised access platform / skid. ATEX Zone 2	PP/GRP/HAST/316SS/PTFE-CS	5000 kg (Flooded) TBC	Nominal 5m W x 2.5m D x 7.0m H TBC	Externally located		Approx 12kW	Alkaline scrubbing fluid NaOH, for fumes such as HCl. Package shall be cabable of operating as an alkaline scrubber using an acid scrubbing fluid	A1
	CU1	1	Acid Gas Scrubber Cooler	Heat exchanger for scrubbing fluid Primary target exit temp <30 C	Hastelloy C-22 process contact & 304/316	SC06 Package	SC06 Package	Externally located				A1
	FN1	1	Acid Gas Scrubber Fan	0 to 425m3/h High pressure centrifugal fan design, External Dp 8kPa.	PP / GRP - (TBC)			Externally located				A1
	FN2	1	Stand-by Acid Gas Scrubber Fan	0 to 425m3/h High pressure centrifugal fan design, External Dp 8kPa.	PP / GRP - (TBC)			Externally located				A1
	P1	1	Acid Gas Scrubber Pump	Centrifugal , sealless mag drive	Glass Filled Furane Resin			Externally located				A1
	P2	1	Acid Gas Scrubber Pump	Centrifugal , sealless mag drive	Glass Filled Furane Resin			Externally located				A1
	PS1	1	Acid Gas Process Scrubber	Packed bed with demister and cleaning sprays, fluid distribution nozzles and piping to BS EN 13121-3	PP / GRP -			Externally located				A1
	ST1	1	Acid Gas Scrubber Storage Tank	Srubbing fluid vessel with appriate nozzles, suppports and manway to BS EN 13121-3	PP / GRP -			Externally located				A1
CATCH TANKS												
-	CT01	1	Relief Catch Tank (Reactor)	8000L Design Temp: -20/200 °C Oper. Temp: -20/130 °C Design P: 0.4 MPa Oper. P: 0.2 MPa	Carbon Steel			Externally located				A1
HYDROGENATION												
-	CT02	1	Relief Catch Tank (Hydrogenation Reactor)	2500L Design Temp: -20/200 °C Oper. Temp: -20/130 °C Design P: 2.4 MPa Oper. P: 1.9 MPa	Carbon steel composite stainless steel			3rd floor				A1
LIQUID NITROGEN SYSTEM												
C01	NA2	1	Nitrogen Accumulator	200 - 300 litres	SS316L			TBC				A1
NS02		1	Nitrogen Supply Package	500 Nm3/h, 6 barg	SS (inner), CS (outer)	50,000 (full skid)	10500 x 6500 (full LN2 tank skid concrete plinth)	Externally located			Existing LN2 tank to be disinvested once new system is fully installed and in operation. New system to supply both existing and extension facilities	A1
NF05 A/B		2	Nitrogen Supply Filter	5 micron	SS/Polypropylene Cartridge			Externally located				A1
NF06 A/B		2	Nitrogen Supply Filter	5 micron	SS/Polypropylene Cartridge			Externally located				A1
HOT HTF DISTRIBUTION												
HTF03	HU1	1	Hot HTF Heater	Welded Plate, 800 kW, $20m^2$	SS316L			Externally located				A1
	CP3	1	Hot HTF Circulation Pump	70m3/hr @ 70m TDH Centrifugal, Mag Drive	SS316L			Externally located				A1
	CP4	1	Hot HTF Circulation Pump	70m3/hr @ 70m TDH Centrifugal, Mag Drive	SS316L			Externally located				A1
PURIFIED WATER GENERATION												
PW01		1	Purified Water Generation Package	$4.0m^3/h$ Ozonated purified water generation system	SS316L			Ground floor				A1
		1	Pre-Treatment Package	Softener + activated carbon filter				Ground floor				A1
	ST2	1	Purified Water Storage Tank	$16m^3$	SS316L			Ground floor				A1
	F1	1	Storage Tank Vent Filter	0.1 micron hydrophobic, ozone-resistance (PTFE)	SS316L			Ground floor				A1
	HC1	1	Purified Water Heat Exchanger	Welded plate, 15kW	SS316L			Ground floor				A1
	P01	1	Cooling Water Pump	Centrifugal Pump - cooling water for trim cooling of PW	SS316L			Ground floor				A1
	HC2	1	Cooling Water Chiller	Provide cooling water from site mains water, 20kW	SS316L			Ground floor				A1
		1	Ozone Generator					Ground floor				A1
		1	Ozone Analyser/ Controller	Maintain target ozone concentration				Ground floor				A1
		1	Ozone Destruct Unit	On purified water storage tank vent				Ground floor				A1
PURIFIED WATER DISTRIBUTION												
	P02	1	Purified Water Supply Pump	Centrifugal Pump $15m^3/h$ @ 60m TDH	SS316L			Ground floor				A1
	P03	1	Spare Purified Water Supply Pump	Centrifugal Pump $15m^3/h$ @ 60m TDH	SS316L			Ground floor				A1
		1	UV De-ozonation Unit		SS316L			Ground floor				A1
AQUEOUS EFFLUENT TANKS												
ET03		1	Trade Effluent Tank	$10m^3$ working volume	Glass-Lined Carbon Steel	7,500 (empty)	2200 D x 3500 H (internal)	Externally located			For extension alone	A1

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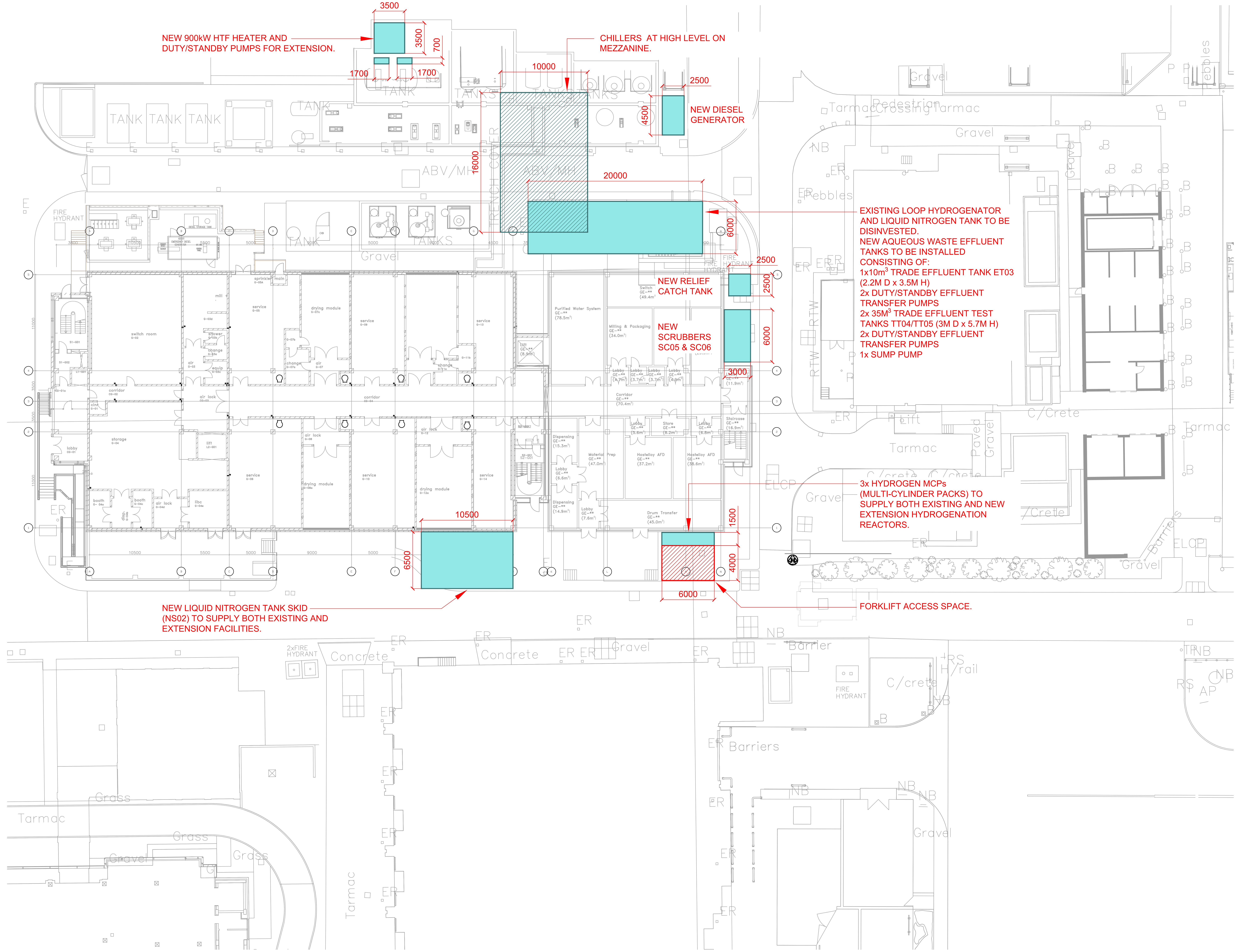
	P1	1	Effluent Transfer Pump	$20 m^3/h$ @ 19.8m TDH Self-priming	Hastelloy-C22			Externally located				A1
	P2	1	Effluent Transfer Pump	$20 m^3/h$ @ 19.8m TDH Self-priming	Hastelloy-C22			Externally located				A1
TT04		1	Trade Effluent Test Tank	$35 m^3$ working volume	Glass-Lined Carbon Steel		3000 D x 5700 H (internal)	Externally located			For extension alone	A1
TT05		1	Trade Effluent Test Tank	$35 m^3$ working volume	Glass-Lined Carbon Steel		3000 D x 5700 H (internal)	Externally located			For extension alone	A1
TT04	P1A	1	Effluent Transfer Pump	$60 m^3/h$ @ 36.4m TDH	Hastelloy-C22			Externally located				A1
TT04	P1B	1	Effluent Transfer Pump	$60 m^3/h$ @ 36.4m TDH	Hastelloy-C22			Externally located				A1
SP02		1	Sump Tank - Effluent Tank Sump		SS316L			Externally located				A1
CONTAINMENT & ISOLATION												
-		1	Solid packing glove box		SS316L			Ground floor				A1
-		1	Solid packing glove box		SS316L			Ground floor				A1
-		1	Crushing packaging isolator		SS316L			Ground floor				A1

Notes:
(1) The selections for equipment such as the condenser and tanks are currently preliminary estimates and will require detailed calculations and assessment in the next phase.
(2) The equipment numbering is provisional/assumed.



Appendix 6 - Plant Setting Out Drawing

ASYMCHEM

NEW PLANT SETTING OUT

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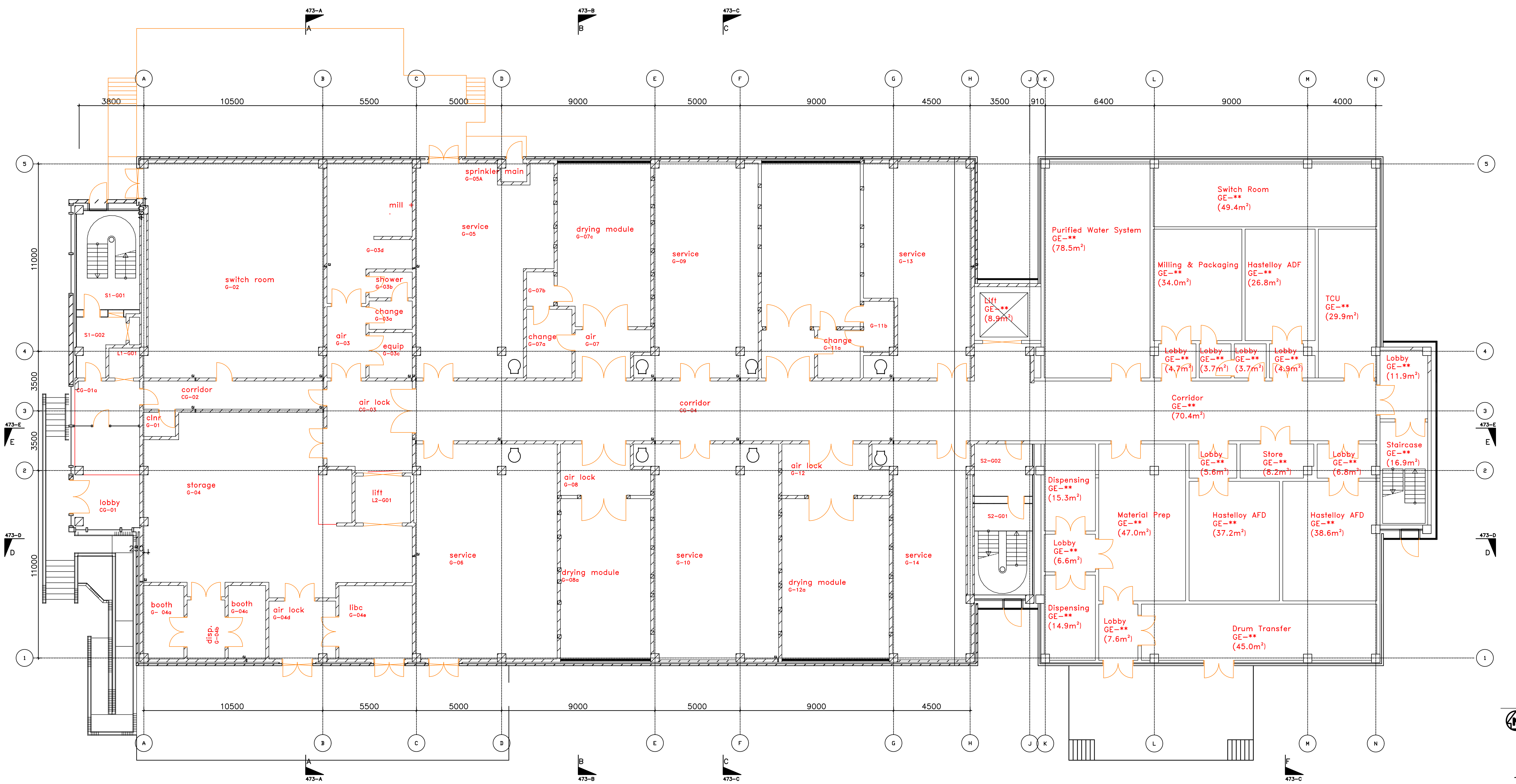
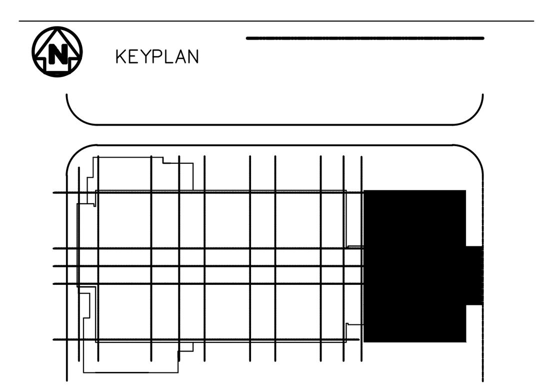
DRAWING No.: 300291-SCI-SI-XX-SK-PR-001

DATE: 07 MAY 26 SCALE: 1:200 REV:P01

Appendix 7 - Architectural Drawings





ASYMCHEM - PDF EXTENSION

PROPOSED GROUND FLOOR

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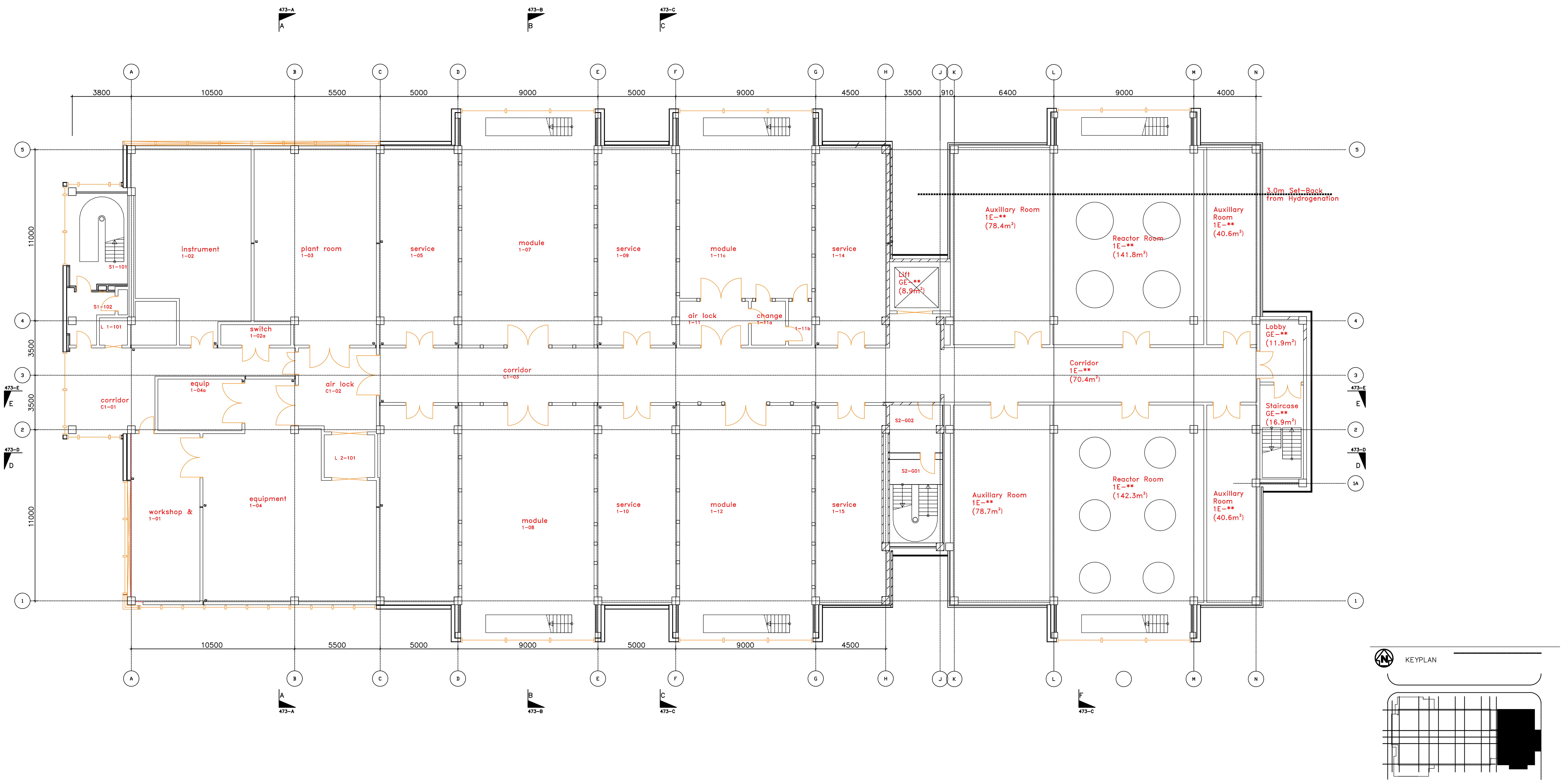
DATE: 20.05.26 SCALE: NTS REV: P01



ASYMCHEM - PDF EXTENSION

PROPOSED FIRST FLOOR

Scitech EKIUM

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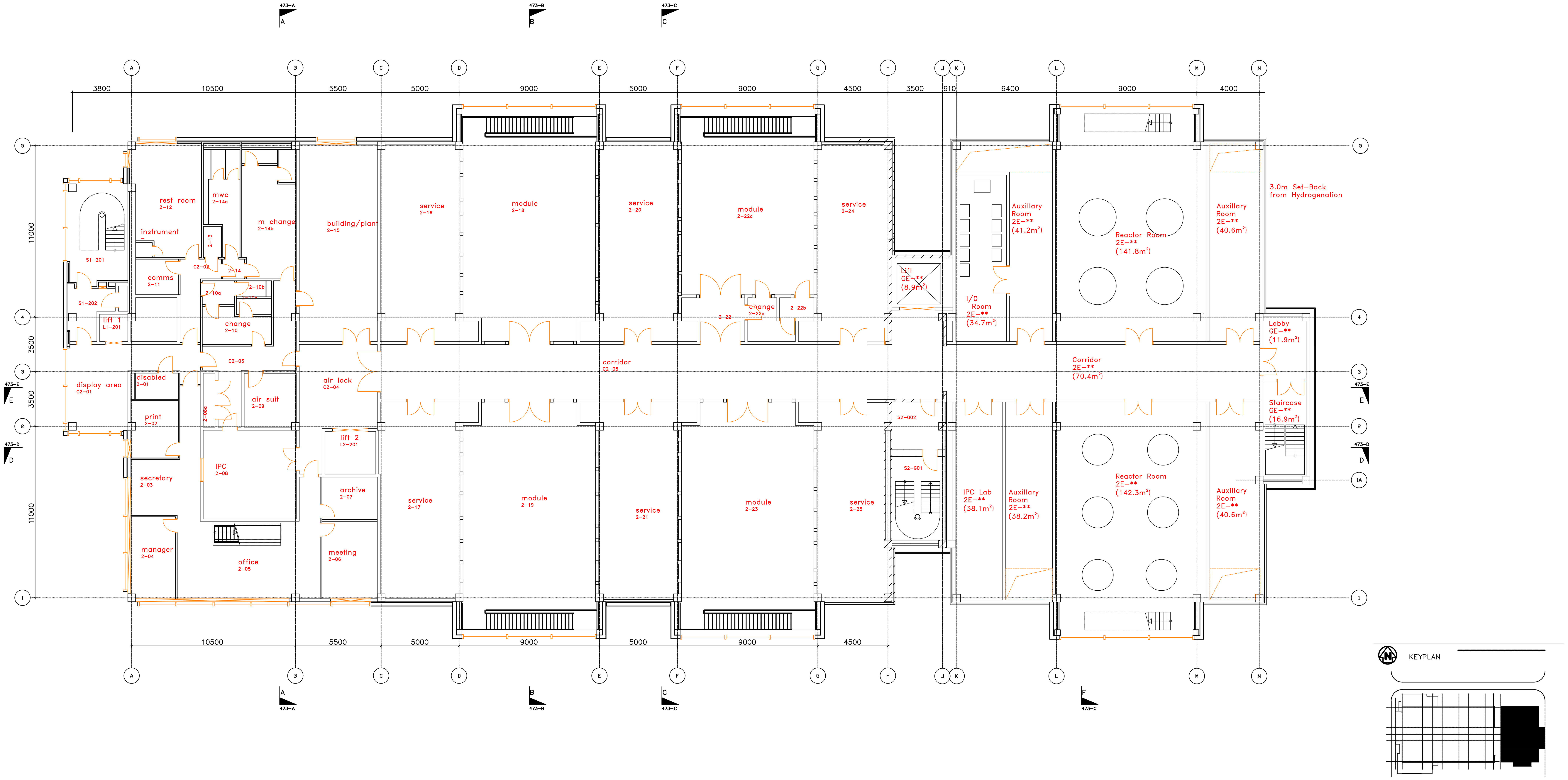
DATE: 20.05.26 SCALE: NTS REV: P01



ASYMCHEM - PDF EXTENSION

PROPOSED SECOND FLOOR

Scitech EKIUM

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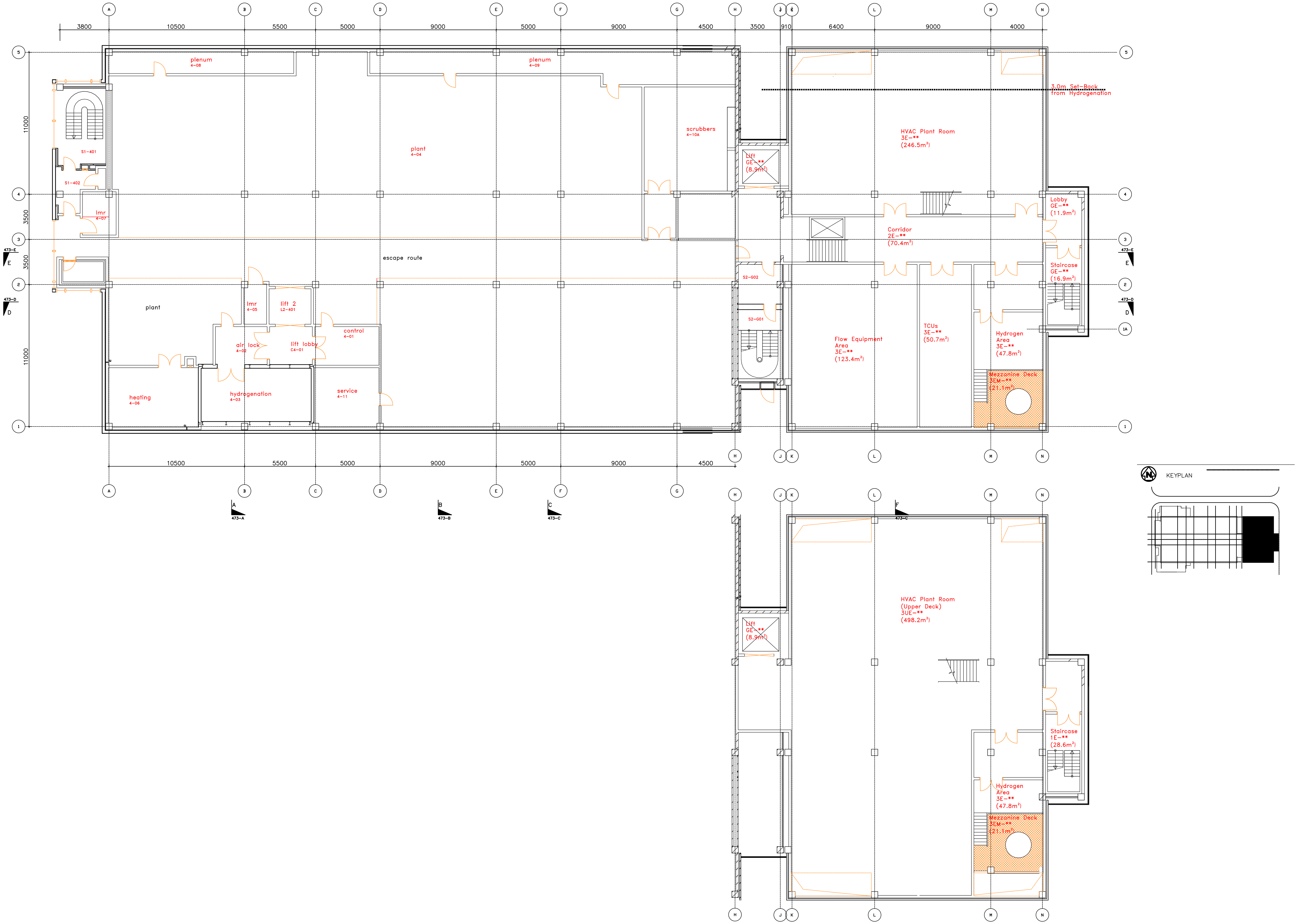
DATE: 20.05.26 SCALE: NTS REV: P01



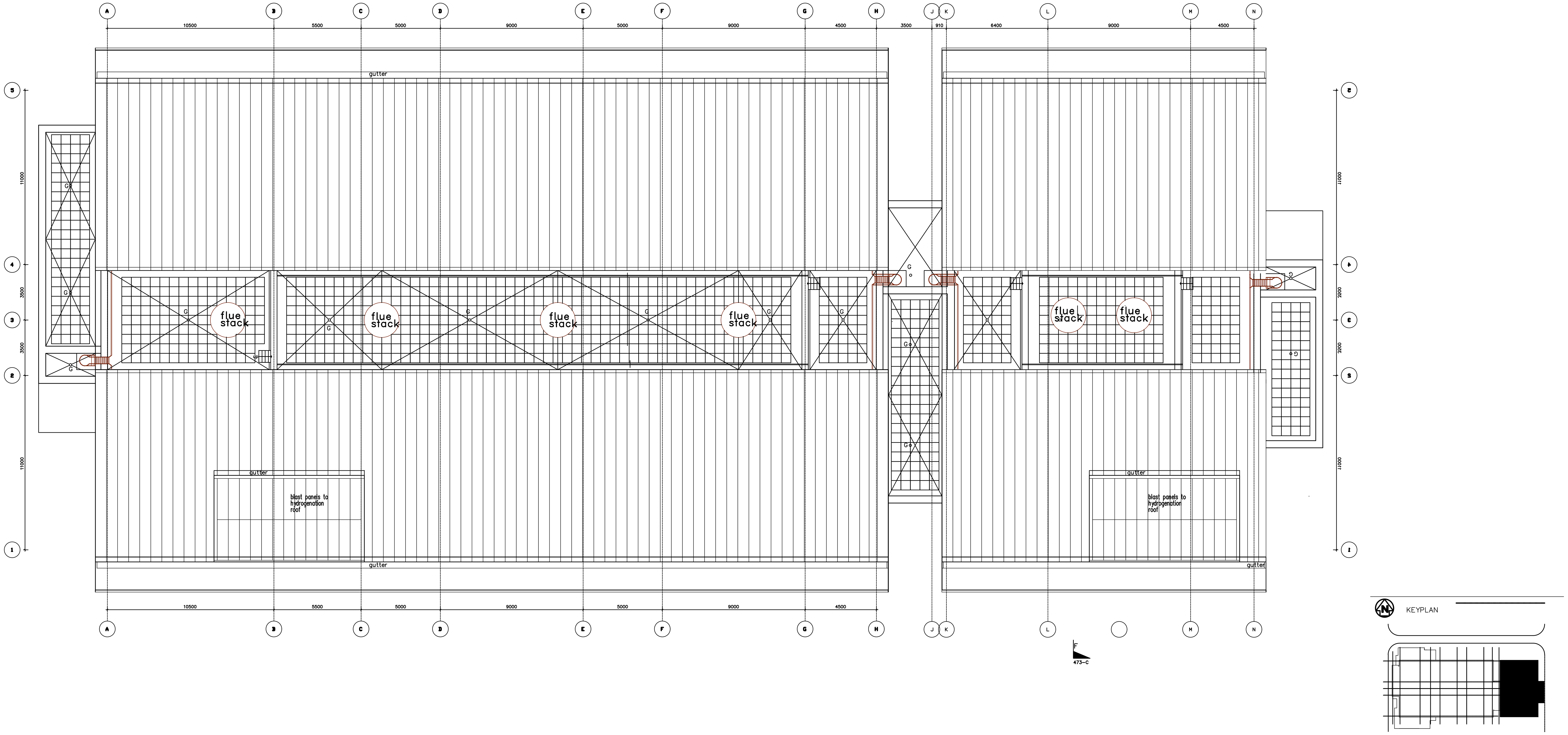
ASYMCHEM - PDF EXTENSION

PROPOSED THIRD FLOOR

Scitech EKIUM

DRAWING No.: 300291-SCI-GA-03-SK-AR-004

DATE: 20.05.26 SCALE: NTS REV: P01



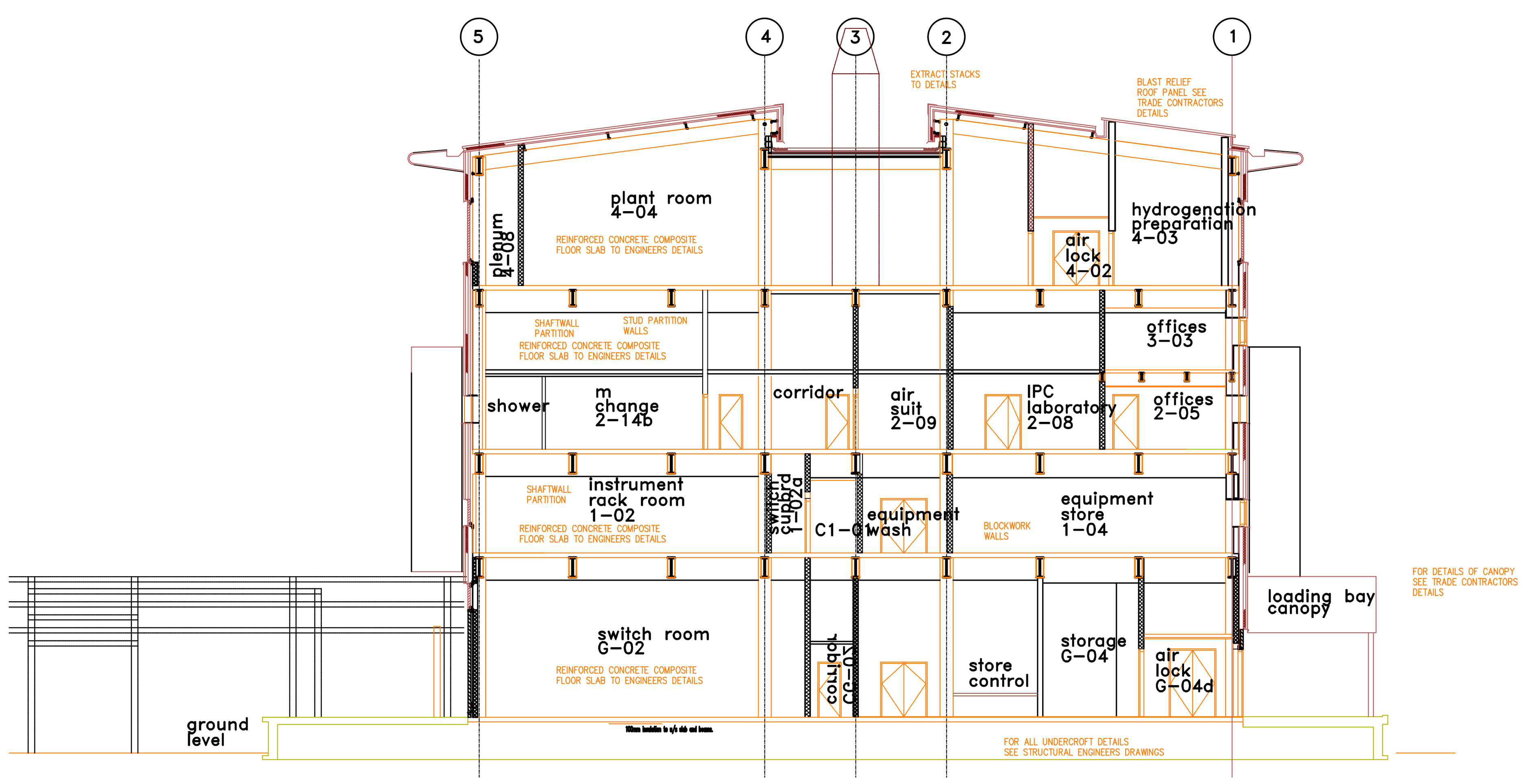
ASYMCHEM - PDF EXTENSION

PROPOSED FOURTH FLOOR

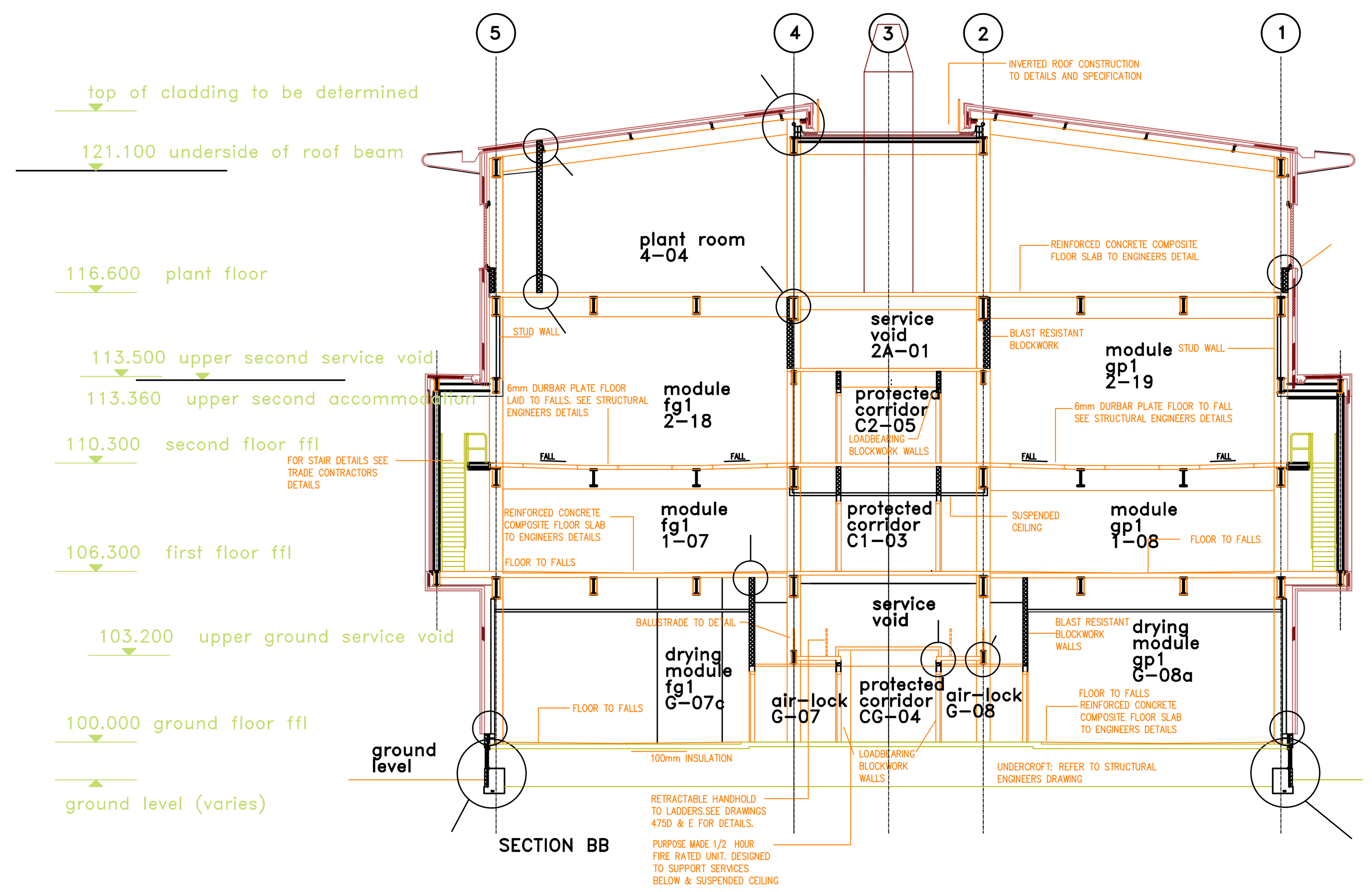
Scitech EKIUM

DRAWING No.: 300291-SCI-GA-04-SK-AR-005

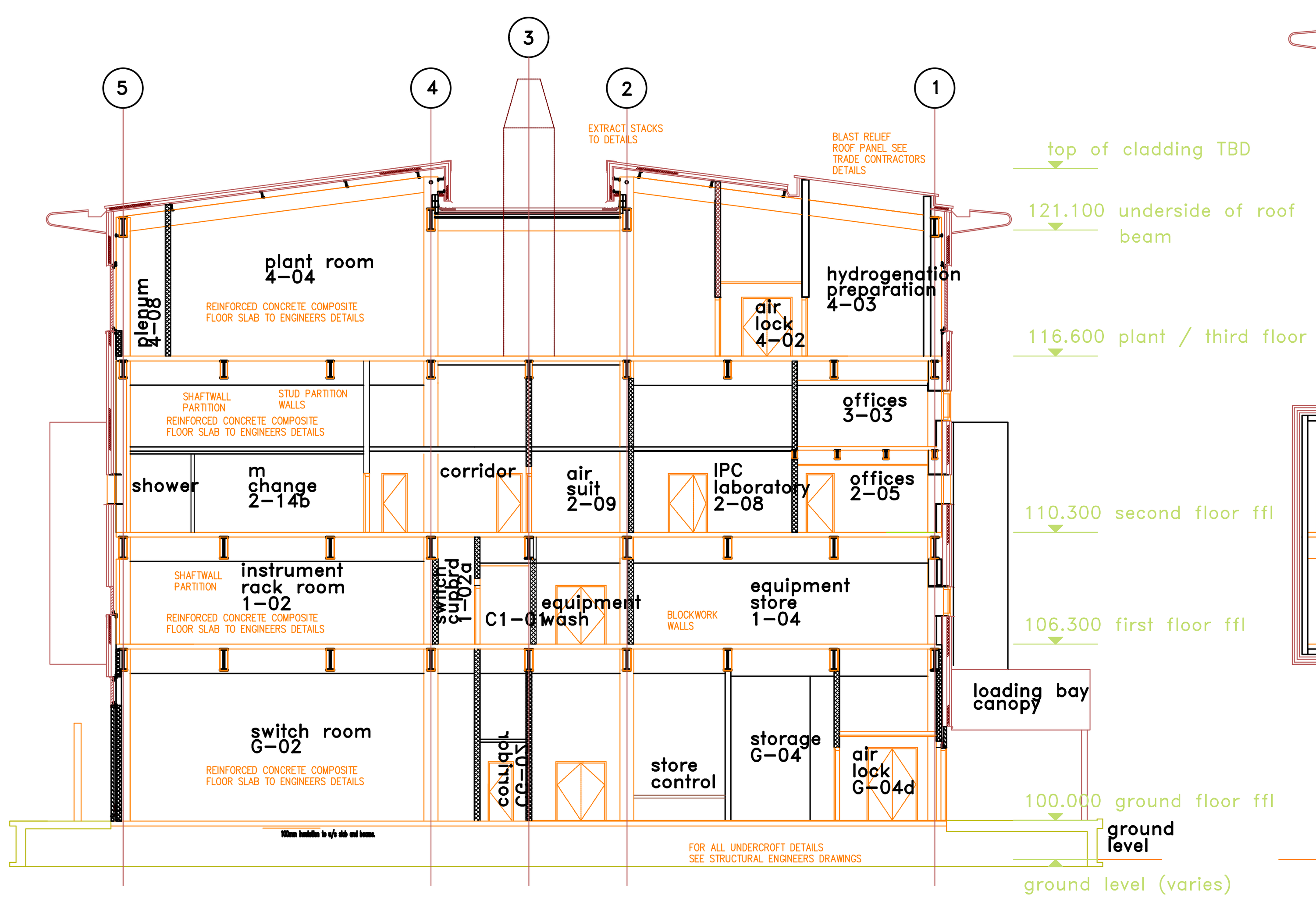
DATE: 20.05.26 SCALE: NTS REV: P01



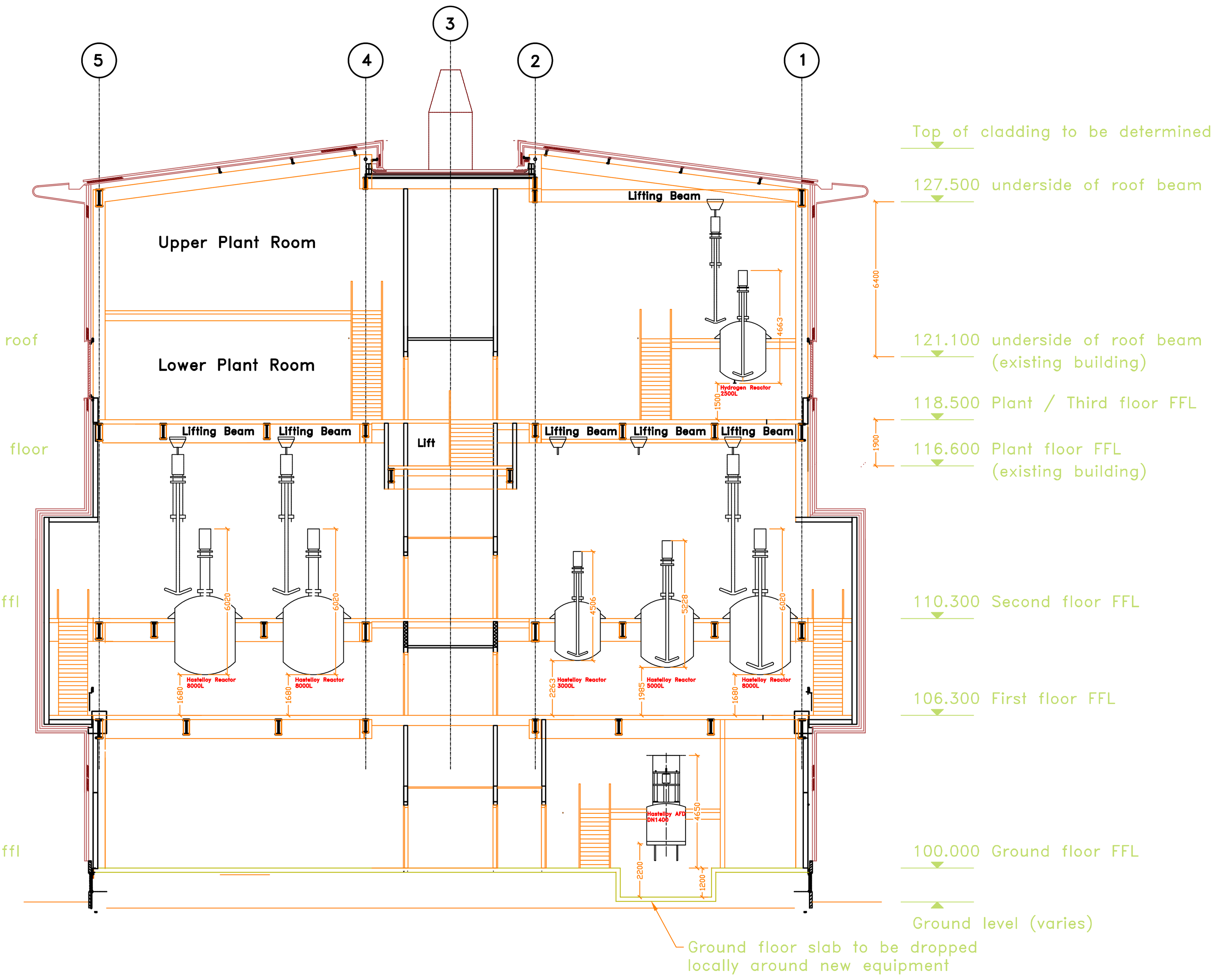
TYPICAL SECTION THROUGH OFFICE AREA



TYPICAL SECTION THROUGH PROCESS MODULES



TYPICAL SECTION THROUGH OFFICE AREA
EXISTING BUILDING



TYPICAL SECTION THROUGH REACTOR SPACE
PROPOSED EXTENSION

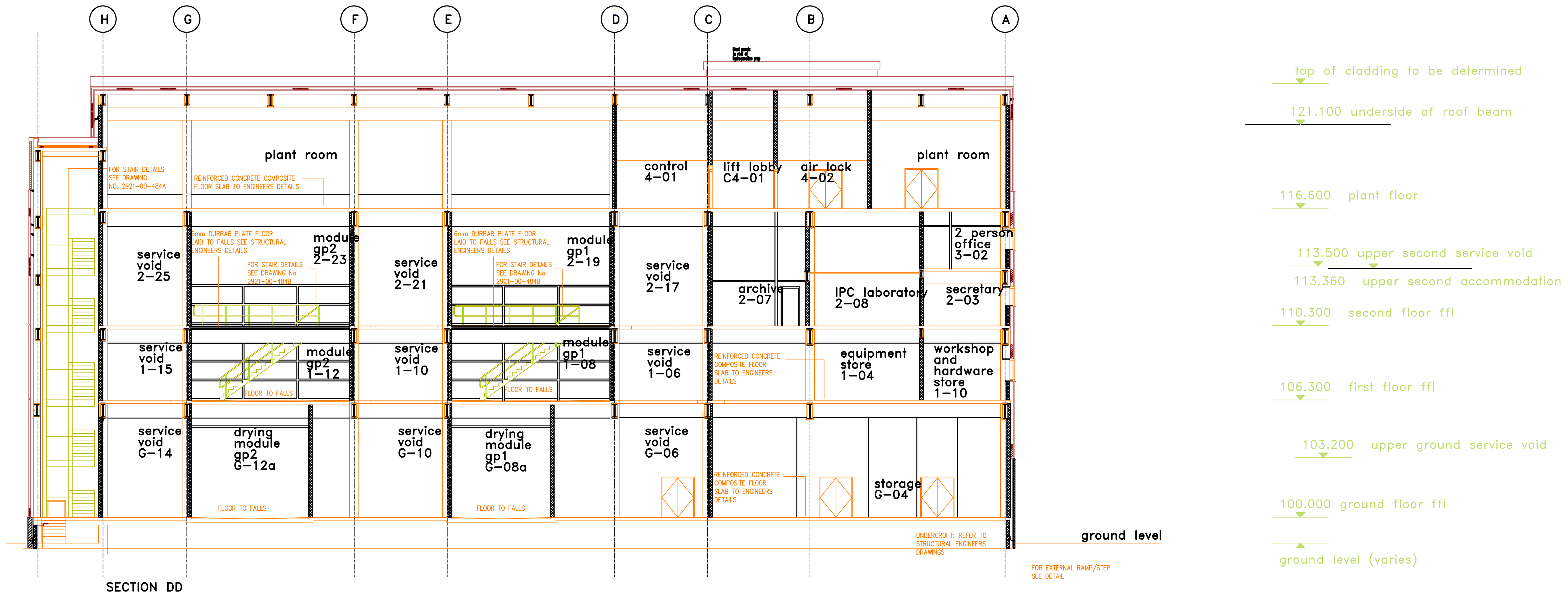
ASYMCHEM - PDF EXTENSION

PROPOSED SECTIONS

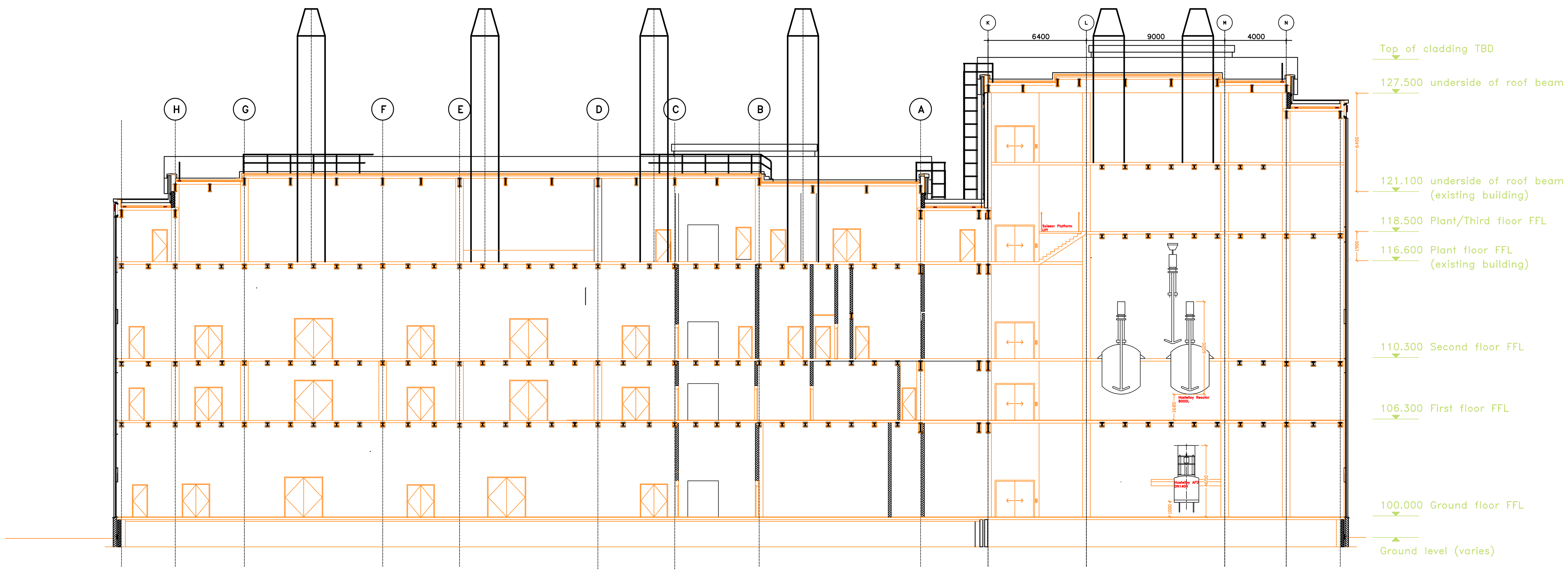
Scitech EKIUM

DRAWING No.: 300291-SCI-SE-ZZ-SK-AR-201

DATE: 20.05.26 SCALE: NTS REV: P01



TYPICAL LONGITUDINAL SECTION THROUGH PROCESS ROOMS



TYPICAL LONGITUDINAL SECTION THROUGH CENTRAL CORRIDOR

ASYMCHEM - PDF EXTENSION

PROPOSED SECTIONS

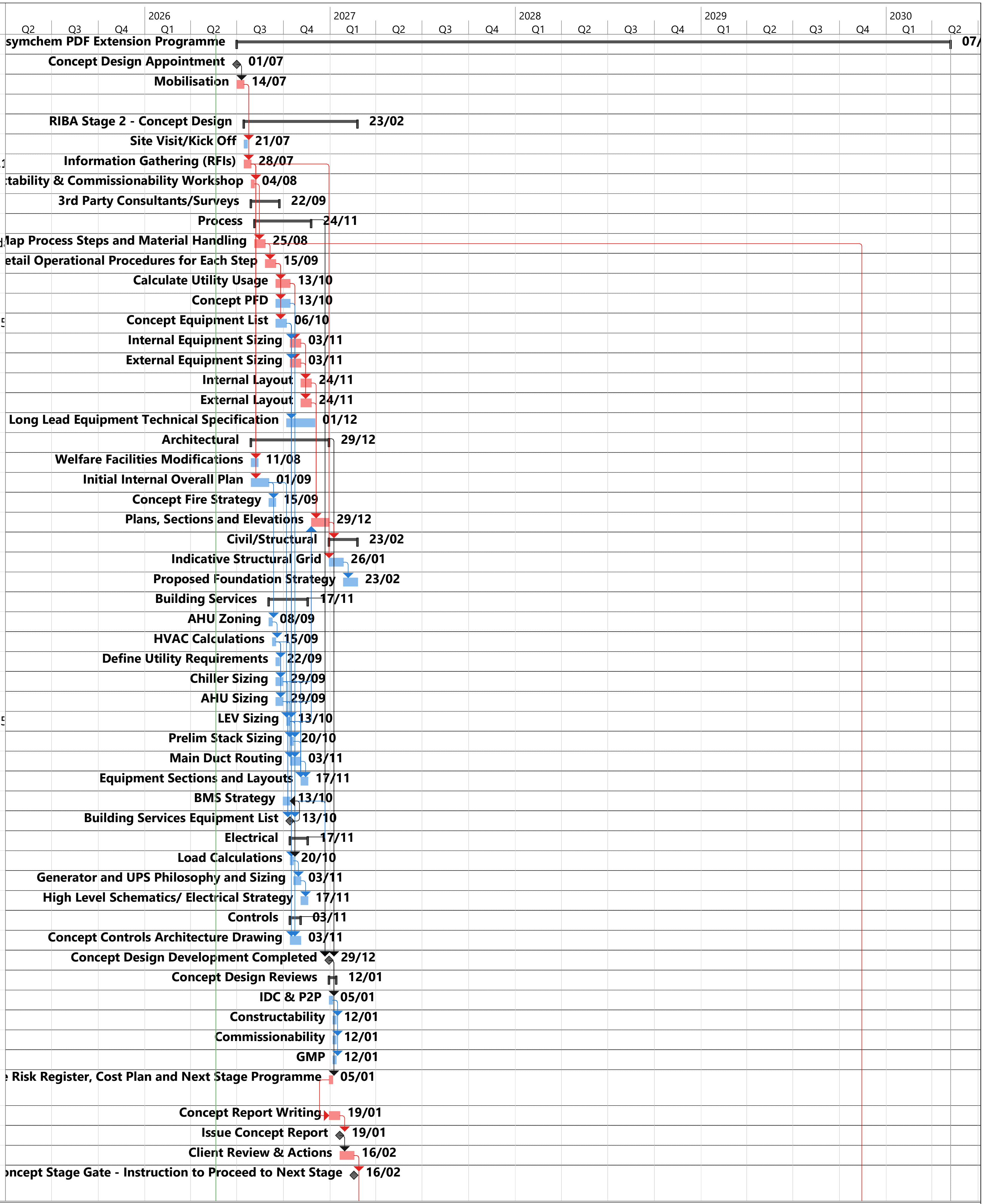
Scitech EKIUM

DRAWING No.: 300291-SCI-SE-ZZ-SK-AR-202

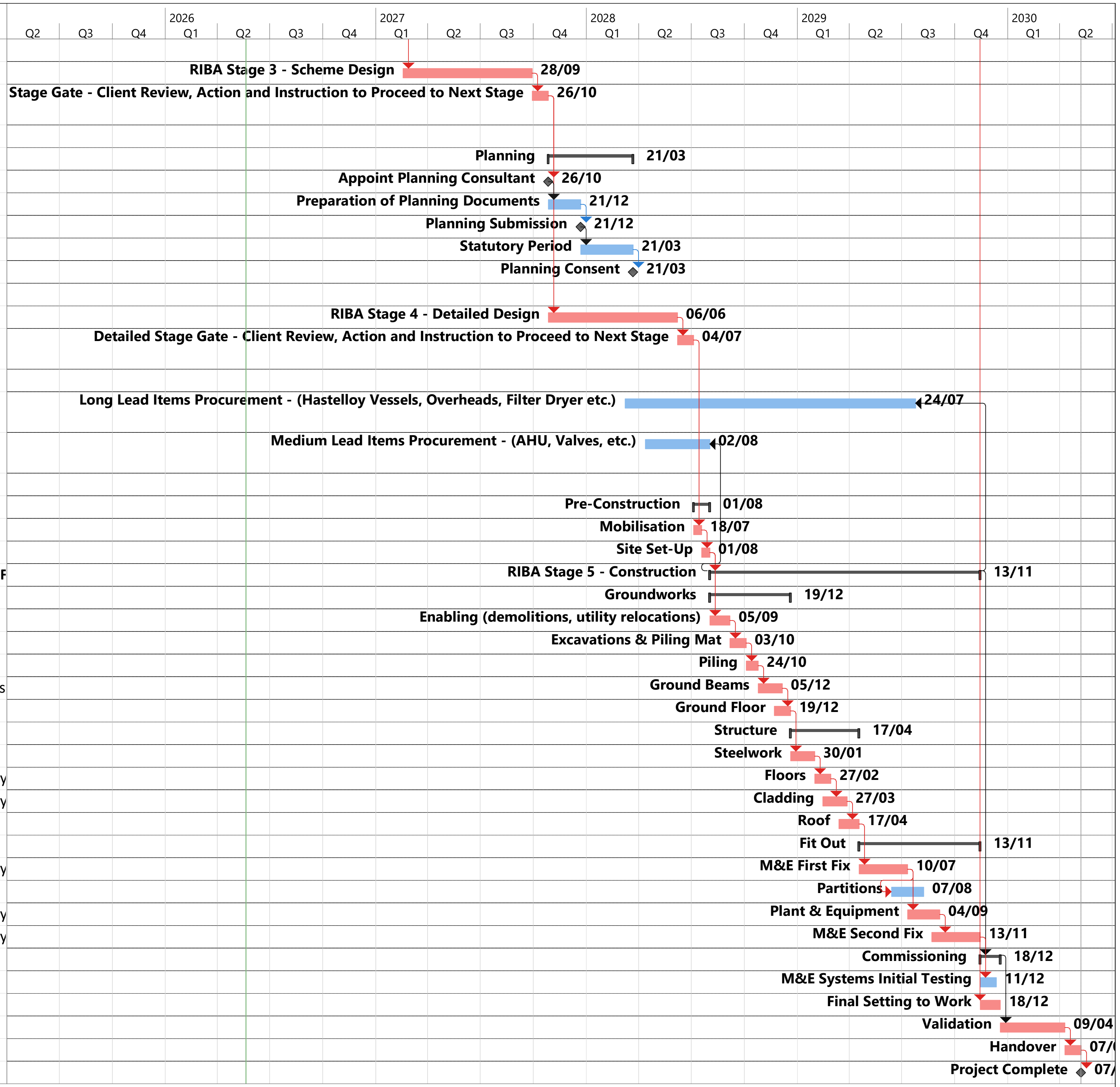
DATE: 20.05.26 SCALE: NTS REV: P01

Appendix 8 - Next Stage Programme

ID	Task Name	Duration	Start	Finish	Predecessors	Successors
1	**1 300291 Asymchem PDF Extension Programme**	**1005 days**	**Wed 01/07/26**	**Tue 07/05/30**		
2	1.1 Concept Design Appointment	0 days	Wed 01/07/26	Wed 01/07/26		3
3	1.2 Mobilisation	10 days	Wed 01/07/26	Tue 14/07/26	2	6,7
4						
5	**1.3 RIBA Stage 2 - Concept Design**	**160 days**	**Wed 15/07/26**	**Tue 23/02/27**		
6	**1.3.1 Site Visit/Kick Off**	**5 days**	**Wed 15/07/26**	**Tue 21/07/26**	**3**	
7	**1.3.2 Information Gathering (RFIs)**	**10 days**	**Wed 15/07/26**	**Tue 28/07/26**	**3**	**10,11,12,13,1**
8	**1.3.3 Constructability & Commissionability Worl**	**5 days**	**Wed 29/07/26**	**Tue 04/08/26**	**7**	**25**
9	**1.3.4 3rd Party Consultants/Surveys**	**40 days**	**Wed 29/07/26**	**Tue 22/09/26**		
24	**1.3.5 Process**	**80 days**	**Wed 05/08/26**	**Tue 24/11/26**		**61**
25	1.3.5.1 Map Process Steps and Material Handli	15 days	Wed 05/08/26	Tue 25/08/26	8	26,111FS-5 d
26	1.3.5.2 Detail Operational Procedures for Each	15 days	Wed 26/08/26	Tue 15/09/26	25	27,28,29
27	1.3.5.3 Calculate Utility Usage	20 days	Wed 16/09/26	Tue 13/10/26	26	30,31
28	1.3.5.4 Concept PFD	20 days	Wed 16/09/26	Tue 13/10/26	26	60
29	1.3.5.5 Concept Equipment List	15 days	Wed 16/09/26	Tue 06/10/26	26	30,31,34,49,5
30	1.3.5.6 Internal Equipment Sizing	15 days	Wed 14/10/26	Tue 03/11/26	29,27	32
31	1.3.5.7 External Equipment Sizing	15 days	Wed 14/10/26	Tue 03/11/26	29,27	33
32	1.3.5.8 Internal Layout	15 days	Wed 04/11/26	Tue 24/11/26	30	39
33	1.3.5.9 External Layout	15 days	Wed 04/11/26	Tue 24/11/26	31	39
34	**1.3.6 Long Lead Equipment Technical Specificati**	**40 days**	**Wed 07/10/26**	**Tue 01/12/26**	**29**	
35	**1.3.7 Architectural**	**110 days**	**Wed 29/07/26**	**Tue 29/12/26**		**61**
36	1.3.7.1 Welfare Facilities Modifications	10 days	Wed 29/07/26	Tue 11/08/26	7	
37	1.3.7.2 Initial Internal Overall Plan	25 days	Wed 29/07/26	Tue 01/09/26	7	38,49,44
38	1.3.7.3 Concept Fire Strategy	10 days	Wed 02/09/26	Tue 15/09/26	37	
39	1.3.7.4 Plans, Sections and Elevations	25 days	Wed 25/11/26	Tue 29/12/26	32,33,48,49,47	40
40	1.3.8 Civil/Structural	40 days	Wed 30/12/26	Tue 23/02/27	39	
41	1.3.8.1 Indicative Structural Grid	20 days	Wed 30/12/26	Tue 26/01/27	7	42
42	1.3.8.2 Proposed Foundation Strategy	20 days	Wed 27/01/27	Tue 23/02/27	41	
43	**1.3.9 Building Services**	**55 days**	**Wed 02/09/26**	**Tue 17/11/26**		**61**
44	1.3.9.1 AHU Zoning	5 days	Wed 02/09/26	Tue 08/09/26	37	45
45	1.3.9.2 HVAC Calculations	5 days	Wed 09/09/26	Tue 15/09/26	44	48,51,46,47
46	1.3.9.3 Define Utility Requirements	5 days	Wed 16/09/26	Tue 22/09/26	45	
47	1.3.9.4 Chiller Sizing	10 days	Wed 16/09/26	Tue 29/09/26	45	39,52,54
48	1.3.9.5 AHU Sizing	10 days	Wed 16/09/26	Tue 29/09/26	45	39,52,50,54
49	1.3.9.6 LEV Sizing	5 days	Wed 07/10/26	Tue 13/10/26	29,37	39,51,52,50,5
50	1.3.9.7 Prelim Stack Sizing	5 days	Wed 14/10/26	Tue 20/10/26	48,49	52
51	1.3.9.8 Main Duct Routing	15 days	Wed 14/10/26	Tue 03/11/26	49,45	52
52	1.3.9.9 Equipment Sections and Layouts	10 days	Wed 04/11/26	Tue 17/11/26	47,48,49,51,5(	
53	1.3.9.10 BMS Strategy	10 days	Wed 30/09/26	Tue 13/10/26	54FF	61
54	1.3.9.11 Building Services Equipment List	0 days	Tue 13/10/26	Tue 13/10/26	47,48,49	56,53FF
55	**1.3.10 Electrical**	**25 days**	**Wed 14/10/26**	**Tue 17/11/26**		**61**
56	1.3.10.1 Load Calculations	5 days	Wed 14/10/26	Tue 20/10/26	29,54	57
57	1.3.10.2 Generator and UPS Philosophy and Siz	10 days	Wed 21/10/26	Tue 03/11/26	56	58
58	1.3.10.3 High Level Schematics/ Electrical Strat	10 days	Wed 04/11/26	Tue 17/11/26	57	
59	**1.3.11 Controls**	**15 days**	**Wed 14/10/26**	**Tue 03/11/26**		**61**
60	1.3.11.1 Concept Controls Architecture Drawin	15 days	Wed 14/10/26	Tue 03/11/26	28,29	
61	**1.3.12 Concept Design Development Completed**	**0 days**	**Tue 29/12/26**	**Tue 29/12/26**	**24,35,43,55,5**	**67,63**
62	**1.3.13 Concept Design Reviews**	**10 days**	**Wed 30/12/26**	**Tue 12/01/27**		
63	1.3.13.1 IDC & P2P	5 days	Wed 30/12/26	Tue 05/01/27	61	64,65,66
64	1.3.13.2 Constructability	5 days	Wed 06/01/27	Tue 12/01/27	63	
65	1.3.13.3 Commissionability	5 days	Wed 06/01/27	Tue 12/01/27	63	
66	1.3.13.4 GMP	5 days	Wed 06/01/27	Tue 12/01/27	63	
67	**1.3.14 Finalise Risk Register, Cost Plan and Next Stage Programme**	**5 days**	**Wed 30/12/26**	**Tue 05/01/27**	**61**	**68SS**
68	**1.3.15 Concept Report Writing**	**15 days**	**Wed 30/12/26**	**Tue 19/01/27**	**67SS**	**69**
69	**1.3.16 Issue Concept Report**	**0 days**	**Tue 19/01/27**	**Tue 19/01/27**	**68**	**70**
70	**1.3.17 Client Review & Actions**	**1 mon**	**Wed 20/01/27**	**Tue 16/02/27**	**69**	**73,71**
71	**1.3.18 Concept Stage Gate - Instruction to Proceed to Next Stage**	**0 days**	**Tue 16/02/27**	**Tue 16/02/27**	**70**	



ID	Task Name	Duration	Start	Finish	Predecessors	Successors
72						
73	**1.4 RIBA Stage 3 - Scheme Design**	**8 mons**	**Wed 17/02/27**	**Tue 28/09/27**	**70**	**74**
74	1.5 Scheme Stage Gate - Client Review, Action and Instruction to Proceed to Next Stage	1 mon	Wed 29/09/27	Tue 26/10/27	73	83,77
75						
76	**1.6 Planning**	**105 days**	**Tue 26/10/27**	**Tue 21/03/28**		
77	1.6.1 Appoint Planning Consultant	0 days	Tue 26/10/27	Tue 26/10/27	74	78
78	1.6.2 Preparation of Planning Documents	40 days	Wed 27/10/27	Tue 21/12/27	77	79
79	1.6.3 Planning Submission	0 days	Tue 21/12/27	Tue 21/12/27	78	80
80	1.6.4 Statutory Period	65 days	Wed 22/12/27	Tue 21/03/28	79	81
81	1.6.5 Planning Consent	0 days	Tue 21/03/28	Tue 21/03/28	80	
82						
83	**1.7 RIBA Stage 4 - Detailed Design**	**8 mons**	**Wed 27/10/27**	**Tue 06/06/28**	**74**	**84**
84	1.8 Detailed Stage Gate - Client Review, Action and Instruction to Proceed to Next Stage	1 mon	Wed 07/06/28	Tue 04/07/28	83	90
85						
86	**1.9 Long Lead Items Procurement - (Hastelloy Vessels, Overheads, Filter Dryer etc.)**	**18 mons**	**Wed 08/03/28**	**Tue 24/07/29**	**92FF-4 mons**	
87	**1.10 Medium Lead Items Procurement - (AHU, Valves, etc.)**	**4 mons**	**Wed 12/04/28**	**Wed 02/08/28**	**92SF**	
88						
89	**1.11 Pre-Construction**	**20 days**	**Wed 05/07/28**	**Tue 01/08/28**		
90	1.11.1 Mobilisation	10 days	Wed 05/07/28	Tue 18/07/28	84	91
91	1.11.2 Site Set-Up	10 days	Wed 19/07/28	Tue 01/08/28	90	92,94
92	**1.12 RIBA Stage 5 - Construction**	**335 days**	**Wed 02/08/28**	**Tue 13/11/29**	**91**	**109,87SF,86F**
93	**1.12.1 Groundworks**	**100 days**	**Wed 02/08/28**	**Tue 19/12/28**		
94	1.12.1.1 Enabling (demolitions, utility relocatio	25 days	Wed 02/08/28	Tue 05/09/28	91	95
95	1.12.1.2 Excavations & Piling Mat	20 days	Wed 06/09/28	Tue 03/10/28	94	96
96	1.12.1.3 Piling	15 days	Wed 04/10/28	Tue 24/10/28	95	97
97	1.12.1.4 Ground Beams	30 days	Wed 25/10/28	Tue 05/12/28	96	98FS-10 days
98	1.12.1.5 Ground Floor	20 days	Wed 22/11/28	Tue 19/12/28	97FS-10 days	100
99	**1.12.2 Structure**	**85 days**	**Wed 20/12/28**	**Tue 17/04/29**		
100	1.12.2.1 Steelwork	30 days	Wed 20/12/28	Tue 30/01/29	98	101
101	1.12.2.2 Floors	20 days	Wed 31/01/29	Tue 27/02/29	100	102FS-10 day
102	1.12.2.3 Cladding	30 days	Wed 14/02/29	Tue 27/03/29	101FS-10 days	103FS-10 day
103	1.12.2.4 Roof	25 days	Wed 14/03/29	Tue 17/04/29	102FS-10 days	105
104	**1.12.3 Fit Out**	**150 days**	**Wed 18/04/29**	**Tue 13/11/29**		
105	1.12.3.1 M&E First Fix	60 days	Wed 18/04/29	Tue 10/07/29	103	106FS-20 day
106	1.12.3.2 Partitions	40 days	Wed 13/06/29	Tue 07/08/29	105FS-20 days	
107	1.12.3.3 Plant & Equipment	40 days	Wed 11/07/29	Tue 04/09/29	105	108FS-10 day
108	1.12.3.4 M&E Second Fix	60 days	Wed 22/08/29	Tue 13/11/29	107FS-10 days	110FS-10 day
109	**1.13 Commissioning**	**25 days**	**Wed 14/11/29**	**Tue 18/12/29**	**92**	**112**
110	1.13.1 M&E Systems Initial Testing	20 days	Wed 14/11/29	Tue 11/12/29	108FS-10 days	
111	1.13.2 Final Setting to Work	25 days	Wed 14/11/29	Tue 18/12/29	25FS-5 days	
112	**1.14 Validation**	**4 mons**	**Wed 19/12/29**	**Tue 09/04/30**	**109**	**113**
113	**1.15 Handover**	**1 mon**	**Wed 10/04/30**	**Tue 07/05/30**	**112**	**114**
114	**1.16 Project Complete**	**0 days**	**Tue 07/05/30**	**Tue 07/05/30**	**113**	



Appendix 9 - Feasibility Risk Register

Client	Asymchem
Project Name	PDF Extension Feasibility
Scitech Project No	300291
Document Ref	300291-PM-RA-0001

Revision No.	A1
Revision Date	21-May-26
By	GS
Reason	Feasibility Design

RISK ENTRY / PRE-ACTION ASSESSMENT (expected outcome without mitigation)

Status	Date	REF NO.	RISK	RISK IMPACT	RAISED BY	CATEGORY (from list)	LIKELIHOOD A (SCORE 1-10)	INDEX A/10	PROGRAMME IMPACT (Weeks)	FACTORED PROGRAMME (Weeks)	CRITICALITY (from list)	COST IMPACT	FACTORED COST	RESPONSE (from list)	ACTION OWNER	PROPOSED MITIGATION ACTION
									182	131.8		£2,636,000	£ 1,757,800			
Open	14/04/2026	1	BREEAM requirements and design-stage implementation	> Missed early-stage opportunities > Project cost increases > Programme delay	MC	Statutory	8	0.8	8	6.4	High	£300,000	£ 240,000	Avoid	GS/CC	> Include allowance in OOM > Include outcome of BREEAM pre-assessment report 12/05/26 in next design stages
Open	14/04/2026	2	Existing electrical infrastructure may not support extension (LV panel is from 1998), requiring major upgrades.	> Construction cost increase > Feasibility viability impacted > Cost of operational downtime	CG	Extg Site	7	0.7	2	1.4	High	£150,000	£ 105,000	Accept	CG/CC	> Vendor unable to confirm feasibility of modifying existing panel without intrusive survey. Intrusive survey quotation received, requiring full site shutdown. > Modification of existing panel not preferred. Current feasibility-stage options include: 1) Install new panel between existing panel and transformer 2) Split electrical supply arrangement 3) Replace existing main panel with new panel > Feasibility study to proceed on basis of provisional allowance, with preferred solution to be confirmed during next design stage. >Order-of-magnitude (OOM) allowance / contingency to be included within cost plan.
Open	14/04/2026	3	Limited available space for utilities.	> Increased design & construction scope and cost > Layout iterations and potential re-work > Sub-optimal layout if existing space used	GJ	Extg Site	8	0.8	12	9.6	High	£125,000	£ 100,000	Accept	ST	> Constraints to be clearly highlighted in feasibility report > Develop detailed construction plan > Follow RIBA design stages for increasing definition and accuracy of layout > Potential for process scheduling to reduce extra utility usage in next stages
Open	14/04/2026	4	Landlord / site-wide constraints limit design and construction options.	> Delays for approvals > Design limitations > Subsequent additional costs	GS	Extg Site	7	0.7	4	2.8	Medium	£40,000	£ 28,000	Mitigate	ST/CC	> Ongoing engagement with landlord in next stages
Open	14/04/2026	5	Communication barriers between UK and China teams.	> Incorrect assumptions & information > Slower information provision & decisions	GS	Project Org	8	0.8	1	0.8	Low	£10,000	£ 8,000	Mitigate	CC	> Centralise communications, RFI, decision making, meetings etc. via CC > Asymchem to devise strategy to reduce impact of communication barriers
Open	14/04/2026	6	Late RFI responses.	> Delivery programme slip > Assumptions made to progress design resulting in potential rework	GS	Project Org	7	0.7	2	1.4	Low	£10,000	£ 7,000	Mitigate	GS/CC	> Weekly RFI tracking + escalation + highlight of critical RFIs > Maintain design momentum using assumptions when required > Stage gate approach to design
Open	20/04/2026	7	Delay in supply chain	> Feasibility programme completion delay	MC	Supply Chain	8	0.8	3	2.4	Medium	£1,000	£ 800	Accept	AW	> Pull estimates from previous projects to produce cost plan > Engage suppliers as early as possible
Open	20/04/2026	8	Building height increase	> Existing stacks need to be taller > Planning permission delay or denied	JB	Technical	10	1.0	16	16	High	£100,000	£ 100,000	Accept	CC	> Pre-planning to provide feedback on building height > Engage with stack consultant in next stage > Engage with planning consultant in next stage > Stack height & associated surveys increase to be incorporated into cost plan if required
Open	20/04/2026	9	Increase of equipment lead times	> Delay to construction	MC	Supply Chain	4	0.4	10	4	Medium	£20,000	£ 8,000	Mitigate	AW	> Early order placement > Identification of long lead items and incorporate into procurement programme > Supplier vetting.
Open	20/04/2026	10	Hydrogenation building cannot be removed	> Redesign of the extension > Increase construction cost as rear wall will need to be blast proof re-inforced concrete	GJ	Technical	5	0.5	2	1	Medium	£200,000	£ 100,000	Accept	CC	> Feasibility design to be based around the removal of hydrogenation building > Two options included for keeping hydrogenation building. > Incorporate selected option into next stage cost plan
Open	20/04/2026	11	Unforseen planning permission constraints	> Prolonged planning discussions > Constraints to building design resulting in redesign > Additional cost of planning conditions	GJ	Statutory	5	0.5	16	8	High	£200,000	£ 100,000	Accept	CC	> Pre-planning to provide early feedback on what to provide for planning application
Open	20/04/2026	12	Inadequate welfare and office facilities	> Modifications required to existing buildings > Increase to construction costs	GJ	Technical	5	0.5	4	2	Medium	£150,000	£ 75,000	Accept	CC	> Staturory requirements reviewed and incorporated into the Feasibility design > Identification of welfare and office requirements at next stage

POST-ACTION ASSESSMENT (expected outcome with listed mitigation action completed)

REF NO.	LIKELIHOOD A (SCORE 1-10)	INDEX A/10	PROGRAMME IMPACT (Weeks)	FACTORED PROGRAMME (Weeks)	CRITICALITY (from list)	COST IMPACT	FACTORED COST	NOTES OR FURTHER ACTION
			107	60.6		£1,126,000	£ 626,600	
1	2	0.2	8	1.6	Low	£20,000	£ 4,000	> Discovery Park (Paul Bax) clarified 'Very-Good' Required for site, however limitations may apply as safety takes precedence. > BREEAM pre-assessment workshop held 11/05/26 and report issued 12/05/26. > Risk reduced but remains live through next design stages pending implementation of identified BREEAM requirements and confirmation of achievable rating.
2	7	0.7	2	1.4	Low	£0	£ -	> Panel vendor (Andy Thompson, AVK) contacted in Feasibility > Cost impact reduces due to incorporation in OOM
3	4	0.4	0	0	Medium	£125,000	£ 50,000	> Utilities requirements confiremd with client (current + future demand) via Feasibility RFIs > Available space validated during site review & on drawings (incl. photos + dimensions) and informed Feasibility layout
4	3	0.3	4	1.2	Low	£40,000	£ 12,000	> Discovery Park (Paul Bax) engaged throughout Feasibility study and site constraints discussed/identified
5	2	0.2	1	0.2	Low	£10,000	£ 2,000	
6	10	1.0	1	1	Low	£10,000	£ 10,000	> Design assumptions made at Feasibility study and recorded in report > To be addressed in next design phases.
7	6	0.6	3	1.8	Low	£1,000	£ 600	> Partially realised by delayed cost receipt for Hastelloy Vessels. To be proceed with informed estimation for Feasibility. > Alternative supplier/ engagement strategy with De Dietrich to be explored in next stage.
8	10	1.0	2	2	Medium	£40,000	£ 40,000	> Building height increase incorporated into final layouts and Feasibility report.
9	2	0.2	10	2	Medium	£20,000	£ 4,000	
10	2	0.2	2	0.4	Low	£0	£ -	> Cost impact reduces due to hyrdrogenation building removal incorporation in OOM > Feasibility constructability review 14/05/26 validates approach
11	5	0.5	2	1	Medium	£200,000	£ 100,000	> Risk to be updated after pre-app feedback. > Programme mitigation not cost.
12	5	0.5	4	2	Medium	£150,000	£ 75,000	> Risk to be updated after decision made at next stage.

Client	**Asymchem**	**Revision No.**	**A1**
Project Name	**PDF Extension Feasibility**	**Revision Date**	**21-May-26**
Scitech Project No	**300291**	**By**	**GS**
Document Ref	**300291-PM-RA-0001**	**Reason**	**Feasibility Design**

RISK ENTRY / PRE-ACTION ASSESSMENT (expected outcome without mitigation)

Status	Date	REF NO.	RISK	RISK IMPACT	RAISED BY	CATEGORY (from list)	LIKELIHOOD A (SCORE 1-10)	INDEX A /10	PROGRAMME IMPACT (Weeks)	FACTORED PROGRAMME (Weeks)	CRITICALITY (from list)	COST IMPACT	FACTORED COST	RESPONSE (from list)	ACTION OWNER	PROPOSED MITIGATION ACTION
Open	20/04/2026	13	Desired construction completion date of Q3 2028 is tight timeline	> Overall programme increased	GJ	Scope	9	0.9	26	23.4	High	£0	£ -	Mitigate	CC	> Compress overall programme by DNB contract employed at Stage 4 > Programme reduction strategies during construction e.g. 24/7 working
Open	20/04/2026	14	Asbestos	> Delay to construction > Cost of removal	GS	Extg Site	8	0.8	6	4.8	Medium	£100,000	£ 80,000	Avoid	ST	> R&D Survey before construction starts (Stage 3)
Closed	21/04/2026	15	Future obselescence of main switchboards	> Replacement of board > Full shutdown of site - just asymchem building or entirety of discovery park	CG	Extg Site										> Covered by Risk Register Item #2
Open	20/04/2026	16	Cost increase of equipment during design phase	> Increase capital cost > Project viability	MC	Supply Chain	7	0.7	8	5.6	High	£80,000	£ 56,000	Mitigate	AW	> Cost plans developed in-line with RIBA stage progression with contingencies and assumptions appropriate to design maturity and improving with accuracy as project progresses > Identify long lead/high-risk equipment early
Open	20/04/2026	17	Force Majeure	> Delay to project progression at any stage > Associated increase in project costs	GS	External	7	0.7	12	8.4	High	£300,000	£ 210,000	Mitigate	GS/CC	> Maintain and regular review of appropriate design/procurement/construction programme > Project lifecycle split into stage gates (i.e. RIBA stages) controlled project development with clear go/no-go > Monitor supply chaing/procurment risks throughout project lifecycle
Open	24/04/2026	18	Unclear purchasing strategy (responsibly party, timing etc.)	> Delay to procurement > Missed orders > Delay to construction programme	CC	Project Org	8	0.8	12	9.6	High	£80,000	£ 64,000	Avoid	ST/CC	> Early identification of long lead items > Procurement strategy to be detailed in design and construction programme > Deliverable at concept stage that breaksdown who owns what > Asymchem to have internal discussion/meeting to devise procurement strategy
Open	24/04/2026	19	Equipment scope change post-concept	> Redesign/rework cost > Delay to overall programme	CC	Scope	7	0.7	4	2.8	Medium	£40,000	£ 28,000	Mitigate	ST/CC	> Stage gate approach > Asymchem to ensure agreement and alignment with CN stakeholders > ST to follow formal change control process (EWN/CO) > Client progress meetings and workshops to inform design (20%, 50% and 90% reviews in concept/scheme/detailed stages)
Open	29/04/2026	20	Surface water drainage likely clashes with building	> Reroute required impacting layout > Increased structural and civil scope > Increased programme timeline and project cost	GJ	Extg Site	7	0.7	4	2.8	Medium	£80,000	£ 56,000	Accept	GJ	> Review drainage strategy and co-ordinate with structural/civil consultant in concept stage. > Incorporate into cost plan
Open	29/04/2026	21	HV cable location uncertain	> Reroute or building over required > Increased structural and civil scope > Increased programme timeline and project cost	GJ	Extg Site	7	0.7	4	2.8	Medium	£80,000	£ 56,000	Accept	CG	> Confirm HV cable location early and assess diversion/build-over requirements during concept development.
Open	06/05/2026	22	Potential requirement to cool third floor due to increased reactor sizes and heat gains.	> Increased utility demand, HVAC sizing and infrastructure requirements > Increased project and construction costs	MC	Scope	6	0.6	4	2.4	High	£150,000	£ 90,000	Accept	JB/AW	> Include cooling for entire building in cost plan and value engineer at later stage > Client to confirm cooling requirements
Open	21/04/2026	23	Existing reactor cooling philosophy may not be suitable for proposed larger-scale reactors.	> Later design stages may require redesign of reactor cooling systems, increased utility capacity, revised controls strategy > Additional capital cost and programme delay.	GS	Extg Site	5	0.5	4	2	Medium	£80,000	£ 40,000	Accept	JB	> Feasibility study proceeding on the assumption that the existing crash cooling philosophy is suitable for the proposed larger scale reactors. > Reactor cooling philosophy to be assessed and validated next stage.
Open	21/04/2026	24	Existing emergency power philosophy and generator capacity may not be sufficient for proposed process expansion.	> Additional generator capacity, new essential boards or revised backup philosophy may be required. > Increase in capex and programme.	GS	Extg Site	5	0.5	2	1	Medium	£80,000	£ 40,000	Mitigate	CG	> Feasibility based on installation of a new dedicated generator for the extension. > Cost to be included in OOM
Open	21/04/2026	25	Existing waste handling arrangements and transfer line capacity may not support proposed process expansion.	> If unavailable, increased tanker movements, additional storage tanks, operational constraints. > Higher opex/capex may result.	GS	Extg Site	5	0.5	4	2	Medium	£80,000	£ 40,000	Accept	MC/GJ	> Feasibility assumes existing waste handling arrangements can be maintained or expanded, including potential reinstatement of transfer line to Waste Care. > Waste handling arrangements to be assessed and validated next stage
Open	21/04/2026	26	Existing nitrogen, compressed air, thermal fluid and cooling systems may not have sufficient spare capacity for proposed extension.	> Discovery of insufficient capacity may require new plant, enlarged distribution systems and redesign.	GS	Extg Site	7	0.7	4	2.8	Medium	£80,000	£ 56,000	Mitigate	MC	> Preliminary calcs have estimated new nitrogen, hot thermal fluid heating and distribution new cooling utilities are required. > To be incorporated into next stage design.
Open	21/04/2026	27	Unknown foundation and structural design.	> Later intrusive surveys may identify abnormal ground conditions, foundation complexity or structural constraints increasing cost.	GS	Extg Site	7	0.7	8	5.6	High	£100,000	£ 70,000	Accept	GJ	> Feasibility assumes that similar design to existing site can be used. > Structural/Civil Engineer and Ground Investigation Consultants to be instructed in next stage > To be allowed for in OOM

POST-ACTION ASSESSMENT (expected outcome with listed mitigation action completed)

REF NO.	LIKELIHOOD A (SCORE 1-10)	INDEX A /10	PROGRAMME IMPACT (Weeks)	FACTORED PROGRAMME (Weeks)	CRITICALITY (from list)	COST IMPACT	FACTORED COST	NOTES OR FURTHER ACTION
13	9	0.9	16	14.4	High	£0	£ -	> To be re-assessed in later stages
14	5	0.5	0	0	Medium	£50,000	£ 25,000	
15								
16	7	0.7	2	1.4	Low	£0	£ -	> Cost impact reduces due to incorporation in OOM
17	7	0.7	12	8.4	High	£300,000	£ 210,000	
18	4	0.4	4	1.6	Low	£20,000	£ 8,000	
19	4	0.4	4	1.6	Low	£40,000	£ 16,000	
20	7	0.7	4	2.8	Medium	£0	£ -	> Cost impact reduces due to incorporation in OOM
21	7	0.7	4	2.8	Medium	£0	£ -	> Cost impact reduces due to incorporation in OOM
22	6	0.6	4	2.4	Medium	£0	£ -	> Cost impact reduces due to incorporation in OOM
23	5	0.5	4	2	Medium	£0	£ -	> Cost impact reduces due to incorporation in OOM
24	5	0.5	2	1	Low	£0	£ -	> Cost impact reduces due to incorporation in OOM
25	5	0.5	4	2	Medium	£0	£ -	> Cost impact reduces due to incorporation in OOM
26	7	0.7	0	0	Low	£0	£ -	> Cost impact reduces due to incorporation in OOM
27	7	0.7	8	5.6	High	£100,000	£ 70,000	

Appendix 10 - Cost Plan

FEASIBILITY COST PLAN

Client	Asymchem
Project Name:	Extension to PDF building, Sandwich, Kent
Project Number:	300291
Date:	22-May-26
Cost Plan Stage:	Feasibility / RIBA Stage 1
Base Date:	22-May-26

AREA SCHEDULE

Floor	GIA (m²)	Notes
Ground Floor	597	As Area sheets
First Floor	642	As Area sheets
Second Floor	637	As Area sheets
Third Floor + Mezz deck plant	1224	As Area sheets
Total GIFA	**3,099**	

BUILDING WORKS COST BREAKDOWN

Category	Description	Rate £/m² / Rate	GIFA (m²) / Nr	Total £
0	**Facilitating Works**			**£254,960**
0.1	Demolition, site clearance, enabling works	£50	3,099	£154,960
0.2	Removal of LN Tank	£50,000	1	£50,000
0.3	Removal of existing Hydrogenation Plant	£50,000	1	£50,000
1	**Substructure**			**£1,499,680**
1.1	Piles, Foundations, ground floor slab	£400	3,099	£1,239,680
1.2	Slabs for equipment	£1,000	150	£150,000
1.3	Pads for Chiller supports	£2,000	20	£40,000
1.4	External Steps and Ramps	£70,000	1	£70,000
2	**Superstructure**			**£3,942,636**
	Frame, floors, roof, external walls, windows	£1,000	3,099	£3,099,200
2.1	Extra Over for "Blast roof and walling"	£500	500	£250,000
2.2	Staircase / stairways	£80	3,099	£247,936
2.3	Metal structures for chillers etc	£200,000	1	£200,000
2.4	Single Doors and frames	£1,500	12	£18,000
2.5	Double Doors and frames	£2,500	51	£127,500
3	**Internal Finishes**			**£1,270,672**
3.1	Wall Finishes Minimal rooms - opted for 50% of floor space	£200	1,550	£309,920
3.2	Ceiling Finishes	£145	3,099	£449,384
3.3	Floor Finishes - Resin	£165	3,099	£511,368
4	**Fittings & Equipment**			**£154,960**
	General fittings, sanitary fittings - all in the existing Building	£50	3,099	£154,960
5	**BUILDING SERVICES**			**£9,532,716**
5.1	BWIC	3%	6,672,308	£166,808
5.1.1	Fire Stopping	£100,000	1	£100,000
5.2	Mechanical services (HVAC)			£4,165,824
5.2.1	Chillers	£100,000	3	£300,000
5.2.2	Offloading Chillers and AHUS	£1,500	8	£12,000
5.2.3	AHU1	£80,000	1	£80,000
5.2.4	AHU2	£80,000	1	£80,000
5.2.5	AHU3	£80,000	1	£80,000
5.2.6	AHU4	£80,000	1	£80,000
5.2.7	AHU5	£80,000	1	£80,000
5.2.8	AHU6	£80,000	1	£80,000
5.2.9	Pump Sets	£25,000	6	£150,000
5.2.10	Contract Lifts / Offloading chillers	£1,500	3	£4,500
5.2.11	Contract Lifts / Offloading AHUs	£1,500	5	£7,500
5.2.12	Pipework	£350	2,560	£896,000
5.2.13	Drainage	£15	2,560	£38,400
5.2.14	Fume Extract Ductwork, incl. fans	£350	2,560	£896,000
5.2.15	Supply and extract Ductwork	£400	2,560	£1,024,000
5.2.16	Extra Over for flue supports for existing	£8,000	3	£24,000
5.2.17	Works to existing Flues	£10,000	3	£30,000
5.2.18	Safety Showers	£5,000	6	£30,000
5.2.19	Sinks with DCW, DHW, Drainage	£80	1,168	£93,424
5.2.20	Sprinkler System / Extension _ (Incumbent to design)	£100,000	1	£100,000
5.2.21	Steam to LTHW Skid, HX, Expansion, Pumps, Insulation	£80,000	1	£80,000
5.3	Electrical installations			£2,506,484
5.3.1	New Main Panel	£150,000	1	£150,000
5.3.2	Standby generator 300kVA	£175,000	1	£175,000
5.3.3	Small Power	£180	3,099	£557,856
5.3.4	Power to Equipment	£180	3,099	£557,856
5.3.5	Extension to existing Lightning Protection	£20,000	1	£20,000
5.3.6	Distribution Boards	£15,000	6	£90,000
5.3.7	Lifts	£150,000	1	£150,000
5.3.8	Fire and security systems	£100	3,099	£309,920
5.3.9	Communications and IT	£60	3,099	£185,952
5.3.10	BMS Integration of new system into site wide system	£100	3,099	£309,900
5.4	EMS			£627,200
5.4.1	EMS Hardware and Software	£180	2,560	£460,800
5.4.2	EMS Controlled Cabling	£65	2,560	£166,400
5.5	Process Controls			£1,966,400
5.5.1	PCS Hardware and Software	£1,800,000	1	£1,800,000
5.5.2	PCS Controlled Cabling	£65	2,560	£166,400

FEASIBILITY COST PLAN

Client	Asymchem
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Project Number:	300291
Date:	22-May-26
Cost Plan Stage:	Feasibility / RIBA Stage 1
Base Date:	22-May-26

AREA SCHEDULE

Floor	GIA (m²)	Notes
Ground Floor	597	As Area sheets
First Floor	642	As Area sheets
Second Floor	637	As Area sheets
Third Floor + Mezz deck plant	1224	As Area sheets
Total GIFA	**3,099**	

BUILDING WORKS COST BREAKDOWN

Category	Description	Rate £/m² / Rate	GIFA (m²) / Nr	Total £
5.6	**PROCESS EQUIPMENT**			**£23,249,083**
5.6.1	Glass lined reactors 2500l	£150,000	1	£150,000
5.6.2	Glass lined reactor 8000l	£250,000	3	£750,000
5.6.3	Glass lined Reactors 5000l	£180,000	3	£540,000
5.6.4	Allowance for Overhead equipment in Hastelloy	£180,000	7	£1,260,000
5.6.5	Forbes SC05/06 (Tower, Recirc pumps, Pipework, Heat Exchanger, Fan, Ducting, Instrumentation, Dosing, skid)	£99,000	1	£99,000
5.6.6	Laminar Flow Booths	£45,000	2	£90,000
5.6.7	Fume Cupboards - Bench type c/w vented under cupboards	£18,000	1	£18,000
5.6.8	General need pumps, jackets, dispensing diaphragm pumps	£100,000	1	£100,000
5.6.9	Controls & Instrumentation	£900	2,500	£2,250,000
5.6.10	Stainless Steel working platform - Hydrogenation suites	£70,000	4	£280,000
5.6.11	Filter Dryers	£1,500,000	3	£4,500,000
5.6.12	Purified Water System	£1,000,000	1	£1,000,000
5.6.13	Hastelloy Pipework	£1,500	300	£450,000
5.6.14	PTFE lined steel pipework	£900	1,800	£1,620,000
5.6.15	Cryo pipework super insulated	£3,000	160	£480,000
5.6.16	Relief and vent ductwork in GRP	£500	700	£350,000
5.6.17	Liquid Nitrogen system	£50,000	1	£50,000
5.6.18	Carbon Steel	£300	1,350	£405,000
5.6.19	Stainless Steel Pipework	£350	3,150	£1,102,500
5.6.20	10m3 Tank - Glass lined - ref: ET03	£200,000	1	£200,000
5.6.21	35m3 Tank - Glass Lined - Ref TT05 and 06	£350,000	2	£700,000
5.6.22	Hastelloy Pumps to serve above tanks	£40,000	3	£120,000
5.6.23	Powder transfer systems -	£50,000	3	£150,000
5.6.24	Split Butterfly Valves (11 reactors, 3 filter dryers, Dispensing etc)	£60,000	25	£1,500,000
5.6.25	Hastelloy Vessels c/w overhead equipment 5000l	£1,000,000	1	£1,000,000
5.6.26	Hastelloy Vessels c/w overhead equipment 8000l	£1,200,000	1	£1,200,000
5.6.27	Hydrogenator with overhead equipment	£900,000	1	£900,000
5.6.28	Glass lined receivers	£60,000	9	£540,000
5.6.29	HTF Heater 800Kw Stainless steel heat exchanger	£75,000	1	£75,000
5.6.30	Circulation pumps to above	£10,000	2	£20,000
5.6.31	PTFE Transfer Pumps - Diaphragm pumps	£6,000	6	£36,000
5.6.32	Vacuum Pumps to suite	£100,000	6	£600,000
5.6.33	2 relief tanks , Carbon Steel composite Stainless steel, 2500l and 8000l	£50,000	2	£100,000
5.6.34	Isolators (glovebox and crushing)	£40,000	3	£120,000
5.6.35	Mill (Corn)	£150,000	1	£150,000
5.6.36	Equipment Offloading / positioning / fixing in location	1.5%		£343,583
6	**Prefabricated Buildings**			**£0**
7	**Work to Existing Buildings**			**£204,000**
	Increase height of existing roof flues (4 Nr)	£20,000	4	£80,000
	Structural Steel to support to the revised	£10,000	4	£40,000
	Allow for additional louvres at high level to blend n buildings and protect flues	£600	140	£84,000
8	**External Works**			**£1,859,520**
	Site works, landscaping, drainage, services	600	3,099	£1,859,520
	WORKS COST SUBTOTAL			**£41,968,226**

9	Preliminaries	8%		£3,357,458
	Site management, welfare, temporary works			
10	Contractor OH&P	5%		£2,266,284
	BASE CONSTRUCTION COST			**£47,591,969**

OTHER PROJECT COSTS

Design Team Fees	8%	of construction	£3,807,357
Surveys & Investigations	£80,000		£80,000
BREEAM	£150,000		£150,000
Planning & Statutory Fees	£25,000		£25,000
Client Direct Costs	excl.		excl.
Other Project Costs Subtotal			**£4,062,357**

RISKS

Risk Register (factored costs)			£626,600
Risk Register Subtotal			**£626,600**

GENERAL RISK & CONTINGENCY ALLOCATIONS AT FEASIBILITY STAGE

Design Development Allowance	15%	Feasibility stage	£7,748,149
Construction Risk and equipment Risk	25%		£12,913,581
Client Contingency	10%		£5,165,433
Risk & Contingency Subtotal			**£25,827,163**

FEASIBILITY COST PLAN

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Base Date:	22-May-26

AREA SCHEDULE

Floor	GIA (m²)	Notes
Ground Floor	597	As Area sheets
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Third Floor + Mezz deck plant	1224	As Area sheets
Total GIFA	**3,099**	

BUILDING WORKS COST BREAKDOWN

Category	Description	Rate £/m² / Rate	GIFA (m²) / Nr	Total £
COST ESCALATION				
	Tender Date			
	Inflation Rate (annual)			excl.
	Escalation Allowance			excl.
	TOTAL PROJECT COST (Order of Magnitude)			**£78,108,089**

FEASIBILITY COST PLAN

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BUILDING WORKS COST BREAKDOWN

Category	Description	Rate £/m² / Rate	GIFA (m²) / Nr	Total £

	ASSUMPTIONS & CLARIFICATIONS
	This is an order of Magnitude (OOM) cost estimate to act as a guide only to a possible capital project value indication and is based on the design information available. It should not be used for any final Capex application. A more comprehensive estimate will be developed during the Concept Design
	There are nominal risk provisions shown at this stage. The client should adjust this as they see fit.
	No allowance has been made for escalations, fluctuations, exchange rate risk, inflation or other factors.
	No allowance has been made for any works resulting from any existing site obstructions or contamination save for that included in the risk register.
	The estimate excludes any taxes, levies, duties, etc. All costing is net. If such charges are to be applied, these will need to be calculated and added.
	The estimate allows for various surveys to be undertaken and assumes no issues are found. The estimate does not allow for any costs associated with the findings of the surveys.
	Client purchased equipment is excluded from the costs. An allowance for lifting and shift Items purchased By Scitech has been included but no allowance has been allowed for any manoeuvring of client purchases.
	Estimate based on latest revision of drawing information issued prior to the date of the estimate.
	Any client direct costs are excluded including project management, operational costs, consumables

Floor	Measured / Transcribed Area (m²)	Count of Spaces
Ground	596.5	23
First	641.6	9
Second	636.7	11
Third	1224.4	13
	3099.2	

Category Summary (all floors)	
Category	**Area (m²)**
Primary / Operational	1301.3
Circulation	402.1
Ancillary / Support	75.9
Vertical Circulation	76.5
Lobby	67.4
Storage	8.2
Plant / Technical	1167.8
	3099.2

Total Checks	
Ground total	596.5
First total	641.6
Second total	636.7
Third total	1224.4
Grand total	3099.2

Notes / Limitations
Ground floor schedule follows the same method as the first schedule previously issued in chat.
First and second floor values are best-effort transcriptions from the uploaded images and should be che
Third floor image resolution limited a full reliable read; affected rows are highlighted for manual check
Workbook includes blank cost mapping columns for onward NRM / package coding.

Status
Complete transcribed schedule
Best-effort transcription from image
Best-effort transcription from image
Partial – manual check required

ecked against the source PDFs / CAD for issue.
in the Third worksheet.

Floor	Ref	Area Name	Area (m²)	Category
Ground	GF-01	Purified Water System	78.5	Primary / Operational
Ground	GF-02	Switch Room	49.4	Plant / Technical
Ground	GF-03	Milling & Packaging	34.0	Primary / Operational
Ground	GF-04	Hastelloy ADF	26.8	Primary / Operational
Ground	GF-05	TCU	29.9	Plant / Technical
Ground	GF-06	Lift	8.9	Vertical Circulation
Ground	GF-07	Lobby 1	4.7	Lobby
Ground	GF-08	Lobby 2	7.4	Lobby
Ground	GF-09	Lobby 3	4.9	Lobby
Ground	GF-10	Corridor	70.4	Circulation
Ground	GF-10a	corridor	20.0	circulation
Ground	GF-11	Dispensing 1	15.3	Primary / Operational
Ground	GF-12	Lobby 4	6.6	Lobby
Ground	GF-13	Dispensing 2	14.9	Primary / Operational
Ground	GF-14	Lobby 5	7.6	Lobby
Ground	GF-15	Material Prep	47.0	Primary / Operational
Ground	GF-16	Lobby 6	5.6	Lobby
Ground	GF-17	Store	8.2	Storage
Ground	GF-18	Lobby 7	6.8	Lobby
Ground	GF-19	Staircase	16.9	Vertical Circulation
Ground	GF-19a	lobby	11.9	Lobby
Ground	GF-20	Hastelloy AFD 1	37.2	Primary / Operational
Ground	GF-21	Hastelloy AFD 2	38.6	Primary / Operational
Ground	GF-22	Drum Transfer	45.0	Primary / Operational

Total Check				**596.5**

Floor	Ref	Area Name	Area (m^2)	Category
First	1F-01	Auxiliary Room 1	78.4	Primary / Operational
First	1F-02	Reactor Room 1	141.8	Primary / Operational
First	1F-03	Auxiliary Room 2	40.6	Primary / Operational
First	1F-04	Corridor	70.4	Circulation
First	1F-04a	Corridor	20.0	Circulation
First	1F-05	Staircase	16.9	Vertical Circulation
First	1F-05a	lobby	11.9	Circulation
First	1F-06	Auxiliary Room 3	78.7	Primary / Operational
First	1F-07	Reactor Room 2	142.3	Primary / Operational
First	1F-08	Auxiliary Room 4	40.6	Primary / Operational

Total Check	**641.6**

Floor	Ref	Area Name	Area (m^2)	Category
Second	2F-01	I/O Room	34.7	Ancillary / Support
Second	2F-10	Auxillary	41.2	Ancillary / Support
Second	2F-02	Reactor Room 1	141.8	Primary / Operational
Second	2F-03	Auxiliary Room 1	40.6	Primary / Operational
Second	2F-04	Corridor	70.4	Circulation
Second	2F-04a	Corridor	20.0	Circulation
Second	2F-05	Staircase	16.9	Vertical Circulation
Second	2F-05a	lobby	11.9	lobby
Second	2F-06	IPC Lab	38.1	Primary / Operational
Second	2F-07	Auxiliary Room 2	38.2	Primary / Operational
Second	2F-08	Reactor Room 2	142.3	Primary / Operational
Second	2F-09	Auxiliary Room 3	40.6	Primary / Operational

Total Check	**636.7**

Floor	Ref	Area Name	Area (m²)	Category
Third	3F-01	HVAC Plant room	246.5	Plant / Technical
Third	3F-02	Central Circulation / Landing	70.4	Circulation
Third	3F-02a	Central Circulation / Landing	20.0	Circulation
Third	3F-03	Staircase	16.9	Vertical Circulation
Third	3F-03a	lobby	11.9	Plant / Technical
Third	3F-04	Flow Equipment	123.4	Plant / Technical
Third	3F-05	TCU's	50.7	Plant / Technical
Third	3F-06	Hydrogen Area	47.8	Plant / Technical
Third	3F-07	Mezz Deck	21.1	Plant / Technical
Third	3F-08	Upper Deck Plant Room	498.2	Plant / Technical
Third	3F-08a	Upper Deck Plant Room	20.0	Plant / Technical
Third	3F-09	Staircase	28.6	Circulation
Third	3F-10	Hydrogen Area	47.8	Plant / Technical
Third	3F-11	Mezz Deck	21.1	Plant / Technical

Total Check	**1224.4**